



130 E. Main Street
Caledonia, MI 49316
Phone: 616-891-5873
Fax: 616-891-5720
www.EIILLC.com

April 4, 2025

Director – Enforcement and Compliance Assurance Division
U.S. EPA Region 5
77 West Jackson Boulevard
Chicago, IL 60604-3507

RE: Wexford County Landfill
Permit Number MI-ROP-N3862-2022
Submittal of Landfill Gas Collection and Control System (GCCS) Design Plan
40 CFR 63 Subpart AAAA & 40 CFR 62 Subpart OOO

To whom it may concern:

On behalf of the Wexford County Landfill, located in Manton, Michigan, Environmental Information Logistics, LLC (EIL) is submitting the attached Landfill GCCS Design Plan. Pursuant to 40 CFR 63.1959(b)(2)(i) and 40 CFR 62.16724(d), a design plan must be submitted within 12 months after the facility NMOC emission rate is calculated to be equal to or greater than 34 Mg per year. The facility intends to operate the system in compliance with applicable regulations within 30 months, as required by 40 CFR 62.16714(b) and (c) and 40 CFR 63.1959(b)(2)(ii).

Pursuant to §62.16714(e)(2), once the NMOC emissions rate exceeds 34 Mg/year at a landfill facility, a Collection and Control System Design Plan (Design Plan) must be prepared and submitted within one year, in accordance with §62.16724(d). A copy of the cover page that contains the engineer's seal must be submitted to the Administrator per §62.16724(d)(4).

Additionally, the Federal Plan requires the owner/operator to notify EPA of the completion of Increments of Progress. The preparation of the design plan and submittal with the engineer's seal to the Administrator represents the first Increment of Progress under §62.16712(a)(1). Please see Attachment 1 for this required information.

This submittal fulfills the regulatory obligations for Final Control Plan and Increments of Progress notification under 40 CFR 62 Subpart OOO. A full copy of the Final Control Plan has been certified and submitted to Michigan EGLE electronically via the MiEnviro Electronic Reporting Portal, since the plan has also been prepared to meet the requirements of the revised Landfill NESHAP (40 CFR 63 Subpart AAAA), which EGLE has delegated authority to administer.

Please note that Increments of Progress 2 (Award Contract), 3 (Initiate On-Site Construction) and 4 (Complete On-Site Construction) do not apply to this facility, since Wexford County Landfill has voluntarily operated an existing gas collection and control system since 2014. A notification for Increment of Progress 5 (Achieve Final Compliance) will be submitted to the Administrator 30 months after March 13, 2025, when the facility was informed that NMOC emissions exceeded 34 Mg/year.

A copy of the GCCS Design Plan is included for your review and approval. If there are any questions regarding the submittal of this GCCS Design Plan, please contact Anthony Testa at (248) 697-4848.

Sincerely,
Environmental Information Logistics, LLC.

A handwritten signature in black ink, appearing to read 'Benjamin Kotrba', with a stylized flourish at the end.

Benjamin Kotrba
Environmental Scientist

Cc: USEPA Region 5 (Submitted Electronically)
Antonio Testa (GFL Environmental)
Tami Craig (GFL Environmental)
Chris Gee (GFL Environmental)
Dana Oleniacz (EIL)

Landfill Gas Collection and Control System Design Plan

Wexford County Landfill

APRIL 2, 2025

TETRA TECH PROJECT #209-4251206

PREPARED FOR

Wexford County Landfill

990 North Mackinaw Trail,
Manton, Wexford County, Michigan 49663

PREPARED BY

Tetra Tech
1250 East Diehl Road, Suite 103
Naperville, Illinois 60563

DESIGN PLAN CERTIFICATION

The material and data in this report were prepared under the supervision and direction of the undersigned.



04/02/2025

Khaled Mahmood, P.E.
Certifying Engineer
MI P.E. License No. 6201048110

Date

04/02/2025

Jakub Sowa
Project Manager – Biogas Engineering

Date

April 2, 2025

TABLE OF CONTENTS

1.0	INTRODUCTION	1-1
2.0	EXISTING SITE CONDITIONS.....	2-4
2.1	Landfill Description	2-4
2.2	Existing Landfill Gas Collection and Control System.....	2-4
3.0	FUTURE SITE DEVELOPMENT	3-5
3.1	Landfill Gas Collection System Development Plan	3-5
3.2	Landfill Gas Collection System Expansion Capabilities	3-7
4.0	COMPLIANCE REVIEW AND EVALUATION	4-1
4.1	Compliance with §62.16728(a)(1) AND §63.1962(a)(1)	4-1
4.1.1	Control of Surface Emissions	4-1
4.1.2	Depth of Refuse	4-2
4.1.3	Landfill Gas Generation Rates and Flow Characteristics	4-2
4.1.4	Landfill Cover Properties	4-3
4.1.5	Landfill Gas Control System Expandability.....	4-3
4.1.6	Leachate and Condensate Management	4-3
4.1.7	Accessibility	4-4
4.1.8	Compatibility with Refuse Filing Operations	4-4
4.1.9	Integration with Closure End Use	4-4
4.1.10	Air Intrusion Control	4-4
4.1.11	Corrosion Resistance	4-5
4.1.12	Fill Settlement	4-5
4.1.13	Resistance to Decomposition Heat	4-5
4.1.14	Ability to Isolate Individual Components or Sections for Repair	4-6
4.2	Compliance with §62.16728(a)(2) and §63.1962(a)(2)	4-6
4.3	Compliance with §62.16728(a)(3) AND §63.1962(a)(3)	4-7
4.3.1	Asbestos and Non-degradable Materials	4-7
4.3.2	Non-productive Areas	4-8
4.4	Compliance with §62.16728(b)(1), (2), AND (3) AND §63.1962(b)(1), (2), AND (3)	4-8
4.4.1	Landfill Gas Extraction Component Construction	4-8
4.4.2	Landfill Gas Extraction Component Installation	4-10
4.4.3	Landfill Gas Extraction Component Connections to Landfill Gas Transmission Piping	4-12
4.5	Compliance with §62.16728(c)(1) AND (2) AND §63.1962(c)(1) AND (2)	4-13
4.5.1	Existing Landfill Gas Flow Rate Data	4-13
4.5.2	Future Landfill Gas Flow Rate Estimates	4-13
4.6	Alternatives and Compliance WITH §62.16714(e)(2) and §63.1962(b)(2)	4-14
4.6.1	Submit a Design Plan	4-14
4.6.2	NSPS and NESHAP Alternative Standards and Requirements	4-14
4.6.3	Specifications for Active Collection Systems	4-15
4.6.4	Control Systems	4-16

April 2, 2025

5.0	PROPOSED ALTERNATIVES	5-20
5.1	Collection Device Monitoring	5-20
5.1.1	Flow Meters When No Bypass Is Present	5-20
5.1.2	Monthly Monitoring Methods.....	5-20
5.1.3	Positive Pressure Under a Synthetic Cover	5-21
5.1.4	Early Installation of Collection Devices.....	5-21
5.1.5	Monitoring During Collection Device Extension.....	5-22
5.1.6	Stability Flexibility in Well Placement and Collection System Design Components.....	5-22
5.2	Surface Emissions Monitoring	5-23
5.2.1	Exclusion of Dangerous Areas from SEM Requirements.....	5-23
5.2.2	SEM for Closed Portions of the Landfill	5-23
5.2.3	SEM Reporting Requirements	5-24
5.3	Exclusions of Odor Control Wells Not In Waste or Not Used For Federal Plan/NESHAP Compliance..	5-24
5.4	Regulatory Approvals	5-24

6.0	LIMITATIONS	6-25
------------	--------------------------	-------------

APPENDIX SECTIONS

APPENDICES

Appendix A-1	Calculations - Gas Generation Rate Modeling
Appendix A-2	Calculations - Radius of Influence and Well Spacing Estimates
Appendix A-3	Calculations - Condensate Generation Estimates
Appendix B	Headloss Analysis
Appendix C	Drawings
Appendix D	Surface Emissions Monitoring Plan
Appendix E	Uplift Resistance Calculations
Appendix F	NSPS Determination Letters

April 2, 2025

1.0 INTRODUCTION

1.1 PURPOSE OF DOCUMENT

The purpose of this document is to provide the Gas Collection and Control System (GCCS) GCCS Design Plan to meet the requirements of the New Source Performance Standard (NSPS) 40 CFR 62 Subpart OOO – Federal Plan Requirements for Municipal Solid Waste Landfills That Commenced Construction On or Before July 17, 2014 and Have Not Been Modified or Reconstructed Since July 17, 2014, as finalized in May 21, 2021 and effective June 21, 2021, as well as 40 CFR Subpart AAAA – National Emissions Standards for Hazardous Air Pollutants (NESHAP): Municipal Solid Waste Landfills, revised and promulgated on March 26, 2020, as effective September 28, 2021, and to provide the Administrator with the design standards and calculations used to prepare this GCCS Design Plan.

1.2 APPLICABILITY

Federal Plan Subpart OOO

§62.16714(a) Each owner or operator of an MSW landfill having a design capacity greater than or equal to 2.5 million megagrams by mass and 2.5 million cubic meters by volume must collect and control MSW landfill emissions at each MSW landfill that meets the following conditions:

(3) The landfill has an NMOC emission rate greater than or equal to 34 megagrams per year or Tier 4 SEM shows a surface emission concentration of 500 parts per million methane or greater.

Wexford County Landfill's capacity is greater than 2.5 million megagrams by mass and 2.5 million cubic meters by volume. In 2016, Wexford County Landfill was subject to 40 CFR Part 60, Subpart WWW. In accordance with that rule, the facility conducted a Tier 2 test in 2016. It was recently brought to the site's attention that the Tier 2 test used incorrect default values. When the correct default values were substituted, the site exceeded 50 Mg (the threshold standard to 40 CFR Part 60, Subpart WWW). The GCCS plan would have been submitted within 12 months of the Tier 2 test if the error in the calculation, which was recently brought to the attention of the facility, was discovered at that time. As such, Wexford

April 2, 2025

County Landfill is now submitting the GCCS plan that complies with current applicable rule 40 CFR 62 Subpart OOO

§62.16714(e)(2) *Submit a collection and control system design plan prepared by a professional engineer to the Administrator within 1 year:*

§62.16714(b)(1) *Install and start up a collection and control system that captures the gas generated within the landfill as required within 30 months after:*

(i) The first annual report in which the NMOC emission rate equals or exceeds 34 Mg/yr, unless Tier 2 or Tier 3 sampling demonstrates that the NMOC emission rate is less than 34 Mg/yr, as specified in §62.16714(d)(4).

Sections 3 and 4 of this Plan fulfill the requirements for Wexford County Landfill to prepare a GCCS Design Plan in accordance with §62.16714(a)(3). The Design Plan outlines the methodology employed to design a landfill gas collection and control system that will collect, transport, and dispose of the landfill gas generated in the entire permitted landfill at final grades. In addition, Wexford County's proposed alternatives and variances to the standard, including monitoring, record keeping, and reporting are discussed in Appendix E.

NESHAP Subpart AAAA

§ 63.1935 *Am I subject to this subpart? You are subject to this subpart if you meet the criteria in paragraph (a) or (b) of this section.*

(a) You are subject to this subpart if you own or operate an MSW landfill that has accepted waste since November 8, 1987, or has additional capacity for waste deposition and meets any one of the three criteria in paragraphs (a)(1) through (3) of this section:

(1) Your MSW landfill is a major source as defined in § 63.2 of subpart A.

(2) Your MSW landfill is collocated with a major source as defined in § 63.2 of subpart A.

(3) Your MSW landfill is an area source landfill that has a design capacity equal to or greater than 2.5 million megagrams (Mg) and 2.5 million cubic

April 2, 2025

meters (m³) and has estimated uncontrolled emissions equal to or greater than 50 megagrams per year (Mg/yr) NMOC as calculated according to § 63.1959.

The site was not previously subject to NESHAP Subpart AAAA since in 2016 the facility conducted a Tier 2 test which showed NMOC emissions rate to be less than 50 Mg/yr. It was recently brought to the site's attention that the 2016 Tier 2 test used incorrect default values. When the correct default values were substituted, the site exceeded the NMOC emissions threshold of 50 Mg/yr. As such, Wexford County Landfill is now submitting the GCCS plan that complies with current applicable rule 40 CFR Subpart AAAA.

With the revision of NESHAP Subpart AAAA, major compliance provisions of the NSPS have been incorporated directly into the updated rule. For this reason, the GCCS will be operated in accordance with the Federal Plan Subpart OOO and NESHAP Subpart AAAA as stated below.

The GCCS design outlined in this Plan complies with the specifications for active collection systems as stipulated in §62.16728 of the Federal Plan Subpart OOO and §63.1962 of the NESHAP Subpart AAAA. If future expansions of the GCCS are necessary, they will be designed to comply with the Federal Plan Subpart OOO and NESHAP Subpart AAAA requirements, to accommodate existing site conditions or any approved alternatives.

Furthermore, the Federal Plan Subpart OOO and NESHAP Subpart AAAA specifically require the GCCS to be designed in accordance with general conditions that are contained within these regulations. Throughout this document, the means on how the landfill is meeting or plans on meeting these regulations are described.

April 2, 2025

2.0 EXISTING SITE CONDITIONS

2.1 LANDFILL DESCRIPTION

Wexford County Landfill currently occupies approximately 196.4 acres, located in Cedar Creek Township, Wexford County, Michigan. Of the 196.4 acres of site property, approximately 137.8 acres is permitted for solid waste disposal. The permitted disposal area is divided into two separate non-contiguous areas commonly designated as the “East Unit” and the “West Unit.” The East Unit, comprising approximately 69.3 acres, includes the original landfill and newer cells, designated as Cell B through Cell J. The West Unit, encompassing 68.5 acres, represents a planned expansion area that is currently undeveloped, with nine individual cells designated Cells 1-9, with Cells 7-9 reserved for Type III Industrial Waste. The landfill originally began receiving refuse in 1977. As of 2018 the overall permitted disposal capacity of the entire facility is 23,260,000 cubic yards (11,519,525 cubic yards for East Unit and 11,740,475 cubic yards for West Unit)

2.2 EXISTING LANDFILL GAS COLLECTION AND CONTROL SYSTEM

Historically, non-methane organic compound (NMOC) emissions did not exceed the 50 megagrams per year that mandated installation of an active gas collection and control system (GCCS). However, Wexford County Landfill voluntarily began installation of a GCCS in 2014 to reduce odors. The existing GCCS uses vertical landfill gas (LFG) extraction wells, installed within the waste mass, along with interim horizontal LFG collectors, to extract LFG and convey it through the lateral and header piping network, including isolation valves, to the facility’s utility flare. The horizontal collectors are installed and intended to be an interim measure in areas where the waste mass depth is insufficient to support the installation of a vertical collectors. Horizontal collectors also provide interim odor control as a best management practice measure.

All lateral and header pipes in the piping network configuration were installed below grade and constructed of high-density polyethylene (HDPE). All the condensate formed in the GCCS within the solid waste boundary drains into condensate sumps or driplegs located around the perimeter of the disposal area. From the sumps and driplegs, the liquid is pumped into a forcemain to the facility’s 120,000-gallon leachate storage tank and/or a back-up 20,000-gallon above-ground tank and disposed of along with the leachate from the landfill in accordance with the site’s approved Solid Waste Permit.

Additional information and drawings of the existing GCCS are included in **Appendix C**.

April 2, 2025

3.0 FUTURE SITE DEVELOPMENT

3.1 LANDFILL GAS COLLECTION SYSTEM DEVELOPMENT PLAN

Wexford County Landfill will continue waste filling operations in accordance with the solid waste permit. In conjunction with this report, a phased GCCS design will be implemented to comply with the Federal Requirements and NESHAP regulations. This Plan includes the required updates specified in 40 CFR 62 Subpart OOO (40 CFR 62.16728 which became applicable June 21, 2021) and 40 CFR 63 Subpart AAAA (40 CFR 63.1962 which became applicable September 27, 2021).

Vertical extraction wells will continue to be the primary mechanism for extracting LFG from the refuse mass. Radius of influence (ROI) and well spacing calculations are discussed in **Section 4.2** and **Appendix A-2**.

Instead of vertical extraction wells, caisson wells may be utilized in some areas that will not reach final grade within five years of initial waste deposition. Caisson wells allow for raising of the well and maximizing gas collection as filling operations progress and can be converted to vertical wells once final grade is reached.

In addition to vertical extraction wells and caisson wells, interim horizontal collectors may be implemented to extract LFG, to provide interim odor control, or to provide compliance with 40 CFR §62.16720(b). Interim horizontal collection trenches may also be utilized in areas that are not closed and have not reached final grade. Horizontal collection trenches allow extraction of LFG from areas where waste depths are not sufficient to support the installation of a vertical well. If utilized, interim trenches will be spaced at an appropriate frequency as discussed in **Section 4.2**. The interim collectors may be abandoned when replaced with permanent wells, and they are no longer a viable source of landfill gas collection.

Although no sections of the future disposal area have been designated for the construction of horizontal gas collection trenches, Wexford County Landfill reserves the right to install these components should future operational concerns dictate need.

Each extraction location will be fitted with an adjustable flow control well head. The wellheads will include a sampling port where LFG may be evaluated for methane, oxygen, hydrogen sulfide, carbon dioxide and balance gases (nitrogen, NMOCs, and other compounds). Each wellhead may also have a flow

April 2, 2025

measurement device (pitot tube, orifice plate or other) and a tuning valve to adjust vacuum application and flow out of the well.

Permanent lateral and header pipes are generally installed below ground surface and typically constructed of HDPE pipe. Temporary piping may be installed either above or below ground surface. The LFG will be conveyed through this pipe network to a flare facility. In the future, the LFG may be conveyed to beneficial-use or other process equipment, in accordance with the Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements.

Under the existing site conditions, condensate that forms in the GCCS piping and that is collected within the solid waste boundary may be discharged to the Leachate Collection System (LCS) through dripleg drains with direct connections to LCS cleanout/access risers or to condensate sumps located around the perimeter of the disposal area. Under the future permanent site conditions, condensate drained by gravity to the condensate sumps will be discharged via forcemain(s) to leachate collection and storage locations on the site and ultimately processed with the collected landfill leachate. The collected condensate and collected landfill leachate will co-mingle in storage tanks to be disposed of in accordance with the requirements of the facility solid waste permit.

Installation of additional GCCS components is anticipated to be coordinated with fill development and as otherwise required by the Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements regarding installation of GCCS components stipulated in 40 CFR §62.16728 and 40 CFR §63.1962. Due to operational changes, the GCCS design presented in **Appendix C** may be altered to maintain compliance with the provisions of the Federal Requirements and NESHAP regulations and to accommodate actual field conditions at the time of construction. Alterations may include but not limited to depth of well drilling, additional vertical or interim horizontal wells for best management practices, new technological advances in well construction/methodologies, alternative material for well construction, alternate alignment of the header and lateral piping placement, relocation of the perimeter header outside limits of waste area etc.

April 2, 2025

3.2 LANDFILL GAS COLLECTION SYSTEM EXPANSION CAPABILITIES

The GCCS is designed to be readily expanded as fill operations proceed. Vertical wells will typically be installed in areas that have reached final grade. Caisson wells may be installed either as waste filling is initiated or in areas that have not reached final grade. Vertical wells, caisson wells and/or horizontal collection trenches will be installed as an interim control measure in disposal areas that have been in place for no more than five years, but that are not yet at final refuse grades.

Vertical extraction wells and caisson wells installed prior to reaching final grade will either be extended to the final grade level or abandoned and replaced. This determination will be made based upon the physical condition of the wells, their ability to provide effective LFG extraction, and field conditions at the time of final cap installation.

LFG headers are sized to accommodate the maximum expected LFG flow (refer to **Section 4.1.3** and **Appendix A-1**) and be fitted with flanged tees for expansion as new collectors are installed, in accordance with the Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements. Additionally, the use of HDPE header piping provides for flexible and efficient connections for future expansion of the header piping system.

April 2, 2025

4.0 COMPLIANCE REVIEW AND EVALUATION

The GCCS has been designed to be consistent with the Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements to achieve comprehensive control of both lateral migration and surface emissions of LFG. The following sections of the application demonstrate the compliance with each requirement.

4.1 Compliance with §62.16728(a)(1) AND §63.1962(a)(1)

§62.16728(a)(1) and §63.1962(a)(1) The collection devices within the interior must be certified to achieve comprehensive control of surface gas emissions by a professional engineer. The following issues must be addressed in the design: depths of refuse, refuse gas generation rates and flow characteristics, cover properties, gas system expandability, leachate and condensate management, accessibility, compatibility with filling operations, integration with closure end use, air intrusion control, corrosion resistance, fill settlement, resistance to the refuse decomposition heat and ability to isolate individual components or sections for repair or troubleshooting without shutting down entire collection system.

Issues related to compliance with §62.16728(a)(1) and §63.1962(a)(1) are discussed in the following sections.

Applicable information used in the design of the GCCS is included in **Appendices A (Calculations), Appendix B (Head Loss Analyses), Appendices C (GCCS Design Plans, and Design Documents), Appendix D (Surface Emissions Monitoring Plan) and Appendices E-F (Approved Alternatives to the NSPS).**

4.1.1 Control of Surface Emissions

The GCCS was designed and will be designed to minimize both subsurface lateral migration and surface emissions of LFG from the landfill. System performance depends upon the installation of a satisfactory GCCS system, its proper management, and maintenance of a suitable final refuse cover system. If there is a temporary exceedance in emissions, it will be addressed by an appropriate response, evaluating both the GCCS and temporary or final cover systems. Appropriate action will then be taken to correct the exceedance, as required by the Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements.

April 2, 2025

The Surface Emissions Monitoring Plan is included in **Appendix D**.

4.1.2 Depth of Refuse

Depths of refuse were calculated, at the time of the design of the GCCS, based upon existing topography, final grade and liner grade permit plans, and record documentation of existing landfill liner grades. The landfill surface elevation was determined from aerial survey data at the time of design and used to verify waste depths.

4.1.3 Landfill Gas Generation Rates and Flow Characteristics

In compliance with §62.16714(b)(2)(i) and §63.1962(c), the maximum expected LFG flow rate for each Unit was used for sizing the GCCS in the corresponding unit. The LFG generation rate calculations were performed using the United States Environmental Protection Agency (USEPA) Landfill Gas Emission Model (LandGEM) V3.03 (AP-42 default values $k = 0.04/\text{year}$ and $L_0 = 100 \text{ m}^3/\text{Mg}$), historical waste receipts, the permitted design capacity of the facility, and current operating parameters were utilized to develop the LFG generation rate calculations.

The maximum LFG generation rate for the East Unit was calculated to be 2,522 scfm and is projected to occur in 2037. The maximum LFG generation rate for the West Unit was calculated to be 2,409 scfm and is projected to occur in 2077. The combined maximum LFG generation rate for the entire landfill was calculated to be 2,919 scfm and is projected to occur in 2077.

The proposed GCCS components to be installed at each Unit shall be sized to accommodate the maximum LFG generation for the corresponding Unit to ensure adequate future capacity. As discussed in **Section 3.2** of this Design Plan, the GCCS is designed in a manner to allow expansion of the wellfield, piping, and LFG control device capacity.

The LandGEM emissions model is a “design tool,” which uses the information available to project future operating conditions. This model was developed based upon operating conditions at a cross-section of landfills within the United States and is approved for use by the USEPA. The most up to date information pertaining to the site operating parameters was utilized for this GCCS Design Plan. Actual operating parameters may change in the future and may require changes in the system flow characteristics and process equipment as the system is developed. These changes will be made in accordance with governing federal regulations, including §62.16714.

LFG generation projections are provided in **Appendix A-1**.

April 2, 2025

4.1.4 Landfill Cover Properties

The final cover design, for areas without an existing final cover system, will incorporate a composite geosynthetic/low permeability soil barrier. The primary purpose of the final cover system is to preclude infiltration of precipitation that would generate additional leachate. However, the final cover system design also provides a significant barrier to LFG emission and air infiltration when combined with an active LFG extraction system. The GCCS components will provide for collection of LFG, relieving pressures from LFG buildup from beneath this layer.

4.1.5 Landfill Gas Control System Expandability

Expandability of the GCCS is achieved by installing expansion points, such as tees with blind flanges, as access points to extend the GCCS. These expansion points provide planned access for expansion of the LFG transmission piping and LFG control device facility(s) in the future. Similarly, additional LFG control devices shall be added to expand the extraction capabilities of the GCCS. These flanges or expansion points provide planned access for expansion of the LFG transmission piping and LFG control device facility(s) in the future. In the event that actual LFG flow rates do exceed the capacity of the system additional GCCS components will be designed and installed in accordance with Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements, to supplement the original design.

4.1.6 Leachate and Condensate Management

Leachate management is accomplished using an engineered leachate collection and handling system. Leachate is collected in the LCS of each landfill cell, then is pumped through forcemain pipes to the permitted discharge point for the facility. The LCS is designed according to Subtitle D standards (40 CFR §257 and §258) and is part of the Solid Waste Disposal Area License.

LFG piping grades will be maximized where practical to reduce the impact of differential settlement and promote positive condensate drainage. The proposed transmission header and lateral piping will be sloped at a minimum of three percent (3%) within waste limits and half a percent (0.5%) outside waste limits, unless otherwise denoted, to promote condensate to flow by gravity to engineered low points in the GCCS piping, and subsequent collection of the condensate. Condensate collected at the engineered low points will be handled by pump stations or gravity drain driplegs that will discharge directly into the LCS or other leachate or condensate collection and control structures.

April 2, 2025

4.1.7 Accessibility

Accessibility to the GCCS components is achieved by installing commonly accessed components (such as wellheads and monitoring ports) on relatively flat surfaces of the landfill or near the facility's Road network. Wellheads, piping risers, valves and monitoring ports of the proposed system will be installed above grade and raised as needed to maintain accessibility.

4.1.8 Compatibility with Refuse Filing Operations

At the time of this report, an active GCCS has been constructed and is operating in Cells B-G and J of the Wexford County Landfill. The proposed GCCS has been designed to integrate the existing GCCS components currently operating at the site. All construction is phased so existing collection can remain operational with minimal, localized interruptions.

As refuse filling operations proceed and portions of the site reach final or near-final grades, additional GCCS components will be installed. This method of installation allows GCCS components to be constructed in accordance with §62.16728(a)(1) and §63.1962(a)(1) while minimizing interference of the GCCS with ongoing filling operations.

4.1.9 Integration with Closure End Use

Currently, the post-closure end-use for the site is unspecified and will likely be utilized as open space. Any modifications to the closure end-use must be approved by facility personnel to evaluate the compatibility with the GCCS. Any items of concern related to maintaining and operating the GCCS will be mitigated by either altering the proposed post-closure end-use or by adjusting or modifying the GCCS in accordance with the Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements.

4.1.10 Air Intrusion Control

Oxygen from air intrusion can cause subsurface reactions or increase the heat content of the LFG. Air intrusion is minimized by using soil backfill and hydrated bentonite plugs in the upper zone of the vertical wells and above the horizontal collectors to seal pathways to the extraction zone. Where extraction wells are placed in final cover areas through a geosynthetic component, the penetration is sealed. To accommodate the penetration of the geomembrane component of the final cover system, in areas with a geosynthetic cap, the geomembrane component will be typically fitted to the pipe penetrations utilizing a 'pipe boot'. Potential air intrusion and LFG emissions through the cover system will be controlled through periodic monitoring and adjustment of the GCCS, in coordination with appropriate maintenance of the landfill cover system.

April 2, 2025

4.1.11 Corrosion Resistance

Corrosion resistance of the GCCS is achieved using corrosion resistant materials or materials that have a corrosion resistant coating, in accordance with §62.16728(b)(1) and §63.1962 (b)(1). The primary components used in the construction of the GCCS are HDPE and polyvinyl chloride (PVC). Components will be inspected during routine GCCS monitoring for abrasion, chipping, or other potential deterioration of the components. If damage to the materials is observed that may be detrimental to the performance of the GCCS, the components will be replaced or repaired.

4.1.12 Fill Settlement

Settlement will occur due to overburden and decomposition of the refuse. The settlement of the waste can impact gas extraction wells, pulling them down, causing tipping or tilting, and deflecting the subgrade riser sections. To accommodate refuse settlement, the GCCS components were designed and are to be installed with several features to account for this settlement, including:

- LFG extraction well heads are connected to the LFG transmission piping by an above-ground flexible pipe or hose connection. This allows the LFG piping to accommodate some changes in the orientation of the LFG transmission piping or LFG extraction well while maintaining this connection.
- LFG transmission piping will be sloped at sufficient grades (see **Section 4.1.6**) so that reasonable amounts of differential and total settlement may occur without causing pipe breakage or disrupting the overall flow gradient of the LFG transmission piping.
- HDPE piping will be used for the construction of the header piping and transmission system. HDPE piping is flexible and absorbs differential settlement without breaking or cracking.

4.1.13 Resistance to Decomposition Heat

Resistance of the GCCS to the heat generated from refuse decomposition is achieved through materials tested and proven to withstand temperatures well above those typically found in landfills. Waste and gas temperatures are monitored during construction and operation of the wellfield. Temperatures showing an increasing trend or sudden increase will be investigated and addressed. The GCCS will be inspected during routine LFG system monitoring for heat damage. If heat damage of the GCCS components is observed and is believed to be detrimental to the operation of the GCCS, the cause of the elevated landfill temperature will be investigated and the GCCS will be adjusted or modified to mitigate the effects of the elevated temperatures.

April 2, 2025

4.1.14 Ability to Isolate Individual Components or Sections for Repair

The GCCS has been designed with isolation valves at various points throughout the GCCS. These valves allow for specific portions of the GCCS to be closed off for repair without shutting down the entire GCCS. These valves are frequently inspected to ensure they are properly functioning and are not damaged.

4.2 COMPLIANCE WITH §62.16728(a)(2) AND §63.1962(a)(2)

§62.16728(a)(2) and §63.1962(a)(2) The sufficient density of gas collection devices determined in paragraph (a)(1) of this section must address landfill gas migration issues and augmentation of the collection system through the use of active or passive systems at the landfill perimeter or exterior.

Per the definitions given in §62.16730 and §63.1990, “sufficient density” means “any number, spacing, and combination of collection system components. . . necessary to maintain emission and migration control as determined by measures of performance set forth in this part.” Vertical LFG extraction well spacing was established using the method described in the book “*Methane Generation and Recovery from Landfills*, EMCON, 1982.” The values developed by this method were compared to those developed by the equation from the *NSPS Background Information Document*, 1991 with a more conservative EMCON value utilized for design purposes. This is consistent with spacing criteria used at other landfills and should effectively control surface emissions and subsurface migration of LFG in accordance with NSPS requirements.

Per calculations presented in **Appendix A-2** for the East Unit the ROI for shallow perimeter wells is required to be 120 feet (well spacing 208 feet), and for deeper wells 140 feet (well spacing 242 feet). For the West Unit the ROI for shallow perimeter wells is required to be 125 feet (well spacing 217 feet), and for deeper wells 145 feet (well spacing 251 feet).

This well schedule information is provided in **Appendix C**. While a well schedule is not required per NSPS and NESHAP regulations this well schedule is being included here per Michigan Part 115 regulations. Additionally, this well schedule is only a conceptual design for wells at the Wexford County Landfill's final closure state. Wells may be installed on an interim basis and will be designed per waste depths at time of installation and may vary from the design parameters shown in Appendix C well schedule. Any alterations made to the well schedule provided in Appendix C during construction should not be considered as a departure from or adjustment to any preexisting operating permit issued by EGLE.

April 2, 2025

Additionally, the spacing of vertical and interim horizontal extraction wells is within the guidelines defined in *Table 5-1, Summary of Suggested Collector Density*, *Training Course for Landfill Gas NSPS/EG Regulatory Personnel to Review GCCS Design Submittals*, North Carolina State University, September 1998. This guidance recommends a horizontal collector spacing of 30 feet to 50 feet vertically and 150 feet to 300 feet horizontally for interim horizontal extraction wells, and a vertical well spacing of 250 feet to 300 feet in post-Subtitle D landfills with wet waste.

Wexford County Landfill conducts quarterly surface emissions monitoring in accordance with applicable federal requirements. If the GCCS at the facility does not meet the measures of performance set forth, the GCCS will be adjusted or modified in accordance with the applicable federal requirements. These adjustments or modifications may include the installation of additional collection elements, cap repairs or other actions defined by field conditions at the time of monitoring.

4.3 COMPLIANCE WITH §62.16728(a)(3) AND §63.1962(a)(3)

§62.16728(a)(3) and §63.1962(a)(3) *The placement of gas collection devices determined in paragraph (a)(1) of this section must control all gas producing areas, except as provided by paragraphs (a)(3)(i) and (ii) of this section.*

Issues related to compliance with §62.16728(a)(3) and §63.1962(a)(3) are discussed in the following sections.

4.3.1 Asbestos and Non-degradable Materials

§62.16728(a)(3)(i) and §63.1962(a)(3)(i) *Any segregated area of asbestos or nondegradable material may be excluded from collection if documented as provided under §62.16726(d) [of Subpart OOO, or §63.1983(d) of Subpart AAAA]. The documentation must provide the nature, date of deposition, location and amount of asbestos or nondegradable material deposited in the area, and must be provided to the Administrator upon request.*

Currently, in the East Unit there are no known segregated areas of the landfill that contain only asbestos and/or non-degradable materials. However, future Cells 7-9 of the West Unit are going to accept Type III Industrial Waste only and therefore will generate no LFG. For this reason, no GCCS is planned to be installed in those cells.

April 2, 2025

4.3.2 Non-productive Areas

§62.16728(a)(3)(ii) and §63.1962(a)(3)(ii) Any nonproductive area of the landfill may be excluded from control, provided that the total of all excluded areas can be shown to contribute less than 1 percent of the total amount of NMOC emissions from the landfill. The amount, location, and age of the material must be documented and provided to the Administrator upon request. A separate NMOC emissions estimate shall be made for each section proposed for exclusion, and the sum of all such sections shall be compared to the NMOC emissions estimate for the entire landfill.

Currently, at the East Unit there are no known non-productive areas. However, future Cells 7-9 of the West Unit are going to accept Type III Industrial Waste only and therefore will generate no LFG. For this reason, no GCCS is planned to be installed in those cells.

4.4 COMPLIANCE WITH §62.16728(b)(1), (2), AND (3) AND §63.1962(b)(1), (2), AND (3)

§62.16728(b) and §63.1962 (b) Each owner or operator seeking to comply with §62.16714(b) [§63.1959(b)(2)(ii) for Subpart AAAA] must construct the gas collection devices using the following equipment or procedures:

4.4.1 Landfill Gas Extraction Component Construction

§62.16728(b)(1) and §63.1962 (b)(1) The landfill gas extraction components must be constructed of polyvinyl chloride (PVC), high density polyethylene (HDPE) pipe, fiberglass, stainless steel, or other nonporous corrosion resistant material of suitable dimensions to: convey projected amounts of gases; withstand installation, static, and settlement forces; and withstand planned overburden or traffic loads. The collection system must extend as necessary to comply with emission and migration standards. Collection devices such as wells and horizontal collectors must be perforated to allow gas entry without head loss sufficient to impair performance across the intended extent of control. Perforations must be situated with regard to the need to prevent excessive air infiltration.

Issues related to compliance with §62.16728(b)(1) and §63.1962(b)(1) are discussed in the following sections.

April 2, 2025

4.4.1.1 Materials

The GCCS components will generally be constructed of polyvinyl chloride (PVC) or high-density polyethylene (HDPE). Some components of the collection assemblies or fittings may be constructed of, fiberglass, corrosion-resistant steel, Neoprene (gaskets and seals) and other non-porous corrosion resistant materials.

4.4.1.2 Component Sizing

The piping network at the East Unit will be sized for a maximum estimated LFG generation rate of 2,522 scfm (see **Section 4.1.3**). The piping network at the West Unit will be sized for a maximum estimated LFG generation rate of 2,409 scfm (see **Section 4.1.3**). The blowers and flares will be sized for a combined maximum estimated LFG generation rate of 2,918 scfm (see **Section 4.1.3**). This accounts for the entire MSW disposal capacity, at the final grade development elevations of the landfill. Capacity of the collector and control system components will be phased in as LFG generation and extraction rates increase.

4.4.1.3 Component Loading

The proposed GCCS components within the Facility are consistent with those installed at other landfills that have been in place for extended periods of time (in excess of 15 years) and verified to withstand installation, static, settlement, overburden, and traffic loads. The materials used for the GCCS construction are industry-standard materials that are proven to withstand typical loads within a landfill environment, and static loads such as overburden and traffic loads for the LFG transmission piping are calculated to be less than the allowable loads recommended by the piping manufacturer. In the event damage to GCCS components are identified, the GCCS will be adjusted or modified in accordance with the Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements.

Typically, GCCS components requiring footings or foundations will be placed outside the waste area. Foundations supporting the GCCS components will include consideration for active and passive loading as well as any mechanical loads, and will be designed for the local soil conditions, and tolerances of the equipment. Within the landfill, settlement forces cannot accurately be predicted due to the non-homogeneous nature of the refuse within the landfill. Road crossings will be designed to protect the GCCS piping using casings or mitigated by burial depth. Traffic loading in active areas is less controllable, therefore loading to GCCS piping will be mitigated by burial depth.

April 2, 2025

4.4.1.4 System Expansion

The GCCS shall be expanded as necessary to comply with the Federal Plan Requirements. Expansion of the GCCS will be certified by a professional engineer, and the measures of performance of the LFG system verified as set forth in the Federal Plan Requirements. Wexford County Landfill will conduct monitoring and document compliance of the GCCS, in accordance with the Federal Plan Requirements. If the GCCS at Wexford County Landfill does not meet the measures of performance set forth in the Federal Plan Requirements, the GCCS will be adjusted or modified in accordance with the Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements.

4.4.1.5 Component Perforation

The vertical well and horizontal collection piping will be perforated as shown on the design plan drawings (**Appendix C**), to allow LFG entry without inducing head losses sufficient to impair performance across the intended extent of control. The perforation patterns used for the Wexford County Landfill GCCS design have been successfully used in previous LFG control applications.

4.4.1.6 Air Infiltration

The LFG collection elements were designed to prevent excessive air infiltration through the use of solid pipe and soil backfill near the ground surface for vertical LFG extraction wells and interim/supplemental horizontal collection trenches. Hydrated bentonite plugs and geomembrane seals, if warranted, will be provided around vertical well casings and interim/supplemental horizontal collection trench access piping, where they penetrate the landfill final cover or interim cover systems. Further, air intrusion control will be accomplished through the operational monitoring standards for the LFG collection elements in accordance with NSPS requirements. If the GCCS does not meet the operational monitoring standards, it will be adjusted or modified in accordance with the Federal Plan Subpart OOO Requirements and NESHAP Subpart AAAA Requirements.

4.4.2 Landfill Gas Extraction Component Installation

§62.16728(b)(2) and §63.1962 (b)(2) Vertical wells must be placed so as not to endanger underlying liners and must address the occurrence of water within the landfill. Holes and trenches constructed for piped wells and horizontal collectors must be of sufficient cross-section so as to allow for their proper construction and completion including, for example, centering of pipes and placement of gravel backfill. Collection devices must be designed so as not to allow indirect short circuiting of air into the cover or refuse into the collection

April 2, 2025

system or gas into the air. Any gravel used around pipe perforations should be of a dimension so as not to penetrate or block perforations.

Issues related to compliance with §62.16728(b)(2) and §63.1962 (b)(2) are discussed in the following sections.

4.4.2.1 Component Placement

Future depths of refuse were calculated, at the time of the design of the GCCS, based upon the permit plan final grades and landfill liner grades. Vertical LFG extraction wells will be designed to extend from the landfill surface to within no less than ten (10) feet of the landfill base, unless otherwise designed for specific site considerations.

4.4.2.2 Leachate

The occurrence of leachate within the landfill will be addressed by the LCS as stated in **Section 4.1.6** of this Design Plan. Leachate management within the portions of the landfill with a composite liner will be accomplished through the LCS which includes a leachate drainage layer, collection piping, side slope risers with liquid pumping equipment, and liquid disposal systems.

For these reasons, it is not expected that free liquids will be frequently encountered during the drilling of the vertical LFG extraction wells. If free liquids are encountered, the drilling contractor will attempt to drill through the perched zone of liquids allowing drainage into the underlying waste mass and the LCS. If the zone of perched liquids cannot be penetrated, the well installation may be terminated. If necessary, appropriate measures will be taken to complete the well installation procedure at a nearby location.

If perched liquids are observed within the extraction wells after installation, and it is determined that the liquid level is restrictive to efficient LFG extraction, the leachate level will be reduced. This is typically accomplished by periodic pumping of the liquids using either electric or pneumatic pumping systems. Liquids removed from the well casings will be discharged to the LCS or other leachate management infrastructure.

4.4.2.3 Wells and Trenches

Vertical wells and interim/supplemental horizontal collection trenches, constructed for LFG collection elements, will be of sufficient cross-section to allow for their proper construction and completion, including centering of the pipes and placement of suitable backfill. The wells and interim/supplemental horizontal collection trenches will be constructed under supervision of a construction quality assurance program

April 2, 2025

implemented by Wexford County Landfill and verified to be properly constructed, as indicated on the design plans in **Appendix C**.

4.4.2.4 Component Short Circuiting

To avoid short-circuiting, or self-damage, the LFG collection elements are designed to prevent air infiltration through the cover, refuse contamination of the collection elements, and direct venting of LFG to the atmosphere. Air intrusion control will be verified through monitoring of gas quality at the extraction components, monitoring of surface emission levels and maintenance of the landfill cover in accordance with NSPS or NESHAP requirements. Separation of the collection elements from the refuse is accomplished by placing gravel backfill in the annular borehole space around extraction wells casings and gravel backfill or other suitable materials for the interim/supplemental horizontal collection trench pipes, providing a filter pack between the refuse and the LFG collection elements. Direct venting of the LFG to the atmosphere is avoided by operating the GCCS under a controlled application of vacuum and the quarterly monitoring of surface emissions (see **Section 4.2**).

4.4.2.5 Backfill

Gravel or other suitable backfill material of sufficient size is specified to prevent penetration or blockages of the LFG collector pipe perforations. Gravel (non-calcareous) to be utilized in vertical wells and horizontal collector trenches will be a nominal 1-inch to 3-inch particle size, as indicated on the design plans. Other suitable material, such as tire chips, may be utilized in interim/supplemental horizontal collection trenches.

4.4.3 Landfill Gas Extraction Component Connections to Landfill Gas Transmission Piping

§62.16728(b)(3) and §63.1962 (b)(3) Collection devices may be connected to the collection header pipes below or above the landfill surface. The connector assembly must include a positive closing throttle valve, any necessary seals and couplings, access couplings and at least one sampling port. The collection devices must be constructed of PVC, HDPE, fiberglass, stainless steel, or other non-porous material of suitable thickness.

The collection devices are connected to the collection header pipes using lateral piping. The lateral piping is connected to the header piping either above or below the landfill surface, as required by field conditions at the time of installation. The connector assemblies (wellheads) will be located above grade. These assemblies include a positive closing throttle valve, necessary seals and couplings, access ports and couplings, and a minimum of one sampling port. The collection devices will be constructed of PVC, HDPE,

April 2, 2025

fiberglass, corrosion-resistant steel, and other non-porous materials of suitable thickness. The GCCS components are designed to withstand anticipated installation, static, settlement, overburden, and traffic loads.

4.5 COMPLIANCE WITH §62.16728(c)(1) AND (2) AND §63.1962(c)(1) AND (2)

§62.16728(c) and §63.1962(c) *Each owner or operator seeking to comply with §62.16714(c) [of Subpart OOO, or §63.1959(b)(2)(iii) of Subpart AAAA] must convey the landfill gas to a control system in compliance with §62.16714(c) [of Subpart OOO, or §63.1959(b)(2)(iii) of Subpart AAAA] through the collection header pipe(s). The gas mover equipment must be sized to handle the maximum gas generation flow rate expected over the intended use period of the gas moving equipment using the following procedures:*

Issues related to compliance with §62.16728(c) and §63.1962(c) are discussed in the following sections.

4.5.1 Existing Landfill Gas Flow Rate Data

§62.16728(c)(1) and §63.1962(c)(1) *For existing collection systems, the flow data must be used to project the maximum flow rate. If no flow data exists, the procedures in paragraph (c)(2) of this section shall be used.*

The estimated total LFG generation rate for year 2025 is approximately 1,494 scfm. At the time of this report, an active GCCS has been constructed and is operating at the facility. The current LFG extraction rate is influenced by the percentage of waste covered by an active GCCS and the extent of final cover installation, which affects the collection efficiency. The current extraction rate is generally consistent with LandGEM projections.

4.5.2 Future Landfill Gas Flow Rate Estimates

§62.16728(c)(2) and §63.1962(c)(2) *For new collection systems, the maximum flow rate must be in accordance with §62.16720(a)(1) [of Subpart OOO], or §63.1960(a)(1) [of Subpart AAAA].*

The maximum LFG generation rate for the East Unit was calculated to be 2,522 scfm and is projected to occur in 2037. The maximum LFG generation rate for the West Unit was calculated to be 2,409 scfm and is projected to occur in 2077. The combined maximum LFG generation rate for the entire landfill was

April 2, 2025

calculated to be 2,918 scfm and is projected to occur in 2077. Refer to **Section 4.1.3** and **Appendix A-1**.

4.6 ALTERNATIVES AND COMPLIANCE WITH §62.16714(e)(2) AND §63.1962(b)(2)

4.6.1 Submit a Design Plan

***§62.16714(e)(2) and §63.1959(b)(2)(i)** If the calculated NMOC emission rate is equal to or greater than 34 megagrams per year using Tier 1, 2, or 3 procedures, the owner or operator must either: Submit a collection and control system design plan prepared by a professional engineer to the Administration within 1 year as specified in §62.16724(d), except for exemptions allowed under §62.16711(g)(3); calculate NMOC emissions using a higher tier in §62.16718; or conduct a surface emission monitoring demonstration using the procedures specified in §62.16718(a)(6).*

***§62.16714(e)(2)** Submit a collection and control system design plan prepared by a professional engineer to the Administrator within 1 year*

In 2016, Wexford County Landfill was subject to 40 CFR Part 60, Subpart WWW. In accordance with that rule, the facility conducted a Tier 2 test in 2016. It was recently brought to the site's attention that the Tier 2 test used incorrect default values. When the correct default values were substituted, the site exceeded 50 Mg (the threshold standard to 40 CFR Part 60, Subpart WWW). The GCCS plan would have been submitted within 12 months of the Tier 2 test if the error in the calculation, which was recently brought to the attention of the facility, was discovered at that time. As such, Wexford County Landfill is now submitting the GCCS plan that complies with current applicable rule Federal 40 CFR 62 Subpart OOO and NESHAP 40 CFR Subpart AAAA

4.6.2 NSPS and NESHAP Alternative Standards and Requirements

The cited sections of Subpart OOO and Subpart AAAA both allow for alternatives to the operational standards, test methods, procedures, compliance requirements, monitoring, and record keeping, and reporting provisions to be requested in the Plan for MSW landfills:

***§62.724(d)(2) and §63.1981(d)(2)** The collection and control system design plan must include any alternatives to the operational standards, test methods, procedures, compliance measures,*

April 2, 2025

monitoring, record keeping or reporting provisions of §62.16716 through 62.16726 [of Subpart OOO, or §63.1957 through 62.1983 of Subpart AAAA] proposed by the owner or operator.

Alternatives to the NSPS or NESHAP are proposed in **Appendix E**. Wexford County Landfill will submit any additional alternative requests under separate cover as future needs dictate. Approval of such future requests will be incorporated in **Appendix E**, requiring only that section of this GCCS design plan to be updated.

4.6.3 Specifications for Active Collection Systems

As stated in **Sections 4.1 through 4.5** of this design plan, the GCCS proposed at the Wexford County Landfill complies with the specifications for active collection systems as stipulated in §62.16728 and §63.1962 of the NSPS. If future expansions of the GCCS are necessary, they will be designed to comply with the NSPS requirements or any approved alternatives.

§62.16714(b)(1) Collection system. *Install and start up a collection and control system that captures the gas generated within the landfill as required by paragraphs (b)(1) through (3) and (c) of this section within 30 months after:*

(ii) The first annual report in which the NMOC emission rate equals or exceeds 50 megagrams per year submitted under previously applicable regulations 40 CFR part 60, subpart WWW, 40 CFR part 62, subpart GGG, or a state plan implementing 40 CFR part 60, subpart Cc for a legacy controlled landfill or a landfill in the closed landfill subcategory...

§62.16714(b)(2) Active. *An active collection system must:*

(i) Be designed to handle the maximum expected gas flow rate from the entire area of the landfill that warrants control over the intended use period of the gas control system equipment:

(ii) Collect gas from each area, cell, or group of cells in the landfill in which the initial solid waste has been placed for a period of 5 years or more if active; or 2 years or more if closed or at final grade.

The GCCS will be constructed within the prescribed schedule under §62.16714(b)(2)(ii). Future expansions to the GCCS will proceed in accordance with the schedules under paragraphs (i) and (ii) of this section.

§62.16714 (b)(2)(iii) and §63.1959(b)(2)(ii)(B)(3) *Collect gas at a sufficient extraction rate*

April 2, 2025

§62.16714 (b)(2)(iv) and §63.1959(b)(2)(ii)(B)(4) Be designed to minimize off-site migration of gas

In compliance with §62.16714(b)(2)(ii)(A)(3-4) and §63.1959(b)(2)(ii)(B)(3-4), the proposed GCCS is designed to extract LFG at a sufficient rate to minimize the subsurface lateral migration and surface emissions of LFG. This is achieved by sizing and installing sufficient collection elements, transmission piping, blower(s), and LFG destruction equipment for the estimated maximum flow rate of LFG.

The GCCS is designed to collect LFG at a sufficient rate, which per the definitions in §62.16730 and §63.1990 means to maintain a negative gauge pressure (vacuum) at all wellheads without causing air infiltration. Application of a negative gauge pressure and minimization of air infiltration will be verified by monitoring the static pressure and nitrogen or oxygen concentrations of the LFG at the wellheads.

Each extraction point will be monitored monthly in accordance with §62.16716 (b) and (c) and §63.1958(b) and (c). Monitoring will be performed for pressure, temperature, oxygen and/or nitrogen, at a minimum.

Verification of the GCCS's ability to minimize off-site subsurface LFG migration is achieved through routine perimeter monitoring for combustible gases at the site, in accordance with applicable the Federal Plan Requirements.

Wexford County Landfill will monitor the GCCS wellheads, after installation, for static pressure and for LFG quality in accordance with NSPS requirements and will continue to monitor the perimeter LFG monitoring locations to detect the presence of off-site LFG migration. If off-site LFG migration is detected, the Wexford County Landfill will take the necessary actions in accordance with NSPS or NESHAP requirements.

4.6.4 Control Systems

The existing control systems (one utility flare) has a maximum rated capacity of approximately 750 scfm. Control unit operates in accordance with the work practice requirement operational standard given by §63.1958(e)(1).

Additional control devices will be installed as future LFG recovery rates dictate need. All devices will be designed and operated in accordance with the applicable citations in this section.

April 2, 2025

4.6.4.1 Flare Units

The required operational performance of the open flare is stipulated by applicable regulations of §62.16714(c) and §63.1959(b)(iii), which allow collected LFG to be controlled by:

§62.16714(c)(1) and §63.1959(b)(iii)(A) *A non-enclosed flare designed and operated in accordance with the parameters established in 40 CFR 60.18 [of Subpart OOO, or §63.11(b) of Subpart AAAA], except as noted in §63.1959 of Subpart AAAA; or*

§62.16714(c)(1) and §63.1959(b)(iii)(A) *A control system designed and operated to reduce NMOC by 98 weight percent; or when an enclosed combustion device is used for control, to either reduce NMOC by 98 weight percent or reduce the outlet NMOC concentration to less than 20 parts-per- million by volume, dry basis as hexane at 3-percent oxygen or less; or*

§62.16714(c)(3) and §63.1959(b)(2)(iii)(C) *Route the collected gas to a treatment system that processes the collected gas for subsequent sale or beneficial use such as fuel for combustion, production of vehicle fuel, production of high-Btu gas for pipeline injection, or use as a raw material in a chemical manufacturing process. Venting of treated landfill gas to the ambient air is not allowed. If the treated landfill gas cannot be routed for subsequent sale or beneficial use, then the treated landfill gas must be controlled according to either paragraph (c)(1) or (2) of this section [either paragraph (b)(2)(iii)(A) or (B) of this section for Subpart AAAA].*

The existing utility flare unit has a maximum rated capacity of approximately 750 scfm, with an operating range of 75 – 750 scfm. The utility flare is designed and operated in accordance with 40 CFR §60.18. Per §62.16714(c)(1) and §63.1959(b)(iii)(A). The utility flare will be operated within the parameter ranges established by the manufacturer of this unit. The utility flare is designed to reduce the concentration of the NMOC present in the LFG by at least 98 percent by weight. This destruction is supported by design, initial performance testing per 40 CFR §60.18, and operation of the utility flare in compliance with the requirements of 40 CFR §60.18 for non-assisted flares.

The existing blower which conveys LFG to the flare is rated for 600 scfm. Total inlet vacuum rating is 30 in. w.c. and a discharge pressure is 15 in. w.c.

Additionally, Wexford County Landfill is currently in a process of procuring a higher capacity blower and flare. The utility flare will have a maximum rated capacity of approximately 1,600 scfm, with an operating range of 160 – 1,600 scfm. The blower which will convey LFG to the flare is rated for 1,600 scfm. Total inlet vacuum rating is 50 in. w.c. and a max. discharge pressure of 3 psi.

April 2, 2025

The applicable regulations list operation and monitoring requirements for a non-enclosed flare as follows:

§63.1961(c) *Each owner or operator seeking to comply with §63.1959(b)(2)(iii) using a non-enclosed flare must install, calibrate and operate according to the manufacture's specifications for the following equipment:*

§63.1961(c)(1) *A heat sensing device such as an ultraviolet beam sensor or thermocouple, at the pilot light or the flame itself to indicate the presence of a flame*

(i) *Install, calibrate, and maintain a gas flow rate measuring device that records the flow to the control device at least once every 15 minutes; and*

(ii) *Secure the bypass line in the closed position with a car-seal or lock-and-key type configuration. A visual inspection of the seal or closure mechanism must be performed at least once every month to ensure that the valve is maintained in the closed position and that the gas flow is not diverted through the bypass line.*

LFG flows conveyed to each control device are recorded at a minimum every 15 minutes. The utility flare temperature is monitored and recorded continuously via a device with a minimum accuracy of + 1 percent of the temperature being measured and expressed.

The control devices are constantly monitored for the presence of a flame and/or temperature, indicating that combustion is occurring. Monitoring for the presence of a flame at the control devices is accomplished by an ultraviolet flame scanner, thermo-couple, or comparable device. If a flame is not detected, indicating that the combustion process has been disrupted, the monitoring system will automatically:

1. Cut power to the LFG mover(s), and;
2. Initiate the closure of either an electric or pneumatic-activated valve at the inlet to the mover(s).

There is no bypass around the backup control device.

Stopping the mover(s) will cause the LFG extraction process to cease. Closing the inlet valve will eliminate the potential for direct venting of LFG through the control device system.

This process will be initiated automatically, in the event of flame failure at a control device, without the need for operator intervention.

April 2, 2025

4.6.4.2 LFG Treatment System

As set forth in §62.16714(c)(3) and §63.1959(b)(2)(iii)(C), the owner may:

Route the collected gas to a treatment system that processes the collected gas for subsequent sale or beneficial use such as fuel for combustion... Venting of treated landfill gas to the ambient air is not allowed. If the treated landfill gas cannot be routed for subsequent sale or beneficial use, then the treated landfill gas must be controlled according to §62.16714 (c)(1) or (c)(2)/ §63.1959(b)(2)(iii)(A) or (B).

Currently Wexford County Landfill has no associated beneficial use facility to route the LFG to. Shall a beneficial use facility be developed at Wexford County Landfill in the future, the beneficial use facility will be maintained and operated in accordance with requirements of §62.16714 (c)(3) and §63.1959(b)(2)(iii)(C).

April 2, 2025

5.0 PROPOSED ALTERNATIVES

The following Federal Plan Subpart OOO and NESHAP Subpart AAAA citation allows for alternatives to the operational standards, test methods, procedures, compliance requirements, monitoring, record keeping, and reporting provisions to be requested in the design plan:

§62.16724(d)(2) and §63.1981(d)(2) *The collection and control system design plan must include any alternatives to the operational standards, test methods, procedures, compliance measures, monitoring, recordkeeping or reporting provisions of §62.16716 through §62.16716/§63.1957 through §63.1983 proposed by the owner or operator.*

The following sections describe proposed alternatives to the Federal Plan Subpart OOO and NESHAP Subpart AAAA.

5.1 COLLECTION DEVICE MONITORING

The following alternatives to the Federal Plan and NESHAP relate to collection system monitoring requirements.

5.1.1 Flow Meters When No Bypass Is Present

Both the Federal Plan Subpart OOO [§62.16722(b)(2), (c)(2), and (g)] and the NESHAP Subpart AAAA [§63.1961(b)(2), (c)(2), and (g)] contain a requirement that the owner or operator calibrate, maintain, and operate a device that records flow to the control device or treatment system, as well as bypass of the control device or treatment system, if applicable. The requirement to monitor bypass flow is not a separate requirement and does not apply to closed loop control systems; most GCCS are not designed to bypass emissions to atmosphere. EPA included the phrase "as applicable" in the regulations to acknowledge not every system will have a bypass. If the system is designed such that there is no physical means to bypass the control device or treatment system, only the flow to the control device or treatment system needs to be monitored, including flow that is re-routed from one device to another.

5.1.2 Monthly Monitoring Methods

The requirements of §62.16722(a)(2) and §63.1961(a)(2) allow for the monitoring of nitrogen or oxygen concentrations in the landfill gas monthly. §62.16722(a)(2)(i)-(ii) and §63.1961(a)(2)(i)-(ii), allow for the use of EPA Method 3C to measure the nitrogen levels and the use of either EPA Method 3A, 3C, or ASTM D6522-11 to establish the oxygen content. In accordance with the general state-of-the-practice

April 2, 2025

procedures, the landfill proposes to use a portable monitoring instrument (e.g., Landtec GEM 2000, Landtec GEM 5000, LMS, Envision, or equivalent instrument) to perform this monitoring. The monitoring equipment will be calibrated in accordance with manufacturer's recommendations to ensure accurate measurement of all parameters for which it is used to monitor. [The alternative has been approved by the EPA for NSPS Subpart WWW at other sites. An EPA Determination Letter Dated May 14, 2001 can be found in Appendix F-1. The Federal Plan Subpart OOO and NESHAP Subpart AAAA requirements are the same as the NSPS Subpart WWW requirement].

5.1.3 Positive Pressure Under a Synthetic Cover

In areas of a landfill where a geomembrane or synthetic cover is being used, §62.16716(b)(2) and §63.1958(b)(2) allows the owner or operator to develop acceptable pressure limits in the design plan. Calculations provided in Appendix E indicate a positive pressure of 18" W.C. or less is acceptable in all areas where a geomembrane or synthetic cover is being used as part of the final cover system. For this reason, incidents of positive pressure less than 18" W.C. in areas where a geomembrane or synthetic cover is installed will not be recorded as exceedances.

5.1.4 Early Installation of Collection Devices

The requirements of §62.16714(a)(2) and §63.1960(b) state that each collection device shall be installed no later than 60 days after the date on which the initial solid waste has been in place for a period of five years or more in active areas, or two years or more if closed or at final grade. However, there may be occasions when the landfill will install collection devices prior to the onset of Federal Plan Subpart OOO and NESHAP Subpart AAAA requirements.

Any collection device installed prior to the requirements of Federal Plan Subpart OOO and NESHAP Subpart AAAA will not be subject to the operational, monitoring, and/or recordkeeping requirements of the Federal Plan Subpart OOO and NESHAP Subpart AAAA until the age of the initial waste placed in the affected area reaches five years old if active, or two years if closed or at final grade. EPA Region IV has prepared correspondence [letter dated May 31, 2007 for NSPS Subpart WWW landfill which is consistent with Federal Plan Subpart OOO and NESHAP Subpart AAAA requirements found in Appendix F-2] regarding this matter.

April 2, 2025

5.1.5 Monitoring During Collection Device Extension

During filling operations, vertical extraction wells periodically need to be “raised” and/or temporarily disconnected (i.e., the well casing extended 15 to 25 feet vertically) in order to not be buried under lifts of trash. The time frame between when a well is raised, and when the waste height and/or final cover is high enough to safely access the sample ports, can often range from a few weeks to a few months. This can result in missed monthly readings at the well, since the well casing is too high for the technician to safely reach.

Since the Federal Plan Subpart OOO and NESHAP Subpart AAAA allows for exclusion of surface monitoring in “dangerous areas” of the site, it is reasonable to request an alternative to monitoring wells that are deemed dangerous for personnel to access (i.e., raised, active and construction areas). As such, the site proposes that monthly readings be taken only at wells that can be safely accessed. This request is in accordance with §62.16724(d)(2) and §63.1981(d)(2), which allows the facility to propose alternatives to the monitoring procedures in the Federal Plan and NESHAP. Any times that a collection device is not monitored due to unsafe conditions, will be noted in the Federal Plan Subpart OOO and NESHAP Subpart AAAA reports.

5.1.6 Stability Flexibility in Well Placement and Collection System Design Components

In the future, it is expected that the industry will develop more efficient methods of collecting landfill gas. Therefore, the design for the landfill gas collection system may be modified in the future to incorporate the new technologies. The drawings contained in this application portray the approximate location and design of the vertical extraction wells and header and lateral transmission pipelines to be installed in the future. Wexford County Landfill may be required to modify this design based on development of the landfill or to incorporate new technologies, while maintaining operational and criteria for the Federal Plan Subpart OOO and NESHAP Subpart AAAA requirements. Modifications that may be required will be provided to Agency in the form of a revised gas collection system and control system design plan.

April 2, 2025

5.2 SURFACE EMISSIONS MONITORING

The following variances are related to Surface Emission Monitoring (SEM) events.

5.2.1 Exclusion of Dangerous Areas from SEM Requirements

Areas with steep slopes or other dangerous areas are excluded from the SEM requirements under the Federal Plan Subpart OOO and NESHAP Subpart AAAA:

§62.16716(d) and §63.1958(d)(1) Operational Standards for Collection and Control Systems:
“...A surface monitoring design plan shall be developed...Areas with steep slopes or other dangerous areas may be excluded from surface testing.

Areas with steep slopes or other dangerous areas are excluded from the SEM requirements under the Federal Plan Subpart OOO and NESHAP Subpart AAAA:

- a. Roads;
- b. Working areas and/or the working face;
- c. Truck traffic areas;
- d. steep and dangerous slopes;
- e. Icy, snow covered, and/or extremely muddy side slopes;
- f. Areas where the landfill cover material has been exposed for the express purpose of installing, expanding, replacing, or repairing components of the LFG, leachate, or gas condensate collection and removal systems.

5.2.2 SEM for Closed Portions of the Landfill

Any portions of the landfill that have been certified closed or have been closed and capped in accordance with the cover conditions contained according to the Federal Plan Subpart OOO and NESHAP Subpart AAAA or Subtitle D will be treated as a closed landfill for SEM events. These closed portions of the landfill will be monitored in accordance with the following section of the Federal Plan Subpart OOO and NESHAP Subpart AAAA:

§62.16722(f) and §63.1961(f) *Any closed landfill that has no monitored exceedances of the operational standard in three consecutive quarterly monitoring periods may skip to annual monitoring. Any reading of 500 ppm or more above background detected during the annual monitoring returns the frequency for the landfill to quarterly monitoring.*

April 2, 2025

The EPA Region IV has prepared correspondence [letter dated July 12, 2004 for NSPS Subpart WWW landfill which is consistent with Federal Plan Subpart OOO and NESHAP Subpart AAAA requirements found in Appendix F-4] regarding this matter.

5.2.3 SEM Reporting Requirements

§62.16722(f) and §63.1961(f) requires the locations of each SEM exceedance of the 500 parts per million methane concentration to be marked and recorded. The locations to be recorded in latitude and longitude coordinates using an instrument with an accuracy of at least 4 meters and the coordinates must be in decimal degrees with at least five decimal places.

Wexford County Landfill is requesting that each exceedance location may alternatively marked and recorded in state plane coordinates system using an instrument with an accuracy of at least 4 meters. The coordinates will be in decimal degrees with at least five decimal places.

5.3 EXCLUSION OF ODOR CONTROL WELLS NOT IN WASTE OR NOT USED FOR FEDERAL PLAN/NESHAP COMPLIANCE

Any wells placed outside limits of waste are not subject to NSPS/NESHAP operation, monitoring, record keeping, and reporting requirements as they are not interior wells as defined in Federal Plan Subpart OOO and NESHAP Subpart AAAA. As such, the existing and future LFG extraction wells installed outside limits of waste for migration/odor control purposes will be excluded from the Federal Plan Subpart OOO and NESHAP Subpart AAAA operation, monitoring, record keeping, and reporting requirements.

5.4 REGULATORY APPROVALS

For reports, notifications, and submittals required by this regulation to be submitted to EGLE or USEPA for approval, if Wexford County Landfill receives no response from EGLE or USEPA within 40 days of submittal, Wexford County Landfill will assume the submittal has been approved.

April 2, 2025

6.0 LIMITATIONS

The work product included in the attached was undertaken in full conformity with generally accepted professional consulting principles and practices and to the fullest extent as allowed by law we expressly disclaim all warranties, express or implied, including warranties of merchantability or fitness for a particular purpose. The work product was completed in full conformity with the contract with our client and this document is solely for the use and reliance of our client (unless previously agreed upon that a third party could rely on the work product) and any reliance on this work product by an unapproved outside party is at such party's risk.

The work product herein (including opinions, conclusions, suggestions, etc.) was based on the situations and circumstances as found at the time, location, scope and goal of our performance and thus should be relied upon and used by our client recognizing these considerations and limitations. Tetra Tech shall not be liable for the consequences of any change in environmental standards, practices, or regulations following the completion of our work and there is no warrant to the veracity of information provided by third parties, or the partial utilization of this work product.

APPENDIX A-1

CALCULATIONS - GAS GENERATION RATE MODELING

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill - Gas Model Update
DESCRIPTION: Landfill Gas Modeling Projections
Prepared By: JPS Checked By: JPS Approved By: KM

PROJECT NO: 4251206
SHEET: 1
OF: 3
DATE: 4/2/2025

Given:

Landfill gas generation projections have been made utilizing the USEPA's Landfill Gas Emission Model (LandGEM) V3.03. LandGEM is based on a first-order decomposition rate equation for quantifying emissions from the decomposition of land filled waste in municipal solid waste (MSW) landfills. The software provides a relatively simple approach to estimating landfill gas emissions. Model defaults are based on empirical data from U.S. landfills. Field test data can also be used in place of model defaults when available. Further guidance on EPA test methods, Clean Air Act (CAA) regulations, and other guidance regarding landfill gas emissions and control technology requirements can be found at <http://www.epa.gov/ttnatw01/landfill/landflpg.html>.

First-Order Decomposition Rate Equation:

$$Q_{CH_4} = \sum_{i=1}^n \sum_{j=0.1}^1 kL_o \left[\frac{M_i}{10} \right] (e^{-kt_{ij}})$$

Where:

Q_{CH_4} = annual methane generation in the year of the calculation ($m^3/year$)

i = 1-year time increment

n = (year of the calculation) - (initial year of waste acceptance)

j = 0.1-year time increment

k = methane generation rate ($year^{-1}$)

L_o = potential methane generation capacity (m^3/Mg)

M_i = mass of waste accepted in the i^{th} year (Mg)

t_{ij} = age of the j^{th} section of waste mass M_i accepted in the i^{th} year (*decimal years*, e.g., 3.2 years)

LandGEM is considered a screening tool — the better the input data, the better the estimates. Often, there are limitations with the available data regarding waste quantity and composition, variation in design and operating practices over time, and changes occurring over time that impact the emissions potential. Changes to landfill operation, such as operating under wet conditions through leachate recirculation or other liquid additions, will result in generating more gas at a faster rate.

Site Conditions:

Waste conditions at the Wexford County Landfill have been developed based upon recorded documentation and reports from site operations personnel. Degradable waste mass (MSW and other degradable waste streams) has been utilized in projecting LFG generation rates. Fill placement activities at the Wexford County Landfill began in 1977. Given the projected site life, the landfill will close in 2076.

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill - Gas Model Update
 DESCRIPTION: Landfill Gas Modeling Projections
 Prepared By: JPS Checked By: JPS Approved By: KM

PROJECT NO: 4251206
 SHEET: 2
 OF: 3
 DATE: 4/2/2025

<i>Year</i>	<i>East Unit Total Waste Intake (ton/year)</i>	<i>West Unit Total Waste Intake (ton/year)</i>	<i>Total Waste In Place (ton/year)</i>	<i>East Unit LFG Generation Rate (scfm)</i>	<i>West Unit LFG Generation Rate (scfm)</i>	<i>Total LFG Generation Rate (scfm)</i>
1977	83,828		83,828	0		0
1978	64,400		148,228	40		40
1979	138,340		286,568	70		70
1980	114,859		401,427	133		133
1981	105,238		506,665	183		183
1982	96,642		603,307	226		226
1983	47,834		651,141	264		264
1984	63,988		715,129	277		277
1985	59,485		774,614	296		296
1986	59,596		834,210	313		313
1987	39,908		874,118	330		330
1988	47,512		921,630	336		336
1989	59,566		981,196	346		346
1990	55,117		1,036,313	361		361
1991	85,811		1,122,124	373		373
1992	78,653		1,200,777	399		399
1993	98,544		1,299,321	422		422
1994	122,263		1,421,584	452		452
1995	97,664		1,519,248	493		493
1996	95,009		1,614,257	521		521
1997	89,444		1,703,701	546		546
1998	100,193		1,803,894	568		568
1999	86,259		1,890,153	593		593
2000	146,993		2,037,146	611		611
2001	128,262		2,165,408	658		658
2002	129,132		2,294,540	694		694
2003	129,895		2,424,435	729		729
2004	120,611		2,545,046	762		762
2005	106,394		2,651,440	790		790
2006	36,141		2,687,581	810		810
2007	70,828		2,758,409	796		796
2008	57,179		2,815,588	799		799
2009	47,158		2,862,746	795		795
2010	65,490		2,928,236	786		786
2011	55,595		2,983,831	787		787
2012	119,892		3,103,723	783		783
2013	134,224		3,237,947	810		810
2014	153,860		3,391,807	842		842
2015	162,147		3,553,954	883		883
2016	177,011		3,730,965	926		926
2017	155,881		3,886,846	975		975
2018	187,078		4,073,924	1,012		1,012
2019	199,092		4,273,016	1,062		1,062
2020	226,039		4,499,055	1,116		1,116
2021	202,941		4,701,996	1,180		1,180
2022	201,109		4,903,105	1,231		1,231

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill - Gas Model Update
 DESCRIPTION: Landfill Gas Modeling Projections
 Prepared By: JPS Checked By: JPS Approved By: KM

PROJECT NO: 4251206
 SHEET: 3
 OF: 3
 DATE: 4/2/2025

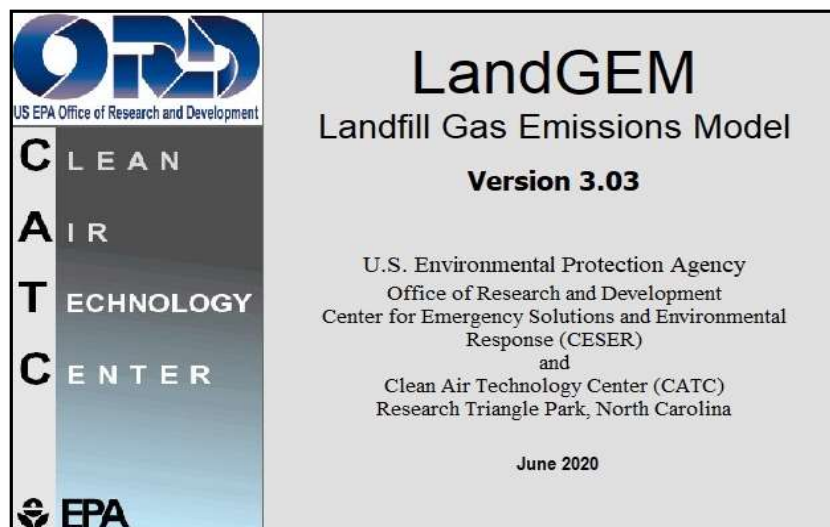
Year	East Unit Total Waste Intake (ton/year)	West Unit Total Waste Intake (ton/year)	Total Waste In Place (ton/year)	East Unit LFG Generation Rate (scfm)	West Unit LFG Generation Rate (scfm)	Total LFG Generation Rate (scfm)
2023	338,298		5,241,403	1,280		1,280
2024	325,394		5,566,797	1,392		1,392
2025	342,467		5,909,264	1,494		1,494
2026	342,467		6,251,731	1,599		1,599
2027	342,467		6,594,198	1,701		1,701
2028	342,467		6,936,665	1,799		1,799
2029	342,467		7,279,132	1,893		1,893
2030	342,467		7,621,599	1,983		1,983
2031	342,467		7,964,066	2,069		2,069
2032	342,467		8,306,533	2,153		2,153
2033	342,467		8,649,000	2,233		2,233
2034	342,467		8,991,467	2,309		2,309
2035	342,467		9,333,934	2,383		2,383
2036	342,467		9,676,401	2,454		2,454
2037		246,550	9,922,951	2,522	0	2,522
2038		246,550	10,169,501	2,423	118	2,542
2039		246,550	10,416,051	2,328	232	2,560
2040		246,550	10,662,601	2,237	341	2,578
2041		246,550	10,909,151	2,149	446	2,596
2042		246,550	11,155,701	2,065	547	2,612
2043		246,550	11,402,251	1,984	644	2,628
2044		246,550	11,648,801	1,906	737	2,643
2045		246,550	11,895,351	1,832	826	2,658
2046		246,550	12,141,901	1,760	912	2,672
2047		246,550	12,388,451	1,691	995	2,686
2048		246,550	12,635,001	1,624	1,074	2,699
2049		246,550	12,881,551	1,561	1,150	2,711
2050		246,550	13,128,101	1,500	1,224	2,723
2051		246,550	13,374,651	1,441	1,294	2,735
2052		246,550	13,621,201	1,384	1,362	2,746
2053		246,550	13,867,751	1,330	1,427	2,757
2054		246,550	14,114,301	1,278	1,489	2,767
2055		246,550	14,360,851	1,228	1,549	2,777
2056		246,550	14,607,401	1,180	1,607	2,786
2057		246,550	14,853,950	1,133	1,662	2,795
2058		246,550	15,100,500	1,089	1,715	2,804
2059		246,550	15,347,050	1,046	1,766	2,812
2060		246,550	15,593,600	1,005	1,815	2,820
2061		246,550	15,840,150	966	1,862	2,828
2062		246,550	16,086,700	928	1,908	2,836
2063		246,550	16,333,250	892	1,951	2,843
2064		246,550	16,579,800	857	1,993	2,850
2065		246,550	16,826,350	823	2,033	2,856
2066		246,550	17,072,900	791	2,072	2,863
2067		246,550	17,319,450	760	2,109	2,869
2068		246,550	17,566,000	730	2,145	2,875

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill - Gas Model Update
 DESCRIPTION: Landfill Gas Modeling Projections
 Prepared By: JPS Checked By: JPS Approved By: KM

PROJECT NO: 4251206
 SHEET: 3
 OF: 3
 DATE: 4/2/2025

Year	East Unit Total Waste Intake (ton/year)	West Unit Total Waste Intake (ton/year)	Total Waste In Place (ton/year)	East Unit LFG Generation Rate (scfm)	West Unit LFG Generation Rate (scfm)	Total LFG Generation Rate (scfm)
2069		246,550	17,812,550	701	2,179	2,880
2070		246,550	18,059,100	674	2,212	2,886
2071		246,550	18,305,650	647	2,243	2,891
2072		246,550	18,552,200	622	2,274	2,896
2073		246,550	18,798,750	598	2,303	2,901
2074		246,550	19,045,300	574	2,331	2,905
2075		246,550	19,291,850	552	2,358	2,910
2076		246,550	19,538,400	530	2,384	2,914
2077			19,538,400	509	2,409	2,918
2078			19,538,400	489	2,314	2,803
2079			19,538,400	470	2,223	2,694
2080			19,538,400	452	2,136	2,588
2081			19,538,400	434	2,053	2,486
2082			19,538,400	417	1,972	2,389
2083			19,538,400	401	1,895	2,295
2084			19,538,400	385	1,820	2,205
2085			19,538,400	370	1,749	2,119
2086			19,538,400	355	1,680	2,036
2087			19,538,400	341	1,615	1,956
2088			19,538,400	328	1,551	1,879
2089			19,538,400	315	1,490	1,806
2090			19,538,400	303	1,432	1,735
2091			19,538,400	291	1,376	1,667
2092			19,538,400	279	1,322	1,601
2093			19,538,400	269	1,270	1,539
2094			19,538,400	258	1,220	1,478
2095			19,538,400	248	1,172	1,420
2096			19,538,400	238	1,126	1,365
2097			19,538,400	229	1,082	1,311
2098			19,538,400	220	1,040	1,260
2099			19,538,400	211	999	1,210
2100			19,538,400	203	960	1,163
2101			19,538,400	195	922	1,117
2102			19,538,400	187	886	1,073
2103			19,538,400	180	851	1,031
2104			19,538,400	173	818	991
2105			19,538,400	166	786	952
2106			19,538,400	160	755	915
2107			19,538,400	153	725	879
2108			19,538,400	147	697	844
2109			19,538,400	142	670	811
2110			19,538,400	136	643	779
2111			19,538,400	131	618	749
2112			19,538,400	126	594	720
2113			19,538,400	121	571	691
2114			19,538,400	116	548	664



Summary Report

Landfill Name or Identifier: Wexford East Unit

Date: Tuesday, April 1, 2025

Description/Comments:

About LandGEM:

First-Order Decomposition Rate Equation:

$$Q_{CH_4} = \sum_{i=1}^n \sum_{j=0.1}^1 k L_o \left(\frac{M_i}{10} \right) e^{-k t_{ij}}$$

Where,

Q_{CH_4} = annual methane generation in the year of the calculation ($m^3/year$)

i = 1-year time increment

n = (year of the calculation) - (initial year of waste acceptance)

j = 0.1-year time increment

k = methane generation rate ($year^{-1}$)

L_o = potential methane generation capacity (m^3/Mg)

M_i = mass of waste accepted in the i^{th} year (Mg)

t_{ij} = age of the j^{th} section of waste mass M_i accepted in the i^{th} year ($decimal\ years$, e.g., 3.2 years)

LandGEM is based on a first-order decomposition rate equation for quantifying emissions from the decomposition of landfilled waste in municipal solid waste (MSW) landfills. The software provides a relatively simple approach to estimating landfill gas emissions. Model defaults are based on empirical data from U.S. landfills. Field test data can also be used in place of model defaults when available. Further guidance on EPA test methods, Clean Air Act (CAA) regulations, and other guidance regarding landfill gas emissions and control technology requirements can be found at <http://www.epa.gov/ttnatw01/landfill/landflpg.html>.

LandGEM is considered a screening tool — the better the input data, the better the estimates. Often, there are limitations with the available data regarding waste quantity and composition, variation in design and operating practices over time, and changes occurring over time that impact the emissions potential. Changes to landfill operation, such as operating under wet conditions through leachate recirculation or other liquid additions, will result in generating more gas at a faster rate. Defaults for estimating emissions for this type of operation are being developed to include in LandGEM along with defaults for conventional landfills (no leachate or liquid additions) for developing emission inventories and determining CAA applicability. Refer to the Web site identified above for future updates.

Input Review

LANDFILL CHARACTERISTICS

Landfill Open Year	1977	
Landfill Closure Year (with 80-year limit)	2036	
Actual Closure Year (without limit)	2036	
Have Model Calculate Closure Year?	No	
Waste Design Capacity	9,676,401	<i>short tons</i>

MODEL PARAMETERS

Methane Generation Rate, k	0.040	<i>year⁻¹</i>
Potential Methane Generation Capacity, L ₀	100	<i>m³/Mg</i>
NMOC Concentration	354	<i>ppmv as hexane</i>
Methane Content	50	<i>% by volume</i>

GASES / POLLUTANTS SELECTED

Gas / Pollutant #1:	Total landfill gas
Gas / Pollutant #2:	Methane
Gas / Pollutant #3:	Carbon dioxide
Gas / Pollutant #4:	NMOC

WASTE ACCEPTANCE RATES

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
1977	76,207	83,828	0	0
1978	58,545	64,400	76,207	83,828
1979	125,764	138,340	134,753	148,228
1980	104,417	114,859	260,516	286,568
1981	95,671	105,238	364,934	401,427
1982	87,856	96,642	460,605	506,665
1983	43,485	47,834	548,461	603,307
1984	58,171	63,988	591,946	651,141
1985	54,077	59,485	650,117	715,129
1986	54,178	59,596	704,195	774,614
1987	36,280	39,908	758,373	834,210
1988	43,193	47,512	794,653	874,118
1989	54,151	59,566	837,845	921,630
1990	50,106	55,117	891,996	981,196
1991	78,010	85,811	942,103	1,036,313
1992	71,503	78,653	1,020,113	1,122,124
1993	89,585	98,544	1,091,615	1,200,777
1994	111,148	122,263	1,181,201	1,299,321
1995	88,785	97,664	1,292,349	1,421,584
1996	86,372	95,009	1,381,135	1,519,248
1997	81,313	89,444	1,467,506	1,614,257
1998	91,085	100,193	1,548,819	1,703,701
1999	78,417	86,259	1,639,904	1,803,894
2000	133,630	146,993	1,718,321	1,890,153
2001	116,602	128,262	1,851,951	2,037,146
2002	117,393	129,132	1,968,553	2,165,408
2003	118,086	129,895	2,085,945	2,294,540
2004	109,646	120,611	2,204,032	2,424,435
2005	96,722	106,394	2,313,678	2,545,046
2006	32,855	36,141	2,410,400	2,651,440
2007	64,389	70,828	2,443,255	2,687,581
2008	51,981	57,179	2,507,645	2,758,409
2009	42,871	47,158	2,559,625	2,815,588
2010	59,536	65,490	2,602,496	2,862,746
2011	50,541	55,595	2,662,033	2,928,236
2012	108,993	119,892	2,712,574	2,983,831
2013	122,022	134,224	2,821,566	3,103,723
2014	139,873	153,860	2,943,588	3,237,947
2015	147,406	162,147	3,083,461	3,391,807
2016	160,919	177,011	3,230,867	3,553,954

WASTE ACCEPTANCE RATES (Continued)

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
2017	141,710	155,881	3,391,786	3,730,965
2018	170,071	187,078	3,533,496	3,886,846
2019	180,993	199,092	3,703,567	4,073,924
2020	205,490	226,039	3,884,560	4,273,016
2021	184,492	202,941	4,090,050	4,499,055
2022	182,827	201,109	4,274,542	4,701,996
2023	307,544	338,298	4,457,368	4,903,105
2024	295,813	325,394	4,764,912	5,241,403
2025	311,334	342,467	5,060,725	5,566,797
2026	311,334	342,467	5,372,058	5,909,264
2027	311,334	342,467	5,683,392	6,251,731
2028	311,334	342,467	5,994,726	6,594,198
2029	311,334	342,467	6,306,059	6,936,665
2030	311,334	342,467	6,617,393	7,279,132
2031	311,334	342,467	6,928,726	7,621,599
2032	311,334	342,467	7,240,060	7,964,066
2033	311,334	342,467	7,551,394	8,306,533
2034	311,334	342,467	7,862,727	8,649,000
2035	311,334	342,467	8,174,061	8,991,467
2036	311,334	342,467	8,485,395	9,333,934
2037	0	0	8,796,728	9,676,401
2038	0	0	8,796,728	9,676,401
2039	0	0	8,796,728	9,676,401
2040	0	0	8,796,728	9,676,401
2041	0	0	8,796,728	9,676,401
2042	0	0	8,796,728	9,676,401
2043	0	0	8,796,728	9,676,401
2044	0	0	8,796,728	9,676,401
2045	0	0	8,796,728	9,676,401
2046	0	0	8,796,728	9,676,401
2047	0	0	8,796,728	9,676,401
2048	0	0	8,796,728	9,676,401
2049	0	0	8,796,728	9,676,401
2050	0	0	8,796,728	9,676,401
2051	0	0	8,796,728	9,676,401
2052	0	0	8,796,728	9,676,401
2053	0	0	8,796,728	9,676,401
2054	0	0	8,796,728	9,676,401
2055	0	0	8,796,728	9,676,401
2056	0	0	8,796,728	9,676,401

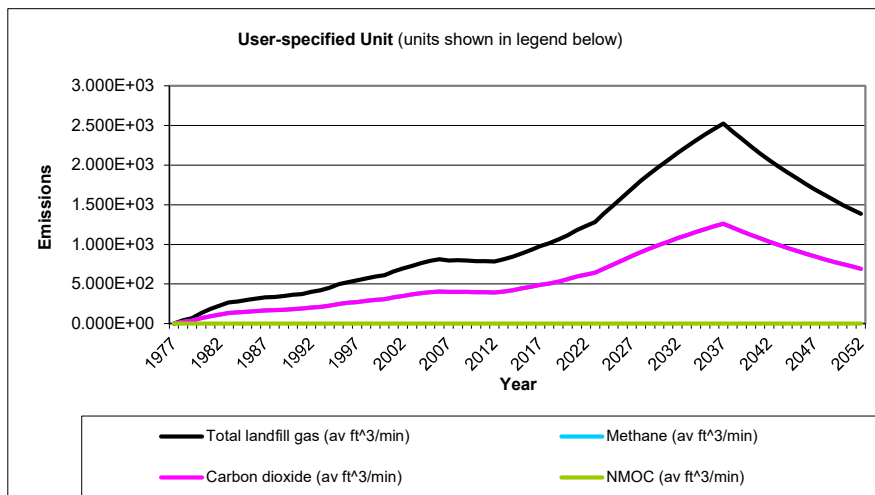
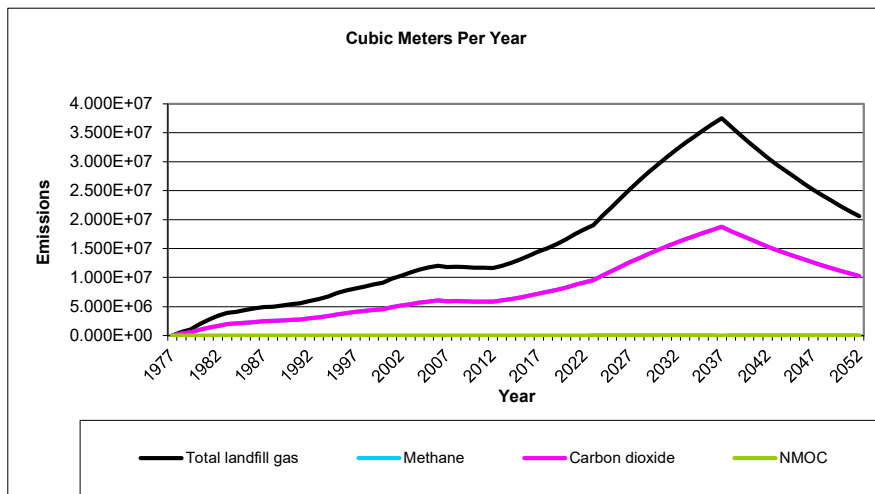
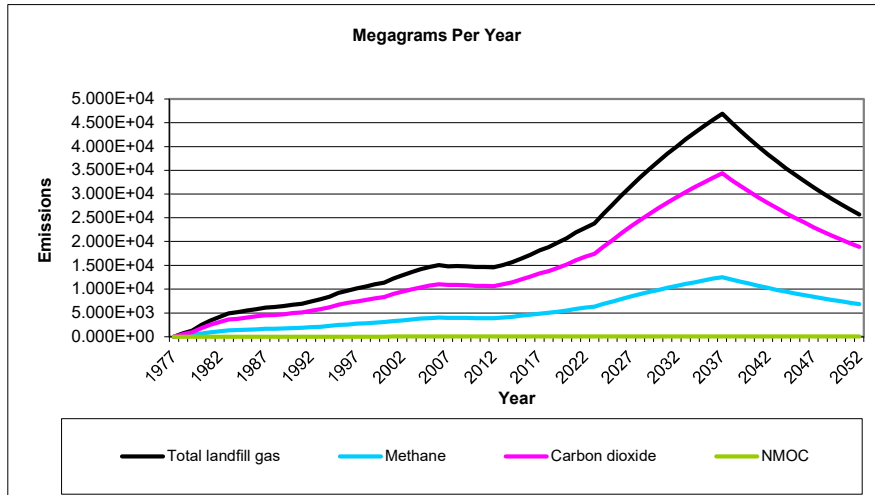
Pollutant Parameters

Gas / Pollutant Default Parameters:				User-specified Pollutant Parameters:	
	Compound	Concentration (ppmv)	Molecular Weight	Concentration (ppmv)	Molecular Weight
Gases	Total landfill gas		0.00		
	Methane		16.04		
	Carbon dioxide		44.01		
	NMOC	4,000	86.18		
Pollutants	1,1,1-Trichloroethane (methyl chloroform) - HAP	0.48	133.41		
	1,1,2,2- Tetrachloroethane - HAP/VOC	1.1	167.85		
	1,1-Dichloroethane (ethylidene dichloride) - HAP/VOC	2.4	98.97		
	1,1-Dichloroethene (vinylidene chloride) - HAP/VOC	0.20	96.94		
	1,2-Dichloroethane (ethylene dichloride) - HAP/VOC	0.41	98.96		
	1,2-Dichloropropane (propylene dichloride) - HAP/VOC	0.18	112.99		
	2-Propanol (isopropyl alcohol) - VOC	50	60.11		
	Acetone	7.0	58.08		
	Acrylonitrile - HAP/VOC	6.3	53.06		
	Benzene - No or Unknown Co-disposal - HAP/VOC	1.9	78.11		
	Benzene - Co-disposal - HAP/VOC	11	78.11		
	Bromodichloromethane - VOC	3.1	163.83		
	Butane - VOC	5.0	58.12		
	Carbon disulfide - HAP/VOC	0.58	76.13		
	Carbon monoxide	140	28.01		
	Carbon tetrachloride - HAP/VOC	4.0E-03	153.84		
	Carbonyl sulfide - HAP/VOC	0.49	60.07		
	Chlorobenzene - HAP/VOC	0.25	112.56		
	Chlorodifluoromethane	1.3	86.47		
	Chloroethane (ethyl chloride) - HAP/VOC	1.3	64.52		
	Chloroform - HAP/VOC	0.03	119.39		
	Chloromethane - VOC	1.2	50.49		
	Dichlorobenzene - (HAP for para isomer/VOC)	0.21	147		
	Dichlorodifluoromethane	16	120.91		
	Dichlorofluoromethane - VOC	2.6	102.92		
	Dichloromethane (methylene chloride) - HAP	14	84.94		
	Dimethyl sulfide (methyl sulfide) - VOC	7.8	62.13		
	Ethane	890	30.07		
	Ethanol - VOC	27	46.08		

Pollutant Parameters (Continued)

[illegible]

Graphs



Results

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
1977	0	0	0	0	0	0
1978	7.478E+02	5.988E+05	4.023E+01	1.998E+02	2.994E+05	2.012E+01
1979	1.293E+03	1.035E+06	6.957E+01	3.454E+02	5.177E+05	3.478E+01
1980	2.476E+03	1.983E+06	1.332E+02	6.615E+02	9.915E+05	6.662E+01
1981	3.404E+03	2.726E+06	1.831E+02	9.092E+02	1.363E+06	9.157E+01
1982	4.209E+03	3.371E+06	2.265E+02	1.124E+03	1.685E+06	1.132E+02
1983	4.906E+03	3.929E+06	2.640E+02	1.311E+03	1.964E+06	1.320E+02
1984	5.141E+03	4.116E+06	2.766E+02	1.373E+03	2.058E+06	1.383E+02
1985	5.510E+03	4.412E+06	2.965E+02	1.472E+03	2.206E+06	1.482E+02
1986	5.825E+03	4.664E+06	3.134E+02	1.556E+03	2.332E+06	1.567E+02
1987	6.128E+03	4.907E+06	3.297E+02	1.637E+03	2.453E+06	1.648E+02
1988	6.244E+03	5.000E+06	3.359E+02	1.668E+03	2.500E+06	1.680E+02
1989	6.423E+03	5.143E+06	3.456E+02	1.716E+03	2.571E+06	1.728E+02
1990	6.702E+03	5.367E+06	3.606E+02	1.790E+03	2.683E+06	1.803E+02
1991	6.931E+03	5.550E+06	3.729E+02	1.851E+03	2.775E+06	1.865E+02
1992	7.425E+03	5.945E+06	3.995E+02	1.983E+03	2.973E+06	1.997E+02
1993	7.835E+03	6.274E+06	4.216E+02	2.093E+03	3.137E+06	2.108E+02
1994	8.407E+03	6.732E+06	4.523E+02	2.246E+03	3.366E+06	2.262E+02
1995	9.168E+03	7.342E+06	4.933E+02	2.449E+03	3.671E+06	2.466E+02
1996	9.680E+03	7.751E+06	5.208E+02	2.586E+03	3.876E+06	2.604E+02
1997	1.015E+04	8.126E+06	5.460E+02	2.711E+03	4.063E+06	2.730E+02
1998	1.055E+04	8.446E+06	5.675E+02	2.818E+03	4.223E+06	2.838E+02
1999	1.103E+04	8.831E+06	5.933E+02	2.946E+03	4.415E+06	2.967E+02
2000	1.137E+04	9.101E+06	6.115E+02	3.036E+03	4.550E+06	3.057E+02
2001	1.223E+04	9.794E+06	6.581E+02	3.267E+03	4.897E+06	3.290E+02
2002	1.290E+04	1.033E+07	6.938E+02	3.445E+03	5.163E+06	3.469E+02
2003	1.354E+04	1.084E+07	7.286E+02	3.617E+03	5.422E+06	3.643E+02
2004	1.417E+04	1.135E+07	7.624E+02	3.785E+03	5.673E+06	3.812E+02
2005	1.469E+04	1.176E+07	7.904E+02	3.924E+03	5.882E+06	3.952E+02
2006	1.506E+04	1.206E+07	8.104E+02	4.024E+03	6.031E+06	4.052E+02
2007	1.480E+04	1.185E+07	7.960E+02	3.952E+03	5.924E+06	3.980E+02
2008	1.485E+04	1.189E+07	7.988E+02	3.966E+03	5.944E+06	3.994E+02
2009	1.477E+04	1.183E+07	7.949E+02	3.946E+03	5.915E+06	3.975E+02
2010	1.462E+04	1.170E+07	7.864E+02	3.904E+03	5.852E+06	3.932E+02
2011	1.463E+04	1.171E+07	7.870E+02	3.907E+03	5.856E+06	3.935E+02
2012	1.455E+04	1.165E+07	7.828E+02	3.886E+03	5.825E+06	3.914E+02
2013	1.505E+04	1.205E+07	8.097E+02	4.020E+03	6.025E+06	4.048E+02
2014	1.566E+04	1.254E+07	8.423E+02	4.182E+03	6.268E+06	4.212E+02
2015	1.641E+04	1.314E+07	8.832E+02	4.385E+03	6.572E+06	4.416E+02
2016	1.722E+04	1.379E+07	9.263E+02	4.599E+03	6.894E+06	4.632E+02
2017	1.812E+04	1.451E+07	9.750E+02	4.840E+03	7.255E+06	4.875E+02
2018	1.880E+04	1.506E+07	1.012E+03	5.022E+03	7.528E+06	5.058E+02
2019	1.973E+04	1.580E+07	1.062E+03	5.271E+03	7.901E+06	5.309E+02
2020	2.074E+04	1.660E+07	1.116E+03	5.539E+03	8.302E+06	5.578E+02
2021	2.194E+04	1.757E+07	1.180E+03	5.860E+03	8.784E+06	5.902E+02
2022	2.289E+04	1.833E+07	1.231E+03	6.114E+03	9.164E+06	6.157E+02
2023	2.379E+04	1.905E+07	1.280E+03	6.353E+03	9.523E+06	6.399E+02
2024	2.587E+04	2.072E+07	1.392E+03	6.910E+03	1.036E+07	6.960E+02
2025	2.776E+04	2.223E+07	1.494E+03	7.415E+03	1.111E+07	7.468E+02
2026	2.973E+04	2.380E+07	1.599E+03	7.940E+03	1.190E+07	7.997E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2027	3.162E+04	2.532E+07	1.701E+03	8.445E+03	1.266E+07	8.505E+02
2028	3.343E+04	2.677E+07	1.799E+03	8.930E+03	1.339E+07	8.993E+02
2029	3.518E+04	2.817E+07	1.893E+03	9.396E+03	1.408E+07	9.463E+02
2030	3.685E+04	2.951E+07	1.983E+03	9.843E+03	1.475E+07	9.913E+02
2031	3.846E+04	3.080E+07	2.069E+03	1.027E+04	1.540E+07	1.035E+03
2032	4.001E+04	3.204E+07	2.153E+03	1.069E+04	1.602E+07	1.076E+03
2033	4.149E+04	3.323E+07	2.233E+03	1.108E+04	1.661E+07	1.116E+03
2034	4.292E+04	3.437E+07	2.309E+03	1.147E+04	1.719E+07	1.155E+03
2035	4.430E+04	3.547E+07	2.383E+03	1.183E+04	1.773E+07	1.192E+03
2036	4.561E+04	3.653E+07	2.454E+03	1.218E+04	1.826E+07	1.227E+03
2037	4.688E+04	3.754E+07	2.522E+03	1.252E+04	1.877E+07	1.261E+03
2038	4.504E+04	3.607E+07	2.423E+03	1.203E+04	1.803E+07	1.212E+03
2039	4.328E+04	3.465E+07	2.328E+03	1.156E+04	1.733E+07	1.164E+03
2040	4.158E+04	3.329E+07	2.237E+03	1.111E+04	1.665E+07	1.119E+03
2041	3.995E+04	3.199E+07	2.149E+03	1.067E+04	1.599E+07	1.075E+03
2042	3.838E+04	3.073E+07	2.065E+03	1.025E+04	1.537E+07	1.033E+03
2043	3.688E+04	2.953E+07	1.984E+03	9.850E+03	1.476E+07	9.920E+02
2044	3.543E+04	2.837E+07	1.906E+03	9.464E+03	1.419E+07	9.531E+02
2045	3.404E+04	2.726E+07	1.832E+03	9.093E+03	1.363E+07	9.158E+02
2046	3.271E+04	2.619E+07	1.760E+03	8.736E+03	1.310E+07	8.799E+02
2047	3.142E+04	2.516E+07	1.691E+03	8.394E+03	1.258E+07	8.454E+02
2048	3.019E+04	2.418E+07	1.624E+03	8.065E+03	1.209E+07	8.122E+02
2049	2.901E+04	2.323E+07	1.561E+03	7.748E+03	1.161E+07	7.804E+02
2050	2.787E+04	2.232E+07	1.500E+03	7.445E+03	1.116E+07	7.498E+02
2051	2.678E+04	2.144E+07	1.441E+03	7.153E+03	1.072E+07	7.204E+02
2052	2.573E+04	2.060E+07	1.384E+03	6.872E+03	1.030E+07	6.921E+02
2053	2.472E+04	1.979E+07	1.330E+03	6.603E+03	9.897E+06	6.650E+02
2054	2.375E+04	1.902E+07	1.278E+03	6.344E+03	9.509E+06	6.389E+02
2055	2.282E+04	1.827E+07	1.228E+03	6.095E+03	9.136E+06	6.139E+02
2056	2.192E+04	1.756E+07	1.180E+03	5.856E+03	8.778E+06	5.898E+02
2057	2.106E+04	1.687E+07	1.133E+03	5.627E+03	8.434E+06	5.667E+02
2058	2.024E+04	1.621E+07	1.089E+03	5.406E+03	8.103E+06	5.444E+02
2059	1.944E+04	1.557E+07	1.046E+03	5.194E+03	7.785E+06	5.231E+02
2060	1.868E+04	1.496E+07	1.005E+03	4.990E+03	7.480E+06	5.026E+02
2061	1.795E+04	1.437E+07	9.658E+02	4.795E+03	7.187E+06	4.829E+02
2062	1.725E+04	1.381E+07	9.279E+02	4.607E+03	6.905E+06	4.639E+02
2063	1.657E+04	1.327E+07	8.915E+02	4.426E+03	6.634E+06	4.458E+02
2064	1.592E+04	1.275E+07	8.565E+02	4.252E+03	6.374E+06	4.283E+02
2065	1.530E+04	1.225E+07	8.230E+02	4.086E+03	6.124E+06	4.115E+02
2066	1.470E+04	1.177E+07	7.907E+02	3.926E+03	5.884E+06	3.953E+02
2067	1.412E+04	1.131E+07	7.597E+02	3.772E+03	5.653E+06	3.798E+02
2068	1.357E+04	1.086E+07	7.299E+02	3.624E+03	5.432E+06	3.650E+02
2069	1.303E+04	1.044E+07	7.013E+02	3.482E+03	5.219E+06	3.506E+02
2070	1.252E+04	1.003E+07	6.738E+02	3.345E+03	5.014E+06	3.369E+02
2071	1.203E+04	9.635E+06	6.474E+02	3.214E+03	4.817E+06	3.237E+02
2072	1.156E+04	9.257E+06	6.220E+02	3.088E+03	4.629E+06	3.110E+02
2073	1.111E+04	8.894E+06	5.976E+02	2.967E+03	4.447E+06	2.988E+02
2074	1.067E+04	8.545E+06	5.742E+02	2.851E+03	4.273E+06	2.871E+02
2075	1.025E+04	8.210E+06	5.516E+02	2.739E+03	4.105E+06	2.758E+02
2076	9.851E+03	7.888E+06	5.300E+02	2.631E+03	3.944E+06	2.650E+02
2077	9.465E+03	7.579E+06	5.092E+02	2.528E+03	3.790E+06	2.546E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2078	9.094E+03	7.282E+06	4.893E+02	2.429E+03	3.641E+06	2.446E+02
2079	8.737E+03	6.996E+06	4.701E+02	2.334E+03	3.498E+06	2.350E+02
2080	8.395E+03	6.722E+06	4.517E+02	2.242E+03	3.361E+06	2.258E+02
2081	8.065E+03	6.458E+06	4.339E+02	2.154E+03	3.229E+06	2.170E+02
2082	7.749E+03	6.205E+06	4.169E+02	2.070E+03	3.103E+06	2.085E+02
2083	7.445E+03	5.962E+06	4.006E+02	1.989E+03	2.981E+06	2.003E+02
2084	7.153E+03	5.728E+06	3.849E+02	1.911E+03	2.864E+06	1.924E+02
2085	6.873E+03	5.504E+06	3.698E+02	1.836E+03	2.752E+06	1.849E+02
2086	6.603E+03	5.288E+06	3.553E+02	1.764E+03	2.644E+06	1.776E+02
2087	6.345E+03	5.080E+06	3.414E+02	1.695E+03	2.540E+06	1.707E+02
2088	6.096E+03	4.881E+06	3.280E+02	1.628E+03	2.441E+06	1.640E+02
2089	5.857E+03	4.690E+06	3.151E+02	1.564E+03	2.345E+06	1.576E+02
2090	5.627E+03	4.506E+06	3.028E+02	1.503E+03	2.253E+06	1.514E+02
2091	5.406E+03	4.329E+06	2.909E+02	1.444E+03	2.165E+06	1.454E+02
2092	5.194E+03	4.159E+06	2.795E+02	1.387E+03	2.080E+06	1.397E+02
2093	4.991E+03	3.996E+06	2.685E+02	1.333E+03	1.998E+06	1.343E+02
2094	4.795E+03	3.840E+06	2.580E+02	1.281E+03	1.920E+06	1.290E+02
2095	4.607E+03	3.689E+06	2.479E+02	1.231E+03	1.845E+06	1.239E+02
2096	4.426E+03	3.544E+06	2.382E+02	1.182E+03	1.772E+06	1.191E+02
2097	4.253E+03	3.405E+06	2.288E+02	1.136E+03	1.703E+06	1.144E+02
2098	4.086E+03	3.272E+06	2.198E+02	1.091E+03	1.636E+06	1.099E+02
2099	3.926E+03	3.144E+06	2.112E+02	1.049E+03	1.572E+06	1.056E+02
2100	3.772E+03	3.020E+06	2.029E+02	1.008E+03	1.510E+06	1.015E+02
2101	3.624E+03	2.902E+06	1.950E+02	9.680E+02	1.451E+06	9.749E+01
2102	3.482E+03	2.788E+06	1.873E+02	9.301E+02	1.394E+06	9.367E+01
2103	3.345E+03	2.679E+06	1.800E+02	8.936E+02	1.339E+06	9.000E+01
2104	3.214E+03	2.574E+06	1.729E+02	8.586E+02	1.287E+06	8.647E+01
2105	3.088E+03	2.473E+06	1.662E+02	8.249E+02	1.236E+06	8.308E+01
2106	2.967E+03	2.376E+06	1.596E+02	7.925E+02	1.188E+06	7.982E+01
2107	2.851E+03	2.283E+06	1.534E+02	7.615E+02	1.141E+06	7.669E+01
2108	2.739E+03	2.193E+06	1.474E+02	7.316E+02	1.097E+06	7.368E+01
2109	2.632E+03	2.107E+06	1.416E+02	7.029E+02	1.054E+06	7.079E+01
2110	2.528E+03	2.025E+06	1.360E+02	6.754E+02	1.012E+06	6.802E+01
2111	2.429E+03	1.945E+06	1.307E+02	6.489E+02	9.726E+05	6.535E+01
2112	2.334E+03	1.869E+06	1.256E+02	6.234E+02	9.345E+05	6.279E+01
2113	2.242E+03	1.796E+06	1.207E+02	5.990E+02	8.978E+05	6.033E+01
2114	2.155E+03	1.725E+06	1.159E+02	5.755E+02	8.626E+05	5.796E+01
2115	2.070E+03	1.658E+06	1.114E+02	5.529E+02	8.288E+05	5.569E+01
2116	1.989E+03	1.593E+06	1.070E+02	5.313E+02	7.963E+05	5.350E+01
2117	1.911E+03	1.530E+06	1.028E+02	5.104E+02	7.651E+05	5.141E+01

Results (Continued)

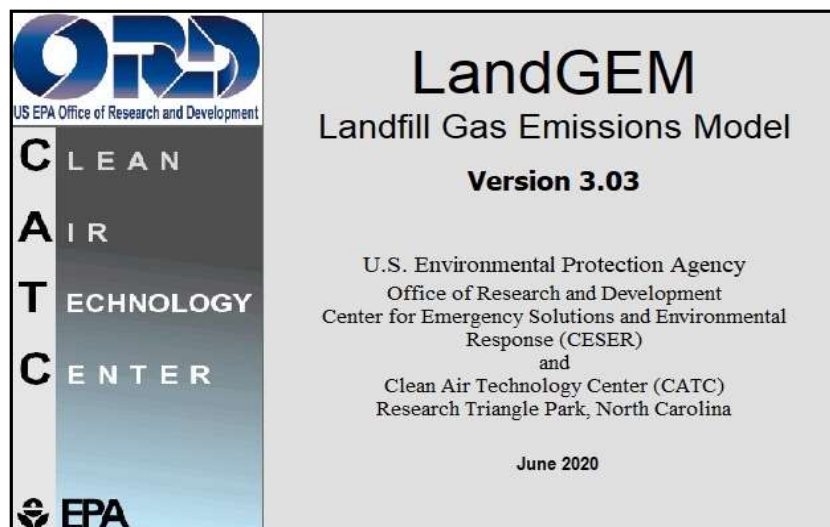
Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
1977	0	0	0	0	0	0
1978	5.481E+02	2.994E+05	2.012E+01	7.598E-01	2.120E+02	1.424E-02
1979	9.476E+02	5.177E+05	3.478E+01	1.314E+00	3.665E+02	2.463E-02
1980	1.815E+03	9.915E+05	6.662E+01	2.516E+00	7.020E+02	4.717E-02
1981	2.495E+03	1.363E+06	9.157E+01	3.459E+00	9.649E+02	6.483E-02
1982	3.085E+03	1.685E+06	1.132E+02	4.277E+00	1.193E+03	8.017E-02
1983	3.596E+03	1.964E+06	1.320E+02	4.985E+00	1.391E+03	9.345E-02
1984	3.768E+03	2.058E+06	1.383E+02	5.223E+00	1.457E+03	9.791E-02
1985	4.038E+03	2.206E+06	1.482E+02	5.599E+00	1.562E+03	1.049E-01
1986	4.269E+03	2.332E+06	1.567E+02	5.918E+00	1.651E+03	1.109E-01
1987	4.491E+03	2.453E+06	1.648E+02	6.226E+00	1.737E+03	1.167E-01
1988	4.576E+03	2.500E+06	1.680E+02	6.344E+00	1.770E+03	1.189E-01
1989	4.707E+03	2.571E+06	1.728E+02	6.526E+00	1.821E+03	1.223E-01
1990	4.912E+03	2.683E+06	1.803E+02	6.810E+00	1.900E+03	1.277E-01
1991	5.080E+03	2.775E+06	1.865E+02	7.043E+00	1.965E+03	1.320E-01
1992	5.442E+03	2.973E+06	1.997E+02	7.544E+00	2.105E+03	1.414E-01
1993	5.742E+03	3.137E+06	2.108E+02	7.961E+00	2.221E+03	1.492E-01
1994	6.162E+03	3.366E+06	2.262E+02	8.542E+00	2.383E+03	1.601E-01
1995	6.719E+03	3.671E+06	2.466E+02	9.316E+00	2.599E+03	1.746E-01
1996	7.094E+03	3.876E+06	2.604E+02	9.836E+00	2.744E+03	1.844E-01
1997	7.437E+03	4.063E+06	2.730E+02	1.031E+01	2.877E+03	1.933E-01
1998	7.731E+03	4.223E+06	2.838E+02	1.072E+01	2.990E+03	2.009E-01
1999	8.083E+03	4.415E+06	2.967E+02	1.121E+01	3.126E+03	2.100E-01
2000	8.330E+03	4.550E+06	3.057E+02	1.155E+01	3.222E+03	2.165E-01
2001	8.964E+03	4.897E+06	3.290E+02	1.243E+01	3.467E+03	2.330E-01
2002	9.451E+03	5.163E+06	3.469E+02	1.310E+01	3.655E+03	2.456E-01
2003	9.925E+03	5.422E+06	3.643E+02	1.376E+01	3.839E+03	2.579E-01
2004	1.038E+04	5.673E+06	3.812E+02	1.440E+01	4.017E+03	2.699E-01
2005	1.077E+04	5.882E+06	3.952E+02	1.493E+01	4.164E+03	2.798E-01
2006	1.104E+04	6.031E+06	4.052E+02	1.531E+01	4.270E+03	2.869E-01
2007	1.084E+04	5.924E+06	3.980E+02	1.503E+01	4.194E+03	2.818E-01
2008	1.088E+04	5.944E+06	3.994E+02	1.509E+01	4.209E+03	2.828E-01
2009	1.083E+04	5.915E+06	3.975E+02	1.501E+01	4.188E+03	2.814E-01
2010	1.071E+04	5.852E+06	3.932E+02	1.485E+01	4.143E+03	2.784E-01
2011	1.072E+04	5.856E+06	3.935E+02	1.486E+01	4.146E+03	2.786E-01
2012	1.066E+04	5.825E+06	3.914E+02	1.478E+01	4.124E+03	2.771E-01
2013	1.103E+04	6.025E+06	4.048E+02	1.529E+01	4.266E+03	2.866E-01
2014	1.147E+04	6.268E+06	4.212E+02	1.591E+01	4.438E+03	2.982E-01
2015	1.203E+04	6.572E+06	4.416E+02	1.668E+01	4.653E+03	3.126E-01
2016	1.262E+04	6.894E+06	4.632E+02	1.749E+01	4.881E+03	3.279E-01
2017	1.328E+04	7.255E+06	4.875E+02	1.841E+01	5.137E+03	3.451E-01
2018	1.378E+04	7.528E+06	5.058E+02	1.910E+01	5.330E+03	3.581E-01
2019	1.446E+04	7.901E+06	5.309E+02	2.005E+01	5.594E+03	3.758E-01
2020	1.520E+04	8.302E+06	5.578E+02	2.107E+01	5.878E+03	3.949E-01
2021	1.608E+04	8.784E+06	5.902E+02	2.229E+01	6.219E+03	4.179E-01
2022	1.678E+04	9.164E+06	6.157E+02	2.326E+01	6.488E+03	4.360E-01
2023	1.743E+04	9.523E+06	6.399E+02	2.417E+01	6.742E+03	4.530E-01
2024	1.896E+04	1.036E+07	6.960E+02	2.629E+01	7.334E+03	4.927E-01
2025	2.034E+04	1.111E+07	7.468E+02	2.821E+01	7.869E+03	5.287E-01
2026	2.179E+04	1.190E+07	7.997E+02	3.020E+01	8.426E+03	5.662E-01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2027	2.317E+04	1.266E+07	8.505E+02	3.212E+01	8.962E+03	6.022E-01
2028	2.450E+04	1.339E+07	8.993E+02	3.397E+01	9.477E+03	6.367E-01
2029	2.578E+04	1.408E+07	9.463E+02	3.574E+01	9.971E+03	6.700E-01
2030	2.701E+04	1.475E+07	9.913E+02	3.744E+01	1.045E+04	7.019E-01
2031	2.819E+04	1.540E+07	1.035E+03	3.908E+01	1.090E+04	7.325E-01
2032	2.932E+04	1.602E+07	1.076E+03	4.065E+01	1.134E+04	7.620E-01
2033	3.041E+04	1.661E+07	1.116E+03	4.216E+01	1.176E+04	7.903E-01
2034	3.146E+04	1.719E+07	1.155E+03	4.361E+01	1.217E+04	8.175E-01
2035	3.246E+04	1.773E+07	1.192E+03	4.501E+01	1.256E+04	8.436E-01
2036	3.343E+04	1.826E+07	1.227E+03	4.635E+01	1.293E+04	8.688E-01
2037	3.436E+04	1.877E+07	1.261E+03	4.763E+01	1.329E+04	8.929E-01
2038	3.301E+04	1.803E+07	1.212E+03	4.577E+01	1.277E+04	8.579E-01
2039	3.172E+04	1.733E+07	1.164E+03	4.397E+01	1.227E+04	8.242E-01
2040	3.047E+04	1.665E+07	1.119E+03	4.225E+01	1.179E+04	7.919E-01
2041	2.928E+04	1.599E+07	1.075E+03	4.059E+01	1.132E+04	7.609E-01
2042	2.813E+04	1.537E+07	1.033E+03	3.900E+01	1.088E+04	7.310E-01
2043	2.703E+04	1.476E+07	9.920E+02	3.747E+01	1.045E+04	7.024E-01
2044	2.597E+04	1.419E+07	9.531E+02	3.600E+01	1.004E+04	6.748E-01
2045	2.495E+04	1.363E+07	9.158E+02	3.459E+01	9.650E+03	6.484E-01
2046	2.397E+04	1.310E+07	8.799E+02	3.323E+01	9.271E+03	6.229E-01
2047	2.303E+04	1.258E+07	8.454E+02	3.193E+01	8.908E+03	5.985E-01
2048	2.213E+04	1.209E+07	8.122E+02	3.068E+01	8.559E+03	5.750E-01
2049	2.126E+04	1.161E+07	7.804E+02	2.947E+01	8.223E+03	5.525E-01
2050	2.043E+04	1.116E+07	7.498E+02	2.832E+01	7.901E+03	5.308E-01
2051	1.963E+04	1.072E+07	7.204E+02	2.721E+01	7.591E+03	5.100E-01
2052	1.886E+04	1.030E+07	6.921E+02	2.614E+01	7.293E+03	4.900E-01
2053	1.812E+04	9.897E+06	6.650E+02	2.512E+01	7.007E+03	4.708E-01
2054	1.741E+04	9.509E+06	6.389E+02	2.413E+01	6.732E+03	4.523E-01
2055	1.672E+04	9.136E+06	6.139E+02	2.319E+01	6.468E+03	4.346E-01
2056	1.607E+04	8.778E+06	5.898E+02	2.228E+01	6.215E+03	4.176E-01
2057	1.544E+04	8.434E+06	5.667E+02	2.140E+01	5.971E+03	4.012E-01
2058	1.483E+04	8.103E+06	5.444E+02	2.056E+01	5.737E+03	3.855E-01
2059	1.425E+04	7.785E+06	5.231E+02	1.976E+01	5.512E+03	3.704E-01
2060	1.369E+04	7.480E+06	5.026E+02	1.898E+01	5.296E+03	3.558E-01
2061	1.316E+04	7.187E+06	4.829E+02	1.824E+01	5.088E+03	3.419E-01
2062	1.264E+04	6.905E+06	4.639E+02	1.752E+01	4.889E+03	3.285E-01
2063	1.214E+04	6.634E+06	4.458E+02	1.684E+01	4.697E+03	3.156E-01
2064	1.167E+04	6.374E+06	4.283E+02	1.618E+01	4.513E+03	3.032E-01
2065	1.121E+04	6.124E+06	4.115E+02	1.554E+01	4.336E+03	2.913E-01
2066	1.077E+04	5.884E+06	3.953E+02	1.493E+01	4.166E+03	2.799E-01
2067	1.035E+04	5.653E+06	3.798E+02	1.435E+01	4.003E+03	2.689E-01
2068	9.943E+03	5.432E+06	3.650E+02	1.378E+01	3.846E+03	2.584E-01
2069	9.553E+03	5.219E+06	3.506E+02	1.324E+01	3.695E+03	2.483E-01
2070	9.178E+03	5.014E+06	3.369E+02	1.272E+01	3.550E+03	2.385E-01
2071	8.818E+03	4.817E+06	3.237E+02	1.223E+01	3.411E+03	2.292E-01
2072	8.473E+03	4.629E+06	3.110E+02	1.175E+01	3.277E+03	2.202E-01
2073	8.140E+03	4.447E+06	2.988E+02	1.129E+01	3.149E+03	2.115E-01
2074	7.821E+03	4.273E+06	2.871E+02	1.084E+01	3.025E+03	2.033E-01
2075	7.514E+03	4.105E+06	2.758E+02	1.042E+01	2.906E+03	1.953E-01
2076	7.220E+03	3.944E+06	2.650E+02	1.001E+01	2.792E+03	1.876E-01
2077	6.937E+03	3.790E+06	2.546E+02	9.617E+00	2.683E+03	1.803E-01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2078	6.665E+03	3.641E+06	2.446E+02	9.240E+00	2.578E+03	1.732E-01
2079	6.403E+03	3.498E+06	2.350E+02	8.878E+00	2.477E+03	1.664E-01
2080	6.152E+03	3.361E+06	2.258E+02	8.530E+00	2.380E+03	1.599E-01
2081	5.911E+03	3.229E+06	2.170E+02	8.195E+00	2.286E+03	1.536E-01
2082	5.679E+03	3.103E+06	2.085E+02	7.874E+00	2.197E+03	1.476E-01
2083	5.457E+03	2.981E+06	2.003E+02	7.565E+00	2.111E+03	1.418E-01
2084	5.243E+03	2.864E+06	1.924E+02	7.268E+00	2.028E+03	1.362E-01
2085	5.037E+03	2.752E+06	1.849E+02	6.983E+00	1.948E+03	1.309E-01
2086	4.840E+03	2.644E+06	1.776E+02	6.710E+00	1.872E+03	1.258E-01
2087	4.650E+03	2.540E+06	1.707E+02	6.447E+00	1.798E+03	1.208E-01
2088	4.467E+03	2.441E+06	1.640E+02	6.194E+00	1.728E+03	1.161E-01
2089	4.292E+03	2.345E+06	1.576E+02	5.951E+00	1.660E+03	1.115E-01
2090	4.124E+03	2.253E+06	1.514E+02	5.718E+00	1.595E+03	1.072E-01
2091	3.962E+03	2.165E+06	1.454E+02	5.493E+00	1.533E+03	1.030E-01
2092	3.807E+03	2.080E+06	1.397E+02	5.278E+00	1.472E+03	9.893E-02
2093	3.658E+03	1.998E+06	1.343E+02	5.071E+00	1.415E+03	9.505E-02
2094	3.514E+03	1.920E+06	1.290E+02	4.872E+00	1.359E+03	9.133E-02
2095	3.376E+03	1.845E+06	1.239E+02	4.681E+00	1.306E+03	8.775E-02
2096	3.244E+03	1.772E+06	1.191E+02	4.498E+00	1.255E+03	8.431E-02
2097	3.117E+03	1.703E+06	1.144E+02	4.321E+00	1.206E+03	8.100E-02
2098	2.995E+03	1.636E+06	1.099E+02	4.152E+00	1.158E+03	7.782E-02
2099	2.877E+03	1.572E+06	1.056E+02	3.989E+00	1.113E+03	7.477E-02
2100	2.764E+03	1.510E+06	1.015E+02	3.833E+00	1.069E+03	7.184E-02
2101	2.656E+03	1.451E+06	9.749E+01	3.682E+00	1.027E+03	6.902E-02
2102	2.552E+03	1.394E+06	9.367E+01	3.538E+00	9.870E+02	6.632E-02
2103	2.452E+03	1.339E+06	9.000E+01	3.399E+00	9.483E+02	6.372E-02
2104	2.356E+03	1.287E+06	8.647E+01	3.266E+00	9.111E+02	6.122E-02
2105	2.263E+03	1.236E+06	8.308E+01	3.138E+00	8.754E+02	5.882E-02
2106	2.175E+03	1.188E+06	7.982E+01	3.015E+00	8.411E+02	5.651E-02
2107	2.089E+03	1.141E+06	7.669E+01	2.897E+00	8.081E+02	5.430E-02
2108	2.007E+03	1.097E+06	7.368E+01	2.783E+00	7.764E+02	5.217E-02
2109	1.929E+03	1.054E+06	7.079E+01	2.674E+00	7.460E+02	5.012E-02
2110	1.853E+03	1.012E+06	6.802E+01	2.569E+00	7.167E+02	4.816E-02
2111	1.780E+03	9.726E+05	6.535E+01	2.468E+00	6.886E+02	4.627E-02
2112	1.711E+03	9.345E+05	6.279E+01	2.372E+00	6.616E+02	4.445E-02
2113	1.644E+03	8.978E+05	6.033E+01	2.279E+00	6.357E+02	4.271E-02
2114	1.579E+03	8.626E+05	5.796E+01	2.189E+00	6.107E+02	4.104E-02
2115	1.517E+03	8.288E+05	5.569E+01	2.103E+00	5.868E+02	3.943E-02
2116	1.458E+03	7.963E+05	5.350E+01	2.021E+00	5.638E+02	3.788E-02
2117	1.400E+03	7.651E+05	5.141E+01	1.942E+00	5.417E+02	3.640E-02



Summary Report

Landfill Name or Identifier: Wexford Landfill - West Unit

Date: Tuesday, April 1, 2025

Description/Comments:

About LandGEM:

First-Order Decomposition Rate Equation:

$$Q_{CH_4} = \sum_{i=1}^n \sum_{j=0.1}^1 k L_o \left(\frac{M_i}{10} \right) e^{-k t_{ij}}$$

Where,

Q_{CH_4} = annual methane generation in the year of the calculation ($m^3/year$)

i = 1-year time increment

n = (year of the calculation) - (initial year of waste acceptance)

j = 0.1-year time increment

k = methane generation rate ($year^{-1}$)

L_o = potential methane generation capacity (m^3/Mg)

M_i = mass of waste accepted in the i^{th} year (Mg)

t_{ij} = age of the j^{th} section of waste mass M_i accepted in the i^{th} year ($decimal\ years$, e.g., 3.2 years)

LandGEM is based on a first-order decomposition rate equation for quantifying emissions from the decomposition of landfilled waste in municipal solid waste (MSW) landfills. The software provides a relatively simple approach to estimating landfill gas emissions. Model defaults are based on empirical data from U.S. landfills. Field test data can also be used in place of model defaults when available. Further guidance on EPA test methods, Clean Air Act (CAA) regulations, and other guidance regarding landfill gas emissions and control technology requirements can be found at <http://www.epa.gov/ttnatw01/landfill/landflpg.html>.

LandGEM is considered a screening tool — the better the input data, the better the estimates. Often, there are limitations with the available data regarding waste quantity and composition, variation in design and operating practices over time, and changes occurring over time that impact the emissions potential. Changes to landfill operation, such as operating under wet conditions through leachate recirculation or other liquid additions, will result in generating more gas at a faster rate. Defaults for estimating emissions for this type of operation are being developed to include in LandGEM along with defaults for conventional landfills (no leachate or liquid additions) for developing emission inventories and determining CAA applicability. Refer to the Web site identified above for future updates.

Input Review

LANDFILL CHARACTERISTICS

Landfill Open Year	2037	
Landfill Closure Year (with 80-year limit)	2076	
Actual Closure Year (without limit)	2076	
Have Model Calculate Closure Year?	No	
Waste Design Capacity	9,861,999	<i>short tons</i>

MODEL PARAMETERS

Methane Generation Rate, k	0.040	<i>year⁻¹</i>
Potential Methane Generation Capacity, L ₀	100	<i>m³/Mg</i>
NMOC Concentration	354	<i>ppmv as hexane</i>
Methane Content	50	<i>% by volume</i>

GASES / POLLUTANTS SELECTED

Gas / Pollutant #1:	Total landfill gas
Gas / Pollutant #2:	Methane
Gas / Pollutant #3:	Carbon dioxide
Gas / Pollutant #4:	NMOC

WASTE ACCEPTANCE RATES

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
2037	224,136	246,550	0	0
2038	224,136	246,550	224,136	246,550
2039	224,136	246,550	448,273	493,100
2040	224,136	246,550	672,409	739,650
2041	224,136	246,550	896,545	986,200
2042	224,136	246,550	1,120,682	1,232,750
2043	224,136	246,550	1,344,818	1,479,300
2044	224,136	246,550	1,568,954	1,725,850
2045	224,136	246,550	1,793,091	1,972,400
2046	224,136	246,550	2,017,227	2,218,950
2047	224,136	246,550	2,241,363	2,465,500
2048	224,136	246,550	2,465,500	2,712,050
2049	224,136	246,550	2,689,636	2,958,600
2050	224,136	246,550	2,913,772	3,205,150
2051	224,136	246,550	3,137,909	3,451,700
2052	224,136	246,550	3,362,045	3,698,250
2053	224,136	246,550	3,586,181	3,944,800
2054	224,136	246,550	3,810,318	4,191,350
2055	224,136	246,550	4,034,454	4,437,900
2056	224,136	246,550	4,258,590	4,684,450
2057	224,136	246,550	4,482,727	4,931,000
2058	224,136	246,550	4,706,863	5,177,549
2059	224,136	246,550	4,931,000	5,424,099
2060	224,136	246,550	5,155,136	5,670,649
2061	224,136	246,550	5,379,272	5,917,199
2062	224,136	246,550	5,603,409	6,163,749
2063	224,136	246,550	5,827,545	6,410,299
2064	224,136	246,550	6,051,681	6,656,849
2065	224,136	246,550	6,275,818	6,903,399
2066	224,136	246,550	6,499,954	7,149,949
2067	224,136	246,550	6,724,090	7,396,499
2068	224,136	246,550	6,948,227	7,643,049
2069	224,136	246,550	7,172,363	7,889,599
2070	224,136	246,550	7,396,499	8,136,149
2071	224,136	246,550	7,620,636	8,382,699
2072	224,136	246,550	7,844,772	8,629,249
2073	224,136	246,550	8,068,908	8,875,799
2074	224,136	246,550	8,293,045	9,122,349
2075	224,136	246,550	8,517,181	9,368,899
2076	224,136	246,550	8,741,317	9,615,449

WASTE ACCEPTANCE RATES (Continued)

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
2077	0	0	8,965,454	9,861,999
2078	0	0	8,965,454	9,861,999
2079	0	0	8,965,454	9,861,999
2080	0	0	8,965,454	9,861,999
2081	0	0	8,965,454	9,861,999
2082	0	0	8,965,454	9,861,999
2083	0	0	8,965,454	9,861,999
2084	0	0	8,965,454	9,861,999
2085	0	0	8,965,454	9,861,999
2086	0	0	8,965,454	9,861,999
2087	0	0	8,965,454	9,861,999
2088	0	0	8,965,454	9,861,999
2089	0	0	8,965,454	9,861,999
2090	0	0	8,965,454	9,861,999
2091	0	0	8,965,454	9,861,999
2092	0	0	8,965,454	9,861,999
2093	0	0	8,965,454	9,861,999
2094	0	0	8,965,454	9,861,999
2095	0	0	8,965,454	9,861,999
2096	0	0	8,965,454	9,861,999
2097	0	0	8,965,454	9,861,999
2098	0	0	8,965,454	9,861,999
2099	0	0	8,965,454	9,861,999
2100	0	0	8,965,454	9,861,999
2101	0	0	8,965,454	9,861,999
2102	0	0	8,965,454	9,861,999
2103	0	0	8,965,454	9,861,999
2104	0	0	8,965,454	9,861,999
2105	0	0	8,965,454	9,861,999
2106	0	0	8,965,454	9,861,999
2107	0	0	8,965,454	9,861,999
2108	0	0	8,965,454	9,861,999
2109	0	0	8,965,454	9,861,999
2110	0	0	8,965,454	9,861,999
2111	0	0	8,965,454	9,861,999
2112	0	0	8,965,454	9,861,999
2113	0	0	8,965,454	9,861,999
2114	0	0	8,965,454	9,861,999
2115	0	0	8,965,454	9,861,999
2116	0	0	8,965,454	9,861,999

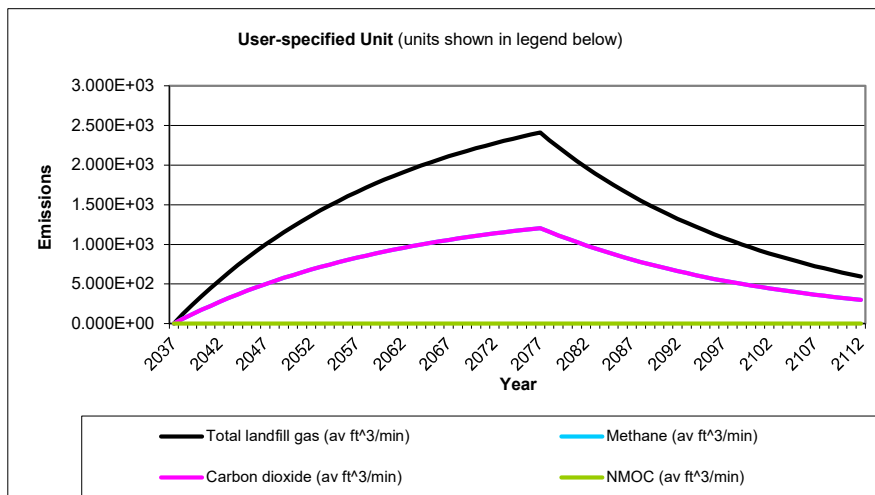
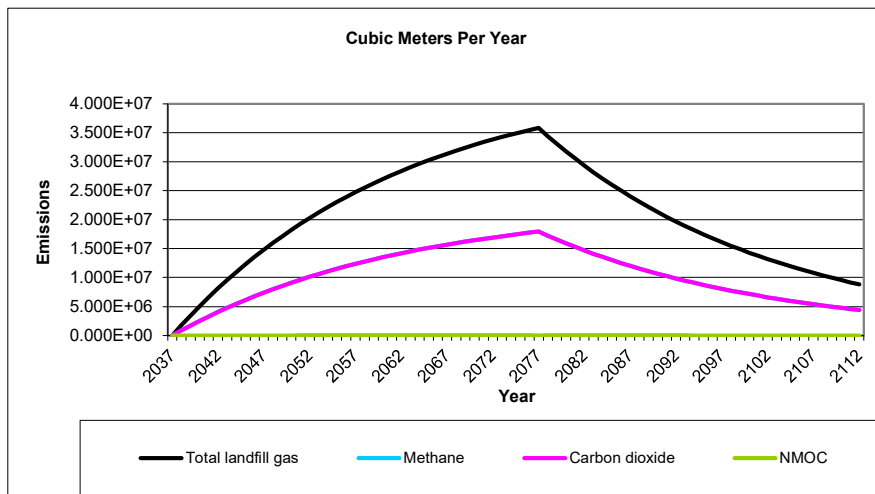
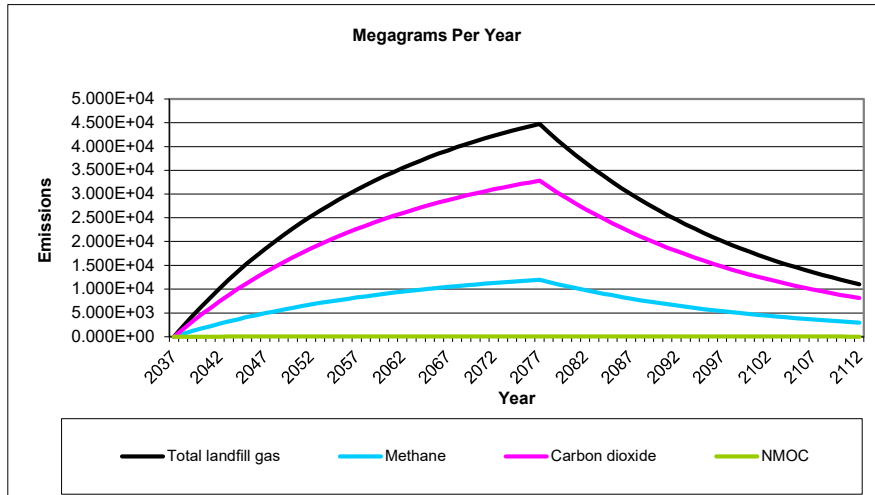
Pollutant Parameters

Gas / Pollutant Default Parameters:				User-specified Pollutant Parameters:	
	Compound	Concentration (ppmv)	Molecular Weight	Concentration (ppmv)	Molecular Weight
Gases	Total landfill gas		0.00		
	Methane		16.04		
	Carbon dioxide		44.01		
	NMOC	4,000	86.18		
Pollutants	1,1,1-Trichloroethane (methyl chloroform) - HAP	0.48	133.41		
	1,1,2,2- Tetrachloroethane - HAP/VOC	1.1	167.85		
	1,1-Dichloroethane (ethylidene dichloride) - HAP/VOC	2.4	98.97		
	1,1-Dichloroethene (vinylidene chloride) - HAP/VOC	0.20	96.94		
	1,2-Dichloroethane (ethylene dichloride) - HAP/VOC	0.41	98.96		
	1,2-Dichloropropane (propylene dichloride) - HAP/VOC	0.18	112.99		
	2-Propanol (isopropyl alcohol) - VOC	50	60.11		
	Acetone	7.0	58.08		
	Acrylonitrile - HAP/VOC	6.3	53.06		
	Benzene - No or Unknown Co-disposal - HAP/VOC	1.9	78.11		
	Benzene - Co-disposal - HAP/VOC	11	78.11		
	Bromodichloromethane - VOC	3.1	163.83		
	Butane - VOC	5.0	58.12		
	Carbon disulfide - HAP/VOC	0.58	76.13		
	Carbon monoxide	140	28.01		
	Carbon tetrachloride - HAP/VOC	4.0E-03	153.84		
	Carbonyl sulfide - HAP/VOC	0.49	60.07		
	Chlorobenzene - HAP/VOC	0.25	112.56		
	Chlorodifluoromethane	1.3	86.47		
	Chloroethane (ethyl chloride) - HAP/VOC	1.3	64.52		
	Chloroform - HAP/VOC	0.03	119.39		
	Chloromethane - VOC	1.2	50.49		
	Dichlorobenzene - (HAP for para isomer/VOC)	0.21	147		
	Dichlorodifluoromethane	16	120.91		
	Dichlorofluoromethane - VOC	2.6	102.92		
	Dichloromethane (methylene chloride) - HAP	14	84.94		
	Dimethyl sulfide (methyl sulfide) - VOC	7.8	62.13		
	Ethane	890	30.07		
	Ethanol - VOC	27	46.08		

Pollutant Parameters (Continued)

Gas / Pollutant Default Parameters:				User-specified Pollutant Parameters:	
	Compound	Concentration (ppmv)	Molecular Weight	Concentration (ppmv)	Molecular Weight
Pollutants	Ethyl mercaptan (ethanethiol) - VOC	2.3	62.13		
	Ethylbenzene - HAP/VOC	4.6	106.16		
	Ethylene dibromide - HAP/VOC	1.0E-03	187.88		
	Fluorotrichloromethane - VOC	0.76	137.38		
	Hexane - HAP/VOC	6.6	86.18		
	Hydrogen sulfide	36	34.08		
	Mercury (total) - HAP	2.9E-04	200.61		
	Methyl ethyl ketone - HAP/VOC	7.1	72.11		
	Methyl isobutyl ketone - HAP/VOC	1.9	100.16		
	Methyl mercaptan - VOC	2.5	48.11		
	Pentane - VOC	3.3	72.15		
	Perchloroethylene (tetrachloroethylene) - HAP	3.7	165.83		
	Propane - VOC	11	44.09		
	t-1,2-Dichloroethene - VOC	2.8	96.94		
	Toluene - No or Unknown Co-disposal - HAP/VOC	39	92.13		
	Toluene - Co-disposal - HAP/VOC	170	92.13		
	Trichloroethylene (trichloroethene) - HAP/VOC	2.8	131.40		
	Vinyl chloride - HAP/VOC	7.3	62.50		
	Xylenes - HAP/VOC	12	106.16		

Graphs



Results

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2037	0	0	0	0	0	0
2038	2.199E+03	1.761E+06	1.183E+02	5.875E+02	8.806E+05	5.917E+01
2039	4.313E+03	3.453E+06	2.320E+02	1.152E+03	1.727E+06	1.160E+02
2040	6.343E+03	5.079E+06	3.413E+02	1.694E+03	2.540E+06	1.706E+02
2041	8.294E+03	6.641E+06	4.462E+02	2.215E+03	3.321E+06	2.231E+02
2042	1.017E+04	8.142E+06	5.471E+02	2.716E+03	4.071E+06	2.735E+02
2043	1.197E+04	9.584E+06	6.439E+02	3.197E+03	4.792E+06	3.220E+02
2044	1.370E+04	1.097E+07	7.370E+02	3.659E+03	5.485E+06	3.685E+02
2045	1.536E+04	1.230E+07	8.265E+02	4.103E+03	6.150E+06	4.132E+02
2046	1.696E+04	1.358E+07	9.124E+02	4.530E+03	6.790E+06	4.562E+02
2047	1.849E+04	1.481E+07	9.950E+02	4.940E+03	7.404E+06	4.975E+02
2048	1.997E+04	1.599E+07	1.074E+03	5.333E+03	7.994E+06	5.371E+02
2049	2.138E+04	1.712E+07	1.150E+03	5.712E+03	8.562E+06	5.752E+02
2050	2.274E+04	1.821E+07	1.224E+03	6.075E+03	9.106E+06	6.119E+02
2051	2.405E+04	1.926E+07	1.294E+03	6.425E+03	9.630E+06	6.470E+02
2052	2.531E+04	2.027E+07	1.362E+03	6.760E+03	1.013E+07	6.808E+02
2053	2.652E+04	2.123E+07	1.427E+03	7.083E+03	1.062E+07	7.133E+02
2054	2.768E+04	2.216E+07	1.489E+03	7.392E+03	1.108E+07	7.445E+02
2055	2.879E+04	2.305E+07	1.549E+03	7.690E+03	1.153E+07	7.745E+02
2056	2.986E+04	2.391E+07	1.607E+03	7.976E+03	1.196E+07	8.033E+02
2057	3.089E+04	2.473E+07	1.662E+03	8.251E+03	1.237E+07	8.310E+02
2058	3.188E+04	2.553E+07	1.715E+03	8.515E+03	1.276E+07	8.575E+02
2059	3.283E+04	2.629E+07	1.766E+03	8.768E+03	1.314E+07	8.831E+02
2060	3.374E+04	2.702E+07	1.815E+03	9.012E+03	1.351E+07	9.076E+02
2061	3.462E+04	2.772E+07	1.862E+03	9.246E+03	1.386E+07	9.312E+02
2062	3.546E+04	2.839E+07	1.908E+03	9.471E+03	1.420E+07	9.539E+02
2063	3.627E+04	2.904E+07	1.951E+03	9.687E+03	1.452E+07	9.756E+02
2064	3.704E+04	2.966E+07	1.993E+03	9.895E+03	1.483E+07	9.965E+02
2065	3.779E+04	3.026E+07	2.033E+03	1.009E+04	1.513E+07	1.017E+03
2066	3.851E+04	3.084E+07	2.072E+03	1.029E+04	1.542E+07	1.036E+03
2067	3.920E+04	3.139E+07	2.109E+03	1.047E+04	1.569E+07	1.054E+03
2068	3.986E+04	3.192E+07	2.145E+03	1.065E+04	1.596E+07	1.072E+03
2069	4.050E+04	3.243E+07	2.179E+03	1.082E+04	1.621E+07	1.089E+03
2070	4.111E+04	3.292E+07	2.212E+03	1.098E+04	1.646E+07	1.106E+03
2071	4.170E+04	3.339E+07	2.243E+03	1.114E+04	1.669E+07	1.122E+03
2072	4.226E+04	3.384E+07	2.274E+03	1.129E+04	1.692E+07	1.137E+03
2073	4.280E+04	3.427E+07	2.303E+03	1.143E+04	1.714E+07	1.151E+03
2074	4.332E+04	3.469E+07	2.331E+03	1.157E+04	1.735E+07	1.165E+03
2075	4.383E+04	3.509E+07	2.358E+03	1.171E+04	1.755E+07	1.179E+03
2076	4.431E+04	3.548E+07	2.384E+03	1.183E+04	1.774E+07	1.192E+03
2077	4.477E+04	3.585E+07	2.409E+03	1.196E+04	1.792E+07	1.204E+03
2078	4.301E+04	3.444E+07	2.314E+03	1.149E+04	1.722E+07	1.157E+03
2079	4.133E+04	3.309E+07	2.223E+03	1.104E+04	1.655E+07	1.112E+03
2080	3.971E+04	3.179E+07	2.136E+03	1.061E+04	1.590E+07	1.068E+03
2081	3.815E+04	3.055E+07	2.053E+03	1.019E+04	1.527E+07	1.026E+03
2082	3.665E+04	2.935E+07	1.972E+03	9.790E+03	1.468E+07	9.860E+02
2083	3.522E+04	2.820E+07	1.895E+03	9.407E+03	1.410E+07	9.474E+02
2084	3.384E+04	2.709E+07	1.820E+03	9.038E+03	1.355E+07	9.102E+02
2085	3.251E+04	2.603E+07	1.749E+03	8.683E+03	1.302E+07	8.745E+02
2086	3.123E+04	2.501E+07	1.680E+03	8.343E+03	1.251E+07	8.402E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2087	3.001E+04	2.403E+07	1.615E+03	8.016E+03	1.201E+07	8.073E+02
2088	2.883E+04	2.309E+07	1.551E+03	7.701E+03	1.154E+07	7.756E+02
2089	2.770E+04	2.218E+07	1.490E+03	7.399E+03	1.109E+07	7.452E+02
2090	2.662E+04	2.131E+07	1.432E+03	7.109E+03	1.066E+07	7.160E+02
2091	2.557E+04	2.048E+07	1.376E+03	6.831E+03	1.024E+07	6.879E+02
2092	2.457E+04	1.967E+07	1.322E+03	6.563E+03	9.837E+06	6.609E+02
2093	2.361E+04	1.890E+07	1.270E+03	6.305E+03	9.451E+06	6.350E+02
2094	2.268E+04	1.816E+07	1.220E+03	6.058E+03	9.081E+06	6.101E+02
2095	2.179E+04	1.745E+07	1.172E+03	5.821E+03	8.725E+06	5.862E+02
2096	2.094E+04	1.677E+07	1.126E+03	5.592E+03	8.383E+06	5.632E+02
2097	2.012E+04	1.611E+07	1.082E+03	5.373E+03	8.054E+06	5.411E+02
2098	1.933E+04	1.548E+07	1.040E+03	5.162E+03	7.738E+06	5.199E+02
2099	1.857E+04	1.487E+07	9.991E+02	4.960E+03	7.435E+06	4.995E+02
2100	1.784E+04	1.429E+07	9.599E+02	4.766E+03	7.143E+06	4.799E+02
2101	1.714E+04	1.373E+07	9.223E+02	4.579E+03	6.863E+06	4.611E+02
2102	1.647E+04	1.319E+07	8.861E+02	4.399E+03	6.594E+06	4.430E+02
2103	1.582E+04	1.267E+07	8.513E+02	4.227E+03	6.335E+06	4.257E+02
2104	1.520E+04	1.217E+07	8.180E+02	4.061E+03	6.087E+06	4.090E+02
2105	1.461E+04	1.170E+07	7.859E+02	3.902E+03	5.848E+06	3.929E+02
2106	1.403E+04	1.124E+07	7.551E+02	3.749E+03	5.619E+06	3.775E+02
2107	1.348E+04	1.080E+07	7.255E+02	3.602E+03	5.399E+06	3.627E+02
2108	1.296E+04	1.037E+07	6.970E+02	3.460E+03	5.187E+06	3.485E+02
2109	1.245E+04	9.967E+06	6.697E+02	3.325E+03	4.984E+06	3.348E+02
2110	1.196E+04	9.576E+06	6.434E+02	3.194E+03	4.788E+06	3.217E+02
2111	1.149E+04	9.201E+06	6.182E+02	3.069E+03	4.600E+06	3.091E+02
2112	1.104E+04	8.840E+06	5.940E+02	2.949E+03	4.420E+06	2.970E+02
2113	1.061E+04	8.493E+06	5.707E+02	2.833E+03	4.247E+06	2.853E+02
2114	1.019E+04	8.160E+06	5.483E+02	2.722E+03	4.080E+06	2.741E+02
2115	9.791E+03	7.840E+06	5.268E+02	2.615E+03	3.920E+06	2.634E+02
2116	9.407E+03	7.533E+06	5.061E+02	2.513E+03	3.767E+06	2.531E+02
2117	9.039E+03	7.238E+06	4.863E+02	2.414E+03	3.619E+06	2.431E+02
2118	8.684E+03	6.954E+06	4.672E+02	2.320E+03	3.477E+06	2.336E+02
2119	8.344E+03	6.681E+06	4.489E+02	2.229E+03	3.341E+06	2.245E+02
2120	8.016E+03	6.419E+06	4.313E+02	2.141E+03	3.210E+06	2.157E+02
2121	7.702E+03	6.168E+06	4.144E+02	2.057E+03	3.084E+06	2.072E+02
2122	7.400E+03	5.926E+06	3.981E+02	1.977E+03	2.963E+06	1.991E+02
2123	7.110E+03	5.693E+06	3.825E+02	1.899E+03	2.847E+06	1.913E+02
2124	6.831E+03	5.470E+06	3.675E+02	1.825E+03	2.735E+06	1.838E+02
2125	6.563E+03	5.256E+06	3.531E+02	1.753E+03	2.628E+06	1.766E+02
2126	6.306E+03	5.050E+06	3.393E+02	1.684E+03	2.525E+06	1.696E+02
2127	6.059E+03	4.852E+06	3.260E+02	1.618E+03	2.426E+06	1.630E+02
2128	5.821E+03	4.661E+06	3.132E+02	1.555E+03	2.331E+06	1.566E+02
2129	5.593E+03	4.479E+06	3.009E+02	1.494E+03	2.239E+06	1.505E+02
2130	5.374E+03	4.303E+06	2.891E+02	1.435E+03	2.151E+06	1.446E+02
2131	5.163E+03	4.134E+06	2.778E+02	1.379E+03	2.067E+06	1.389E+02
2132	4.960E+03	3.972E+06	2.669E+02	1.325E+03	1.986E+06	1.334E+02
2133	4.766E+03	3.816E+06	2.564E+02	1.273E+03	1.908E+06	1.282E+02
2134	4.579E+03	3.667E+06	2.464E+02	1.223E+03	1.833E+06	1.232E+02
2135	4.400E+03	3.523E+06	2.367E+02	1.175E+03	1.761E+06	1.184E+02
2136	4.227E+03	3.385E+06	2.274E+02	1.129E+03	1.692E+06	1.137E+02
2137	4.061E+03	3.252E+06	2.185E+02	1.085E+03	1.626E+06	1.093E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2138	3.902E+03	3.125E+06	2.099E+02	1.042E+03	1.562E+06	1.050E+02
2139	3.749E+03	3.002E+06	2.017E+02	1.001E+03	1.501E+06	1.009E+02
2140	3.602E+03	2.884E+06	1.938E+02	9.621E+02	1.442E+06	9.690E+01
2141	3.461E+03	2.771E+06	1.862E+02	9.244E+02	1.386E+06	9.310E+01
2142	3.325E+03	2.663E+06	1.789E+02	8.882E+02	1.331E+06	8.945E+01
2143	3.195E+03	2.558E+06	1.719E+02	8.533E+02	1.279E+06	8.594E+01
2144	3.069E+03	2.458E+06	1.651E+02	8.199E+02	1.229E+06	8.257E+01
2145	2.949E+03	2.362E+06	1.587E+02	7.877E+02	1.181E+06	7.933E+01
2146	2.833E+03	2.269E+06	1.524E+02	7.568E+02	1.134E+06	7.622E+01
2147	2.722E+03	2.180E+06	1.465E+02	7.272E+02	1.090E+06	7.324E+01
2148	2.616E+03	2.094E+06	1.407E+02	6.987E+02	1.047E+06	7.036E+01
2149	2.513E+03	2.012E+06	1.352E+02	6.713E+02	1.006E+06	6.760E+01
2150	2.415E+03	1.933E+06	1.299E+02	6.449E+02	9.667E+05	6.495E+01
2151	2.320E+03	1.858E+06	1.248E+02	6.197E+02	9.288E+05	6.241E+01
2152	2.229E+03	1.785E+06	1.199E+02	5.954E+02	8.924E+05	5.996E+01
2153	2.141E+03	1.715E+06	1.152E+02	5.720E+02	8.574E+05	5.761E+01
2154	2.058E+03	1.648E+06	1.107E+02	5.496E+02	8.238E+05	5.535E+01
2155	1.977E+03	1.583E+06	1.064E+02	5.280E+02	7.915E+05	5.318E+01
2156	1.899E+03	1.521E+06	1.022E+02	5.073E+02	7.604E+05	5.109E+01
2157	1.825E+03	1.461E+06	9.818E+01	4.874E+02	7.306E+05	4.909E+01
2158	1.753E+03	1.404E+06	9.433E+01	4.683E+02	7.020E+05	4.717E+01
2159	1.685E+03	1.349E+06	9.063E+01	4.500E+02	6.745E+05	4.532E+01
2160	1.619E+03	1.296E+06	8.708E+01	4.323E+02	6.480E+05	4.354E+01
2161	1.555E+03	1.245E+06	8.367E+01	4.154E+02	6.226E+05	4.183E+01
2162	1.494E+03	1.196E+06	8.038E+01	3.991E+02	5.982E+05	4.019E+01
2163	1.435E+03	1.149E+06	7.723E+01	3.834E+02	5.747E+05	3.862E+01
2164	1.379E+03	1.104E+06	7.420E+01	3.684E+02	5.522E+05	3.710E+01
2165	1.325E+03	1.061E+06	7.129E+01	3.540E+02	5.305E+05	3.565E+01
2166	1.273E+03	1.019E+06	6.850E+01	3.401E+02	5.097E+05	3.425E+01
2167	1.223E+03	9.795E+05	6.581E+01	3.267E+02	4.898E+05	3.291E+01
2168	1.175E+03	9.411E+05	6.323E+01	3.139E+02	4.706E+05	3.162E+01
2169	1.129E+03	9.042E+05	6.075E+01	3.016E+02	4.521E+05	3.038E+01
2170	1.085E+03	8.687E+05	5.837E+01	2.898E+02	4.344E+05	2.919E+01
2171	1.042E+03	8.347E+05	5.608E+01	2.784E+02	4.173E+05	2.804E+01
2172	1.002E+03	8.020E+05	5.388E+01	2.675E+02	4.010E+05	2.694E+01
2173	9.622E+02	7.705E+05	5.177E+01	2.570E+02	3.853E+05	2.589E+01
2174	9.245E+02	7.403E+05	4.974E+01	2.469E+02	3.701E+05	2.487E+01
2175	8.883E+02	7.113E+05	4.779E+01	2.373E+02	3.556E+05	2.390E+01
2176	8.534E+02	6.834E+05	4.592E+01	2.280E+02	3.417E+05	2.296E+01
2177	8.200E+02	6.566E+05	4.412E+01	2.190E+02	3.283E+05	2.206E+01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2037	0	0	0	0	0	0
2038	1.612E+03	8.806E+05	5.917E+01	2.235E+00	6.235E+02	4.189E-02
2039	3.161E+03	1.727E+06	1.160E+02	4.382E+00	1.222E+03	8.214E-02
2040	4.649E+03	2.540E+06	1.706E+02	6.445E+00	1.798E+03	1.208E-01
2041	6.078E+03	3.321E+06	2.231E+02	8.427E+00	2.351E+03	1.580E-01
2042	7.452E+03	4.071E+06	2.735E+02	1.033E+01	2.882E+03	1.937E-01
2043	8.772E+03	4.792E+06	3.220E+02	1.216E+01	3.393E+03	2.280E-01
2044	1.004E+04	5.485E+06	3.685E+02	1.392E+01	3.883E+03	2.609E-01
2045	1.126E+04	6.150E+06	4.132E+02	1.561E+01	4.354E+03	2.926E-01
2046	1.243E+04	6.790E+06	4.562E+02	1.723E+01	4.807E+03	3.230E-01
2047	1.355E+04	7.404E+06	4.975E+02	1.879E+01	5.242E+03	3.522E-01
2048	1.463E+04	7.994E+06	5.371E+02	2.029E+01	5.660E+03	3.803E-01
2049	1.567E+04	8.562E+06	5.752E+02	2.173E+01	6.062E+03	4.073E-01
2050	1.667E+04	9.106E+06	6.119E+02	2.311E+01	6.447E+03	4.332E-01
2051	1.763E+04	9.630E+06	6.470E+02	2.444E+01	6.818E+03	4.581E-01
2052	1.855E+04	1.013E+07	6.808E+02	2.572E+01	7.174E+03	4.820E-01
2053	1.943E+04	1.062E+07	7.133E+02	2.694E+01	7.516E+03	5.050E-01
2054	2.028E+04	1.108E+07	7.445E+02	2.812E+01	7.845E+03	5.271E-01
2055	2.110E+04	1.153E+07	7.745E+02	2.925E+01	8.161E+03	5.483E-01
2056	2.188E+04	1.196E+07	8.033E+02	3.034E+01	8.464E+03	5.687E-01
2057	2.264E+04	1.237E+07	8.310E+02	3.139E+01	8.756E+03	5.883E-01
2058	2.336E+04	1.276E+07	8.575E+02	3.239E+01	9.036E+03	6.071E-01
2059	2.406E+04	1.314E+07	8.831E+02	3.335E+01	9.305E+03	6.252E-01
2060	2.473E+04	1.351E+07	9.076E+02	3.428E+01	9.564E+03	6.426E-01
2061	2.537E+04	1.386E+07	9.312E+02	3.517E+01	9.812E+03	6.593E-01
2062	2.599E+04	1.420E+07	9.539E+02	3.603E+01	1.005E+04	6.753E-01
2063	2.658E+04	1.452E+07	9.756E+02	3.685E+01	1.028E+04	6.907E-01
2064	2.715E+04	1.483E+07	9.965E+02	3.764E+01	1.050E+04	7.055E-01
2065	2.770E+04	1.513E+07	1.017E+03	3.840E+01	1.071E+04	7.198E-01
2066	2.822E+04	1.542E+07	1.036E+03	3.913E+01	1.092E+04	7.334E-01
2067	2.873E+04	1.569E+07	1.054E+03	3.983E+01	1.111E+04	7.466E-01
2068	2.921E+04	1.596E+07	1.072E+03	4.050E+01	1.130E+04	7.592E-01
2069	2.968E+04	1.621E+07	1.089E+03	4.115E+01	1.148E+04	7.713E-01
2070	3.013E+04	1.646E+07	1.106E+03	4.177E+01	1.165E+04	7.830E-01
2071	3.056E+04	1.669E+07	1.122E+03	4.237E+01	1.182E+04	7.942E-01
2072	3.097E+04	1.692E+07	1.137E+03	4.294E+01	1.198E+04	8.049E-01
2073	3.137E+04	1.714E+07	1.151E+03	4.349E+01	1.213E+04	8.152E-01
2074	3.175E+04	1.735E+07	1.165E+03	4.402E+01	1.228E+04	8.252E-01
2075	3.212E+04	1.755E+07	1.179E+03	4.453E+01	1.242E+04	8.347E-01
2076	3.247E+04	1.774E+07	1.192E+03	4.502E+01	1.256E+04	8.439E-01
2077	3.281E+04	1.792E+07	1.204E+03	4.549E+01	1.269E+04	8.527E-01
2078	3.152E+04	1.722E+07	1.157E+03	4.370E+01	1.219E+04	8.192E-01
2079	3.029E+04	1.655E+07	1.112E+03	4.199E+01	1.171E+04	7.871E-01
2080	2.910E+04	1.590E+07	1.068E+03	4.034E+01	1.126E+04	7.562E-01
2081	2.796E+04	1.527E+07	1.026E+03	3.876E+01	1.081E+04	7.266E-01
2082	2.686E+04	1.468E+07	9.860E+02	3.724E+01	1.039E+04	6.981E-01
2083	2.581E+04	1.410E+07	9.474E+02	3.578E+01	9.983E+03	6.707E-01
2084	2.480E+04	1.355E+07	9.102E+02	3.438E+01	9.591E+03	6.444E-01
2085	2.383E+04	1.302E+07	8.745E+02	3.303E+01	9.215E+03	6.192E-01
2086	2.289E+04	1.251E+07	8.402E+02	3.174E+01	8.854E+03	5.949E-01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2087	2.199E+04	1.201E+07	8.073E+02	3.049E+01	8.507E+03	5.716E-01
2088	2.113E+04	1.154E+07	7.756E+02	2.930E+01	8.173E+03	5.491E-01
2089	2.030E+04	1.109E+07	7.452E+02	2.815E+01	7.853E+03	5.276E-01
2090	1.951E+04	1.066E+07	7.160E+02	2.704E+01	7.545E+03	5.069E-01
2091	1.874E+04	1.024E+07	6.879E+02	2.598E+01	7.249E+03	4.870E-01
2092	1.801E+04	9.837E+06	6.609E+02	2.496E+01	6.965E+03	4.680E-01
2093	1.730E+04	9.451E+06	6.350E+02	2.399E+01	6.692E+03	4.496E-01
2094	1.662E+04	9.081E+06	6.101E+02	2.305E+01	6.429E+03	4.320E-01
2095	1.597E+04	8.725E+06	5.862E+02	2.214E+01	6.177E+03	4.150E-01
2096	1.534E+04	8.383E+06	5.632E+02	2.127E+01	5.935E+03	3.988E-01
2097	1.474E+04	8.054E+06	5.411E+02	2.044E+01	5.702E+03	3.831E-01
2098	1.416E+04	7.738E+06	5.199E+02	1.964E+01	5.479E+03	3.681E-01
2099	1.361E+04	7.435E+06	4.995E+02	1.887E+01	5.264E+03	3.537E-01
2100	1.308E+04	7.143E+06	4.799E+02	1.813E+01	5.057E+03	3.398E-01
2101	1.256E+04	6.863E+06	4.611E+02	1.742E+01	4.859E+03	3.265E-01
2102	1.207E+04	6.594E+06	4.430E+02	1.673E+01	4.669E+03	3.137E-01
2103	1.160E+04	6.335E+06	4.257E+02	1.608E+01	4.485E+03	3.014E-01
2104	1.114E+04	6.087E+06	4.090E+02	1.545E+01	4.310E+03	2.896E-01
2105	1.071E+04	5.848E+06	3.929E+02	1.484E+01	4.141E+03	2.782E-01
2106	1.029E+04	5.619E+06	3.775E+02	1.426E+01	3.978E+03	2.673E-01
2107	9.882E+03	5.399E+06	3.627E+02	1.370E+01	3.822E+03	2.568E-01
2108	9.495E+03	5.187E+06	3.485E+02	1.316E+01	3.672E+03	2.467E-01
2109	9.122E+03	4.984E+06	3.348E+02	1.265E+01	3.528E+03	2.371E-01
2110	8.765E+03	4.788E+06	3.217E+02	1.215E+01	3.390E+03	2.278E-01
2111	8.421E+03	4.600E+06	3.091E+02	1.168E+01	3.257E+03	2.188E-01
2112	8.091E+03	4.420E+06	2.970E+02	1.122E+01	3.129E+03	2.103E-01
2113	7.774E+03	4.247E+06	2.853E+02	1.078E+01	3.007E+03	2.020E-01
2114	7.469E+03	4.080E+06	2.741E+02	1.035E+01	2.889E+03	1.941E-01
2115	7.176E+03	3.920E+06	2.634E+02	9.949E+00	2.776E+03	1.865E-01
2116	6.895E+03	3.767E+06	2.531E+02	9.559E+00	2.667E+03	1.792E-01
2117	6.624E+03	3.619E+06	2.431E+02	9.184E+00	2.562E+03	1.721E-01
2118	6.365E+03	3.477E+06	2.336E+02	8.824E+00	2.462E+03	1.654E-01
2119	6.115E+03	3.341E+06	2.245E+02	8.478E+00	2.365E+03	1.589E-01
2120	5.875E+03	3.210E+06	2.157E+02	8.145E+00	2.272E+03	1.527E-01
2121	5.645E+03	3.084E+06	2.072E+02	7.826E+00	2.183E+03	1.467E-01
2122	5.423E+03	2.963E+06	1.991E+02	7.519E+00	2.098E+03	1.409E-01
2123	5.211E+03	2.847E+06	1.913E+02	7.224E+00	2.015E+03	1.354E-01
2124	5.007E+03	2.735E+06	1.838E+02	6.941E+00	1.936E+03	1.301E-01
2125	4.810E+03	2.628E+06	1.766E+02	6.669E+00	1.860E+03	1.250E-01
2126	4.622E+03	2.525E+06	1.696E+02	6.407E+00	1.788E+03	1.201E-01
2127	4.440E+03	2.426E+06	1.630E+02	6.156E+00	1.717E+03	1.154E-01
2128	4.266E+03	2.331E+06	1.566E+02	5.915E+00	1.650E+03	1.109E-01
2129	4.099E+03	2.239E+06	1.505E+02	5.683E+00	1.585E+03	1.065E-01
2130	3.938E+03	2.151E+06	1.446E+02	5.460E+00	1.523E+03	1.023E-01
2131	3.784E+03	2.067E+06	1.389E+02	5.246E+00	1.464E+03	9.833E-02
2132	3.635E+03	1.986E+06	1.334E+02	5.040E+00	1.406E+03	9.448E-02
2133	3.493E+03	1.908E+06	1.282E+02	4.843E+00	1.351E+03	9.077E-02
2134	3.356E+03	1.833E+06	1.232E+02	4.653E+00	1.298E+03	8.721E-02
2135	3.224E+03	1.761E+06	1.184E+02	4.470E+00	1.247E+03	8.379E-02
2136	3.098E+03	1.692E+06	1.137E+02	4.295E+00	1.198E+03	8.051E-02
2137	2.976E+03	1.626E+06	1.093E+02	4.127E+00	1.151E+03	7.735E-02

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2138	2.860E+03	1.562E+06	1.050E+02	3.965E+00	1.106E+03	7.432E-02
2139	2.748E+03	1.501E+06	1.009E+02	3.809E+00	1.063E+03	7.140E-02
2140	2.640E+03	1.442E+06	9.690E+01	3.660E+00	1.021E+03	6.860E-02
2141	2.536E+03	1.386E+06	9.310E+01	3.516E+00	9.810E+02	6.591E-02
2142	2.437E+03	1.331E+06	8.945E+01	3.379E+00	9.426E+02	6.333E-02
2143	2.341E+03	1.279E+06	8.594E+01	3.246E+00	9.056E+02	6.085E-02
2144	2.250E+03	1.229E+06	8.257E+01	3.119E+00	8.701E+02	5.846E-02
2145	2.161E+03	1.181E+06	7.933E+01	2.997E+00	8.360E+02	5.617E-02
2146	2.077E+03	1.134E+06	7.622E+01	2.879E+00	8.032E+02	5.397E-02
2147	1.995E+03	1.090E+06	7.324E+01	2.766E+00	7.717E+02	5.185E-02
2148	1.917E+03	1.047E+06	7.036E+01	2.658E+00	7.414E+02	4.982E-02
2149	1.842E+03	1.006E+06	6.760E+01	2.553E+00	7.124E+02	4.786E-02
2150	1.770E+03	9.667E+05	6.495E+01	2.453E+00	6.844E+02	4.599E-02
2151	1.700E+03	9.288E+05	6.241E+01	2.357E+00	6.576E+02	4.418E-02
2152	1.634E+03	8.924E+05	5.996E+01	2.265E+00	6.318E+02	4.245E-02
2153	1.569E+03	8.574E+05	5.761E+01	2.176E+00	6.070E+02	4.079E-02
2154	1.508E+03	8.238E+05	5.535E+01	2.091E+00	5.832E+02	3.919E-02
2155	1.449E+03	7.915E+05	5.318E+01	2.009E+00	5.604E+02	3.765E-02
2156	1.392E+03	7.604E+05	5.109E+01	1.930E+00	5.384E+02	3.617E-02
2157	1.337E+03	7.306E+05	4.909E+01	1.854E+00	5.173E+02	3.476E-02
2158	1.285E+03	7.020E+05	4.717E+01	1.781E+00	4.970E+02	3.339E-02
2159	1.235E+03	6.745E+05	4.532E+01	1.712E+00	4.775E+02	3.208E-02
2160	1.186E+03	6.480E+05	4.354E+01	1.645E+00	4.588E+02	3.083E-02
2161	1.140E+03	6.226E+05	4.183E+01	1.580E+00	4.408E+02	2.962E-02
2162	1.095E+03	5.982E+05	4.019E+01	1.518E+00	4.235E+02	2.846E-02
2163	1.052E+03	5.747E+05	3.862E+01	1.459E+00	4.069E+02	2.734E-02
2164	1.011E+03	5.522E+05	3.710E+01	1.401E+00	3.910E+02	2.627E-02
2165	9.712E+02	5.305E+05	3.565E+01	1.346E+00	3.756E+02	2.524E-02
2166	9.331E+02	5.097E+05	3.425E+01	1.294E+00	3.609E+02	2.425E-02
2167	8.965E+02	4.898E+05	3.291E+01	1.243E+00	3.467E+02	2.330E-02
2168	8.613E+02	4.706E+05	3.162E+01	1.194E+00	3.332E+02	2.238E-02
2169	8.276E+02	4.521E+05	3.038E+01	1.147E+00	3.201E+02	2.151E-02
2170	7.951E+02	4.344E+05	2.919E+01	1.102E+00	3.075E+02	2.066E-02
2171	7.639E+02	4.173E+05	2.804E+01	1.059E+00	2.955E+02	1.985E-02
2172	7.340E+02	4.010E+05	2.694E+01	1.018E+00	2.839E+02	1.907E-02
2173	7.052E+02	3.853E+05	2.589E+01	9.777E-01	2.728E+02	1.833E-02
2174	6.776E+02	3.701E+05	2.487E+01	9.394E-01	2.621E+02	1.761E-02
2175	6.510E+02	3.556E+05	2.390E+01	9.025E-01	2.518E+02	1.692E-02
2176	6.255E+02	3.417E+05	2.296E+01	8.671E-01	2.419E+02	1.625E-02
2177	6.009E+02	3.283E+05	2.206E+01	8.331E-01	2.324E+02	1.562E-02

APPENDIX A-2

CALCULATIONS - RADIUS OF INFLUENCE AND WELL SPACING ESTIMATES

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4251206

DESCRIPTION: Vertical Well ROI Calculation CALC NO: SHEET 1 OF 6

PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 4/2/2025

Required: Determine the radius of influence (ROI) for vertical LFG extraction wells, using both EMCON and NSPS methods for the average flow rate condition.

Purpose: The ROI is calculated in the design of an active gas extraction system to properly locate extraction wells. The location of the extraction well will dictate the well depth, once the refuse depth is known. A detail of each well can then be created with respect to total well depth, depth to slotted pipe, applied vacuum and rate of extraction.

Although horizontal collectors are not modeled directly by this analysis, the effected flow patterns developed by horizontal collectors are analogous to those created by vertical extraction wells. The net result is that vertical well spacing criteria can be translated directly to horizontal collectors in a similar environment.

Method: The following methods were used to estimate the theoretical ROI for an LFG extraction well.

A) EMCON Method (from *Methane Generation and Recovery from Landfills*, EMCON, 1982, pg. 81)

$$Q_w = \frac{k \pi R^2 t D r}{C}$$

where: Q_w = individual extraction well LFG flow rate [L/s]
 k = conversion factor (1.157×10^{-8}) [(L/s)/(mL/day)]
 R = radius of influence [m]
 t = perforated pipe length [m]
 D = in-place refuse density [kg/m³]
 r = methane production rate [mL/kg/day]
 C = fractional methane concentration [-]

1. Noting that the methane production rate (r) divided by the fractional methane concentration (C) is equal to the LFG production rate (G), and solving for the ROI yields:

$$R = \left(\frac{Q_w}{k \pi t D G} \right)^{1/2}$$

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4251206
DESCRIPTION: Vertical Well ROI Calculation CALC NO: _____ SHEET 2 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 4/2/2025

2. Converting from metric to English units yields the following conversion factors (allowing input in English units):

Q_w (from cfm to L/s):

$$\frac{1 \text{ ft}^3}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ sec}} \times \frac{28.317 \text{ L}}{\text{ft}^3} = 0.47195$$

t (from ft to m):

$$1 \text{ ft} \times \frac{0.3048 \text{ m}}{\text{ft}} = 0.3048$$

D (from lb/cy to kg/m³):

$$\frac{1 \text{ lb}}{1 \text{ cy}} \times \frac{0.4536 \text{ kg}}{1 \text{ lb}} \times \frac{1 \text{ cy}}{27 \text{ ft}^3} \times \left(\frac{1 \text{ ft}}{0.3048 \text{ m}} \right)^3 = 0.5932$$

In addition, converting G from flow per volume to flow per mass equals:

G (from cfm/cy to mL/kg/day):

$$\begin{aligned} & \frac{1 \text{ ft}^3}{\text{min yd}^3} \times \left(\frac{1}{D} \times \frac{1 \text{ yd}^3}{1 \text{ lb}} \right) \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{24 \text{ hr}}{1 \text{ day}} \times \frac{28.317 \text{ L}}{\text{ft}^3} \times \frac{1 \text{ lb}}{0.4536 \text{ kg}} \times \frac{1000 \text{ mL}}{\text{L}} \\ &= \frac{89,895,238}{D} \end{aligned}$$

Therefore:

$$R = \left[\frac{0.47195 Q_w}{1.157 \times 10^{-8} \pi (0.3048 t) (0.5932 D) \left(\frac{89,895,238}{D} \times G \right)} \right]^{1/2}$$

Where: R is in meters

Q_w is in cfm

t is in feet

D is in lb/cy

G is in cfm/cy

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4251206
DESCRIPTION: Vertical Well ROI Calculation CALC NO: SHEET 3 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 4/2/2025

Then:

$$R = 0.8938 \left(\frac{Q_w}{t G} \right)^{1/2}$$

Converting results from meters to feet (1 ft = 0.3048 m):

$$0.3048R = 0.8938 \left(\frac{Q_w}{t G} \right)^{1/2}$$

$$R = 2.932 \left(\frac{Q_w}{t G} \right)^{1/2}$$

B) NSPS Method (from EPA *NSPS Bid*, 1991, pg. G-1)

$$R = \left(\frac{Q_w DC}{\pi L \rho Q_{gen} \eta} \right)^{1/2}$$

Where: ρ = in-place density of refuse [kg/m³]
 Q_w = LFG flow rate per well [m³/s]
DC = design capacity of landfill [kg]
L = perforated pipe length [m]
 Q_{gen} = peak LFG generation rate [m³/s]
 η = system collection efficiency [%]
R = radius of influence [m]

1. Converting from metric to English units yields the following conversion factors (allowing input in English units):

Q_w (from cfm to m³/s):

$$\frac{1 \text{ ft}^3}{1 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ sec}} \times \left(\frac{0.3048 \text{ m}}{\text{ft}} \right)^3 = 4.719 \times 10^{-4}$$

DC (from tons to kg):

$$1 \text{ ton} \times \frac{2,000 \text{ lb}}{1 \text{ ton}} \times \frac{0.4536 \text{ kg}}{1 \text{ lb}} = 907.2$$

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4251206

DESCRIPTION: Vertical Well ROI Calculation CALC NO: _____ SHEET 4 OF 6

PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 4/2/2025

L (from ft to m):

$$1 \text{ ft} \times \frac{0.3048 \text{ m}}{\text{ft}} = 0.3048$$

ρ (from lb/cy to kg/m³):

$$\frac{1 \text{ lb}}{\text{yd}^3} \times \frac{0.4536 \text{ kg}}{1 \text{ lb}} \times \frac{1 \text{ yd}^3}{27 \text{ ft}^3} \times \left(\frac{1 \text{ ft}}{0.3048 \text{ m}} \right)^3 = 0.5933$$

Converting results from meters to feet (1 ft = 0.3048 m):

$$0.3048R = \left[\frac{(4.719 \times 10^{-4}) Q_w (907.2) DC}{\pi (0.3048L) (0.5933\rho) (4.719 \times 10^{-4} Q_{\text{gen}}) \eta} \right]^{1/2}$$

$$R = \frac{1}{0.3048} \left[\frac{(4.719 \times 10^{-4}) Q_w (907.2) DC}{\pi (0.3048L) (0.5933\rho) (4.719 \times 10^{-4} Q_{\text{gen}}) \eta} \right]^{1/2}$$

Example Calculation:

Estimate the ROI for the following average LFG flow rate condition at the XYZ Landfill

Given:

Design Capacity	=	2.5 x 10 ⁷ tons
Collection efficiency	=	70%
Average LFG generation rate	=	2,469 scfm
In-place refuse density	=	1,400 lb/cy
Depth of well	=	100 ft
Well flow rate	=	26 scfm

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4251206
DESCRIPTION: Vertical Well ROI Calculation CALC NO: _____ SHEET 5 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 4/2/2025

A) EMCON Method

$$R = 2.932 \left(\frac{26 \text{ scfm}}{(100 \text{ ft}) \left(\frac{2,469 \text{ scfm}}{25,000,000 \text{ tons}} \right) \left(\frac{1 \text{ ton}}{2,000 \text{ lb}} \right) \left(\frac{1,400 \text{ lb}}{1 \text{ cy}} \right)} \right)^{1/2}$$

R = 180 ft

B) NSPS Method

$$R = \frac{1}{0.3048} \left[\frac{(4.719 \times 10^{-4})(26 \text{ scfm})(907.2)(25,000,000 \text{ tons})}{\pi (0.3048)(100 \text{ ft})(0.5933)(1,400 \text{ lb / cy})(4.719 \times 10^{-4})(2,469 \text{ scfm})(0.70)} \right]^{1/2}$$

R = 215 ft

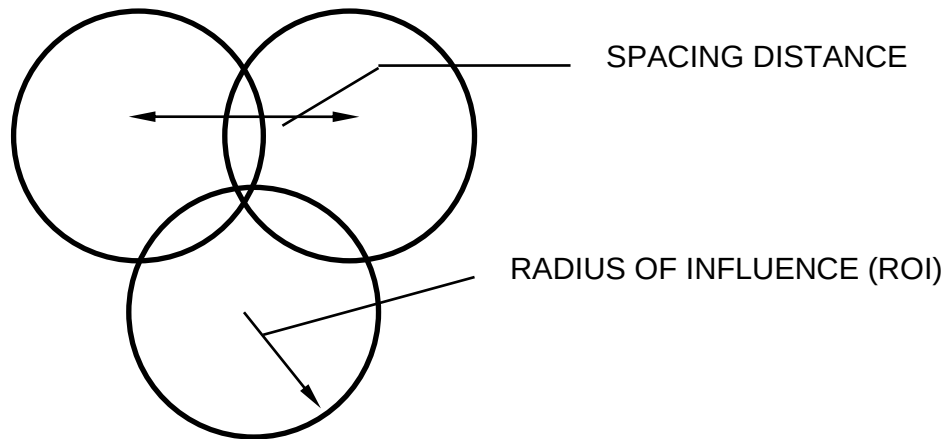
COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4251206
DESCRIPTION: Vertical Well ROI Calculation CALC NO: SHEET 6 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 4/2/2025

Well Spacing

Utilizing the more conservative (EMCON) value for ROI of R = 180 ft for this extraction well, and applying the criteria for well spacing of:

$$\text{SPACING} = \sqrt{3} \text{ ROI}$$



$$\underline{\underline{\text{SPACING} = 312 \text{ ft}}}$$

Site Specific Calculations:

ROI calculations for the facility are shown on the following spreadsheet. These calculations were performed for a series of typical well depths, ranging from 40 feet to 100 feet and will be utilized as the basis for determining well placement across the selected area of the Project.

**Landfill Gas Extraction Well Radius of Influence
Based on Volume of Affected Refuse
Wexford County Landfill - East Unit**

Well Number ^(c)	Well Depth (ft)	LFG Flow Rate Condition	LFG Flow Rate (scfm)	Radius of Influence (ft) ^{(a)(b)}	
				NSPS	EMCON
A	40	Low	10	104	99
		Average	15	127	120
		High	20	148	140
B	60	Low	20	121	114
		Average	25	135	128
		High	30	148	140
C	80	Low	30	128	121
		Average	40	148	140
		High	50	165	157
D	100	Low	50	148	140
		Average	60	162	153
		High	70	175	166

Total East Unit Waste Design Tonnage	9,676,401 tons	LFG Generation Rate	2,522 scfm
Total East Unit Waste Design Capacity	11,519,525 cy	Refuse Density	1,680 lb/cy
Total Landfill Waste Design Capacity	23,260,000 cy	NSPS Collection Efficiency	90 %
ROI of Perimeter Wells =<		120 feet	Well Spacing => 208 feet
ROI of Interior Wells =<		140 feet	Well Spacing => 242 feet

Notes:

- a) Radius of influence based on the estimated capacity of the facility and the anticipated LFG generation rate.
- b) Calculations assume 20 feet of solid well casing from ground surface to start of perforations.
- c) Well number is general for the typical depth shown.

Landfill Gas Extraction Well Radius of Influence
Based on Volume of Affected Refuse
Wexford County Landfill - West Unit

Well Number ^(c)	Well Depth (ft)	LFG Flow Rate Condition	LFG Flow Rate (scfm)	Radius of Influence (ft) ^{(a)(b)}	
				NSPS	EMCON
A	40	Low	10	108	102
		Average	15	130	125
		High	20	153	145
B	60	Low	20	125	118
		Average	25	139	132
		High	30	153	145
C	80	Low	30	132	125
		Average	40	151	145
		High	50	171	162
D	100	Low	50	153	145
		Average	60	167	159
		High	70	180	171

Total West Unit Waste Design Tonnage	9,861,999 tons	LFG Generation Rate	2,409 scfm
Total West Unit Waste Design Capacity	11,740,475 cy	Refuse Density	1,680 lb/cy
Total Landfill Waste Design Capacity	23,260,000 cy	NSPS Collection Efficiency	90 %
ROI of Perimeter Wells =<		125 feet	Well Spacing => 217 feet
ROI of Interior Wells =<		145 feet	Well Spacing => 251 feet

Notes:

- a) Radius of influence based on the estimated capacity of the facility and the anticipated LFG generation rate.
b) Calculations assume 20 feet of solid well casing from ground surface to start of perforations.
c) Well number is general for the typical depth shown.

APPENDIX A-3

CALCULATIONS - CONDENSATE GENERATION ESTIMATES

COMPUTATION SHEET

PROJECT TITLE: Wexford County - East Unit Landfill
 DESCRIPTION: Landfill Gas Design Plan
Landfill Gas Condensate Generation - Worst Case Conditions
 Prepared By: JPS Checked By: JPS Approved By: KM

PROJECT NO: 209-4251206
 SHEET: 1
 OF: 1
 DATE: 4/2/2025

Assumptions:

1. LFG temperature at the wellhead is the warmest gas.
2. LFG traveling in a buried header will try to stabilize its temperature consistent with the ground temperature.
3. Ground temp. below the frost line remains constant year round at 59 °F
4. LFG header pipe is installed below frost depth.
5. The LFG temperature (just before the blower) depends on the distance traveled in buried header pipe and the thermal conductivity of the header pipe.
6. Work at the blower compresses the LFG, but no additional condensate is generated due to the heat of compression.

Condensate Generation (gallons/day) = (Y_{wellhead} - Y_{blower inlet}) * LFG Flow * (18 / (380 * 8.35)) * 60min/hr * 24 hrs/day

Wellhead Conditions		
P (psia)	VP (psia)	Y (moles)
14.34	0.950	0.066

Blower Inlet Conditions			
	P (psia)	VP (psia)	Y (moles)
59 °F	12.90	0.217	0.017
60 °F	12.90	0.256	0.020
70 °F	12.90	0.363	0.028
80 °F	12.90	0.507	0.039
90 °F	12.90	0.699	0.054
Y(moles) = VP/P			

Temperature vs. Vapor Pressure	
LFG Temperature (°F)	Vapor Pressure (psia)
0	0.090
35	0.100
40	0.122
45	0.147
50	0.178
55	0.217
60	0.256
65	0.310
70	0.363
75	0.435
80	0.507
85	0.603
90	0.699
95	0.825
100	0.950
105	1.115
110	1.280
115	1.485
120	1.690

LFG Flow Data		
Typical Pressure at Wellhead	-10	in. w.c.
Blower Inlet Pressure	-50	in. w.c.
Typical Wellhead Temperature	100	°F
Total Site LFG Flow	2,522	scfm

Daily Condensate Generation Rates	
59 °F	1,018
60 °F	956
70 °F	785
80 °F	555
90 °F	-248

Summary: Anticipated condensate generation rate will be 1,018 gallons per day under the noted conditions.

COMPUTATION SHEET

PROJECT TITLE: Wexford County - West Unit Landfill
 DESCRIPTION: Landfill Gas Design Plan
Landfill Gas Condensate Generation - Worst Case Conditions
 Prepared By: JPS Checked By: JPS Approved By: KM

PROJECT NO: 209-4251206
 SHEET: 1
 OF: 1
 DATE: 4/2/2025

Assumptions:

1. LFG temperature at the wellhead is the warmest gas.
2. LFG traveling in a buried header will try to stabilize its temperature consistent with the ground temperature.
3. Ground temp. below the frost line remains constant year round at 59 °F
4. LFG header pipe is installed below frost depth.
5. The LFG temperature (just before the blower) depends on the distance traveled in buried header pipe and the thermal conductivity of the header pipe.
6. Work at the blower compresses the LFG, but no additional condensate is generated due to the heat of compression.

Condensate Generation (gallons/day) = (Y_{wellhead} - Y_{blower inlet}) * LFG Flow * (18/(380*8.35)) * 60min/hr * 24 hrs/day

Wellhead Conditions		
P (psia)	VP (psia)	Y (moles)
14.34	0.950	0.066

Blower Inlet Conditions			
	P (psia)	VP (psia)	Y (moles)
59 °F	12.90	0.217	0.017
60 °F	12.90	0.256	0.020
70 °F	12.90	0.363	0.028
80 °F	12.90	0.507	0.039
90 °F	12.90	0.699	0.054
Y(moles) = VP/P			

Temperature vs. Vapor Pressure	
LFG Temperature (°F)	Vapor Pressure (psia)
0	0.090
35	0.100
40	0.122
45	0.147
50	0.178
55	0.217
60	0.256
65	0.310
70	0.363
75	0.435
80	0.507
85	0.603
90	0.699
95	0.825
100	0.950
105	1.115
110	1.280
115	1.485
120	1.690

LFG Flow Data		
Typical Pressure at Wellhead	-10	in. w.c.
Blower Inlet Pressure	-50	in. w.c.
Typical Wellhead Temperature	100	°F
Total Site LFG Flow	2,409	scfm

Daily Condensate Generation Rates	
59 °F	973
60 °F	913
70 °F	750
80 °F	530
90 °F	-237

Summary: Anticipated condensate generation rate will be 973 gallons per day under the noted conditions.

APPENDIX B

HEADLOSS ANALYSIS

KYGAS HEADLOSS CALCULATIONS

KYGAS was developed by Dr. Don J. Wood and Dr. James E. Funk at the University of Kentucky. The program was modeled after KYPIPE, which models water distribution systems. KYGAS is used to determine head losses, system pressures, and velocities in piping systems controlled under vacuum. KYGAS operates under the assumptions that all flow in the piping system is steady, one-dimensional, isothermal flow for an ideal gas. The program uses the Darcy-Weisbach equation for head losses related to incompressible flow and the Ideal Gas Law for pressure-temperature-density relationships.

KYGAS has several useful options to develop a LFG system. The program allows the user to model any type of piping system material or configuration to coincide with field conditions. The program includes tabular and graphic interfaces for the input of information regarding the system. Multiple blower locations may be used in the program to simulate actual field conditions. In addition, blower performance curves may be entered into the program for comparison to operational and actual field conditions. KYGAS is capable of running multiple scenarios for any piping configuration including looped header systems. LFG flow units and pressure values at the wells are user specified for comparison to values obtained in the field.

The following parameters are required for operation of KYGAS.

- Pipe inside diameter
- Pipe length
- Minor loss coefficient
- Roughness within the pipe
- LFG flow rate into the system at each well or node
- LFG operating temperature (90°F)
- Specific gravity of the LFG (0.943)
- Ratio of specific heats (1.300)
- Absolute viscosity of LFG (2.82×10^{-6} lb*sec/ft²)

The design process begins with the development of the LFG flow rates for use in the program. The LFG flow is determined from the USEPA LandGEM LFG model. The LFG model is developed based on site specific information relating to waste inflow, composition, landfill capacity, and site life. The second step involves evaluating the conceptual design of the LFG system to be installed. Based on the conceptual design and a general understanding of the planned phasing of the landfill, the total LFG flow for a landfill is divided into various amounts to simulate varied flow rates at the wellheads over the life of the site.

As mentioned before, KYGAS allows the user to input information in either a tabular or graphical method. The graphical method is called KYCAD. KYCAD allows the user to input the piping network to scale in an AutoCAD format. The user constructs the system in the computer by drawing the system as it will appear in the field, and then adjusts the necessary pipe lengths and locations of wells (nodes) as required. Site specific conditions are considered when laying out a system. No elevations are used for the various nodes in this analysis. It is assumed that all LFG flow will proceed through the system regardless of node elevations.

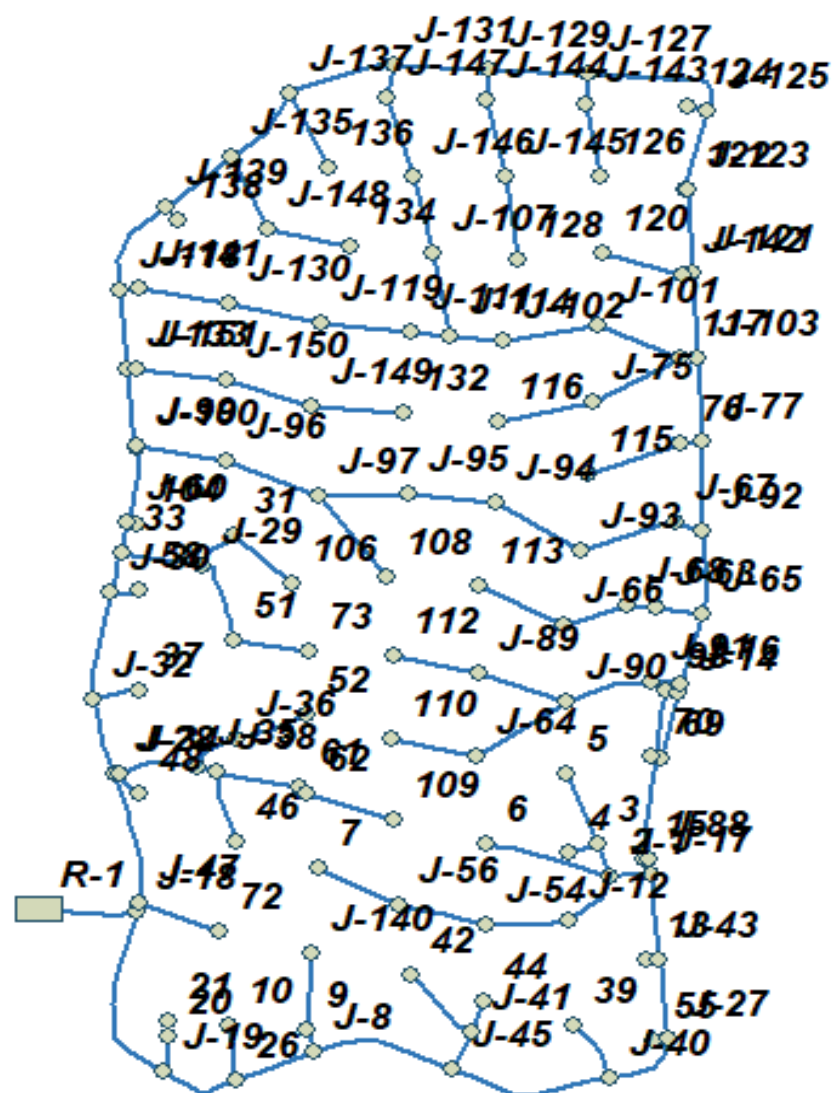
Once all of the required information is in the program, the user can begin to evaluate the system. Evaluation of the system is an iterative process. The initial design is based on the engineer's previous design experience for similar sized systems. Once the results of the initial model are reviewed, the iterative process begins by balancing the system to control LFG velocity, pressure loss, and pipe diameter for various parts of the system. The initial flow rates and their input locations into the system remain unchanged throughout this process. The main factor adjusted for every iteration is the pipe diameter. The inside pipe diameter determines the LFG velocity and pressure drop in each pipe. Once the velocities in the system piping and the vacuum pressure remaining at the furthest node meet design requirements, the engineer may proceed with developing and finalizing the system for construction.

The design criteria for the wellfield header system are:

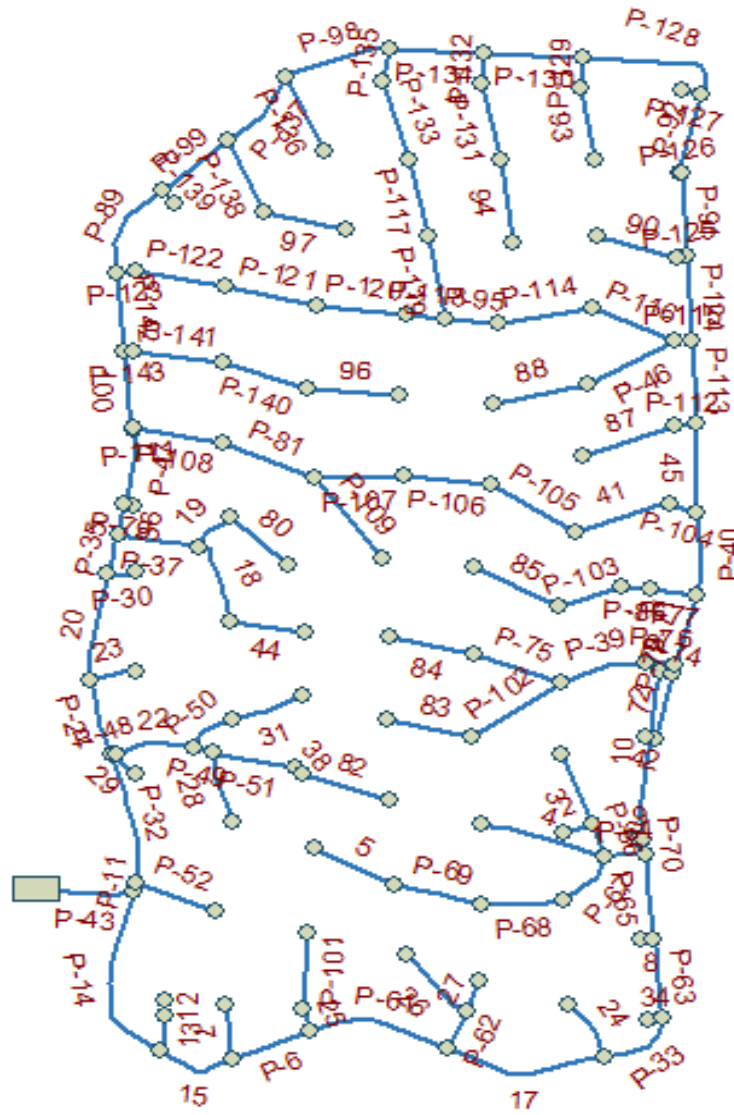
<i>Maximum Counter-Current Velocity</i>	<i>25 fps</i>
<i>Maximum Counter-Current Pressure Drop</i>	<i>1" w.c. per 100 feet of pipe, or 0.36 psig per 1000 feet of pipe</i>

The results of the KYGAS analysis indicate a total system pressure drop of approximately 2.54 in. w.c. (Junction 7) in the east unit, and 1.1 in. w.c. (Junction 41) in the west unit. Please refer to the attached KYGAS output files. The maximum pressure drop is 0.1 psig per 1000 feet of pipe (Pipe P-43) in the east unit, and 0.04 psig per 1000 feet of pipe (Pipe P-28) in the west unit. The only east unit header section (Pipe P-43) where flow velocity exceeds 25 fps at 33.44 fps provides concurrent LFG flow. All values are within allowable operational ranges.

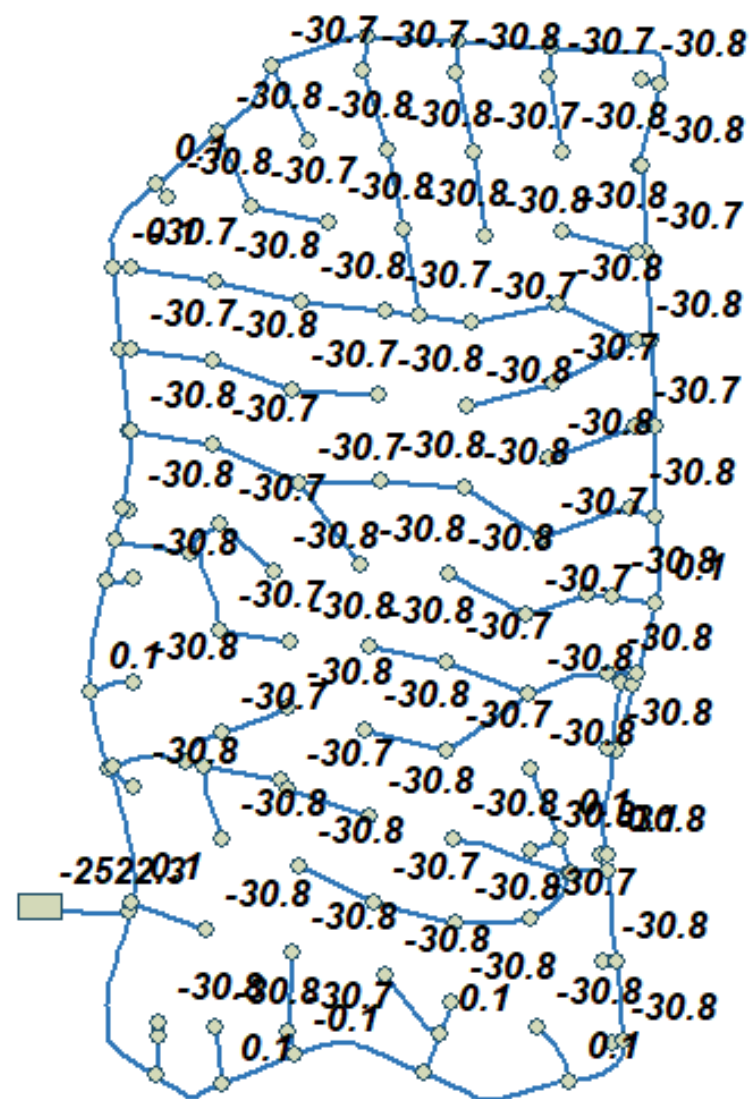
Wexford County Landfill (East Unit) – Final Development
 Landfill Gas Collection and Control System
 Junction Name



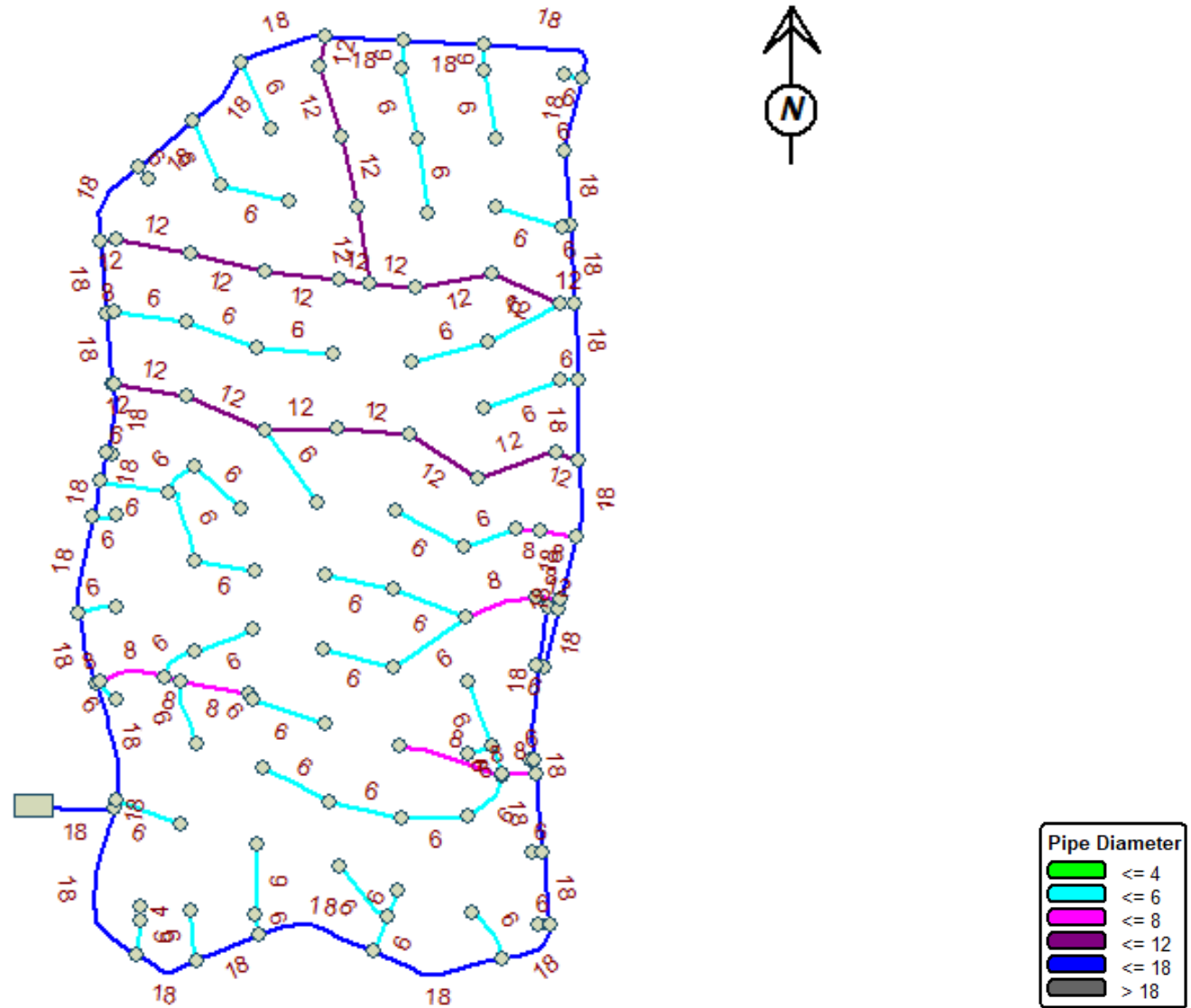
Wexford County Landfill (East Unit) – Final Development
Landfill Gas Collection and Control System
Pipe Name



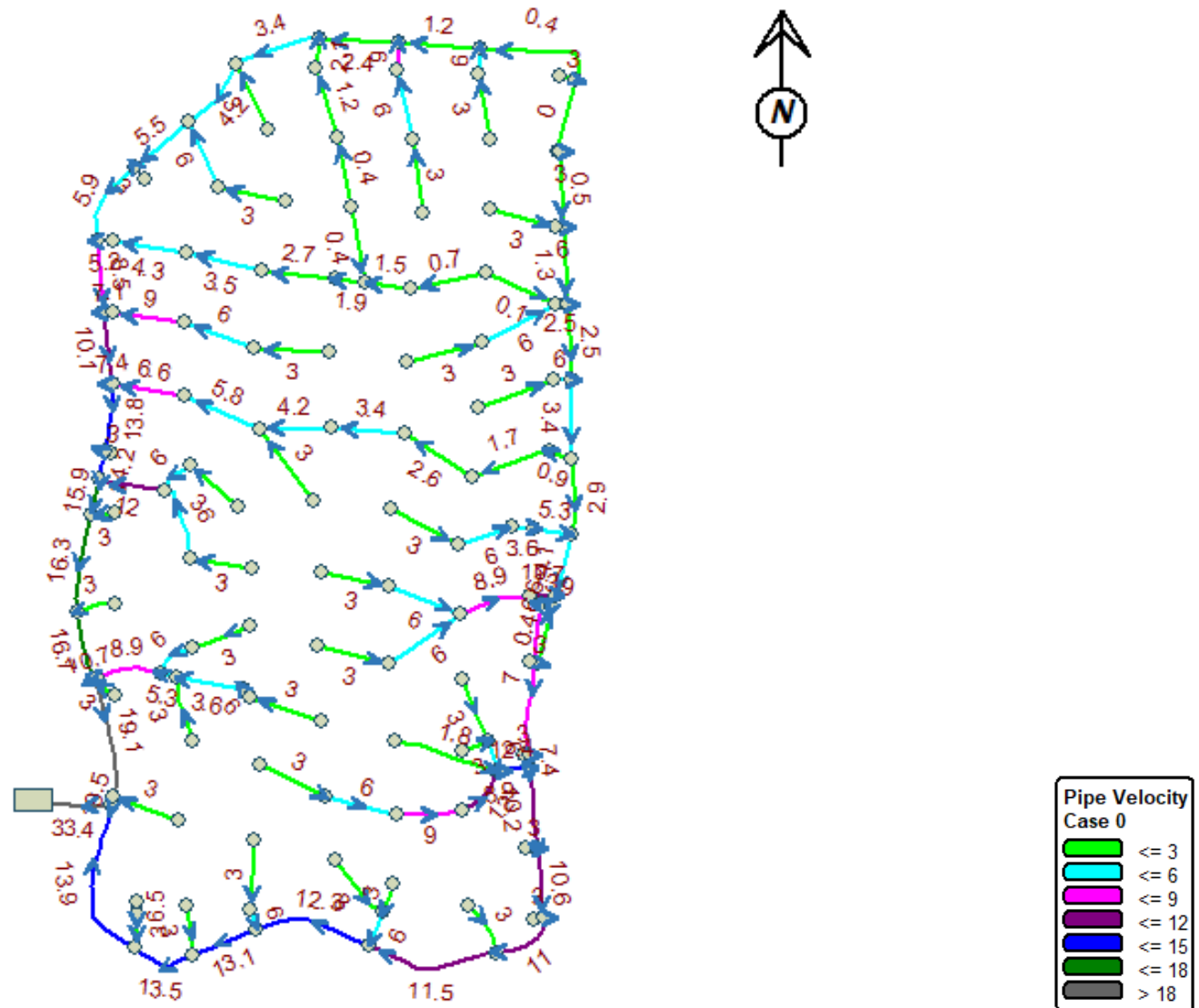
Wexford County Landfill (East Unit) – Final Development
Landfill Gas Collection and Control System
Junction LFG Demand (scfm)



Wexford County Landfill (East Unit) – Final Development
Landfill Gas Collection and Control System
Pipe Diameter (in.)



Wexford County Landfill (East Unit) – Final Development
 Landfill Gas Collection and Control System
 Pipe Velocity (fps)



Junction Available Vacuum (in. w.c.)



```

* * * * * K Y G A S * * * * *
*
* Gas Network Analysis Software (KYGAS)
*
* CopyRighted by KYPIPE LLC (www.kypipe.com)
* Version: 12.009b 03-25-2024
* Company: TETRAWIN Serial #: 580297
* Interface: KYnetic
* Licensed for Pipe2024
*
* * * * *

```

```

INPUT DATA FILE NAME FOR THIS SIMULATION =
c:\users\JARRET~1.DUT\DOCUME~1\WVWC
-2~3.KYP\wvwc-202.DAT
OUTPUT DATA FILE NAME FOR THIS SIMULATION =
c:\users\JARRET~1.DUT\DOCUME~1\WVWC
-2~3.KYP\wvwc-202.OT2

```

```

DATE FOR THIS COMPUTER RUN      : 3-21-2025
START TIME FOR THIS COMPUTER RUN : 9:56:46:58

```

SUMMARY OF DISTRIBUTION SYSTEM CHARACTERISTICS:

```

NUMBER OF PIPES      = 133
NUMBER OF JUNCTION NODES = 129

UNITS SPECIFIED      = ENGLISH

```

```

A CONSTANT DENSITY FLUID IS SPECIFIED - DENSITY = .07POUNDS/CUBIC
FOOT
ABSOLUTE VISCOSITY      = .280E-06 POUND SECONDS/SQUARE
FOOT

```

```

USER SPEC. FLOW UNITS (USFU)      = SCF/MIN
USER SPEC. PRESSURE UNITS(USPU)   = INCHES OF WATER (GAUGE)

```

----- SIMULATION DESCRIPTION -----

----- SUMMARY OF PIPE NETWORK GEOMETRIC AND OPERATING DATA -----

PIPE NAME	NODE #1	NODE #2	LENGTH (FT.)	DIAM. (IN.)	ROUGHNESS (MILLIFEET)	SUM-M FACT.	PUMP ID	ELEVATION CHANGE
--------------	------------	------------	-----------------	----------------	--------------------------	----------------	------------	---------------------

1	2	J-1	3.0	7.5	.100	1.8	0	.0
2	3	4	60.2	5.8	.100	1.8	0	.0
3	3	5	62.5	5.8	.100	1.8	0	.0
4	2	6	248.2	7.5	.100	1.8	0	.0
5	7	J-56	157.4	5.8	.100	1.8	0	.0
7	10	26	15.6	5.8	.100	1.8	0	.0
8	13	J-43	24.8	5.8	.100	1.8	0	.0
9	15	J-88	13.8	5.8	.100	1.8	0	.0
10	J-88	98	78.6	15.8	.100	1.8	0	.0
12	20	21	.5	3.9	.100	1.8	0	.0
13	20	J-19	10.7	5.8	.100	1.8	0	.0
15	J-19	26	111.4	15.8	.100	1.8	0	.0
17	J-45	J-40	177.2	15.8	.100	1.8	0	.0
18	J-29	51	225.3	5.8	.100	3.6	0	.0
19	31	J-29	63.5	5.8	.100	1.8	0	.0
20	J-30	J-32	80.9	15.8	.100	1.8	0	.0
22	J-34	J-35	157.1	7.5	.100	1.8	0	.0
23	37	J-32	94.5	5.8	.100	1.8	0	.0
24	39	J-40	71.2	5.8	.100	1.8	0	.0
25	9	J-8	16.2	5.8	.100	1.8	0	.0
26	42	J-41	119.4	5.8	.100	1.8	0	.0
27	44	J-41	27.1	5.8	.100	1.8	0	.0
28	46	J-38	40.7	5.8	.100	1.8	0	.0
29	48	J-34	41.2	5.8	.100	1.8	0	.0
31	52	J-36	127.4	5.8	.100	1.8	0	.0
34	55	J-27	25.8	5.8	.100	1.8	0	.0
36	J-60	33	15.9	15.8	.100	1.8	0	.0
38	61	62	12.8	5.8	.100	1.8	0	.0
41	J-67	J-93	190.2	11.2	.100	1.8	0	.0
42	69	70	18.1	5.8	.100	1.8	0	.0
44	73	51	148.5	5.8	.100	1.8	0	.0
45	J-77	J-92	3.9	15.8	.100	1.8	0	.0
72	69	J-14	29.1	15.8	.100	1.8	0	.0
80	31	106	116.1	5.8	.100	1.8	0	.0
82	109	62	176.5	5.8	.100	1.8	0	.0
83	110	J-64	157.2	5.8	.100	1.8	0	.0
84	112	J-89	162.5	5.8	.100	1.8	0	.0
85	113	J-66	175.2	5.8	.100	1.8	0	.0
87	115	76	182.2	5.8	.100	1.8	0	.0
88	116	J-75	177.6	5.8	.100	1.8	0	.0
90	120	J-142	161.7	5.8	.100	1.8	0	.0
93	126	J-143	23.3	5.8	.100	1.8	0	.0
94	128	J-145	28.2	5.8	.100	1.8	0	.0
96	132	J-149	182.0	5.8	.100	1.8	0	.0
97	134	J-148	116.4	5.8	.100	1.8	0	.0
100	J-100	J-133	86.8	15.8	.100	1.8	0	.0
P-101	J-140	9	7.8	5.8	.100	1.8	0	.0
P-102	J-90	J-64	215.0	5.8	.100	1.8	0	.0
P-103	J-66	J-68	123.9	5.8	.100	1.8	0	.0
P-104	J-67	J-92	57.0	11.2	.100	1.8	0	.0
P-105	J-93	J-94	185.6	11.2	.100	1.8	0	.0
P-106	J-94	J-95	168.3	11.2	.100	1.8	0	.0
P-107	J-95	J-97	166.8	11.2	.100	1.8	0	.0

P-108	J-96	J-99	170.4	11.2	.100	1.8	0	.0
P-109	108	J-97	220.0	5.8	.100	1.8	0	.0
P-11	J-18	J-47	20.0	15.8	.100	1.8	0	.0
P-111	J-100	J-99	4.0	11.2	.100	1.8	0	.0
P-112	J-77	76	44.0	5.8	.100	1.8	0	.0
P-113	J-103	J-77	92.9	15.8	.100	1.8	0	.0
P-114	J-102	J-101	184.1	11.2	.100	1.8	0	.0
P-115	J-103	117	32.0	11.2	.100	1.8	0	.0
P-116	117	J-101	184.0	11.2	.100	1.8	0	.0
P-117	J-107	J-146	38.1	11.2	.100	1.8	0	.0
P-118	J-111	J-114	73.1	11.2	.100	1.8	0	.0
P-119	J-114	J-107	187.0	11.2	.100	1.8	0	.0
P-120	J-119	J-111	178.5	11.2	.100	1.8	0	.0
P-121	J-130	J-119	182.7	11.2	.100	1.8	0	.0
P-122	J-141	J-130	179.9	11.2	.100	1.8	0	.0
P-123	J-118	J-141	37.0	11.2	.100	1.8	0	.0
P-124	J-121	J-103	94.4	15.8	.100	1.8	0	.0
P-125	J-121	J-142	18.0	5.8	.100	1.8	0	.0
P-126	J-123	122	3.0	5.8	.100	1.8	0	.0
P-127	J-125	124	42.0	5.8	.100	1.8	0	.0
P-128	J-127	J-125	151.2	15.8	.100	1.8	0	.0
P-129	J-127	J-143	64.0	5.8	.100	1.8	0	.0
P-130	J-129	J-127	99.4	15.8	.100	1.8	0	.0
P-131	J-145	J-144	26.6	5.8	.100	1.8	0	.0
P-132	J-129	J-144	68.0	5.8	.100	1.8	0	.0
P-133	J-146	J-147	39.6	11.2	.100	1.8	0	.0
P-134	J-131	J-129	95.7	15.8	.100	1.8	0	.0
P-135	J-147	J-131	74.0	11.2	.100	1.8	0	.0
P-136	J-137	136	178.0	5.8	.100	1.8	0	.0
P-137	J-135	J-137	92.5	15.8	.100	1.8	0	.0
P-138	J-135	J-148	174.0	5.8	.100	1.8	0	.0
P-139	J-139	138	38.0	5.8	.100	1.8	0	.0
P-14	J-19	J-18	184.5	15.8	.100	1.8	0	.0
P-140	J-149	J-150	177.8	5.8	.100	1.8	0	.0
P-141	J-150	J-151	174.3	5.8	.100	1.8	0	.0
P-142	J-133	J-118	88.1	15.8	.100	1.8	0	.0
P-143	J-133	J-151	20.0	7.5	.100	1.8	0	.0
P-21	J-32	J-28	35.6	15.8	.100	1.8	0	.0
P-30	58	J-30	58.4	5.8	.100	1.8	0	.0
P-32	J-28	J-47	60.5	15.8	.100	1.8	0	.0
P-33	J-40	J-27	86.8	15.8	.100	1.8	0	.0
P-35	J-30	33	17.5	15.8	.100	1.8	0	.0
P-37	J-29	33	163.7	5.8	.100	1.8	0	.0
P-39	J-90	J-91	173.3	7.5	.100	1.8	0	.0
P-40	J-65	J-92	37.0	15.8	.100	1.8	0	.0
P-43	R-1	J-18	194.4	15.8	.100	1.8	0	.0
P-46	J-75	117	187.0	5.8	.100	1.8	0	.0
P-47	J-60	J-100	45.7	15.8	.100	1.8	0	.0
P-48	J-34	J-28	10.7	7.5	.100	1.8	0	.0
P-49	J-35	J-38	40.3	7.5	.100	1.8	0	.0
P-50	J-36	J-35	90.2	5.8	.100	1.8	0	.0
P-51	J-38	61	163.5	7.5	.100	1.8	0	.0
P-52	J-47	72	170.0	5.8	.100	1.8	0	.0
P-6	J-8	26	157.1	15.8	.100	1.8	0	.0

P-61	J-45	J-8	275.1	15.8	.100	1.8	0	.0
P-62	J-41	J-45	35.1	5.8	.100	1.8	0	.0
P-63	J-27	J-43	94.3	15.8	.100	1.8	0	.0
P-64	J-1	J-17	81.1	7.5	.100	1.8	0	.0
P-65	J-17	J-43	62.5	15.8	.100	1.8	0	.0
P-66	J-1	3	75.0	5.8	.100	1.8	0	.0
P-67	J-12	2	118.3	5.8	.100	1.8	0	.0
P-68	J-54	J-12	142.7	5.8	.100	1.8	0	.0
P-69	J-56	J-54	160.9	5.8	.100	1.8	0	.0
P-70	J-88	J-17	11.3	15.8	.100	1.8	0	.0
P-71	J-16	J-65	31.6	15.8	.100	1.8	0	.0
P-73	J-14	J-16	4.6	15.8	.100	1.8	0	.0
P-74	J-14	98	28.0	11.2	.100	1.8	0	.0
P-75	J-89	J-90	177.3	5.8	.100	1.8	0	.0
P-76	J-16	J-91	60.3	7.5	.100	1.8	0	.0
P-77	J-65	J-63	92.0	7.5	.100	1.8	0	.0
P-79	J-60	104	15.0	5.8	.100	1.8	0	.0
P-81	J-97	J-96	192.0	11.2	.100	1.8	0	.0
P-86	J-68	J-63	51.1	7.5	.100	1.8	0	.0
P-89	J-118	J-139	109.8	15.8	.100	1.8	0	.0
P-91	J-123	J-121	92.0	15.8	.100	1.8	0	.0
P-92	J-125	J-123	90.2	15.8	.100	1.8	0	.0
P-95	J-114	J-102	107.8	11.2	.100	1.8	0	.0
P-98	J-137	J-131	108.8	15.8	.100	1.8	0	.0
P-99	J-139	J-135	86.0	15.8	.100	1.8	0	.0

JUNCTION NAME	NODE TITLE	ELEV	DEMAND (USFU)	FPN PRESSURE
2		.00	.00	
3		.00	.00	
4		.00	-30.76	
5		.00	-30.76	
6		.00	-30.76	
7		.00	-30.76	
9		.00	-30.76	
10		.00	-30.76	
13		.00	-30.76	
15		.00	-30.76	
20		.00	.00	
21		.00	-30.76	
26		.00	.00	
31		.00	-30.76	
33		.00	.00	
37		.00	-30.76	
39		.00	-30.76	
42		.00	-30.76	
44		.00	-30.76	
46		.00	-30.76	
48		.00	-30.76	
51		.00	-30.76	
52		.00	-30.76	
55		.00	-30.76	

58	.00	-30.76
61	.00	.00
62	.00	-30.76
69	.00	.00
70	.00	-30.76
72	.00	-30.76
73	.00	-30.76
76	.00	-30.76
98	.00	.00
104	.00	-30.76
106	.00	-30.76
108	.00	-30.76
109	.00	-30.76
110	.00	-30.76
112	.00	-30.76
113	.00	-30.76
115	.00	-30.76
116	.00	-30.76
117	.00	-30.76
120	.00	-30.76
122	.00	-30.76
124	.00	-30.76
126	.00	-30.76
128	.00	-30.76
132	.00	-30.76
134	.00	-30.76
136	.00	-30.76
138	.00	-30.76
J-1	.00	.00
J-100	.00	.00
J-101	.00	-30.76
J-102	.00	-30.76
J-103	.00	.00
J-107	.00	-30.76
J-111	.00	-30.76
J-114	.00	.00
J-118	.00	.00
J-119	.00	-30.76
J-12	.00	-30.76
J-121	.00	.00
J-123	.00	.00
J-125	.00	.00
J-127	.00	.00
J-129	.00	.00
J-130	.00	-30.76
J-131	.00	.00
J-133	.00	.00
J-135	.00	-30.76
J-137	.00	-30.76
J-139	.00	.00
J-14	.00	.00
J-140	.00	-30.76
J-141	.00	-30.76
J-142	.00	-30.76

J-143	.00	-30.76	
J-144	.00	-30.76	
J-145	.00	-30.76	
J-146	.00	-30.76	
J-147	.00	-30.76	
J-148	.00	-30.76	
J-149	.00	-30.76	
J-150	.00	-30.76	
J-151	.00	-30.76	
J-16	.00	.00	
J-17	.00	.00	
J-18	.00	.00	
J-19	.00	.00	
J-27	.00	.00	
J-28	.00	.00	
J-29	.00	.00	
J-30	.00	.00	
J-32	.00	.00	
J-34	.00	.00	
J-35	.00	.00	
J-36	.00	-30.76	
J-38	.00	.00	
J-40	.00	.00	
J-41	.00	.00	
J-43	.00	.00	
J-45	.00	.00	
J-47	.00	.00	
J-54	.00	-30.76	
J-56	.00	-30.76	
J-60	.00	.00	
J-63	.00	-30.76	
J-64	.00	-30.76	
J-65	.00	.00	
J-66	.00	-30.76	
J-67	.00	-30.76	
J-68	.00	.00	
J-75	.00	-30.76	
J-77	.00	.00	
J-8	.00	.00	
J-88	.00	.00	
J-89	.00	-30.76	
J-90	.00	-30.76	
J-91	.00	-30.76	
J-92	.00	.00	
J-93	.00	-30.76	
J-94	.00	-30.76	
J-95	.00	-30.76	
J-96	.00	-30.76	
J-97	.00	-30.76	
J-99	.00	-30.76	
R-1	.00	.00	-50.00

=====

=====

Set = 0

===== RESULTS FOR THIS SIMULATION FOLLOW
=====

Solution was obtained in 17 trials
Flow Accuracy = .1391E-02[< .500E-02]
RV Accuracy = .0000E+00[< .100E-02]

```
-----
-----
      PIPE      NODE      NODE      FLOW      LOSS VELOCITY DENSITY
FRICION AREA  #1      #2      (USFU)    (USPU)    (FT/S)    (#/CF ) FACTOR
RATIO
-----
-----
      1         2         J-1      153.800    .03       8.88     .0710
.0220
      2         3         4       -30.760    .01       3.01     .0710
.0302
      3         3         5       -30.760    .01       3.01     .0710
.0302
      4         2         6       -30.760    .01       1.78     .0710
.0323
      5         7         J-56    30.760    .02       3.01     .0710
.0302
      7        10        26      30.760    .01       3.01     .0710
.0302
      8        13        J-43    30.760    .01       3.01     .0710
.0302
      9        15        J-88    30.760    .01       3.01     .0710
.0302
     10       J-88        98     -525.667    .03       6.97     .0710
.0195
     12        20        21      -30.760    .02       6.52     .0710
.0276
     13        20        J-19    30.760    .00       3.01     .0710
.0302
     15       J-19        26    -1017.827    .13      13.50     .0710
.0172
     17       J-45       J-40     -864.027    .12      11.46     .0710
.0177
     18       J-29        51      -61.520    .12       6.02     .0710
.0255
     19        31       J-29     61.520    .04       6.02     .0710
.0255
```

.0167	20	J-30	J-32	1227.653	.16	16.28	.0710
.0220	22	J-34	J-35	-153.800	.12	8.88	.0710
.0302	23	37	J-32	30.760	.01	3.01	.0710
.0302	24	39	J-40	30.760	.01	3.01	.0710
.0255	25	9	J-8	61.520	.02	6.02	.0710
.0302	26	42	J-41	30.760	.02	3.01	.0710
.0302	27	44	J-41	30.760	.01	3.01	.0710
.0302	28	46	J-38	30.760	.01	3.01	.0710
.0302	29	48	J-34	30.760	.01	3.01	.0710
.0302	31	52	J-36	30.760	.02	3.01	.0710
.0302	34	55	J-27	30.760	.01	3.01	.0710
.0171	36	J-60	33	1073.853	.09	14.24	.0710
.0255	38	61	62	-61.520	.02	6.02	.0710
.0291	41	J-67	J-93	66.174	.01	1.75	.0710
.0302	42	69	70	-30.760	.01	3.01	.0710
.0302	44	73	51	30.760	.02	3.01	.0710
.0228	45	J-77	J-92	253.481	.00	3.36	.0710
.0400	72	69	J-14	30.760	.00	.41	.0710
.0302	80	31	106	-30.760	.02	3.01	.0710
.0302	82	109	62	30.760	.02	3.01	.0710
.0302	83	110	J-64	30.760	.02	3.01	.0710
.0302	84	112	J-89	30.760	.02	3.01	.0710
.0302	85	113	J-66	30.760	.02	3.01	.0710
.0302	87	115	76	30.760	.03	3.01	.0710
.0302	88	116	J-75	30.760	.02	3.01	.0710
.0302	90	120	J-142	30.760	.02	3.01	.0710
.0302	93	126	J-143	30.760	.01	3.01	.0710

.0302	94	128	J-145	30.760	.01	3.01	.0710
.0302	96	132	J-149	30.760	.03	3.01	.0710
.0302	97	134	J-148	30.760	.02	3.01	.0710
.0182	100	J-100	J-133	-761.599	.06	10.10	.0710
.0302	P-101	J-140	9	30.760	.00	3.01	.0710
.0255	P-102	J-90	J-64	-61.520	.10	6.02	.0710
.0255	P-103	J-66	J-68	61.520	.06	6.02	.0710
.0345	P-104	J-67	J-92	-35.414	.00	.94	.0710
.0265	P-105	J-93	J-94	96.934	.01	2.56	.0710
.0248	P-106	J-94	J-95	127.694	.02	3.37	.0710
.0236	P-107	J-95	J-97	158.454	.02	4.19	.0710
.0213	P-108	J-96	J-99	250.734	.05	6.63	.0710
.0302	P-109	108	J-97	30.760	.03	3.01	.0710
.0162	P-11	J-18	J-47	-1473.733	.17	19.54	.0710
.0208	P-111	J-100	J-99	-281.494	.02	7.44	.0710
.0255	P-112	J-77	76	-61.520	.03	6.02	.0710
.0243	P-113	J-103	J-77	191.961	.00	2.55	.0710
.0375	P-114	J-102	J-101	-26.756	.00	.71	.0710
.0265	P-115	J-103	117	-96.284	.00	2.54	.0710
.0046	P-116	117	J-101	-4.004	.00	.11	.0710
.0439	P-117	J-107	J-146	16.094	.00	.43	.0710
.0285	P-118	J-111	J-114	-72.183	.00	1.91	.0710
.0456	P-119	J-114	J-107	-14.666	.00	.39	.0710
.0261	P-120	J-119	J-111	-102.943	.01	2.72	.0710
.0245	P-121	J-130	J-119	-133.703	.02	3.53	.0710
.0234	P-122	J-141	J-130	-164.463	.03	4.35	.0710
.0225	P-123	J-118	J-141	-195.223	.02	5.16	.0710

.0289	P-124	J-121	J-103	95.677	.00	1.27	.0710
.0255	P-125	J-121	J-142	-61.520	.02	6.02	.0710
.0302	P-126	J-123	122	-30.760	.00	3.01	.0710
.0302	P-127	J-125	124	-30.760	.01	3.01	.0710
.0417	P-128	J-127	J-125	-27.363	.00	.36	.0710
.0255	P-129	J-127	J-143	-61.520	.04	6.02	.0710
.0295	P-130	J-129	J-127	-88.883	.00	1.18	.0710
.0255	P-131	J-145	J-144	61.520	.02	6.02	.0710
.0233	P-132	J-129	J-144	-92.280	.09	9.03	.0710
.0320	P-133	J-146	J-147	46.854	.00	1.24	.0710
.0247	P-134	J-131	J-129	-181.163	.00	2.40	.0710
.0280	P-135	J-147	J-131	77.614	.00	2.05	.0710
.0302	P-136	J-137	136	-30.760	.02	3.01	.0710
.0217	P-137	J-135	J-137	-320.296	.01	4.25	.0710
.0255	P-138	J-135	J-148	-61.520	.08	6.02	.0710
.0302	P-139	J-139	138	-30.760	.01	3.01	.0710
.0171	P-14	J-19	J-18	1048.587	.17	13.90	.0710
.0255	P-140	J-149	J-150	61.520	.09	6.02	.0710
.0233	P-141	J-150	J-151	92.280	.18	9.03	.0710
.0188	P-142	J-133	J-118	-638.559	.05	8.47	.0710
.0230	P-143	J-133	J-151	-123.040	.03	7.11	.0710
.0166	P-21	J-32	J-28	1258.413	.13	16.69	.0710
.0302	P-30	58	J-30	30.760	.01	3.01	.0710
.0162	P-32	J-28	J-47	1442.973	.20	19.13	.0710
.0179	P-33	J-40	J-27	-833.267	.08	11.05	.0710
.0167	P-35	J-30	33	-1196.893	.11	15.87	.0710
.0220	P-37	J-29	33	123.040	.28	12.04	.0710

	P-39	J-90	J-91	153.800	.13	8.88	.0710
.0220							
	P-40	J-65	J-92	-218.067	.00	2.89	.0710
.0236							
	P-43	R-1	J-18	-2522.320	.95	33.44	.0710
.0149							
	P-46	J-75	117	61.520	.09	6.02	.0710
.0255							
	P-47	J-60	J-100	-1043.093	.10	13.83	.0710
.0171							
	P-48	J-34	J-28	184.560	.05	10.66	.0710
.0212							
	P-49	J-35	J-38	-92.280	.02	5.33	.0710
.0246							
	P-50	J-36	J-35	61.520	.05	6.02	.0710
.0255							
	P-51	J-38	61	-61.520	.02	3.55	.0710
.0270							
	P-52	J-47	72	-30.760	.02	3.01	.0710
.0302							
	P-6	J-8	26	987.067	.14	13.09	.0710
.0173							
	P-61	J-45	J-8	925.547	.17	12.27	.0710
.0175							
	P-62	J-41	J-45	61.520	.03	6.02	.0710
.0255							
	P-63	J-27	J-43	-802.507	.07	10.64	.0710
.0180							
	P-64	J-1	J-17	215.320	.15	12.44	.0710
.0205							
	P-65	J-17	J-43	771.747	.06	10.23	.0710
.0181							
	P-66	J-1	3	-61.520	.04	6.02	.0710
.0255							
	P-67	J-12	2	123.040	.22	12.04	.0710
.0220							
	P-68	J-54	J-12	92.280	.15	9.03	.0710
.0233							
	P-69	J-56	J-54	61.520	.08	6.02	.0710
.0255							
	P-70	J-88	J-17	556.427	.02	7.38	.0710
.0193							
	P-71	J-16	J-65	-310.347	.01	4.12	.0710
.0218							
	P-73	J-14	J-16	-494.907	.02	6.56	.0710
.0198							
	P-74	J-14	98	525.667	.10	13.89	.0710
.0184							
	P-75	J-89	J-90	61.520	.09	6.02	.0710
.0255							
	P-76	J-16	J-91	-184.560	.09	10.66	.0710
.0212							
	P-77	J-65	J-63	-92.280	.03	5.33	.0710
.0246							

P-79	J-60	104	-30.760	.01	3.01	.0710	
.0302	P-81	J-97	J-96	219.974	.05	5.81	.0710
.0219	P-86	J-68	J-63	61.520	.01	3.55	.0710
.0270	P-89	J-118	J-139	-443.336	.03	5.88	.0710
.0202	P-91	J-123	J-121	34.157	.00	.45	.0710
.0385	P-92	J-125	J-123	3.397	.00	.05	.0710
.0393	P-95	J-114	J-102	-57.516	.00	1.52	.0710
.0303	P-98	J-137	J-131	-258.776	.01	3.43	.0710
.0227	P-99	J-139	J-135	-412.576	.02	5.47	.0710
.0205	R-1	R-1	R-1	-2522.320	.00	.01	.0710
.0369							

JUNCTION	NODE	DEMAND	PRESSURE	PRESSURE	PRESSURE	
DENSITY						
NAME	TITLE	(USFU)	(USPU)	(PSIA)	(PSIG)	
#/CF						

2		.00	-47.93	12.97	-1.73	
.071						
3		.00	-47.92	12.97	-1.73	
.071						
4		-30.76	-47.91	12.97	-1.73	
.071						
5		-30.76	-47.91	12.97	-1.73	
.071						
6		-30.76	-47.92	12.97	-1.73	
.071						
7		-30.76	-47.46	12.98	-1.71	
.071						
9		-30.76	-48.59	12.94	-1.75	
.071						
10		-30.76	-48.75	12.94	-1.76	
.071						
13		-30.76	-48.16	12.96	-1.74	
.071						
15		-30.76	-48.08	12.96	-1.73	
.071						
20		.00	-48.87	12.93	-1.76	
.071						

.071	21	-30.76	-48.86	12.93	-1.76
.071	26	.00	-48.75	12.94	-1.76
.071	31	-30.76	-47.96	12.97	-1.73
.071	33	.00	-48.29	12.95	-1.74
.071	37	-30.76	-48.54	12.94	-1.75
.071	39	-30.76	-48.31	12.95	-1.74
.071	42	-30.76	-48.39	12.95	-1.75
.071	44	-30.76	-48.40	12.95	-1.75
.071	46	-30.76	-48.48	12.95	-1.75
.071	48	-30.76	-48.63	12.94	-1.75
.071	51	-30.76	-47.88	12.97	-1.73
.071	52	-30.76	-48.44	12.95	-1.75
.071	55	-30.76	-48.23	12.96	-1.74
.071	58	-30.76	-48.38	12.95	-1.75
.071	61	.00	-48.47	12.95	-1.75
.071	62	-30.76	-48.45	12.95	-1.75
.071	69	.00	-47.96	12.97	-1.73
.071	70	-30.76	-47.95	12.97	-1.73
.071	72	-30.76	-48.86	12.93	-1.76
.071	73	-30.76	-47.86	12.97	-1.73
.071	76	-30.76	-47.89	12.97	-1.73
.071	98	.00	-48.05	12.96	-1.73
.071	104	-30.76	-48.19	12.96	-1.74
.071	106	-30.76	-47.94	12.97	-1.73
.071	108	-30.76	-47.95	12.97	-1.73
.071	109	-30.76	-48.42	12.95	-1.75
.071	110	-30.76	-47.59	12.98	-1.72

.071	112	-30.76	-47.61	12.98	-1.72
.071	113	-30.76	-47.80	12.97	-1.72
.071	115	-30.76	-47.87	12.97	-1.73
.071	116	-30.76	-47.80	12.97	-1.72
.071	117	-30.76	-47.92	12.97	-1.73
.071	120	-30.76	-47.87	12.97	-1.73
.071	122	-30.76	-47.91	12.97	-1.73
.071	124	-30.76	-47.91	12.97	-1.73
.071	126	-30.76	-47.87	12.97	-1.73
.071	128	-30.76	-47.80	12.97	-1.72
.071	132	-30.76	-47.72	12.97	-1.72
.071	134	-30.76	-47.84	12.97	-1.73
.071	136	-30.76	-47.91	12.97	-1.73
.071	138	-30.76	-47.96	12.97	-1.73
.071	J-1	.00	-47.96	12.97	-1.73
.071	J-100	.00	-48.10	12.96	-1.74
.071	J-101	-30.76	-47.92	12.97	-1.73
.071	J-102	-30.76	-47.92	12.97	-1.73
.071	J-103	.00	-47.92	12.97	-1.73
.071	J-107	-30.76	-47.92	12.97	-1.73
.071	J-111	-30.76	-47.92	12.97	-1.73
.071	J-114	.00	-47.92	12.97	-1.73
.071	J-118	.00	-47.99	12.96	-1.73
.071	J-119	-30.76	-47.93	12.97	-1.73
.071	J-12	-30.76	-47.71	12.97	-1.72
.071	J-121	.00	-47.92	12.97	-1.73
.071	J-123	.00	-47.92	12.97	-1.73

J-125	.00	-47.92	12.97	-1.73
.071 J-127	.00	-47.92	12.97	-1.73
.071 J-129	.00	-47.92	12.97	-1.73
.071 J-130	-30.76	-47.95	12.97	-1.73
.071 J-131	.00	-47.92	12.97	-1.73
.071 J-133	.00	-48.04	12.96	-1.73
.071 J-135	-30.76	-47.95	12.97	-1.73
.071 J-137	-30.76	-47.93	12.97	-1.73
.071 J-139	.00	-47.97	12.97	-1.73
.071 J-14	.00	-47.96	12.97	-1.73
.071 J-140	-30.76	-48.59	12.94	-1.75
.071 J-141	-30.76	-47.98	12.96	-1.73
.071 J-142	-30.76	-47.90	12.97	-1.73
.071 J-143	-30.76	-47.88	12.97	-1.73
.071 J-144	-30.76	-47.83	12.97	-1.73
.071 J-145	-30.76	-47.81	12.97	-1.73
.071 J-146	-30.76	-47.92	12.97	-1.73
.071 J-147	-30.76	-47.92	12.97	-1.73
.071 J-148	-30.76	-47.86	12.97	-1.73
.071 J-149	-30.76	-47.75	12.97	-1.72
.071 J-150	-30.76	-47.83	12.97	-1.73
.071 J-151	-30.76	-48.01	12.96	-1.73
.071 J-16	.00	-47.94	12.97	-1.73
.071 J-17	.00	-48.11	12.96	-1.74
.071 J-18	.00	-49.05	12.93	-1.77
.071 J-19	.00	-48.88	12.93	-1.76
.071 J-27	.00	-48.24	12.96	-1.74
.071				

.071	J-28	.00	-48.69	12.94	-1.76
.071	J-29	.00	-48.00	12.96	-1.73
.071	J-30	.00	-48.39	12.95	-1.75
.071	J-32	.00	-48.55	12.94	-1.75
.071	J-34	.00	-48.63	12.94	-1.75
.071	J-35	.00	-48.51	12.95	-1.75
.071	J-36	-30.76	-48.46	12.95	-1.75
.071	J-38	.00	-48.49	12.95	-1.75
.071	J-40	.00	-48.32	12.95	-1.74
.071	J-41	.00	-48.41	12.95	-1.75
.071	J-43	.00	-48.17	12.96	-1.74
.071	J-45	.00	-48.44	12.95	-1.75
.071	J-47	.00	-48.88	12.93	-1.76
.071	J-54	-30.76	-47.56	12.98	-1.72
.071	J-56	-30.76	-47.48	12.98	-1.71
.071	J-60	.00	-48.20	12.96	-1.74
.071	J-63	-30.76	-47.90	12.97	-1.73
.071	J-64	-30.76	-47.62	12.98	-1.72
.071	J-65	.00	-47.93	12.97	-1.73
.071	J-66	-30.76	-47.83	12.97	-1.73
.071	J-67	-30.76	-47.93	12.97	-1.73
.071	J-68	.00	-47.89	12.97	-1.73
.071	J-75	-30.76	-47.83	12.97	-1.73
.071	J-77	.00	-47.92	12.97	-1.73
.071	J-8	.00	-48.61	12.94	-1.75
.071	J-88	.00	-48.09	12.96	-1.74
.071	J-89	-30.76	-47.63	12.98	-1.72

.071	J-90	-30.76	-47.72	12.97	-1.72
.071	J-91	-30.76	-47.85	12.97	-1.73
.071	J-92	.00	-47.93	12.97	-1.73
.071	J-93	-30.76	-47.93	12.97	-1.73
.071	J-94	-30.76	-47.94	12.97	-1.73
.071	J-95	-30.76	-47.96	12.97	-1.73
.071	J-96	-30.76	-48.03	12.96	-1.73
.071	J-97	-30.76	-47.98	12.96	-1.73
.071	J-99	-30.76	-48.08	12.96	-1.73
.071	R-1	.00	-50.00	12.89	-1.80

* This designates the use of default density in a low pressure region

THE NET SYSTEM DEMAND (USFU) = -2522.321

SUMMARY OF INFLOWS (+) AND OUTFLOWS (-) :

NAME	FLOW (USFU)	FPN TITLE
R-1	-2522.3	R-1

SUMMARY OF MINIMUM AND MAXIMUM VELOCITIES (FT/S)

	MINIMUM		MAXIMUM
R-1	.01	P-43	33.44
P-92	.05	P-11	19.54
P-116	.11	P-32	19.13
P-128	.36	P-21	16.69
P-119	.39	20	16.28

SUMMARY OF MINIMUM AND MAXIMUM LOSS/1000. (PSI)

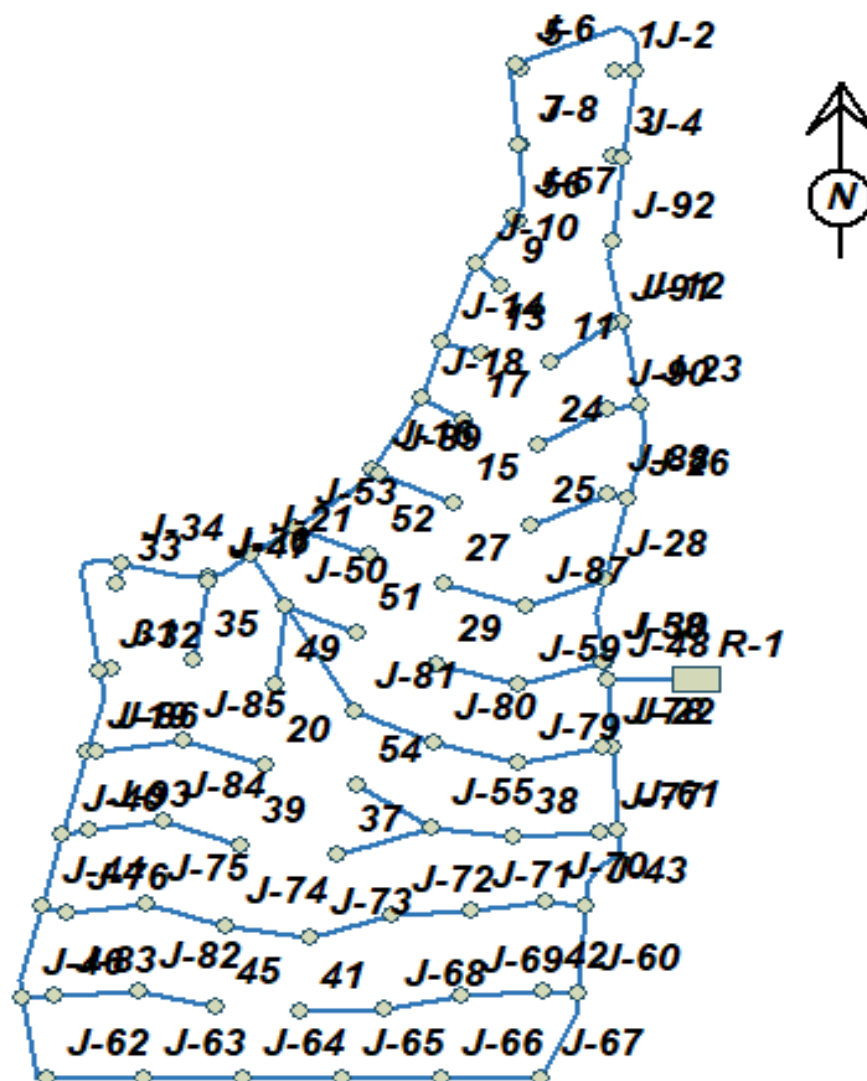
	MINIMUM		MAXIMUM
R-1	.00	P-43	.10
P-116	.00	P-37	.05
P-92	.00	P-67	.05

P-128	.00	P-64	.04
72	.00	P-11	.04

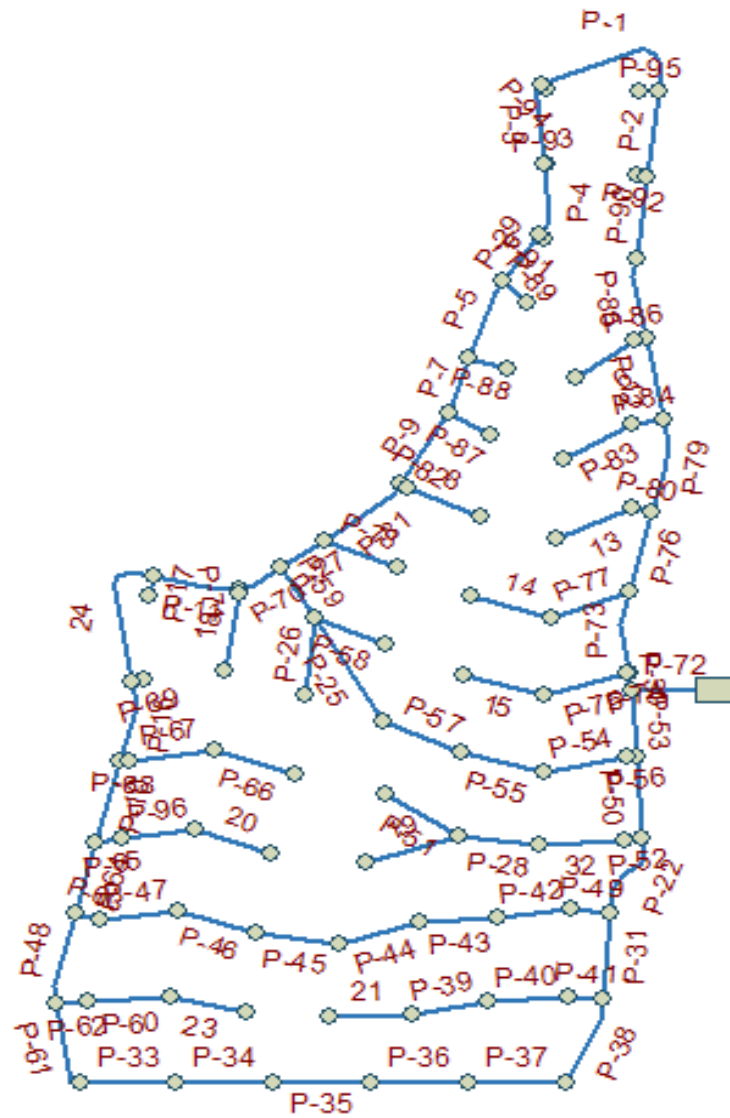
SUMMARY OF MINIMUM,AND,MAXIMUM PRESSURES (USPU)

	MINIMUM		MAXIMUM
R-1	-50.00	7	-47.46
J-18	-49.05	J-56	-47.48
J-47	-48.88	J-54	-47.56
J-19	-48.88	110	-47.59
20	-48.87	112	-47.61

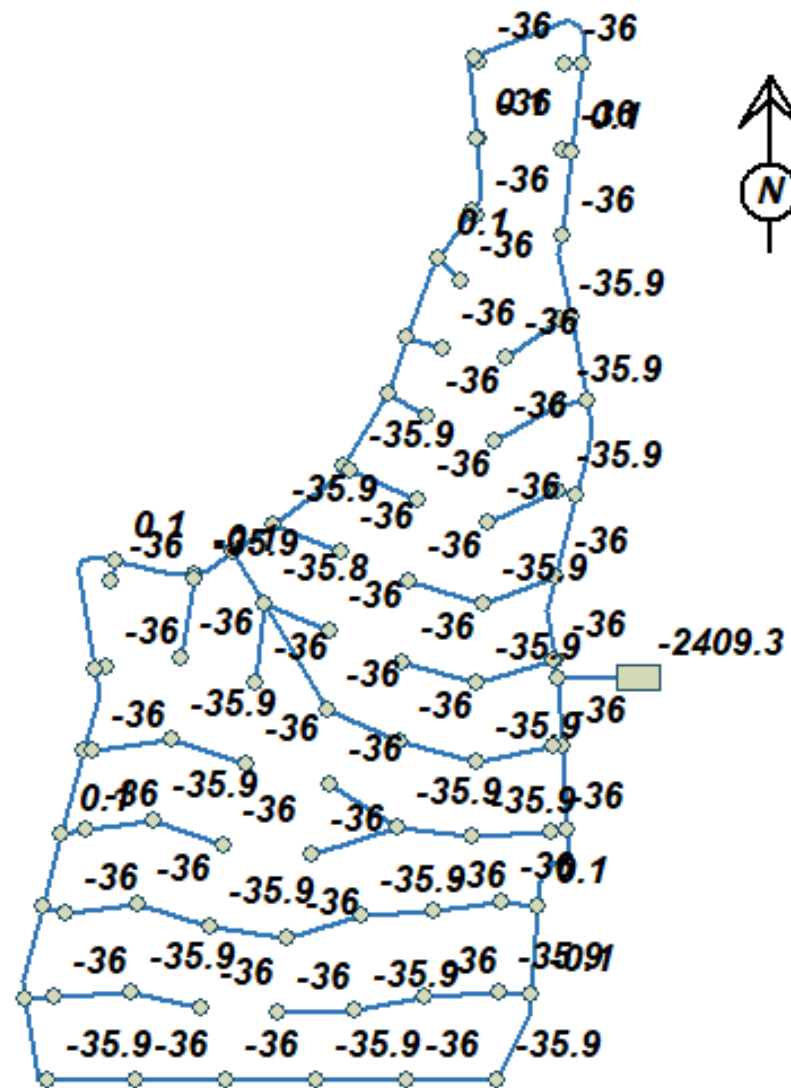
Wexford County Landfill (West Unit) – Final Development
 Landfill Gas Collection and Control System
 Junction Name



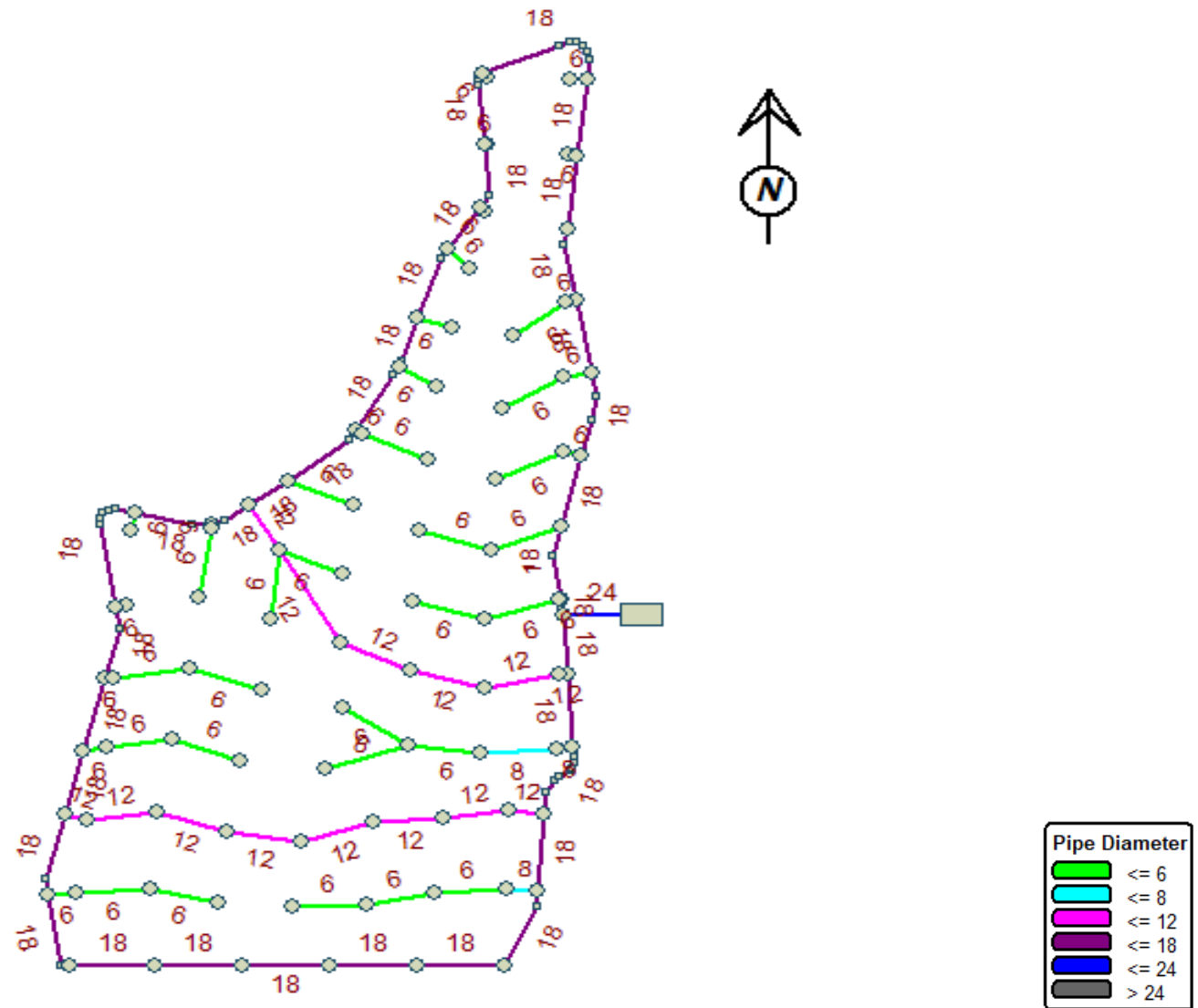
Wexford County Landfill (West Unit) – Final Development
Landfill Gas Collection and Control System
Pipe Name



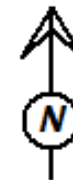
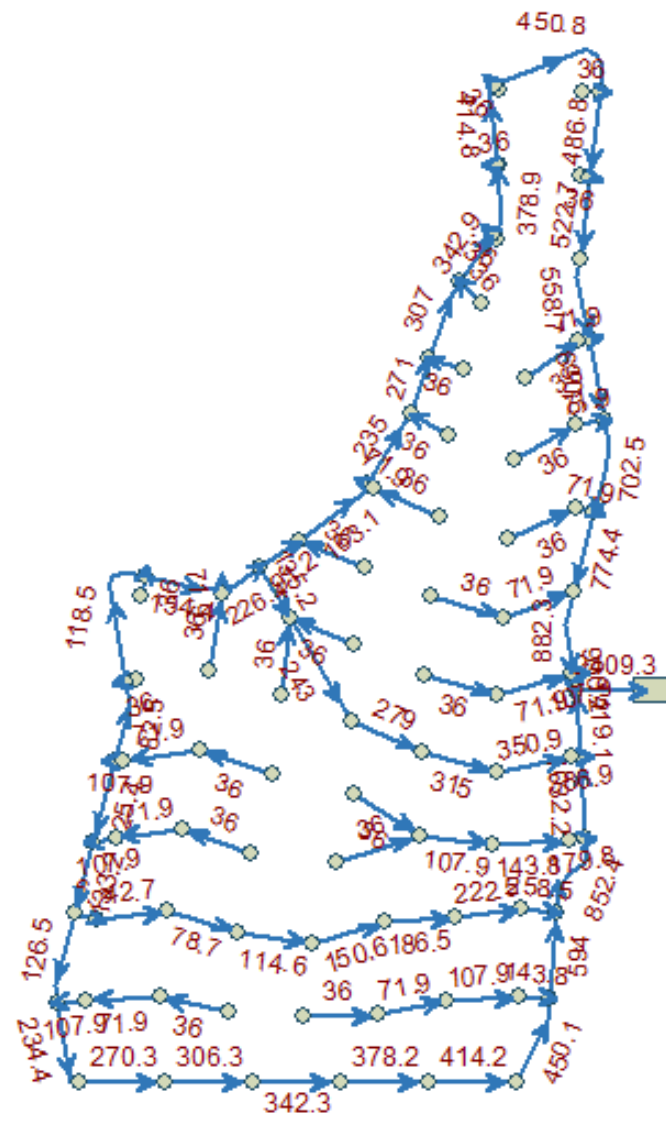
Wexford County Landfill (West Unit) – Final Development
Landfill Gas Collection and Control System
Junction LFG Demand (scfm)



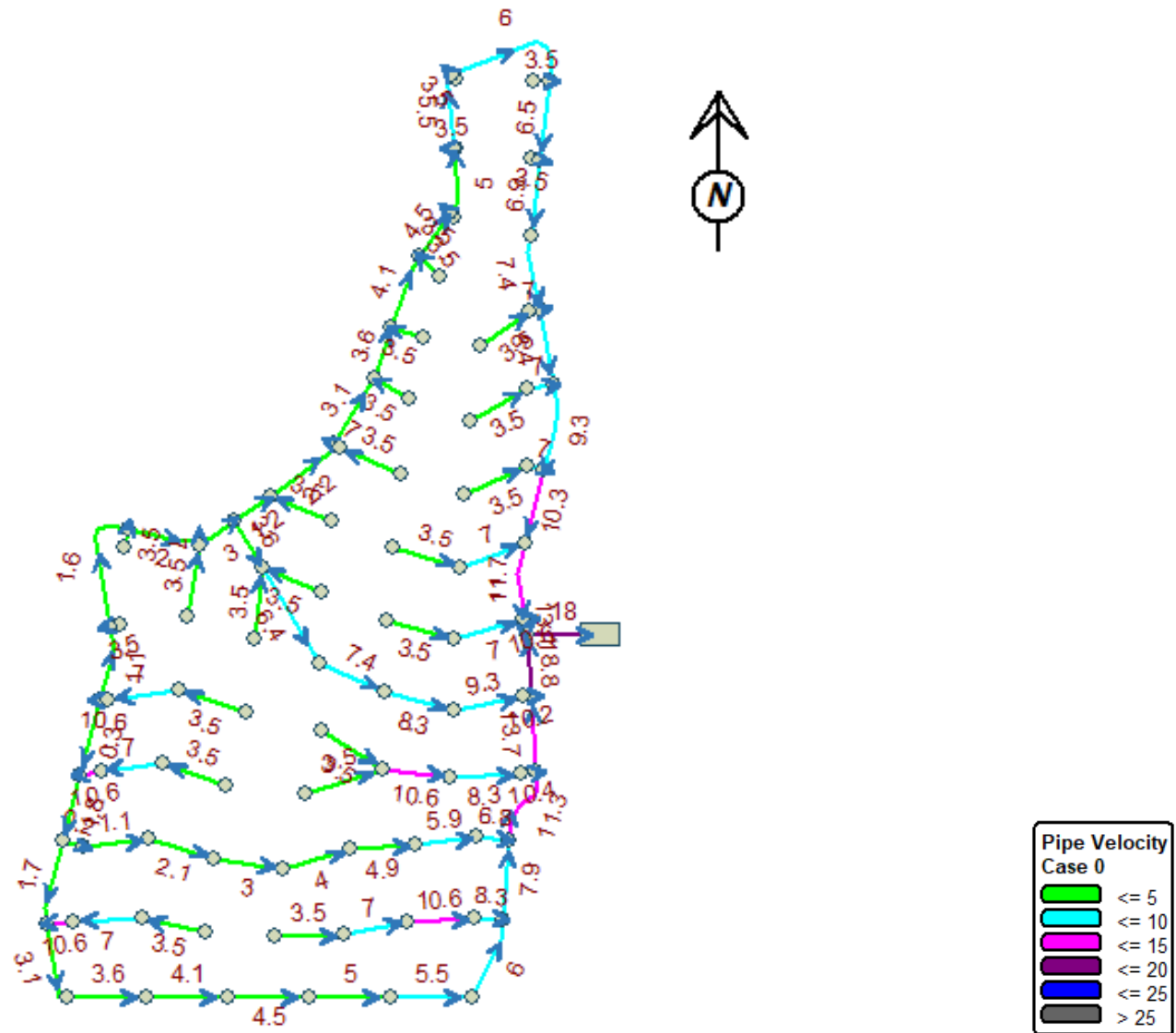
Wexford County Landfill (West Unit) – Final Development
Landfill Gas Collection and Control System
Pipe Diameter (in.)



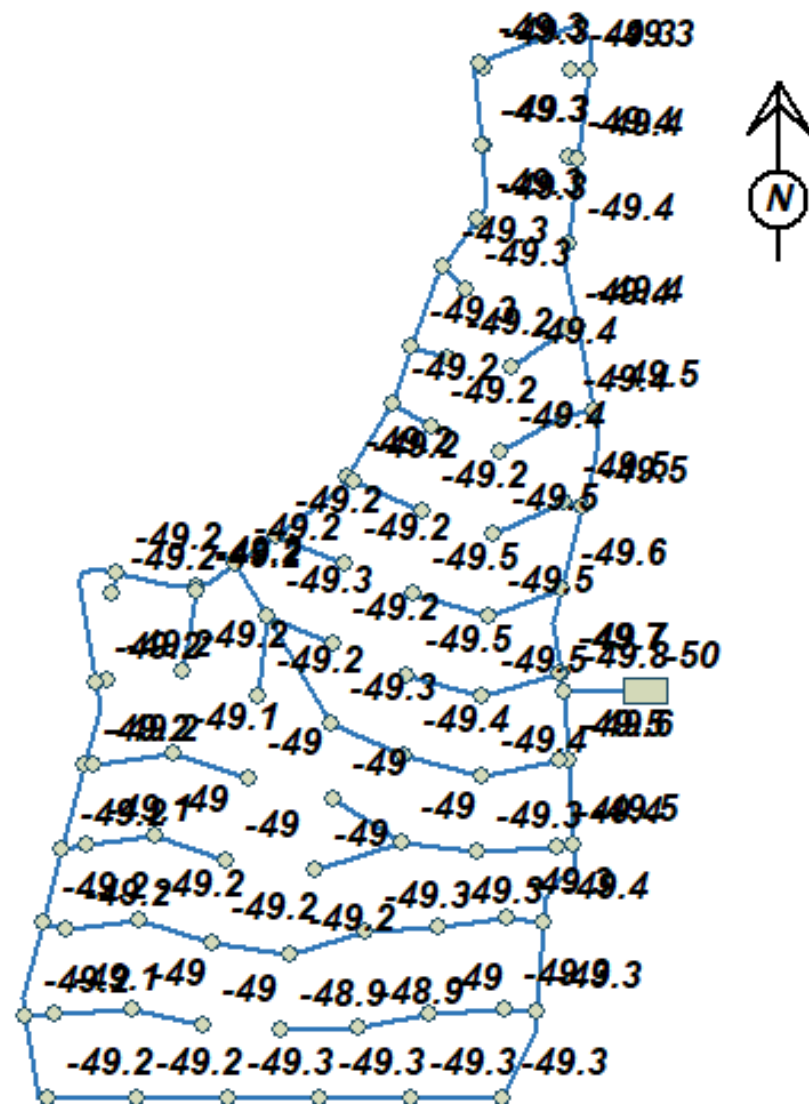
Wexford County Landfill (West Unit) – Final Development
Landfill Gas Collection and Control System
Pipe Flow (scfm)



Wexford County Landfill (West Unit) – Final Development
Landfill Gas Collection and Control System
Pipe Velocity (fps)



Wexford County Landfill (West Unit) – Final Development
Landfill Gas Collection and Control System
Junction Available Vacuum (in. w.c.)



```

* * * * * K Y G A S * * * * *
*
* Gas Network Analysis Software (KYGAS)
*
* CopyRighted by KYPIPE LLC (www.kypipe.com)
* Version: 12.009b 03-25-2024
* Company: TETRAWIN Serial #: 580297
* Interface: KYnetic
* Licensed for Pipe2024
*
* * * * *

```

```

INPUT DATA FILE NAME FOR THIS SIMULATION =
c:\users\JARRET~1.DUT\DOCUME~1\WVWC
-2~4.KYP\wvwc-202.DAT
OUTPUT DATA FILE NAME FOR THIS SIMULATION =
c:\users\JARRET~1.DUT\DOCUME~1\WVWC
-2~4.KYP\wvwc-202.OT2

```

```

DATE FOR THIS COMPUTER RUN      : 3-21-2025
START TIME FOR THIS COMPUTER RUN : 10: 9:37:94

```

SUMMARY OF DISTRIBUTION SYSTEM CHARACTERISTICS:

```

NUMBER OF PIPES      = 97
NUMBER OF JUNCTION NODES = 94

UNITS SPECIFIED      = ENGLISH

```

```

A CONSTANT DENSITY FLUID IS SPECIFIED - DENSITY = .07POUNDS/CUBIC
FOOT
ABSOLUTE VISCOSITY      = .280E-06 POUND SECONDS/SQUARE
FOOT

```

```

USER SPEC. FLOW UNITS (USFU)      = SCF/MIN
USER SPEC. PRESSURE UNITS(USPU)   = INCHES OF WATER (GAUGE)

```

----- SIMULATION DESCRIPTION -----

----- SUMMARY OF PIPE NETWORK GEOMETRIC AND OPERATING DATA -----

PIPE NAME	NODE #1	NODE #2	LENGTH (FT.)	DIAM. (IN.)	ROUGHNESS (MILLIFEET)	SUM-M FACT.	PUMP ID	ELEVATION CHANGE
--------------	------------	------------	-----------------	----------------	--------------------------	----------------	------------	---------------------

6	11	J-91	126.2	5.8	.100	1.8	0	.0
8	15	J-89	150.8	5.8	.100	1.8	0	.0
13	25	J-88	159.3	5.8	.100	1.8	0	.0
14	27	J-87	163.2	5.8	.100	1.8	0	.0
15	29	J-59	167.6	5.8	.100	1.8	0	.0
18	35	J-47	30.3	5.8	.100	1.8	0	.0
19	37	J-55	198.4	5.8	.100	1.8	0	.0
20	39	J-84	158.8	5.8	.100	1.8	0	.0
21	41	J-68	169.7	5.8	.100	1.8	0	.0
23	45	J-82	158.0	5.8	.100	1.8	0	.0
24	J-32	J-34	147.8	15.8	.100	1.8	0	.0
32	38	J-77	176.5	7.5	.100	1.8	0	.0
P-1	J-2	J-6	152.8	15.8	.100	1.8	0	.0
P-10	J-19	J-40	86.5	15.8	.100	1.8	0	.0
P-11	J-34	J-36	83.0	15.8	.100	1.8	0	.0
P-12	J-23	J-12	82.9	15.8	.100	1.8	0	.0
P-16	J-32	J-19	81.2	15.8	.100	1.8	0	.0
P-17	J-34	33	46.0	5.8	.100	1.8	0	.0
P-2	J-4	J-2	87.4	15.8	.100	1.8	0	.0
P-22	J-43	J-61	88.5	15.8	.100	1.8	0	.0
P-25	J-50	J-81	212.4	11.2	.100	1.8	0	.0
P-26	J-50	49	178.0	5.8	.100	1.8	0	.0
P-27	J-53	J-21	48.5	15.8	.100	1.8	0	.0
P-28	J-55	38	161.3	5.8	.100	1.8	0	.0
P-29	J-57	J-10	56.3	15.8	.100	1.8	0	.0
P-3	J-6	J-8	83.5	15.8	.100	1.8	0	.0
P-30	J-48	J-30	16.5	15.8	.100	1.8	0	.0
P-31	J-60	J-43	87.6	15.8	.100	1.8	0	.0
P-33	J-62	J-63	89.3	15.8	.100	1.8	0	.0
P-34	J-63	J-64	89.3	15.8	.100	1.8	0	.0
P-35	J-64	J-65	89.3	15.8	.100	1.8	0	.0
P-36	J-65	J-66	89.3	15.8	.100	1.8	0	.0
P-37	J-66	J-67	89.3	15.8	.100	1.8	0	.0
P-38	J-67	J-60	94.9	15.8	.100	1.8	0	.0
P-39	J-68	J-69	158.9	5.8	.100	1.8	0	.0
P-4	J-8	J-57	75.5	15.8	.100	1.8	0	.0
P-40	J-69	42	167.9	5.8	.100	1.8	0	.0
P-41	J-60	42	72.0	7.5	.100	1.8	0	.0
P-42	J-70	J-71	148.8	11.2	.100	1.8	0	.0
P-43	J-71	J-72	159.6	11.2	.100	1.8	0	.0
P-44	J-72	J-73	171.2	11.2	.100	1.8	0	.0
P-45	J-73	J-74	165.9	11.2	.100	1.8	0	.0
P-46	J-74	J-75	166.3	11.2	.100	1.8	0	.0
P-47	J-75	J-76	163.6	11.2	.100	1.8	0	.0
P-48	J-44	J-46	96.6	15.8	.100	1.8	0	.0
P-49	J-43	J-70	85.0	11.2	.100	1.8	0	.0
P-5	J-10	J-14	84.5	15.8	.100	1.8	0	.0
P-50	J-61	J-22	83.8	15.8	.100	1.8	0	.0
P-51	J-55	54	180.0	5.8	.100	1.8	0	.0
P-52	J-61	J-77	38.0	7.5	.100	1.8	0	.0
P-53	J-22	J-48	67.4	15.8	.100	1.8	0	.0
P-54	J-79	J-78	135.0	11.2	.100	1.8	0	.0
P-55	J-80	J-79	133.3	11.2	.100	1.8	0	.0

P-56	J-22	J-78	21.0	11.2	.100	1.8	0	.0
P-57	J-81	J-80	136.8	11.2	.100	1.8	0	.0
P-58	J-50	51	162.0	5.8	.100	1.8	0	.0
P-59	J-21	J-50	133.0	11.2	.100	1.8	0	.0
P-60	J-82	J-83	168.4	5.8	.100	1.8	0	.0
P-61	J-46	J-62	89.2	15.8	.100	1.8	0	.0
P-62	J-46	J-83	66.0	5.8	.100	1.8	0	.0
P-63	J-76	J-44	51.0	11.2	.100	1.8	0	.0
P-64	J-40	J-44	73.3	15.8	.100	1.8	0	.0
P-65	J-40	J-93	54.0	5.8	.100	1.8	0	.0
P-66	J-85	20	174.9	5.8	.100	1.8	0	.0
P-67	J-86	J-85	171.3	5.8	.100	1.8	0	.0
P-68	J-19	J-86	19.0	5.8	.100	1.8	0	.0
P-69	J-32	31	28.0	5.8	.100	1.8	0	.0
P-7	J-14	J-18	59.1	15.8	.100	1.8	0	.0
P-70	J-36	J-21	45.5	15.8	.100	1.8	0	.0
P-71	J-36	J-47	8.0	5.8	.100	1.8	0	.0
P-72	J-48	R-1	184.0	21.0	.100	1.8	0	.0
P-73	J-30	J-28	86.3	15.8	.100	1.8	0	.0
P-74	J-30	J-58	2.0	5.8	.100	1.8	0	.0
P-75	J-59	J-58	171.4	5.8	.100	1.8	0	.0
P-76	J-28	J-26	82.6	15.8	.100	1.8	0	.0
P-77	J-28	J-87	172.0	5.8	.100	1.8	0	.0
P-78	52	J-53	164.0	5.8	.100	1.8	0	.0
P-79	J-26	J-23	97.3	15.8	.100	1.8	0	.0
P-80	J-26	J-88	43.0	5.8	.100	1.8	0	.0
P-81	J-16	J-53	92.7	15.8	.100	1.8	0	.0
P-82	J-16	J-89	15.0	5.8	.100	1.8	0	.0
P-83	J-90	24	148.4	5.8	.100	1.8	0	.0
P-84	J-23	J-90	64.0	5.8	.100	1.8	0	.0
P-85	J-12	J-92	84.4	15.8	.100	1.8	0	.0
P-86	J-12	J-91	23.0	5.8	.100	1.8	0	.0
P-87	J-18	17	99.0	5.8	.100	1.8	0	.0
P-88	J-14	13	84.0	5.8	.100	1.8	0	.0
P-89	J-10	9	71.0	5.8	.100	1.8	0	.0
P-9	J-18	J-16	85.0	15.8	.100	1.8	0	.0
P-90	J-92	J-4	83.6	15.8	.100	1.8	0	.0
P-91	J-57	56	12.0	5.8	.100	1.8	0	.0
P-92	J-4	3	20.0	5.8	.100	1.8	0	.0
P-93	J-8	7	6.0	5.8	.100	1.8	0	.0
P-94	J-6	5	18.0	5.8	.100	1.8	0	.0
P-95	J-2	1	40.0	5.8	.100	1.8	0	.0
P-96	J-93	J-84	155.0	5.8	.100	1.8	0	.0

JUNCTION NAME	NODE TITLE	ELEV	DEMAND (USFU)	FPN PRESSURE
1		.00	-35.96	
3		.00	-35.96	
5		.00	-35.96	
7		.00	-35.96	
9		.00	-35.96	
11		.00	-35.96	

13	.00	-35.96
15	.00	-35.96
17	.00	-35.96
20	.00	-35.96
24	.00	-35.96
25	.00	-35.96
27	.00	-35.96
29	.00	-35.96
31	.00	-35.96
33	.00	-35.96
35	.00	-35.96
37	.00	-35.96
38	.00	-35.96
39	.00	-35.96
41	.00	-35.96
42	.00	-35.96
45	.00	-35.96
49	.00	-35.96
51	.00	-35.96
52	.00	-35.96
54	.00	-35.96
56	.00	-35.96
J-10	.00	.00
J-12	.00	.00
J-14	.00	.00
J-16	.00	.00
J-18	.00	.00
J-19	.00	.00
J-2	.00	.00
J-21	.00	.00
J-22	.00	.00
J-23	.00	.00
J-26	.00	.00
J-28	.00	-35.96
J-30	.00	.00
J-32	.00	.00
J-34	.00	.00
J-36	.00	.00
J-4	.00	.00
J-40	.00	.00
J-43	.00	.00
J-44	.00	.00
J-46	.00	.00
J-47	.00	-35.96
J-48	.00	.00
J-50	.00	-35.96
J-53	.00	-35.96
J-55	.00	-35.96
J-57	.00	.00
J-58	.00	-35.96
J-59	.00	-35.96
J-6	.00	.00
J-60	.00	.00
J-61	.00	.00

J-62	.00	-35.96	
J-63	.00	-35.96	
J-64	.00	-35.96	
J-65	.00	-35.96	
J-66	.00	-35.96	
J-67	.00	-35.96	
J-68	.00	-35.96	
J-69	.00	-35.96	
J-70	.00	-35.96	
J-71	.00	-35.96	
J-72	.00	-35.96	
J-73	.00	-35.96	
J-74	.00	-35.96	
J-75	.00	-35.96	
J-76	.00	-35.96	
J-77	.00	-35.96	
J-78	.00	-35.96	
J-79	.00	-35.96	
J-8	.00	.00	
J-80	.00	-35.96	
J-81	.00	-35.96	
J-82	.00	-35.96	
J-83	.00	-35.96	
J-84	.00	-35.96	
J-85	.00	-35.96	
J-86	.00	-35.96	
J-87	.00	-35.96	
J-88	.00	-35.96	
J-89	.00	-35.96	
J-90	.00	-35.96	
J-91	.00	-35.96	
J-92	.00	-35.96	
J-93	.00	-35.96	
R-1	.00	.00	-50.00

=====

=====

Set = 0

===== RESULTS FOR THIS SIMULATION FOLLOW

=====

Solution was obtained in 14 trials

Flow Accuracy = .2223E-02[< .500E-02]

RV Accuracy = .0000E+00[< .100E-02]

PIPE FRICTION RATIO	AREA NO.	NODE #1	NODE #2	FLOW (USFU)	LOSS (USPU)	VELOCITY (FT/S)	DENSITY (#/CF)	FACTOR
.0290	6	11	J-91	35.960	.02	3.52	.0710	
.0290	8	15	J-89	35.960	.03	3.52	.0710	
.0290	13	25	J-88	35.960	.03	3.52	.0710	
.0290	14	27	J-87	35.960	.03	3.52	.0710	
.0290	15	29	J-59	35.960	.03	3.52	.0710	
.0290	18	35	J-47	35.960	.01	3.52	.0710	
.0290	19	37	J-55	35.960	.04	3.52	.0710	
.0290	20	39	J-84	35.960	.03	3.52	.0710	
.0290	21	41	J-68	35.960	.03	3.52	.0710	
.0290	23	45	J-82	35.960	.03	3.52	.0710	
.0274	24	J-32	J-34	118.487	.00	1.57	.0710	
.0223	32	38	J-77	143.840	.12	8.31	.0710	
.0202	P-1	J-2	J-6	-450.809	.03	5.98	.0710	
.0420	P-10	J-19	J-40	25.353	.00	.34	.0710	
.0257	P-11	J-34	J-36	154.447	.00	2.05	.0710	
.0188	P-12	J-23	J-12	-630.609	.04	8.36	.0710	
.0302	P-16	J-32	J-19	-82.527	.00	1.09	.0710	
.0290	P-17	J-34	33	-35.960	.01	3.52	.0710	
.0198	P-2	J-4	J-2	-486.769	.03	6.45	.0710	
.0178	P-22	J-43	J-61	852.433	.08	11.30	.0710	
.0215	P-25	J-50	J-81	243.038	.06	6.42	.0710	
.0290	P-26	J-50	49	-35.960	.03	3.52	.0710	
.0293	P-27	J-53	J-21	-91.209	.00	1.21	.0710	

	P-28	J-55	38	107.880	.22	10.56	.0710
.0225							
	P-29	J-57	J-10	-342.929	.01	4.55	.0710
.0214							
	P-3	J-6	J-8	-414.849	.02	5.50	.0710
.0205							
	P-30	J-48	J-30	-990.209	.07	13.13	.0710
.0173							
	P-31	J-60	J-43	593.971	.04	7.88	.0710
.0191							
	P-33	J-62	J-63	270.331	.01	3.58	.0710
.0225							
	P-34	J-63	J-64	306.291	.01	4.06	.0710
.0219							
	P-35	J-64	J-65	342.251	.01	4.54	.0710
.0213							
	P-36	J-65	J-66	378.211	.02	5.01	.0710
.0209							
	P-37	J-66	J-67	414.171	.02	5.49	.0710
.0205							
	P-38	J-67	J-60	450.131	.02	5.97	.0710
.0201							
	P-39	J-68	J-69	71.920	.10	7.04	.0710
.0246							
	P-4	J-8	J-57	-378.889	.02	5.02	.0710
.0209							
	P-40	J-69	42	107.880	.23	10.56	.0710
.0225							
	P-41	J-60	42	-143.840	.06	8.31	.0710
.0223							
	P-42	J-70	J-71	-222.502	.04	5.88	.0710
.0219							
	P-43	J-71	J-72	-186.542	.03	4.93	.0710
.0227							
	P-44	J-72	J-73	-150.582	.02	3.98	.0710
.0238							
	P-45	J-73	J-74	-114.622	.01	3.03	.0710
.0254							
	P-46	J-74	J-75	-78.662	.01	2.08	.0710
.0279							
	P-47	J-75	J-76	-42.702	.00	1.13	.0710
.0327							
	P-48	J-44	J-46	126.491	.00	1.68	.0710
.0269							
	P-49	J-43	J-70	-258.462	.04	6.83	.0710
.0212							
	P-5	J-10	J-14	-306.969	.01	4.07	.0710
.0219							
	P-50	J-61	J-22	1032.233	.12	13.69	.0710
.0172							
	P-51	J-55	54	-35.960	.03	3.52	.0710
.0290							
	P-52	J-61	J-77	-179.800	.07	10.38	.0710
.0213							

P-53	J-22	J-48	1419.111	.20	18.82	.0710
.0163						
P-54	J-79	J-78	350.918	.09	9.27	.0710
.0199						
P-55	J-80	J-79	314.958	.07	8.32	.0710
.0204						
P-56	J-22	J-78	-386.878	.05	10.22	.0710
.0195						
P-57	J-81	J-80	278.998	.06	7.37	.0710
.0209						
P-58	J-50	51	-35.960	.03	3.52	.0710
.0290						
P-59	J-21	J-50	135.158	.01	3.57	.0710
.0245						
P-60	J-82	J-83	71.920	.11	7.04	.0710
.0246						
P-61	J-46	J-62	234.371	.01	3.11	.0710
.0232						
P-62	J-46	J-83	-107.880	.12	10.56	.0710
.0225						
P-63	J-76	J-44	-6.742	.00	.18	.0710
.0562						
P-64	J-40	J-44	133.233	.00	1.77	.0710
.0265						
P-65	J-40	J-93	-107.880	.10	10.56	.0710
.0225						
P-66	J-85	20	-35.960	.03	3.52	.0710
.0290						
P-67	J-86	J-85	-71.920	.11	7.04	.0710
.0246						
P-68	J-19	J-86	-107.880	.06	10.56	.0710
.0225						
P-69	J-32	31	-35.960	.01	3.52	.0710
.0290						
P-7	J-14	J-18	-271.009	.01	3.59	.0710
.0225						
P-70	J-36	J-21	226.367	.00	3.00	.0710
.0234						
P-71	J-36	J-47	-71.920	.02	7.04	.0710
.0246						
P-72	J-48	R-1	2409.320	.23	17.97	.0710
.0155						
P-73	J-30	J-28	-882.329	.09	11.70	.0710
.0177						
P-74	J-30	J-58	-107.880	.04	10.56	.0710
.0225						
P-75	J-59	J-58	71.920	.11	7.04	.0710
.0246						
P-76	J-28	J-26	-774.449	.07	10.27	.0710
.0181						
P-77	J-28	J-87	-71.920	.11	7.04	.0710
.0246						
P-78	52	J-53	35.960	.03	3.52	.0710
.0290						

.0184	P-79	J-26	J-23	-702.529	.06	9.32	.0710
.0246	P-80	J-26	J-88	-71.920	.04	7.04	.0710
.0253	P-81	J-16	J-53	-163.129	.00	2.16	.0710
.0246	P-82	J-16	J-89	-71.920	.03	7.04	.0710
.0290	P-83	J-90	24	-35.960	.03	3.52	.0710
.0246	P-84	J-23	J-90	-71.920	.05	7.04	.0710
.0193	P-85	J-12	J-92	-558.689	.04	7.41	.0710
.0246	P-86	J-12	J-91	-71.920	.03	7.04	.0710
.0290	P-87	J-18	17	-35.960	.02	3.52	.0710
.0290	P-88	J-14	13	-35.960	.02	3.52	.0710
.0290	P-89	J-10	9	-35.960	.02	3.52	.0710
.0232	P-9	J-18	J-16	-235.049	.01	3.12	.0710
.0196	P-90	J-92	J-4	-522.729	.03	6.93	.0710
.0290	P-91	J-57	56	-35.960	.01	3.52	.0710
.0290	P-92	J-4	3	-35.960	.01	3.52	.0710
.0290	P-93	J-8	7	-35.960	.01	3.52	.0710
.0290	P-94	J-6	5	-35.960	.01	3.52	.0710
.0290	P-95	J-2	1	-35.960	.01	3.52	.0710
.0246	P-96	J-93	J-84	-71.920	.10	7.04	.0710
.0374	R-1	R-1	R-1	-2409.320	.00	.01	.0710

JUNCTION		NODE	DEMAND	PRESSURE	PRESSURE	PRESSURE
DENSITY						
NAME	TITLE	(USFU)	(USPU)	(PSIA)	(PSIG)	
#/CF						

1		-35.96	-49.33	12.92	-1.78	
.071						

.071	3	-35.96	-49.36	12.91	-1.78
.071	5	-35.96	-49.30	12.92	-1.78
.071	7	-35.96	-49.29	12.92	-1.78
.071	9	-35.96	-49.25	12.92	-1.78
.071	11	-35.96	-49.38	12.91	-1.78
.071	13	-35.96	-49.23	12.92	-1.78
.071	15	-35.96	-49.18	12.92	-1.77
.071	17	-35.96	-49.22	12.92	-1.78
.071	20	-35.96	-49.02	12.93	-1.77
.071	24	-35.96	-49.40	12.91	-1.78
.071	25	-35.96	-49.47	12.91	-1.79
.071	27	-35.96	-49.46	12.91	-1.78
.071	29	-35.96	-49.50	12.91	-1.79
.071	31	-35.96	-49.21	12.92	-1.78
.071	33	-35.96	-49.21	12.92	-1.78
.071	35	-35.96	-49.20	12.92	-1.78
.071	37	-35.96	-49.01	12.93	-1.77
.071	38	-35.96	-49.26	12.92	-1.78
.071	39	-35.96	-48.99	12.93	-1.77
.071	41	-35.96	-48.90	12.93	-1.76
.071	42	-35.96	-49.27	12.92	-1.78
.071	45	-35.96	-48.97	12.93	-1.77
.071	49	-35.96	-49.21	12.92	-1.78
.071	51	-35.96	-49.22	12.92	-1.78
.071	52	-35.96	-49.20	12.92	-1.78
.071	54	-35.96	-49.01	12.93	-1.77
.071	56	-35.96	-49.27	12.92	-1.78

.071	J-10	.00	-49.26	12.92	-1.78
.071	J-12	.00	-49.44	12.91	-1.78
.071	J-14	.00	-49.25	12.92	-1.78
.071	J-16	.00	-49.24	12.92	-1.78
.071	J-18	.00	-49.24	12.92	-1.78
.071	J-19	.00	-49.22	12.92	-1.78
.071	J-2	.00	-49.34	12.92	-1.78
.071	J-21	.00	-49.23	12.92	-1.78
.071	J-22	.00	-49.57	12.91	-1.79
.071	J-23	.00	-49.48	12.91	-1.79
.071	J-26	.00	-49.54	12.91	-1.79
.071	J-28	-35.96	-49.61	12.91	-1.79
.071	J-30	.00	-49.69	12.90	-1.79
.071	J-32	.00	-49.22	12.92	-1.78
.071	J-34	.00	-49.23	12.92	-1.78
.071	J-36	.00	-49.23	12.92	-1.78
.071	J-4	.00	-49.37	12.91	-1.78
.071	J-40	.00	-49.22	12.92	-1.78
.071	J-43	.00	-49.37	12.91	-1.78
.071	J-44	.00	-49.22	12.92	-1.78
.071	J-46	.00	-49.23	12.92	-1.78
.071	J-47	-35.96	-49.21	12.92	-1.78
.071	J-48	.00	-49.77	12.90	-1.80
.071	J-50	-35.96	-49.25	12.92	-1.78
.071	J-53	-35.96	-49.23	12.92	-1.78
.071	J-55	-35.96	-49.04	12.93	-1.77
.071	J-57	.00	-49.28	12.92	-1.78

	J-58	-35.96	-49.65	12.90	-1.79
.071					
	J-59	-35.96	-49.54	12.91	-1.79
.071					
	J-6	.00	-49.31	12.92	-1.78
.071					
	J-60	.00	-49.33	12.92	-1.78
.071					
	J-61	.00	-49.45	12.91	-1.78
.071					
	J-62	-35.96	-49.23	12.92	-1.78
.071					
	J-63	-35.96	-49.24	12.92	-1.78
.071					
	J-64	-35.96	-49.25	12.92	-1.78
.071					
	J-65	-35.96	-49.27	12.92	-1.78
.071					
	J-66	-35.96	-49.29	12.92	-1.78
.071					
	J-67	-35.96	-49.31	12.92	-1.78
.071					
	J-68	-35.96	-48.93	12.93	-1.77
.071					
	J-69	-35.96	-49.04	12.93	-1.77
.071					
	J-70	-35.96	-49.33	12.92	-1.78
.071					
	J-71	-35.96	-49.30	12.92	-1.78
.071					
	J-72	-35.96	-49.27	12.92	-1.78
.071					
	J-73	-35.96	-49.24	12.92	-1.78
.071					
	J-74	-35.96	-49.23	12.92	-1.78
.071					
	J-75	-35.96	-49.23	12.92	-1.78
.071					
	J-76	-35.96	-49.22	12.92	-1.78
.071					
	J-77	-35.96	-49.38	12.91	-1.78
.071					
	J-78	-35.96	-49.52	12.91	-1.79
.071					
	J-79	-35.96	-49.43	12.91	-1.78
.071					
	J-8	.00	-49.29	12.92	-1.78
.071					
	J-80	-35.96	-49.36	12.91	-1.78
.071					
	J-81	-35.96	-49.31	12.92	-1.78
.071					
	J-82	-35.96	-49.00	12.93	-1.77
.071					

.071	J-83	-35.96	-49.11	12.92	-1.77
.071	J-84	-35.96	-49.02	12.93	-1.77
.071	J-85	-35.96	-49.05	12.93	-1.77
.071	J-86	-35.96	-49.16	12.92	-1.77
.071	J-87	-35.96	-49.49	12.91	-1.79
.071	J-88	-35.96	-49.50	12.91	-1.79
.071	J-89	-35.96	-49.21	12.92	-1.78
.071	J-90	-35.96	-49.43	12.91	-1.78
.071	J-91	-35.96	-49.41	12.91	-1.78
.071	J-92	-35.96	-49.40	12.91	-1.78
.071	J-93	-35.96	-49.12	12.92	-1.77
.071	R-1	.00	-50.00	12.89	-1.80

* This designates the use of default density in a low pressure region

THE NET SYSTEM DEMAND (USFU) = -2409.319

SUMMARY OF INFLOWS(+) .AND. OUTFLOWS(-) :

NAME	FLOW (USFU)	FPN TITLE
R-1	-2409.3	R-1

SUMMARY OF MINIMUM .AND. MAXIMUM VELOCITIES (FT/S)

	MINIMUM		MAXIMUM
R-1	.01	P-53	18.82
P-63	.18	P-72	17.97
P-10	.34	P-50	13.69
P-16	1.09	P-30	13.13
P-47	1.13	P-73	11.70

SUMMARY OF MINIMUM .AND. MAXIMUM LOSS/1000. (PSI)

MINIMUM	MAXIMUM
---------	---------

R-1	.00	P-28	.04
P-63	.00	P-74	.04
P-10	.00	P-62	.04
P-16	.00	P-53	.03
P-27	.00	P-52	.03

SUMMARY OF MINIMUM.AND.MAXIMUM PRESSURES (USPU)

	MINIMUM		MAXIMUM

R-1	-50.00	41	-48.90
J-48	-49.77	J-68	-48.93
J-30	-49.69	45	-48.97
J-58	-49.65	39	-48.99
J-28	-49.61	J-82	-49.00

APPENDIX C

DRAWINGS

NSPS DESIGN PLANS FOR THE

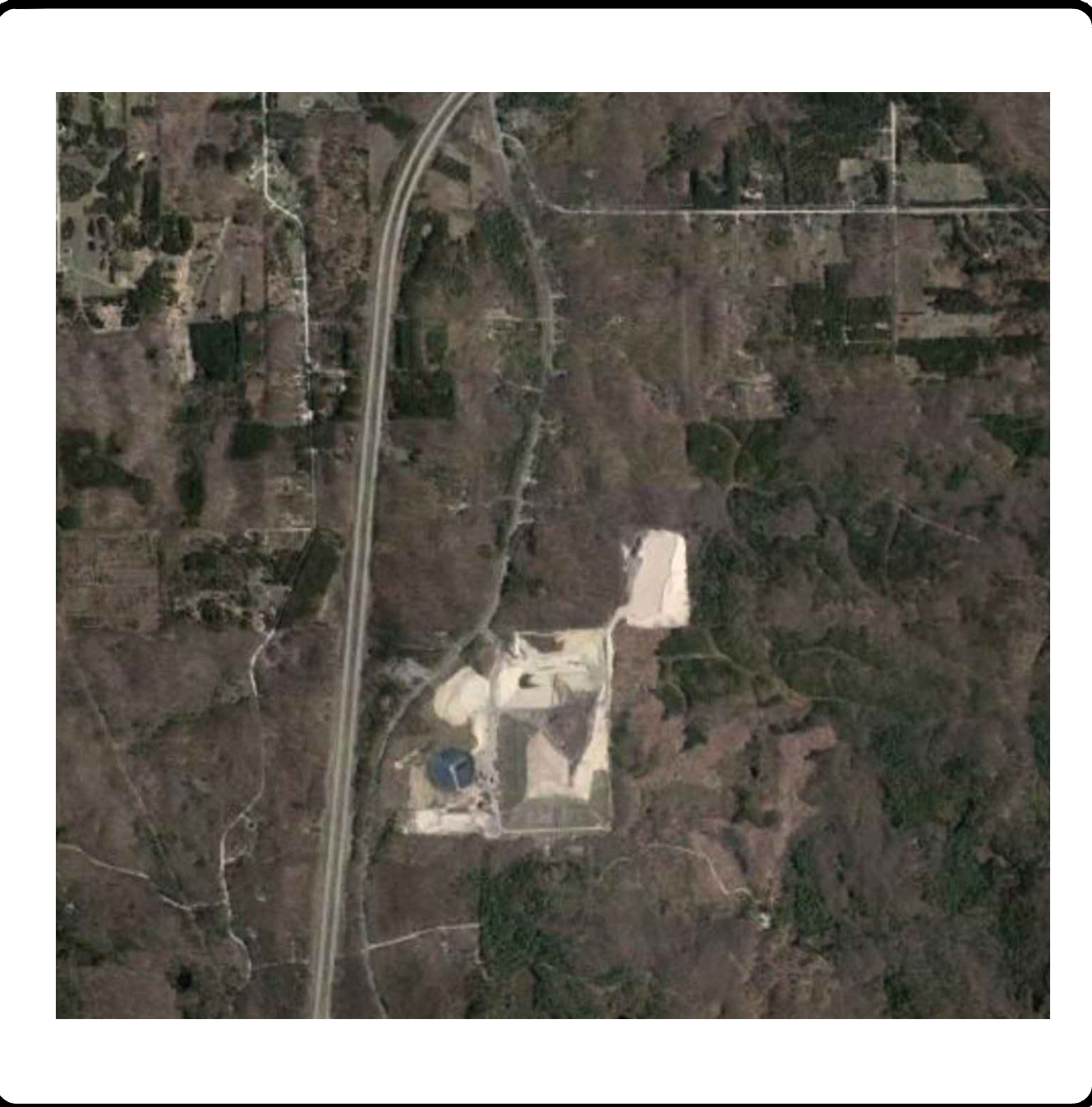
LANDFILL GAS COLLECTION AND CONTROL SYSTEM

PREPARED FOR

WEXFORD COUNTY LANDFILL

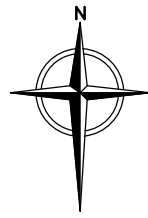
MANTON, MICHIGAN

APRIL 2025



LOCATION MAP

SCALE: 1" = 2000'
SOURCE: GOOGLE EARTH PRO



PREPARED FOR:

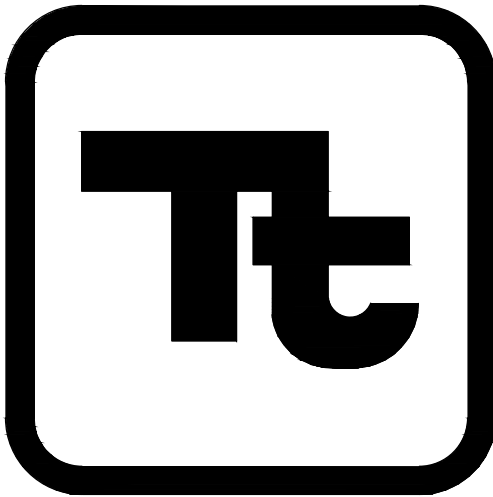


990 NORTH MACKINAW TRAIL,
MANTON, MI 49663

SHEET INDEX

1	EXISTING GCCS CONDITIONS SITE PLAN
2	FINAL DEVELOPMENT SITE PLAN - EAST UNIT
3	FINAL DEVELOPMENT SITE PLAN - WEST UNIT
4	WELL RADIUS OF INFLUENCE PLAN - EAST UNIT
5	WELL RADIUS OF INFLUENCE PLAN - WEST UNIT
DS1	DETAILS
DS2	DETAILS
DS3	DETAILS

PREPARED BY:



TETRA TECH

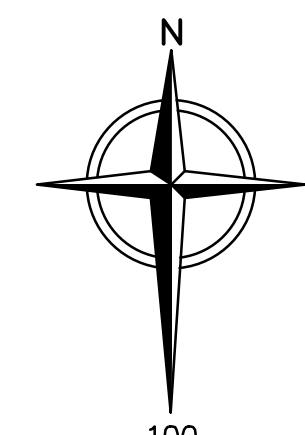
1250 E DIEHL ROAD
SUITE 103
NAPERVILLE, ILLINOIS 60563
Tel: (630) 633-5519
Fax (630) 791-9003



NOTE:

THE EXISTING UTILITIES SHOWN ON THESE DRAWINGS ARE APPROXIMATE, AND UTILITY LINES MAY EXIST WHERE NONE ARE SHOWN. SOME INFORMATION MAY HAVE BEEN DERIVED FROM INFORMATION PROVIDED TO THE ENGINEER BY OTHERS. SUCH INFORMATION MAY BE INCOMPLETE OR MAY BE OBSOLETE BY THE TIME CONSTRUCTION COMMENCES. CONTACT MISS DIG AT 1-800-482-7171 (OR UTILIZE THEIR E-LOCATE SERVICE) AND ANY NON-PARTICIPATING UTILITY COMPANIES AT LEAST 48 HOURS BEFORE CONSTRUCTION. THE CONTRACTOR SHALL EXCAVATE AND VERIFY THE HORIZONTAL AND VERTICAL LOCATIONS OF PERTINENT UTILITIES, LANDFILL LINERS, AND OTHER EXISTING FEATURES IN OR NEAR THE AREA OF WORK, WHETHER INDICATED ON THESE DRAWINGS OR NOT. SHOULD A CONFLICT EXIST, THE CONTRACTOR SHALL NOTIFY THE ENGINEER AS SOON AS POSSIBLE. THE CONTRACTOR SHALL EXERCISE DUE CARE TO AVOID DISTURBING ANY UNDERGROUND UTILITIES. THE CONTRACTOR SHALL COORDINATE ANY POTENTIAL DISRUPTIONS IN UTILITY SERVICE WITH THE UTILITY COMPANIES AFFECTED AT LEAST 24 HOURS PRIOR TO THE DISRUPTION. THE CONTRACTOR SHALL REPAIR DAMAGE TO EXISTING UTILITIES AT THE CONTRACTOR'S EXPENSE.

This drawing represents intellectual property of Tetra Tech. Any modification to the original by other than Tetra Tech personnel violates its original purpose and as such is rendered void. Tetra Tech will not be held liable for any changes made to this document without express written consent of the originator.



0 100 200
SCALE IN FEET

LEGEND

- PROPERTY BOUNDARY
- EXISTING SOLID WASTE BOUNDARY
- EXISTING PHASE BOUNDARY
- 900' EXISTING 10' CONTOUR
- 18" EXISTING 2' CONTOUR
- 18" EXISTING LANDFILL GAS HEADER
- 6" EXISTING LANDFILL GAS HORIZONTAL COLLECTOR
- A-2 A-2 A-2 EXISTING 2" AIRLINE
- FM-3 FM-3 FM-3 EXISTING 3" FORCEMAIN
- FM-306 FM-306 FM-306 EXISTING 3"x6" DUAL CONTAINED FORCEMAIN LINE
- FM-408 FM-408 FM-408 EXISTING 4"x8" DUAL CONTAINED FORCEMAIN LINE
- ⊕ GW- EXISTING LFG EXTRACTION WELL
- ⊕ GW- EXISTING LFG EXTRACTION WELL WITH PUMP
- ⊕ GW- EXISTING LFG EXTRACTION WELL WITH CAISSON
- ⊕ RW- EXISTING REMOTE WELLHEAD
- ⊕ EXISTING HEADER ACCESS RISER
- ⊕ EXISTING ISOLATION VALVE
- ⊕ EXISTING FORCEMAIN ISOLATION VALVE
- ⊕ EXISTING AIRLINE ISOLATION VALVE
- ⊕ PV EXISTING PNEUMATIC AIRLINE ISOLATION VALVE
- FCO EXISTING FORCEMAIN CLEANOUT
- ⊕ EXISTING BLIND FLANGE
- ⊕ EXISTING FLANGE CONNECTION
- ⊕ EXISTING REDUCER FITTING
- ⊕ DL-3 EXISTING CONDENSATE DRIPLEG
- MH EXISTING MANHOLE
- ⊕ CS EXISTING CONDENSATE SUMP
- ⊕ RS EXISTING HORIZONTAL COLLECTOR ROCK SUMP
- ⊕ EXISTING ANTI-SEEP COLLAR
- ⊕ EXISTING FINAL COVER AREA

NOTES:

1. TOPOGRAPHIC SURVEY MAP PROVIDED BY HURON ENGINEERING AND SURVEYING INC., DATE OF SURVEY OCTOBER 3, 2024.
2. 2023 GCCS AS-BUILT INFORMATION PROVIDED BY LANDFILL DRILLING AND PIPING, INC. IN FILE TITLED 'WEXFORD COUNTY RECORD STAKING POINTS COMPLETE' DATED MARCH 13, 2024.

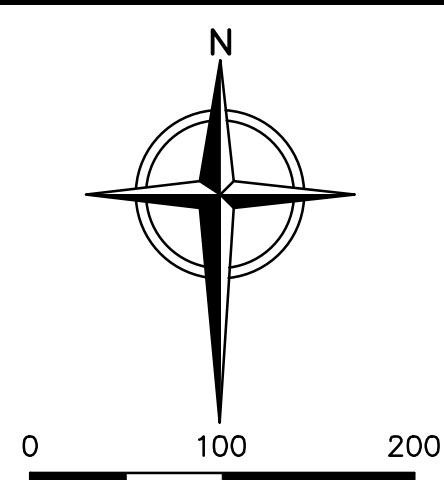


REV	DATE	DESCRIPTION	OWN BY	DES BY	CHK BY	APP BY
04/02/25			JRD	JPS		
			JPS			



GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN
**GAS COLLECTION AND CONTROL SYSTEM
EXISTING GCCS CONDITIONS SITE PLAN**

SHEET NO.
1
PROJECT NO.
209-4251206



- LEGEND**
- PROPERTY BOUNDARY
 - EXISTING SOLID WASTE BOUNDARY
 - EXISTING PHASE BOUNDARY
 - PROPOSED 10' CONTOUR
 - PROPOSED 2' CONTOUR
 - EXISTING LANDFILL GAS HEADER
 - EXISTING LANDFILL GAS HORIZONTAL COLLECTOR
 - EXISTING 3" FORCEMAIN
 - EXISTING 3"x6" DUAL CONTAINED FORCEMAIN LINE
 - EXISTING 4"x8" DUAL CONTAINED FORCEMAIN LINE
 - EXISTING 2" AIRLINE
 - EXISTING LFG EXTRACTION WELL
 - EXISTING LFG EXTRACTION WELL WITH CAISSON
 - EXISTING REMOTE WELLHEAD
 - EXISTING HEADER ACCESS RISER
 - EXISTING ISOLATION VALVE
 - EXISTING BLIND FLANGE
 - EXISTING FLANGE CONNECTION
 - EXISTING REDUCER FITTING
 - EXISTING CONDENSATE DRIPLEG
 - EXISTING MANHOLE
 - EXISTING FORCEMAIN ISOLATION VALVE
 - EXISTING AIRLINE ISOLATION VALVE
 - EXISTING FORCEMAIN CLEANOUT
 - EXISTING FLANGE CONNECTION
 - EXISTING CONDENSATE SUMP
 - EXISTING FINAL COVER AREA
 - PROPOSED LANDFILL GAS HEADER/LATERAL
 - PROPOSED LFG EXTRACTION WELL
 - PROPOSED REMOTE WELLHEAD
 - PROPOSED HEADER ISOLATION VALVE
 - PROPOSED FLANGE CONNECTION
 - PROPOSED CONDENSATE SUMP
 - PROPOSED HEADER HIGH POINT
 - PROPOSED HEADER ACCESS RISER

- NOTES:**
- TOP OF WASTE GRADES PROVIDED BY WEAVER CONSULTANTS GROUP, INC. IN FILE TITLED "WC TOW GRADES".
 - AS-BUILT GAS COLLECTION AND CONTROL SYSTEM DESIGN TAKEN FROM AXIN LAND SURVEYS, INC. IN DRAWING TITLED "GAS LINES 2016 WEXFORD COUNTY LANDFILL" DATED OCTOBER 24, 2016 AND FROM GOSLING CZUBAK ENGINEERING SCIENCES, INC. IN DRAWING TITLED "LANDFILL GAS COLLECTION SYSTEM WEXFORD COUNTY LANDFILL GFL ENVIRONMENTAL" DATED FEBRUARY 3, 2022.
 - LOCATION OF PROPOSED LANDFILL GAS COLLECTION AND CONTROL SYSTEM COMPONENTS IS APPROXIMATE AND MAY VARY TO MEET FIELD CONDITIONS AT THE TIME OF CONSTRUCTION. ALL HORIZONTAL AND VERTICAL DATUM TO BE VERIFIED PRIOR TO CONSTRUCTION.



1" = 100' 0"

FILE: C:\Users\jma8851\OneDrive\Temp\jma8851_2450\WMC-202201-001.dwg, Layout: P001 User: jma8851, Apr 03, 2025, 11:24am

REV	DATE	DESCRIPTION	OWN	DES	CHK	APP
04/02/25		DATE OF ISSUE	JRD	JPS		
		DESIGNED BY	JPS	CHECKED BY	KM	

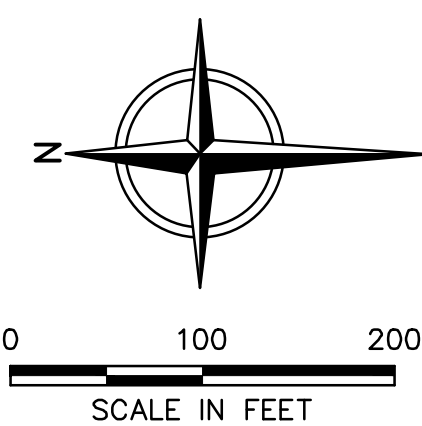


GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

**GAS COLLECTION AND CONTROL SYSTEM
GCCS DESIGN PLAN - EAST UNIT**

SHEET NO.
2

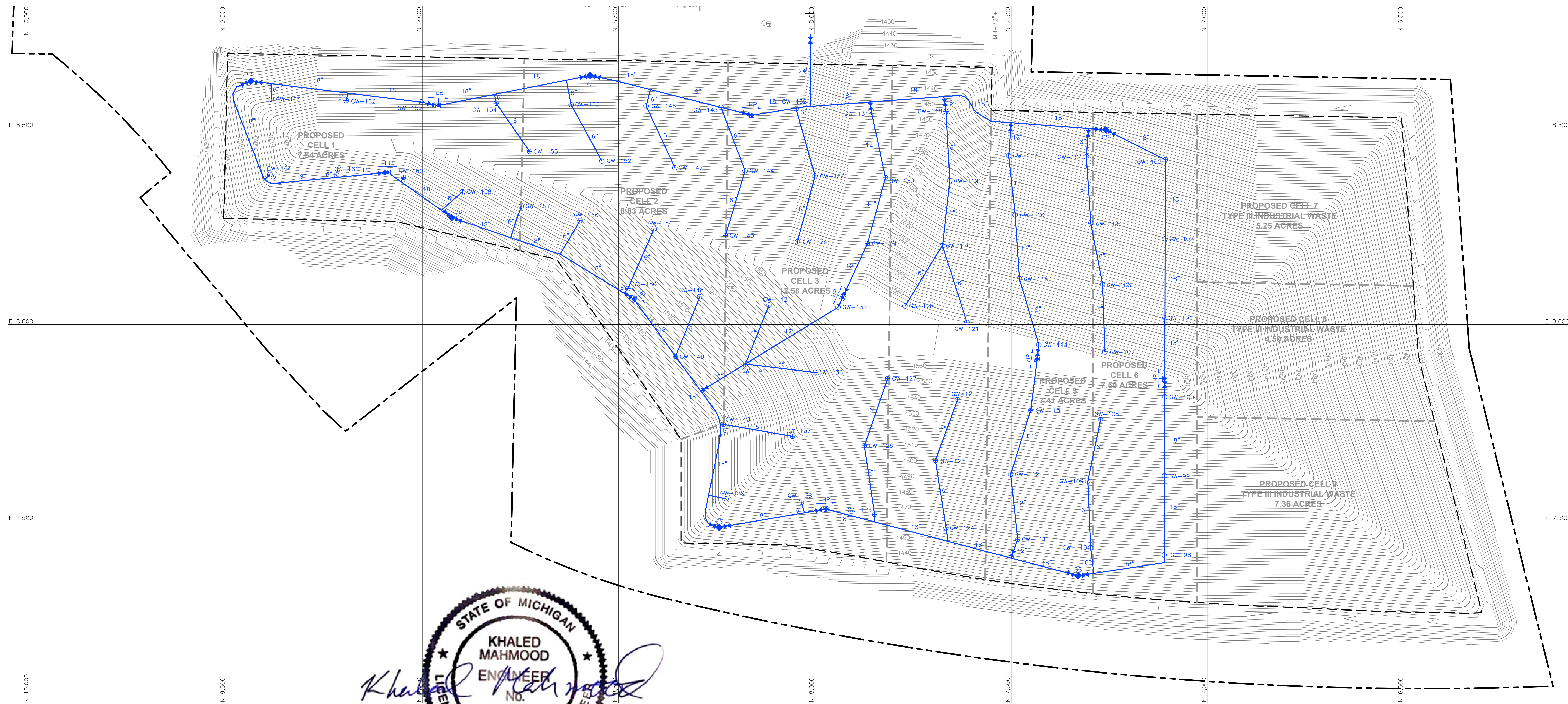
PROJECT NO.
209-4251206



- LEGEND**
- PROPERTY BOUNDARY
 - EXISTING SOLID WASTE BOUNDARY
 - EXISTING PHASE BOUNDARY
 - PROPOSED 10' CONTOUR
 - PROPOSED 2' CONTOUR
 - PROPOSED LANDFILL GAS HEADER/LATERAL
 - PROPOSED LFG EXTRACTION WELL
 - PROPOSED REMOTE WELLHEAD
 - PROPOSED HEADER ISOLATION VALVE
 - PROPOSED FLANGE CONNECTION
 - PROPOSED CONDENSATE SUMP
 - PROPOSED HEADER HIGH POINT
 - PROPOSED HEADER ACCESS RISER

NOTES:

- TOP OF WASTE GRADES PROVIDED BY WEAVER CONSULTANTS GROUP, INC. IN FILE TITLED 'WC TOW GRADES'.
- AS-BUILT GAS COLLECTION AND CONTROL SYSTEM DESIGN TAKEN FROM AXIN LAND SURVEYS, INC. IN DRAWING TITLED 'GAS LINES 2016 WEXFORD COUNTY LANDFILL' DATED OCTOBER 24, 2016 AND FROM GOSLING CZUBAK ENGINEERING SCIENCES, INC. IN DRAWING TITLED 'LANDFILL GAS COLLECTION SYSTEM WEXFORD COUNTY LANDFILL GFL ENVIRONMENTAL' DATED FEBRUARY 3, 2022.
- LOCATION OF PROPOSED LANDFILL GAS COLLECTION AND CONTROL SYSTEM COMPONENTS IS APPROXIMATE AND MAY VARY TO MEET FIELD CONDITIONS AT THE TIME OF CONSTRUCTION. ALL HORIZONTAL AND VERTICAL DATUM TO BE VERIFIED PRIOR TO CONSTRUCTION.



REV	DATE	DESCRIPTION	DWN BY	DES BY	CHK BY	APP BY
04	02/25	DATE OF ISSUE	JRD	JPS	JPS	KM
04	02/25	DRAWN BY	JRD	CHECKED BY	JPS	APPROVED BY
04	02/25	DESIGNED BY	JPS	APPROVED BY	KM	



GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

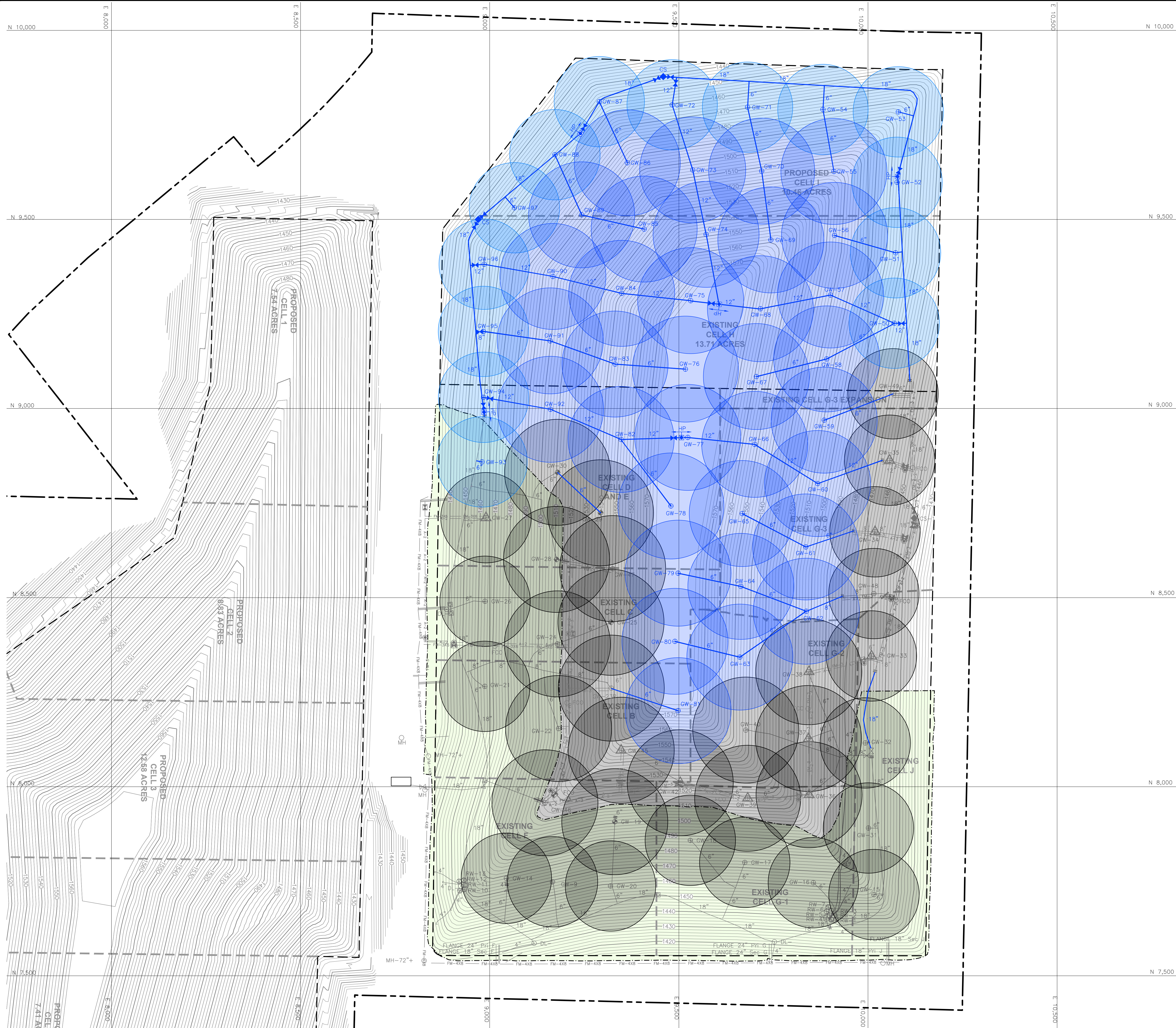
**GAS COLLECTION AND CONTROL SYSTEM
GCCS DESIGN PLAN - WEST UNIT**

SHEET NO.

3

PROJECT NO.
209-4251206

This drawing represents intellectual property of Tetra Tech. Any modification to the original by other than Tetra Tech personnel constitutes its original purpose and as such is not covered by Tetra Tech's intellectual property rights. Tetra Tech will not be held liable for any changes made to this document without express written consent of the originator.



LEGEND

---	PROPERTY BOUNDARY
---	EXISTING SOLID WASTE BOUNDARY
---	EXISTING PHASE BOUNDARY
---	PROPOSED 10' CONTOUR
---	PROPOSED 2' CONTOUR
---	EXISTING LANDFILL GAS HEADER
---	EXISTING LANDFILL GAS HORIZONTAL COLLECTOR
---	EXISTING 3" FORCEMAIN
---	EXISTING 3"x8" DUAL CONTAINED FORCEMAIN LINE
---	EXISTING 4"x8" DUAL CONTAINED FORCEMAIN LINE
---	EXISTING 2" AIRLINE
---	EXISTING LFG EXTRACTION WELL
---	EXISTING LFG EXTRACTION WELL WITH CAISSON
---	EXISTING REMOTE WELLHEAD
---	EXISTING HEADER ACCESS RISER
---	EXISTING ISOLATION VALVE
---	EXISTING BLIND FLANGE
---	EXISTING FLANGE CONNECTION
---	EXISTING REDUCER FITTING
---	EXISTING CONDENSATE DRIPLEG
---	EXISTING MANHOLE
---	EXISTING FORCEMAIN ISOLATION VALVE
---	EXISTING AIRLINE ISOLATION VALVE
---	EXISTING FORCEMAIN CLEANOUT
---	EXISTING FLANGE CONNECTION
---	EXISTING CONDENSATE SUMP
---	EXISTING FINAL COVER AREA
---	PROPOSED LANDFILL GAS HEADER/LATERAL
---	PROPOSED LFG EXTRACTION WELL
---	PROPOSED REMOTE WELLHEAD
---	PROPOSED HEADER ISOLATION VALVE
---	PROPOSED FLANGE CONNECTION
---	PROPOSED CONDENSATE SUMP
---	PROPOSED HEADER HIGH POINT
---	PROPOSED HEADER ACCESS RISER
---	PROPOSED 140' ROI
---	PROPOSED 120' ROI
---	EXISTING 120', 140' ROI

- NOTES:**
1. TOP OF WASTE GRADES PROVIDED BY WEAVER CONSULTANTS GROUP, INC. IN FILE TITLED "WC TOW GRADES".
 2. AS-BUILT GAS COLLECTION AND CONTROL SYSTEM DESIGN TAKEN FROM AXIN LAND SURVEYS, INC. IN DRAWING TITLED "GAS LINES 2016 WEXFORD COUNTY LANDFILL" DATED OCTOBER 24, 2016 AND FROM GOSLING CZUBAK ENGINEERING SCIENCES, INC. IN DRAWING TITLED "LANDFILL GAS COLLECTION SYSTEM WEXFORD COUNTY LANDFILL GFL ENVIRONMENTAL" DATED FEBRUARY 3, 2022.
 3. LOCATION OF PROPOSED LANDFILL GAS COLLECTION AND CONTROL SYSTEM COMPONENTS IS APPROXIMATE AND MAY VARY TO MEET FIELD CONDITIONS AT THE TIME OF CONSTRUCTION. ALL HORIZONTAL AND VERTICAL DATUM TO BE VERIFIED PRIOR TO CONSTRUCTION.



REV	DATE	DESCRIPTION	DWN BY	DES BY	CHK BY	APP BY
04/02/25			JRD	JPS	KM	



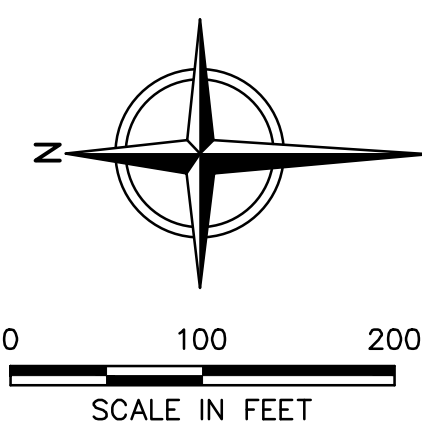
GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

**GAS COLLECTION AND CONTROL SYSTEM
WELL RADIUS OF INFLUENCE PLAN - EAST UNIT**

SHEET NO.
4

PROJECT NO.
209-4251206

This drawing represents intellectual property of Tetra Tech, Inc. and is to be used only for the project and site for which it was prepared. No part of this drawing may be reproduced or transmitted in any form or by any means electronic or mechanical, including photocopying, recording, or by any information storage and retrieval system, without written consent of Tetra Tech, Inc.

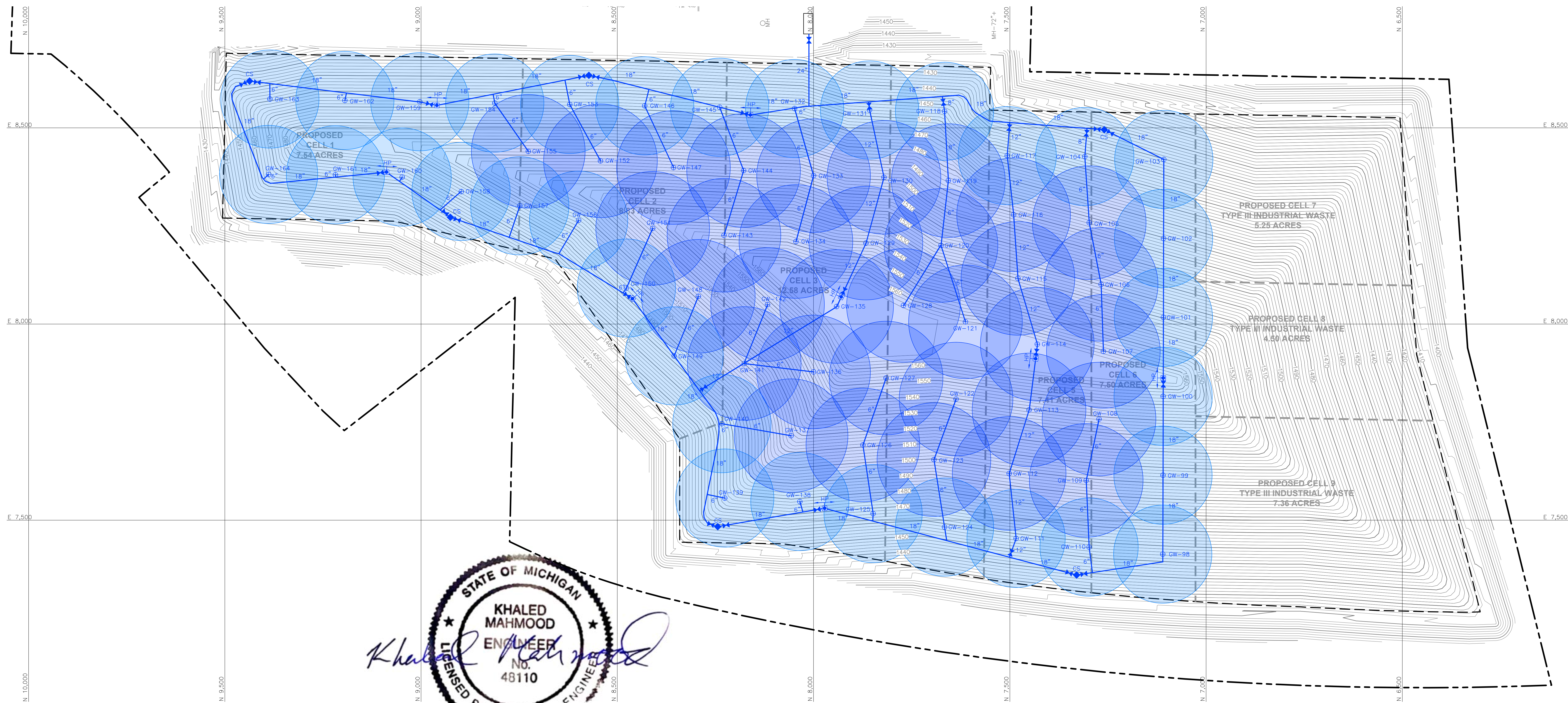


LEGEND

- PROPERTY BOUNDARY
- EXISTING SOLID WASTE BOUNDARY
- EXISTING PHASE BOUNDARY
- PROPOSED 10' CONTOUR
- PROPOSED 2' CONTOUR
- PROPOSED LANDFILL GAS HEADER/LATERAL
- PROPOSED LFG EXTRACTION WELL
- PROPOSED REMOTE WELLHEAD
- PROPOSED HEADER ISOLATION VALVE
- PROPOSED FLANGE CONNECTION
- PROPOSED CONDENSATE SUMP
- PROPOSED HEADER HIGH POINT
- PROPOSED HEADER ACCESS RISER
- PROPOSED 145' ROI
- PROPOSED 125' ROI

NOTES:

- TOP OF WASTE GRADES PROVIDED BY WEAVER CONSULTANTS GROUP, INC. IN FILE TITLED 'WC TOW GRADES'.
- AS-BUILT GAS COLLECTION AND CONTROL SYSTEM DESIGN TAKEN FROM AXIN LAND SURVEYS, INC. IN DRAWING TITLED 'GAS LINES 2016 WEXFORD COUNTY LANDFILL' DATED OCTOBER 24, 2016 AND FROM GOSLING CZUBAK ENGINEERING SCIENCES, INC. IN DRAWING TITLED 'LANDFILL GAS COLLECTION SYSTEM WEXFORD COUNTY LANDFILL GFL ENVIRONMENTAL' DATED FEBRUARY 3, 2022.
- LOCATION OF PROPOSED LANDFILL GAS COLLECTION AND CONTROL SYSTEM COMPONENTS IS APPROXIMATE AND MAY VARY TO MEET FIELD CONDITIONS AT THE TIME OF CONSTRUCTION. ALL HORIZONTAL AND VERTICAL DATUM TO BE VERIFIED PRIOR TO CONSTRUCTION.



Khaled Mahmoud
STATE OF MICHIGAN
KHALED MAHMOOD
ENGINEER
No. 48110
LICENSED PROFESSIONAL ENGINEER

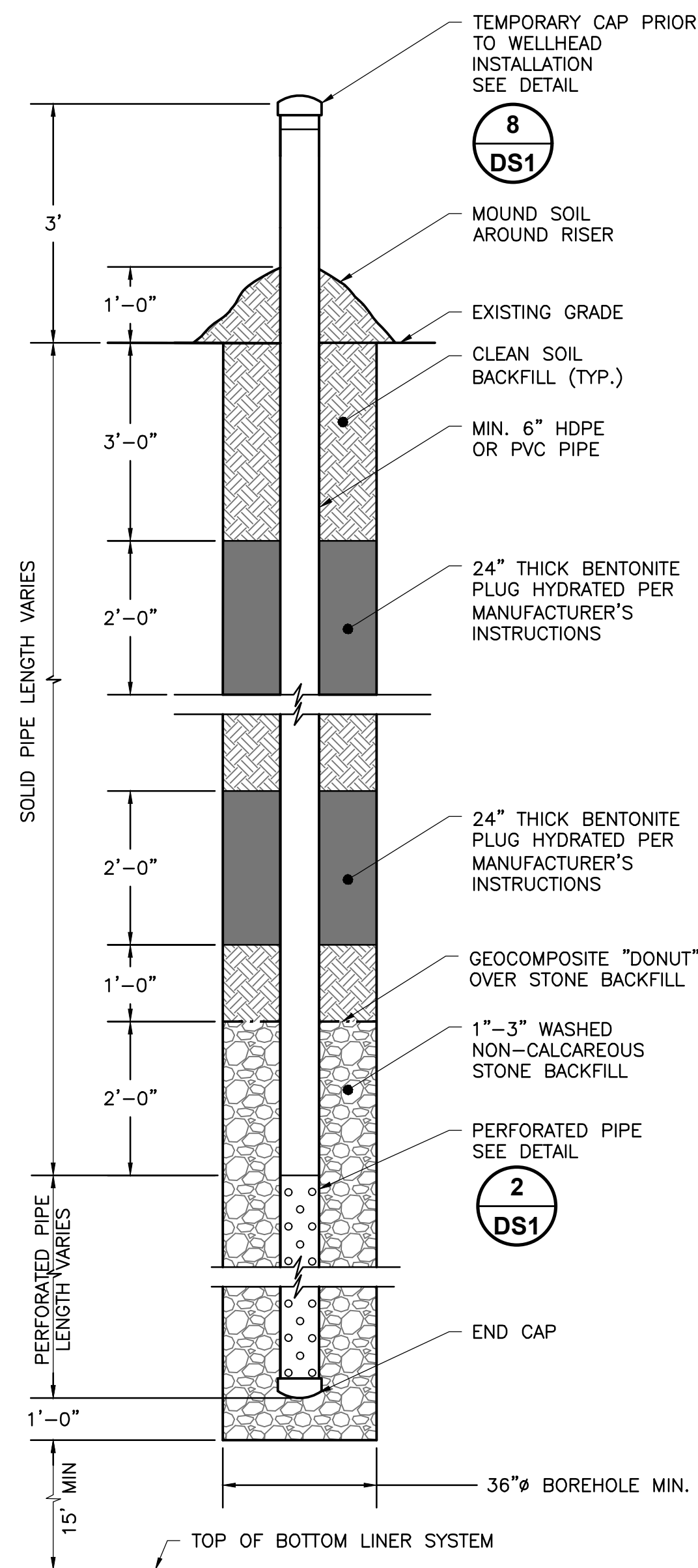
REV	DATE	DESCRIPTION	DWN BY	DES BY	CHK BY	APP BY
04	02/25		JPS	JPS	JPS	JPS



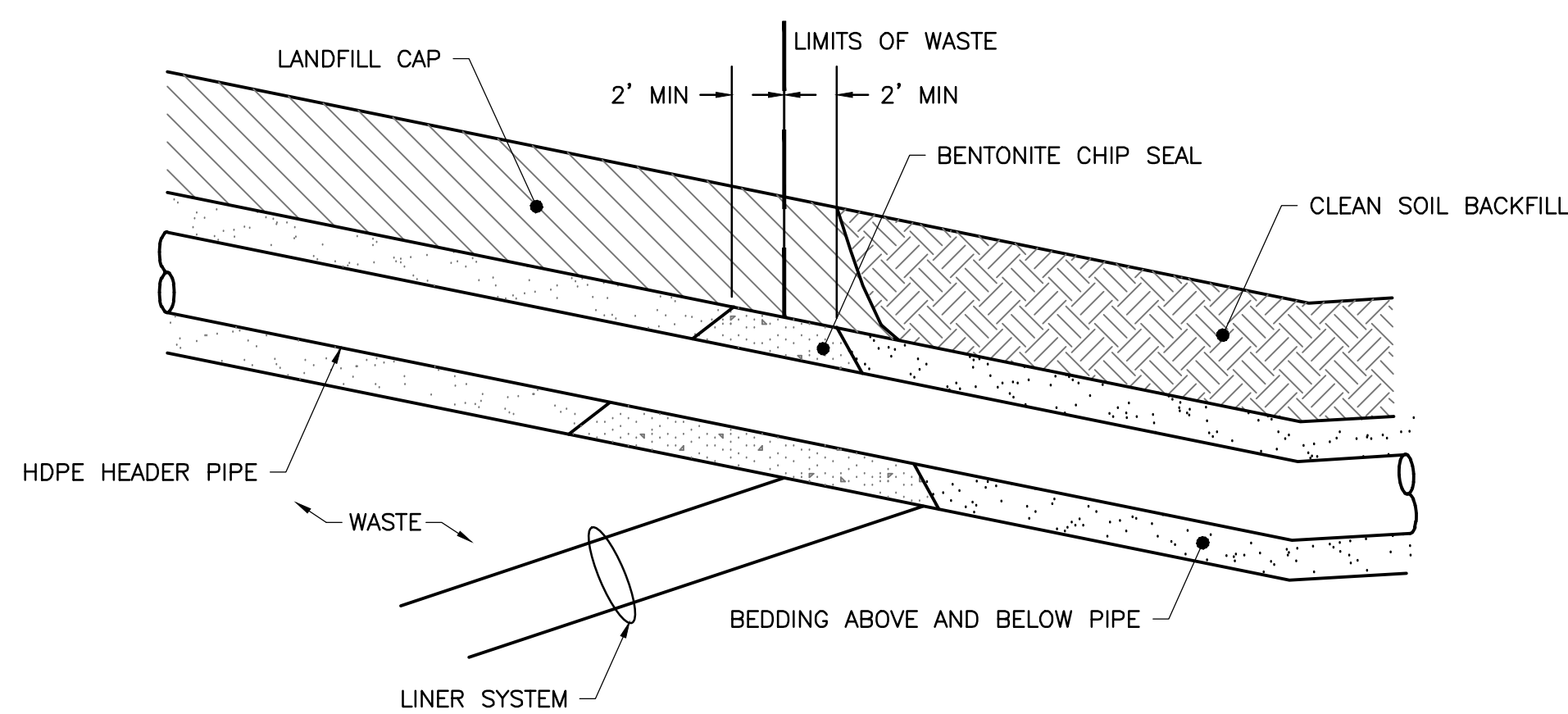
GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN
GAS COLLECTION AND CONTROL SYSTEM
WELL RADIUS OF INFLUENCE PLAN - WEST UNIT

SHEET NO.
5
PROJECT NO.
209-4251206

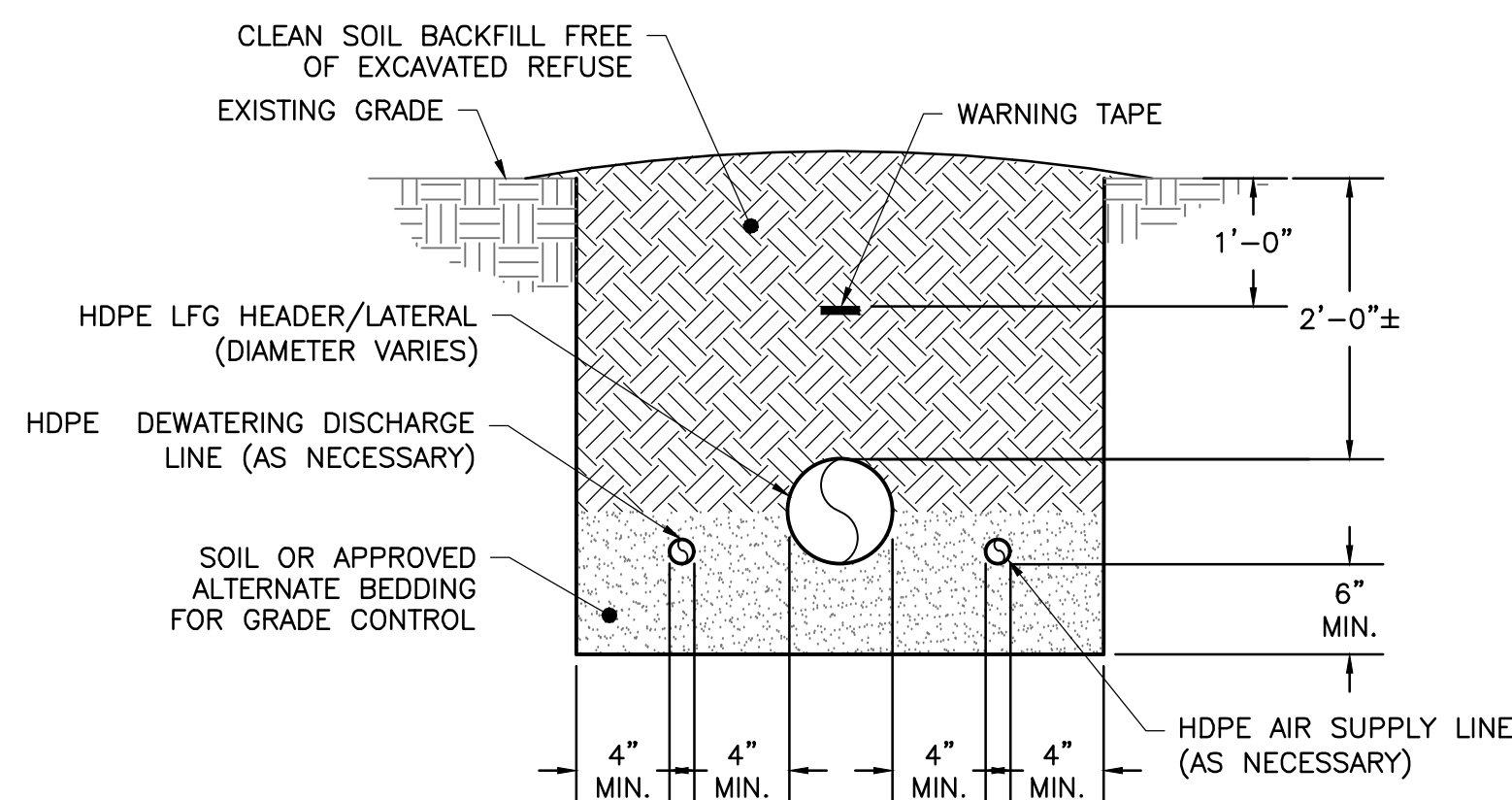
This drawing represents intellectual property of Tetra Tech, Inc. and is not to be reproduced, stored, or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording, or by any information storage and retrieval system, without the prior written consent of Tetra Tech, Inc.



EXTRACTION WELL
DETAIL 1
SCALE: NOT TO SCALE
DS1

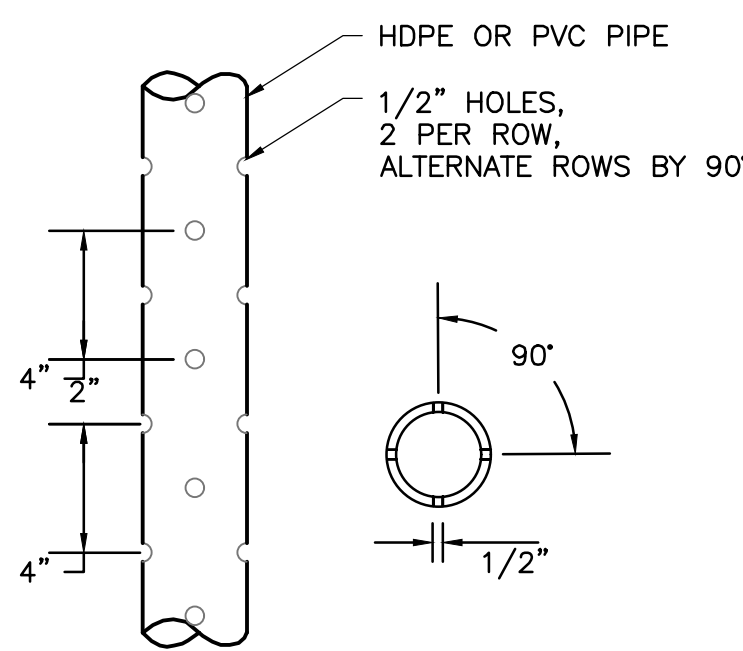


ANTI-SEEP COLLAR
DETAIL 6
SCALE: NOT TO SCALE
DS1



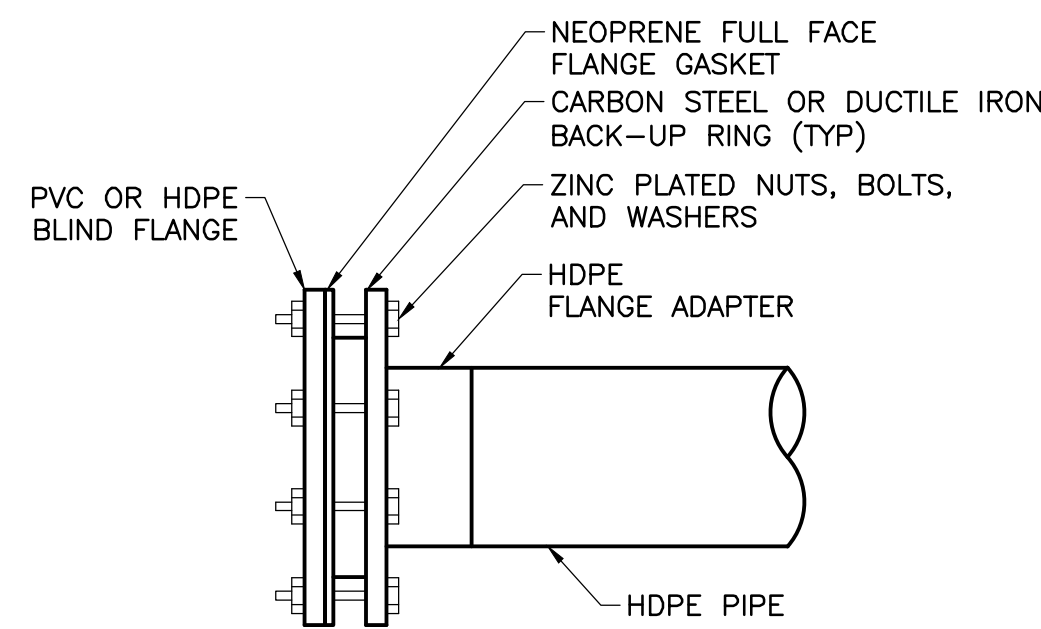
HEADER AND LATERAL PIPELINE TRENCH
DETAIL 9
SCALE: NOT TO SCALE
DS1

- NOTES:
- WARNING TAPE SHALL BE MIN. 3" WIDE AND IMPRINTED WITH "GAS LINE BURIED BELOW".
 - ALL HEADER AND LATERAL SHALL BE INSTALLED AT MIN. 3% SLOPE UNLESS APPROVED IN ADVANCE BY ENGINEER.
 - THE NUMBER AND TYPES OF PIPES INSTALLED IN THE TRENCH MAY VARY.

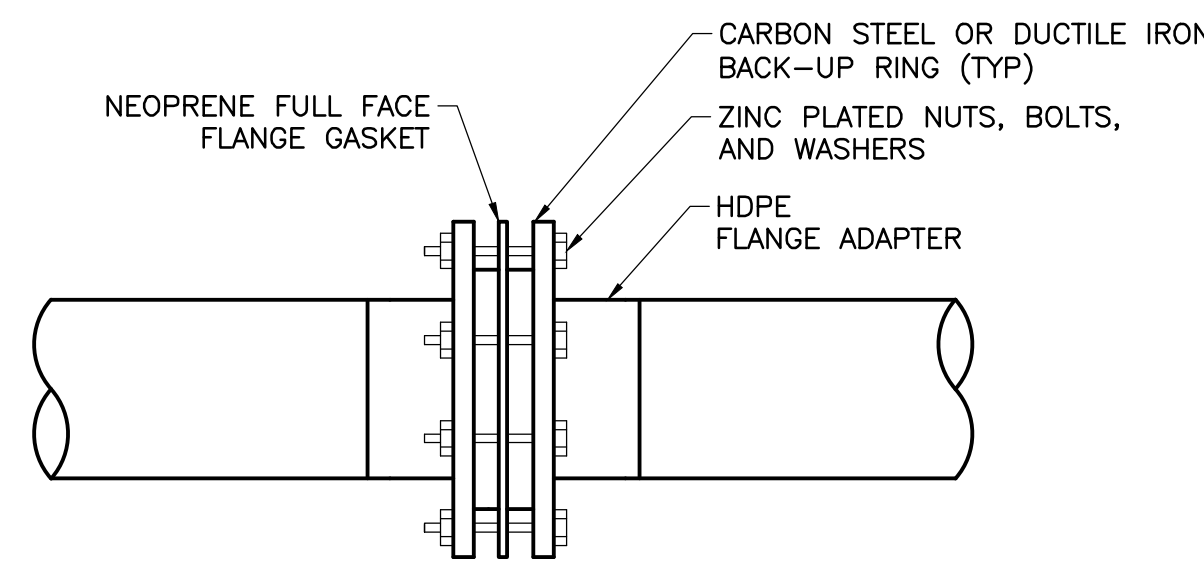


PERFORATED PIPE
DETAIL 2
SCALE: NOT TO SCALE
DS1

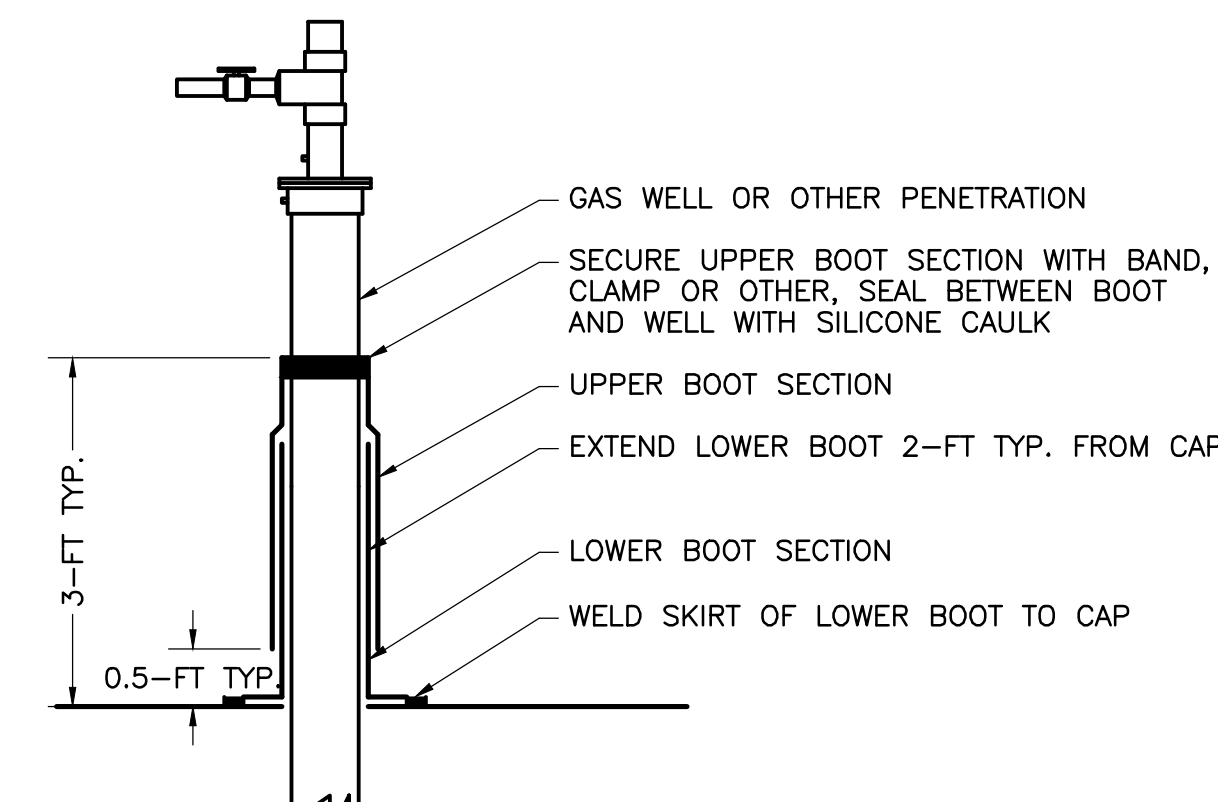
- NOTES:
- PERFORATIONS SPACED 90° APART HORIZONTALLY.
 - PERFORATIONS SPACED 4" APART VERTICALLY.
 - 90° AND 270° ROWS STAGGERED 2" BELOW 0° AND 180° ROWS.
 - ALTERNATE PERFORATED SIZES AND PATTERNS MAY BE UTILIZED.
 - ALTERNATE MATERIALS MAY BE UTILIZED TO ACCOMMODATE FIELD CONDITIONS.



BLIND FLANGE ASSEMBLY
DETAIL 3
SCALE: NOT TO SCALE
DS1

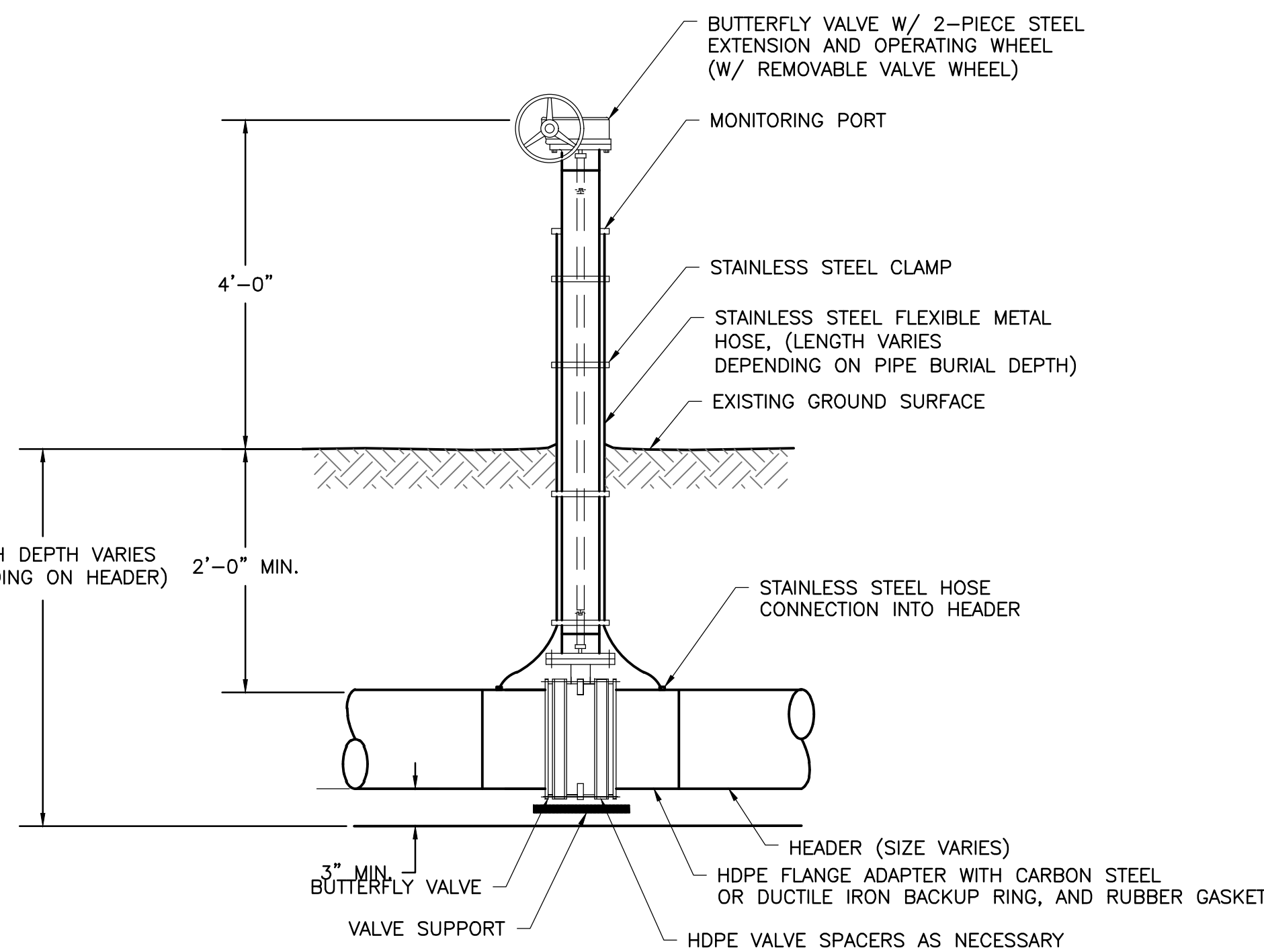


FLANGE CONNECTION
DETAIL 4
SCALE: NOT TO SCALE
DS1

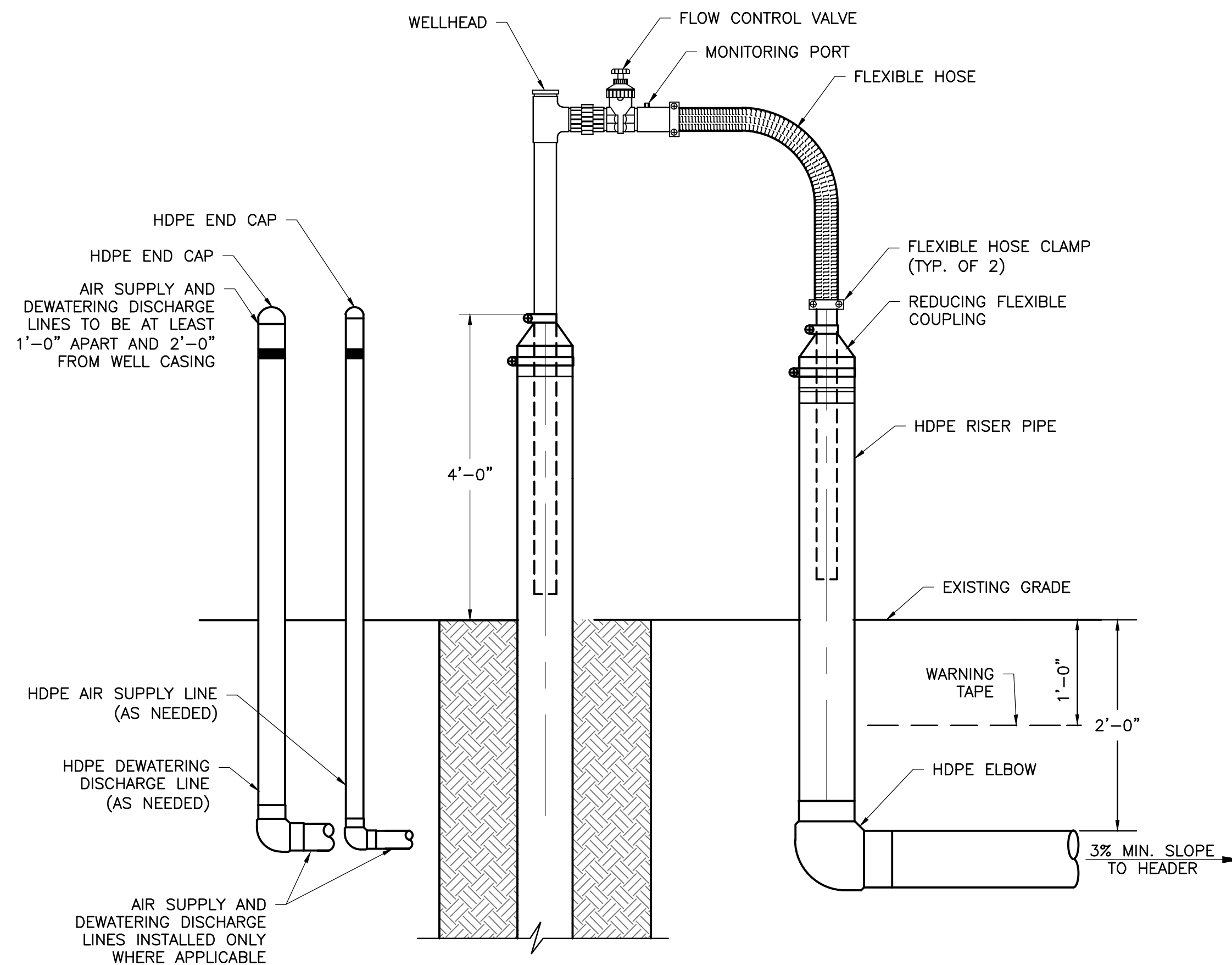


EXTRACTION WELL AND VERTICAL PIPE PENETRATION BOOT
DETAIL 5
SCALE: NOT TO SCALE
DS1

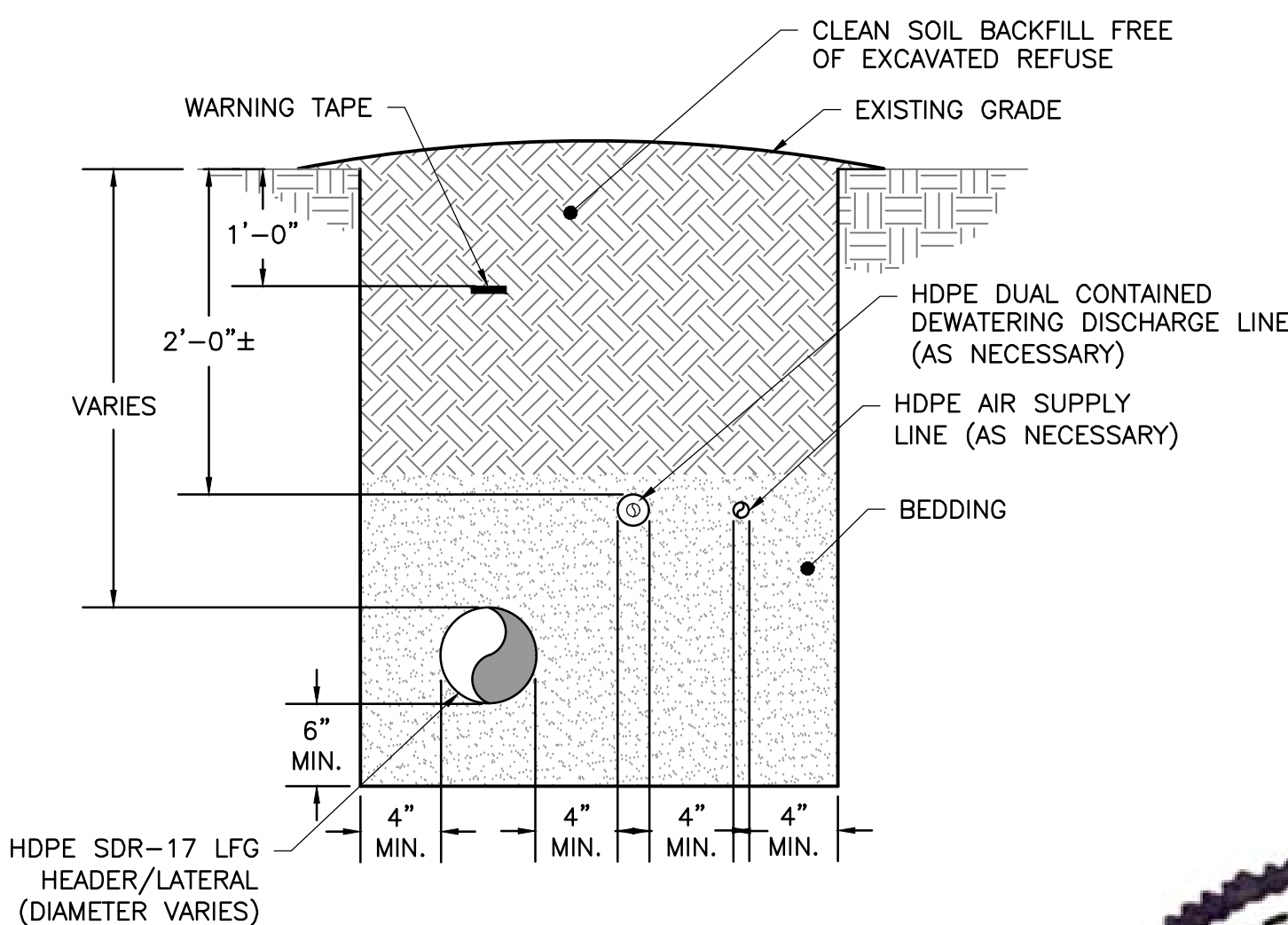
- NOTE:
- WELL SEAL IS TO BE INSTALLED ON WELLS AND VERTICAL PIPE PENETRATIONS IN FINAL CAPPED AREAS.



ISOLATION VALVE
DETAIL 7
SCALE: NOT TO SCALE
DS1



WELLHEAD
DETAIL 8
SCALE: NOT TO SCALE
DS1

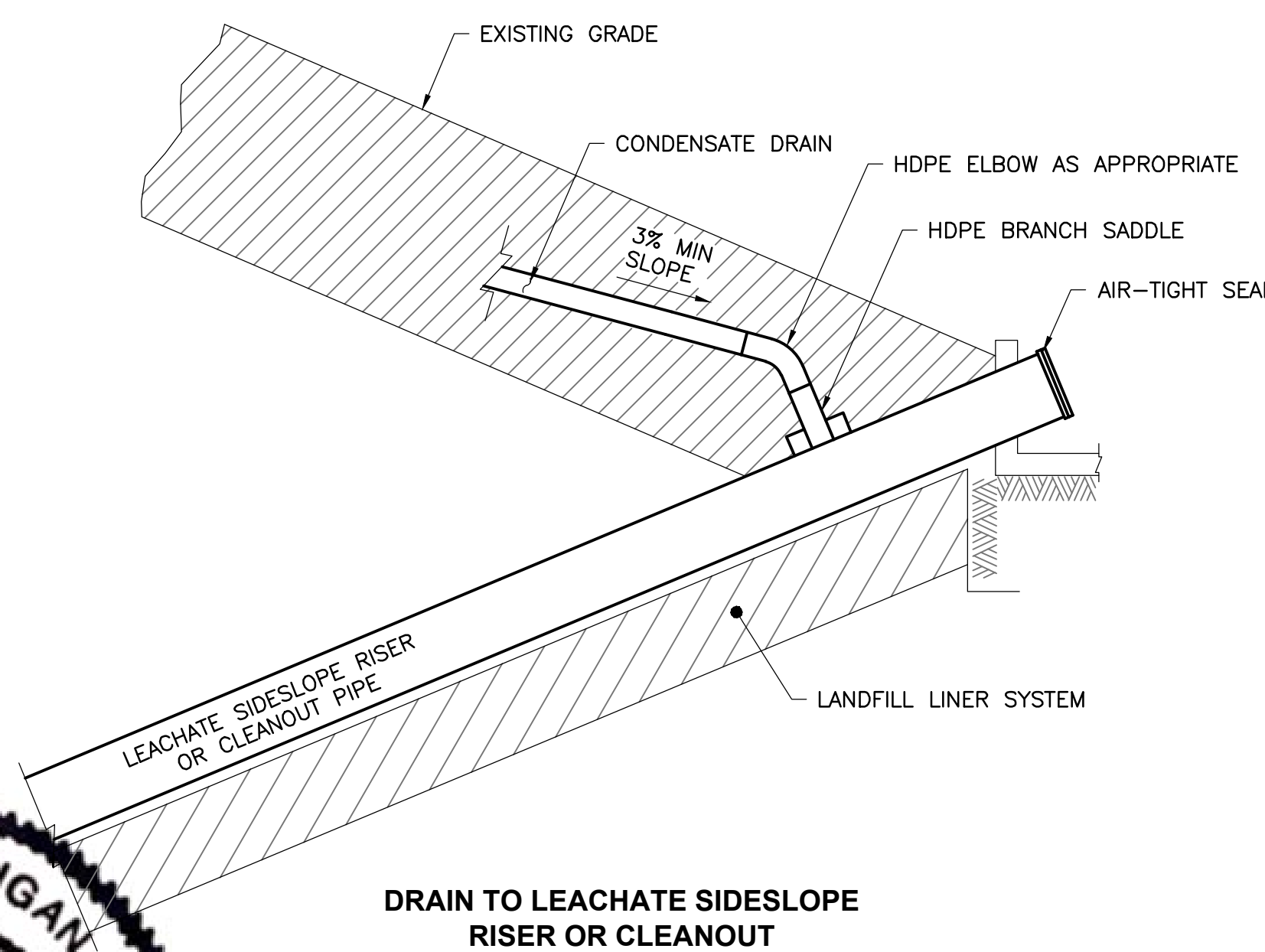


HEADER/LATERAL TRENCH INSTALLED OUTSIDE LIMITS OF WASTE
DETAIL 10
SCALE: NOT TO SCALE
DS1

- NOTES:
- WARNING TAPE SHALL BE MIN. 3" WIDE AND IMPRINTED WITH "GAS LINE BURIED BELOW".
 - ALL HEADER AND LATERAL SHALL BE INSTALLED AT MIN. 0.5% SLOPE UNLESS APPROVED IN ADVANCE BY ENGINEER.
 - THE NUMBER AND TYPES OF PIPES INSTALLED IN THE TRENCH MAY VARY.
 - DEWATERING LINES SHALL BE DUAL CONTAINED WHEN INSTALLED OUTSIDE LIMITS OF WASTE.

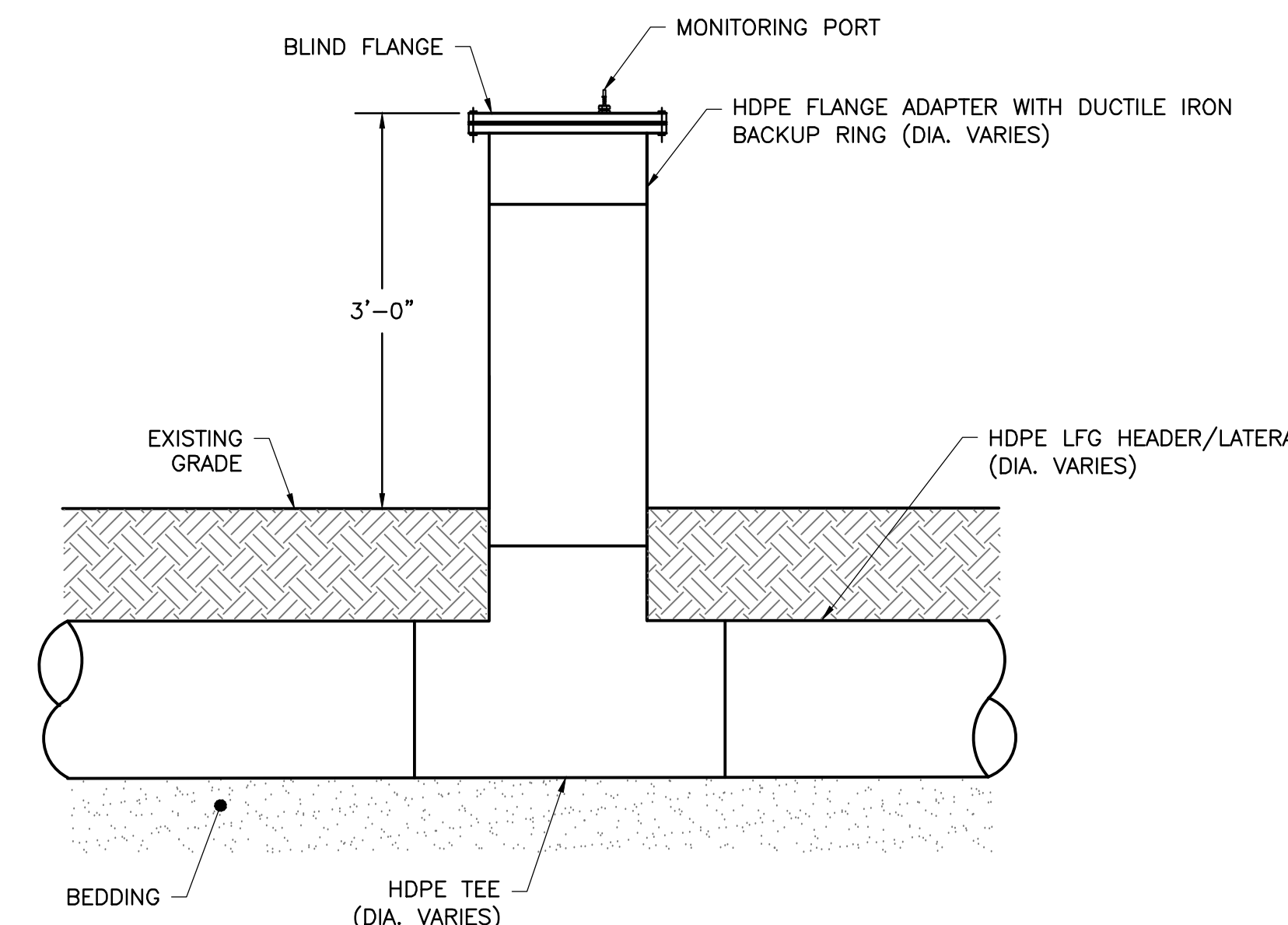
NOTE:

LAYOUT OF COMPONENTS AND DETAILS OF CONSTRUCTION ARE TYPICAL AND MAY VARY TO ACCOMMODATE FIELD CONDITIONS AND AVAILABLE TECHNOLOGY AT THE TIME OF CONSTRUCTION.



DRAIN TO LEACHATE SIDESLOPE RISER OR CLEANOUT
DETAIL 11
SCALE: NOT TO SCALE
DS1

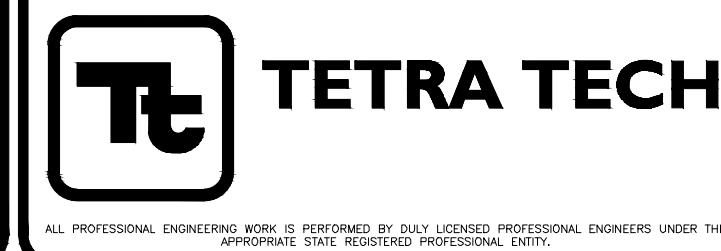
- NOTES:
- FIELD FIT CONDENSATE CONNECTION TO EXISTING LEACHATE RISER. RISER SIZE VARIES (6" TO 18" TYPICAL).
 - MAINTAIN MINIMUM 3% SLOPE INTO CLEANOUT RISER CONNECTION.



HEADER ACCESS RISER
DETAIL 12
SCALE: NOT TO SCALE
DS1



REV	DATE	DESCRIPTION	OWN BY	DES BY	CHK BY	APP BY
04/02/25			JRD	JPS	KM	



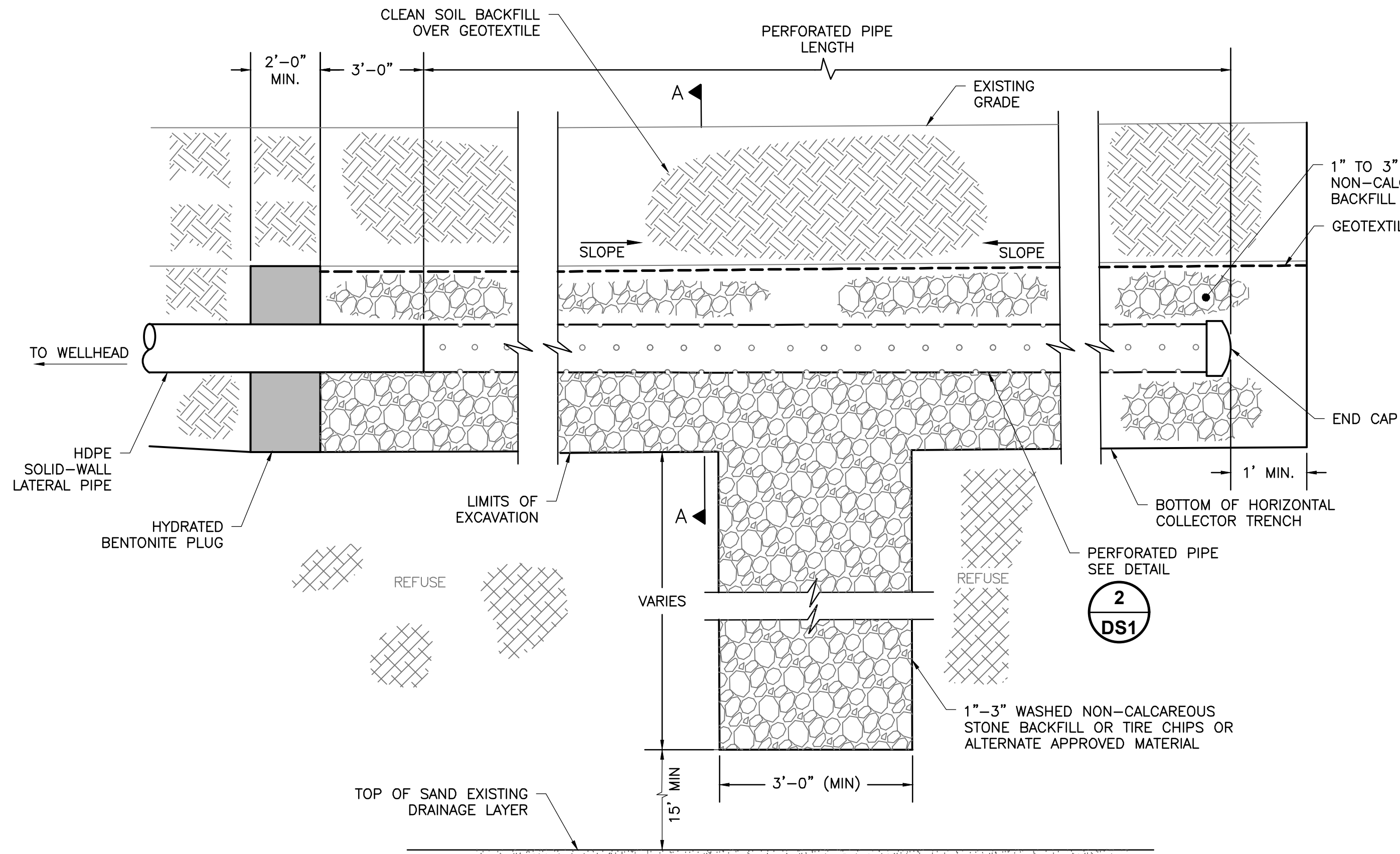
GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

**GAS COLLECTION AND CONTROL SYSTEM
DETAILS**

SHEET NO.

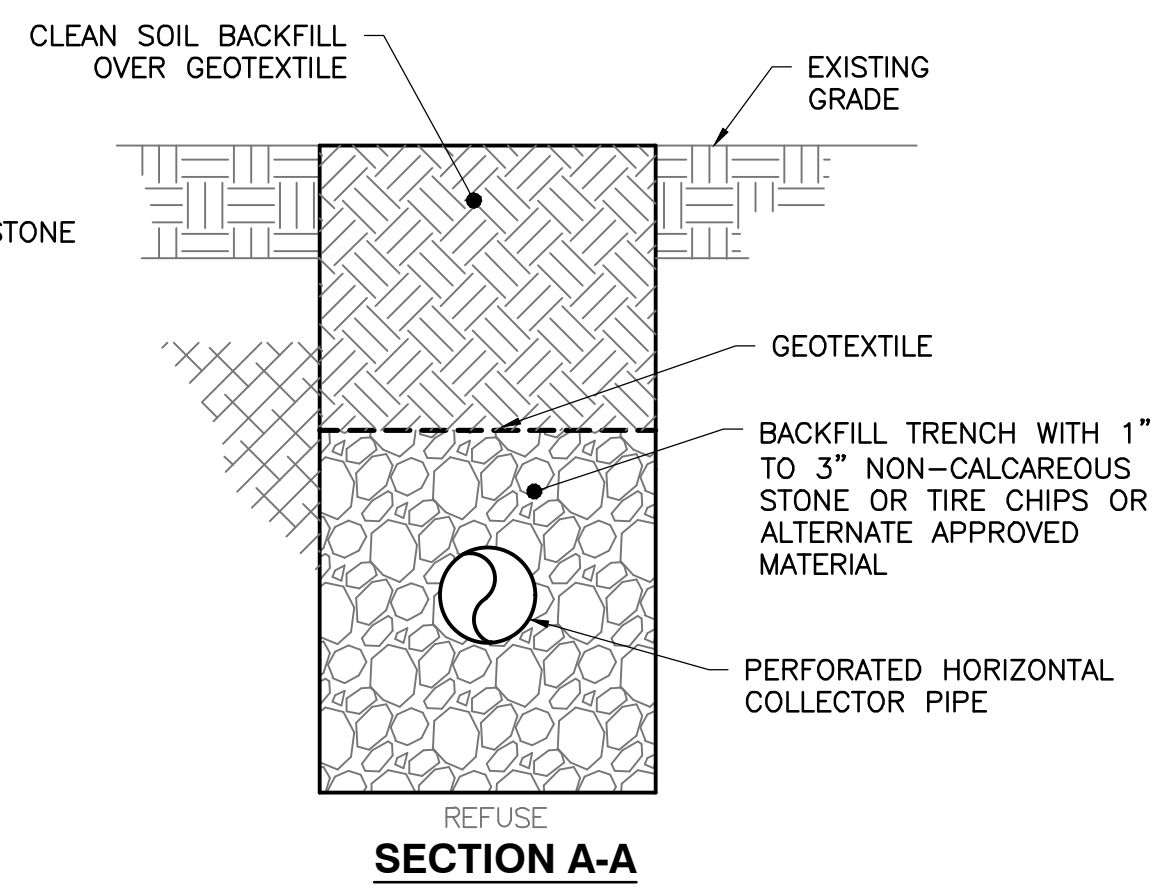
DS1

PROJECT NO.
209-4251206

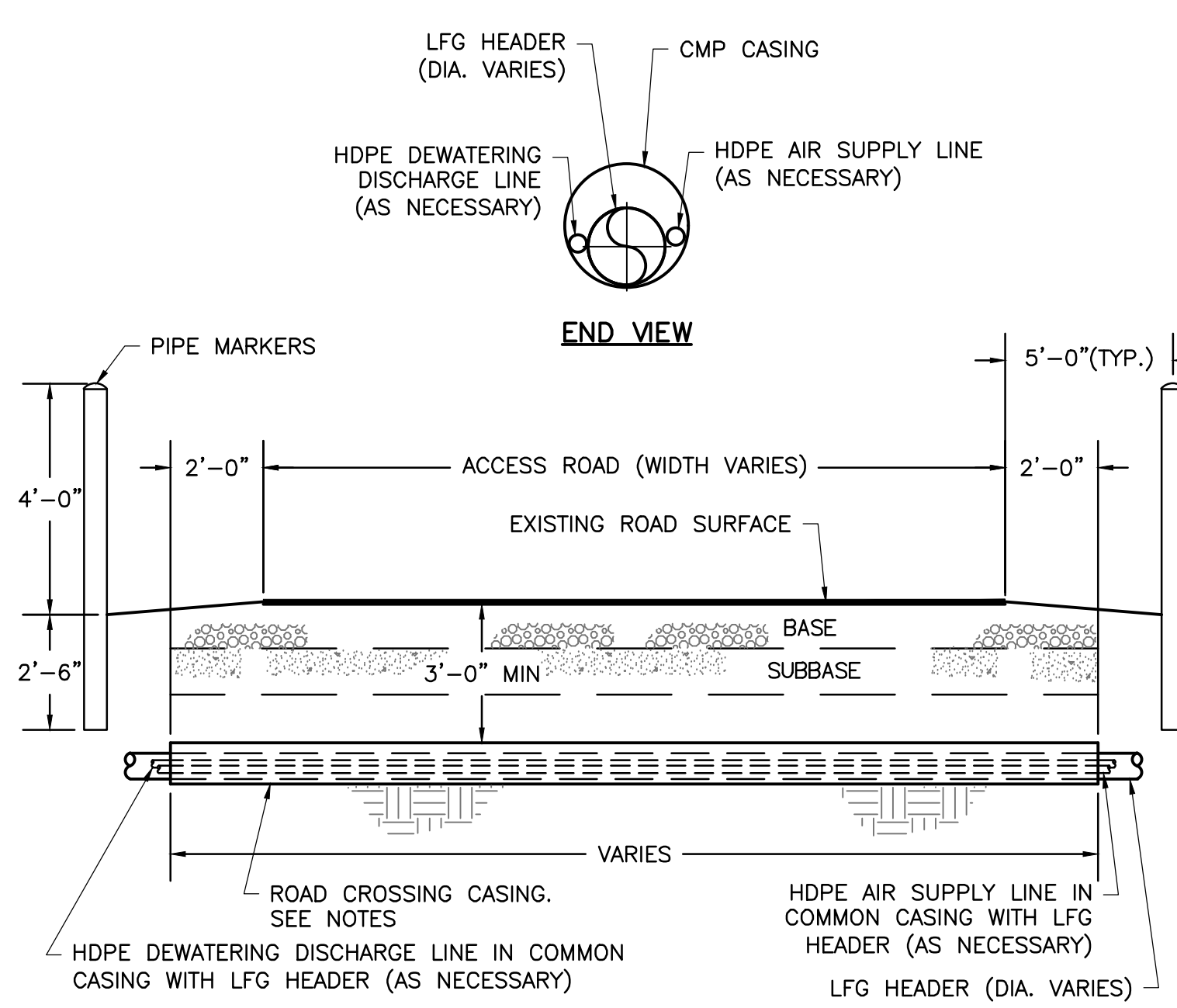


INTERIM HORIZONTAL COLLECTOR

DETAIL 1
SCALE: NOT TO SCALE DS2



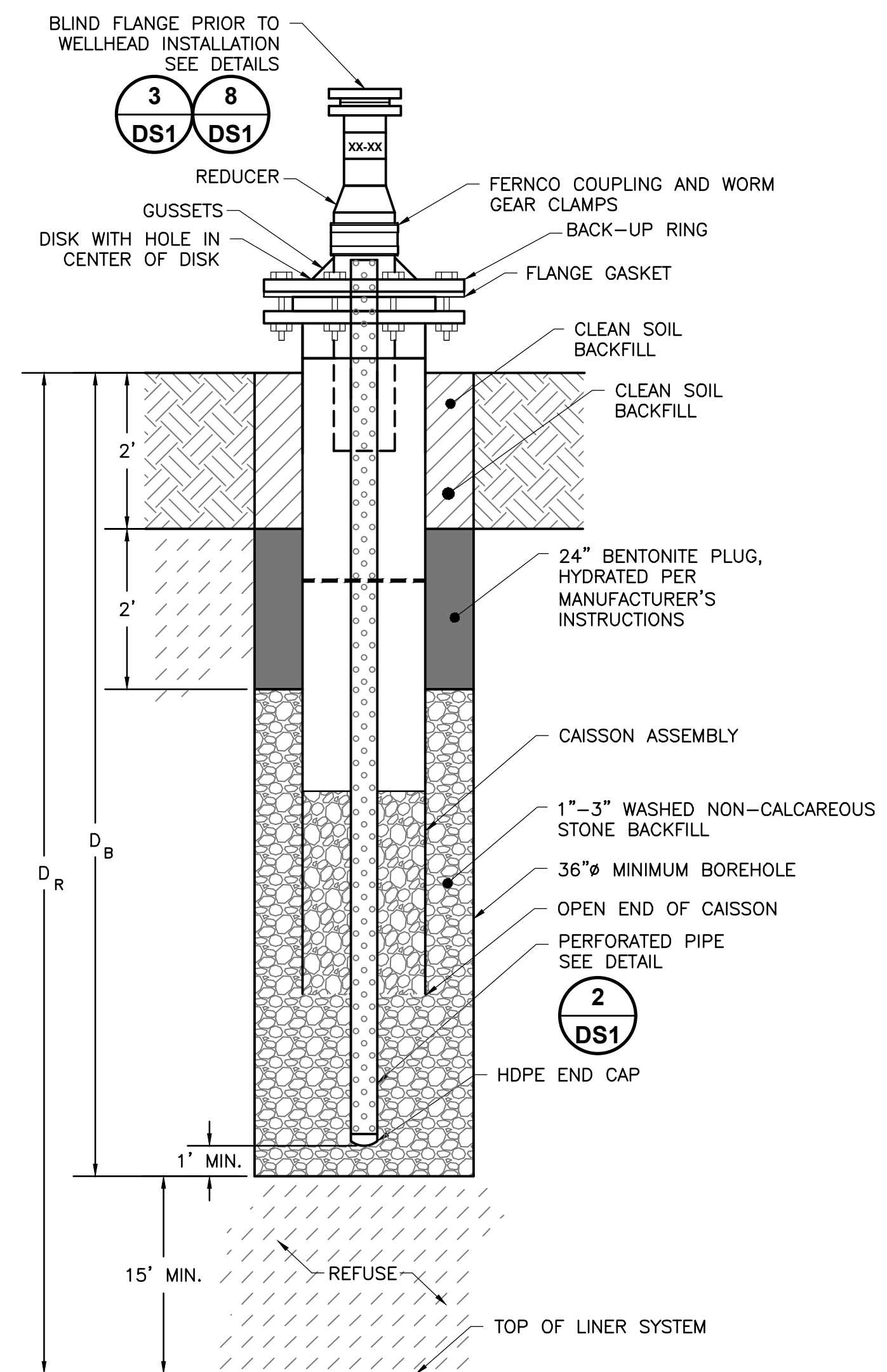
SECTION A-A



TRENCHED ROAD CROSSING

DETAIL 2
SCALE: NOT TO SCALE DS2

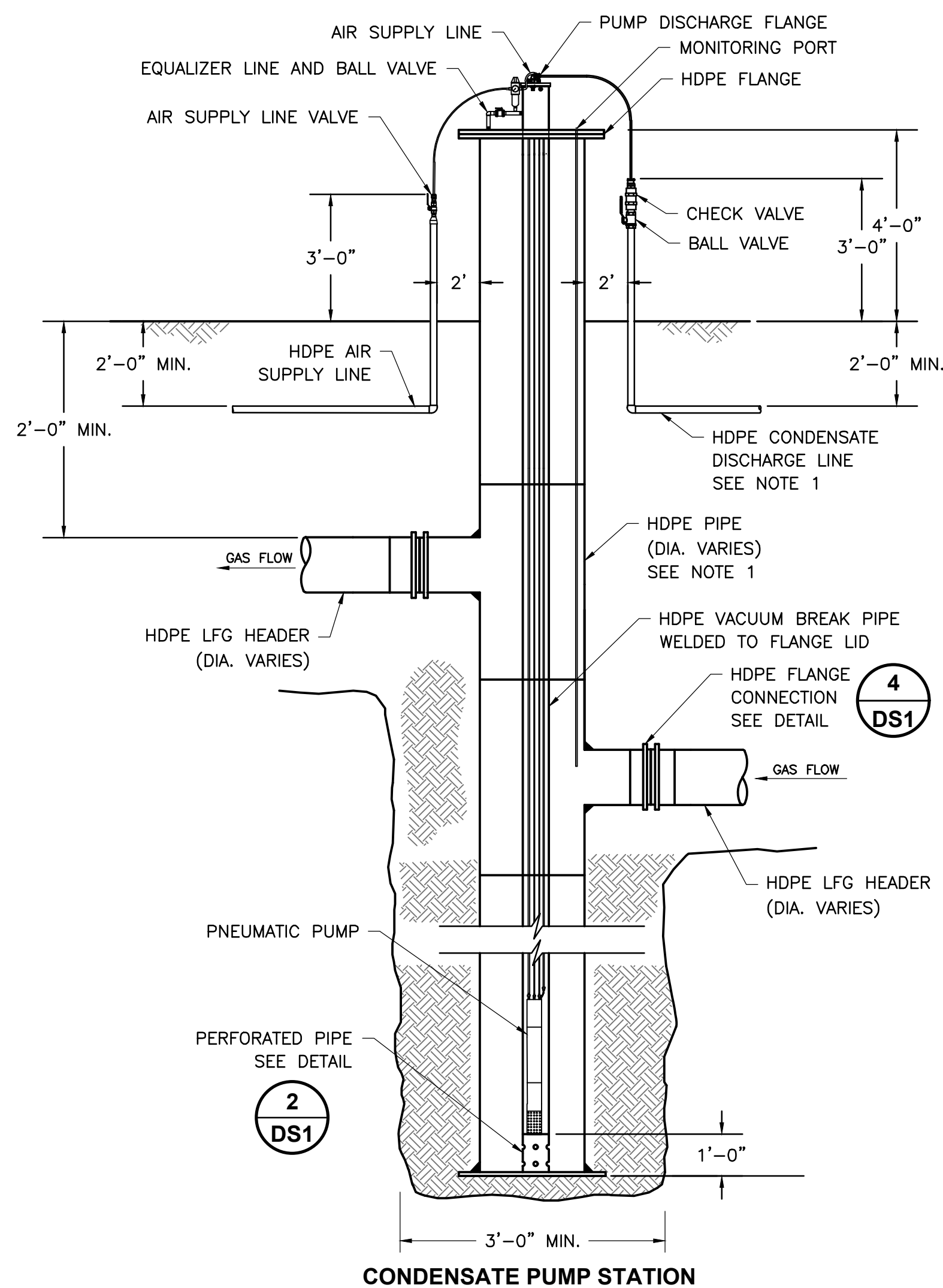
- NOTES:
1. ROAD SHALL BE RESTORED TO MATCH ORIGINAL CONDITIONS.
 2. CMP CASING A MINIMUM OF 5" GREATER THAN THE CUMULATIVE DIAMETER OF THE PIPES ENCASED.
 3. NUMBER AND TYPES OF PIPES ENCASED IN ROAD CROSSING VARIES.



LFG EXTRACTION WELL WITH CAISSON

DETAIL 3
SCALE: NOT TO SCALE DS2

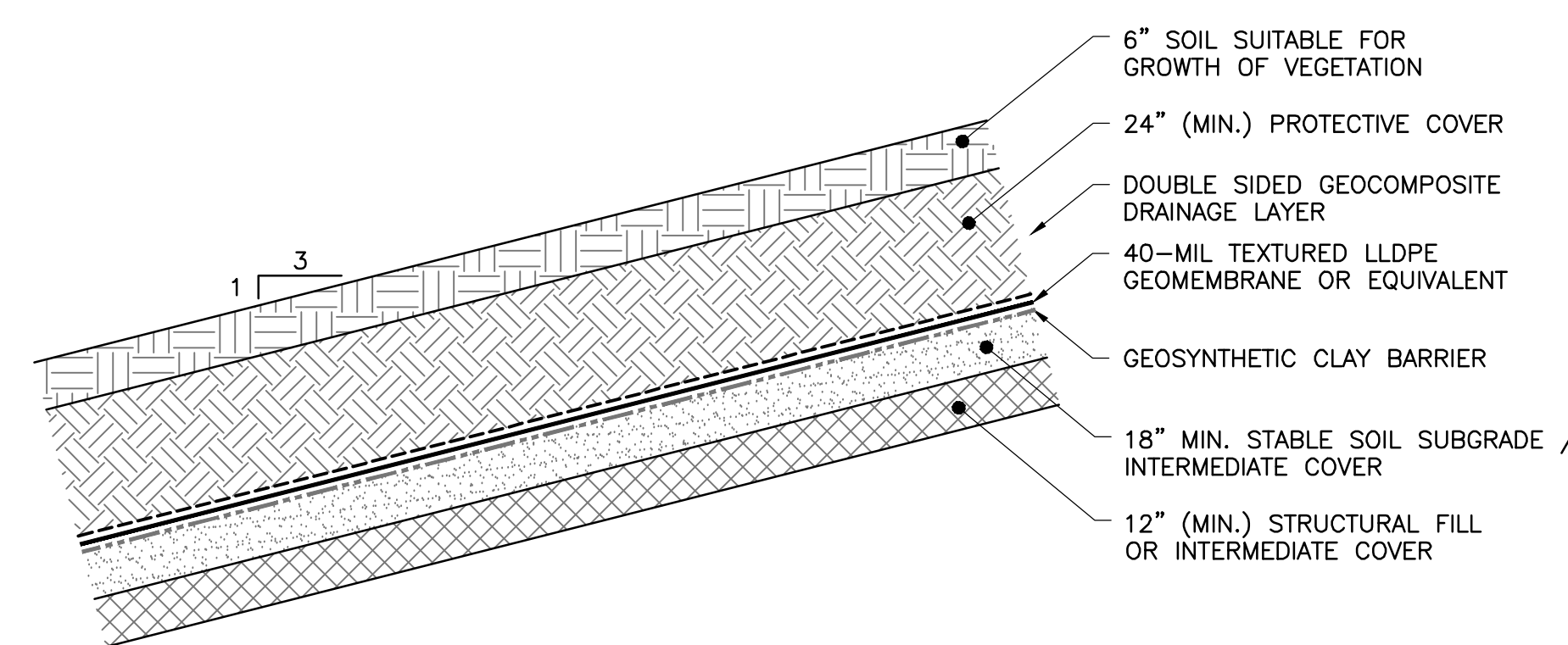
- NOTES:
1. D_r = DEPTH OF REFUSE
D_b = DEPTH OF BORING
 2. MINIMUM 15 FEET SEPARATION BETWEEN BASE OF BORING AND TOP OF LINER SYSTEM.



CONDENSATE PUMP STATION

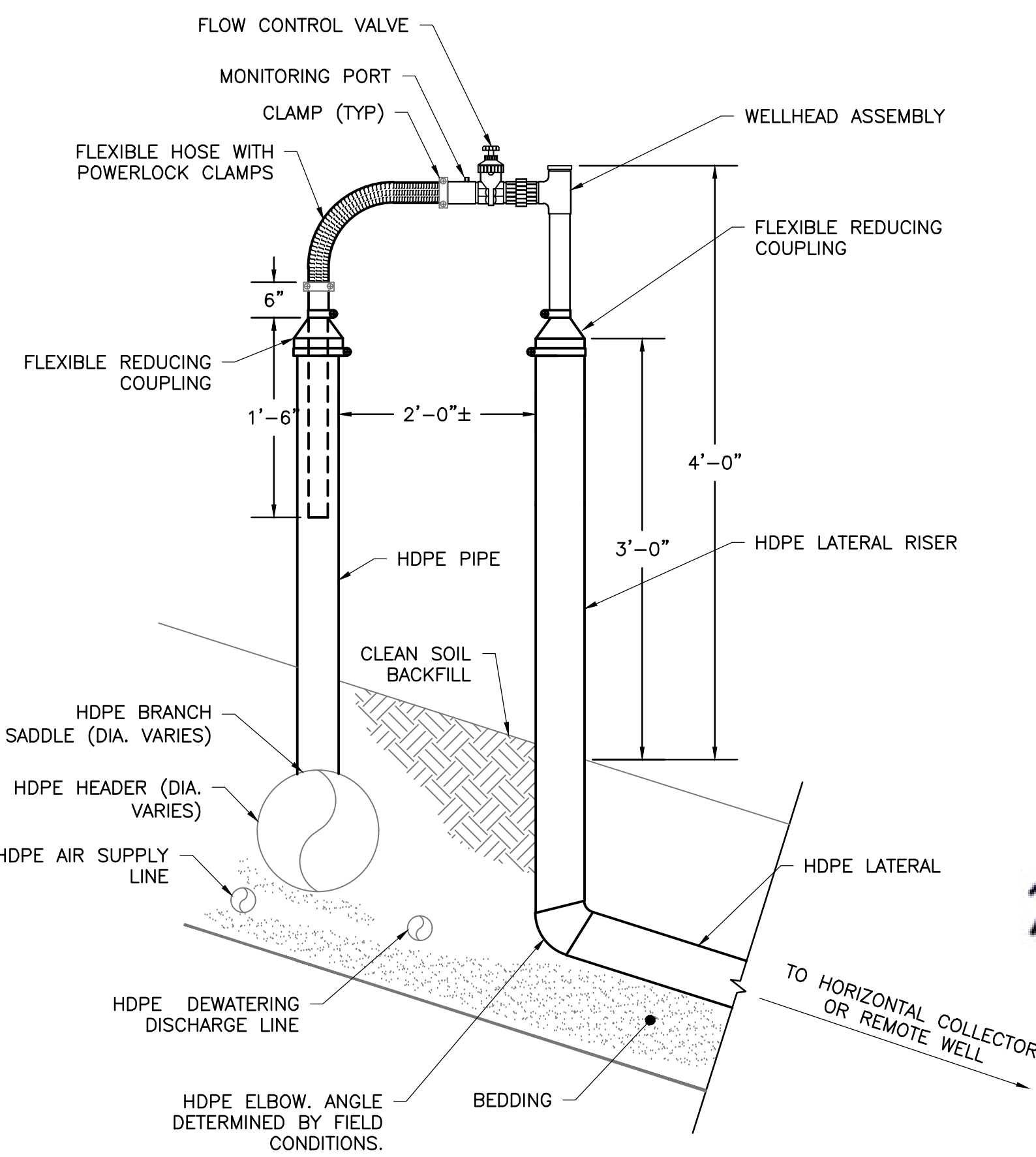
DETAIL 5
SCALE: NOT TO SCALE DS2

- NOTES:
1. CONDENSATE PUMP STATIONS AND CONDENSATE DISCHARGE LINES INSTALLED OUTSIDE OF LIMITS OF WASTE SHALL BE DUAL CONTAINED.



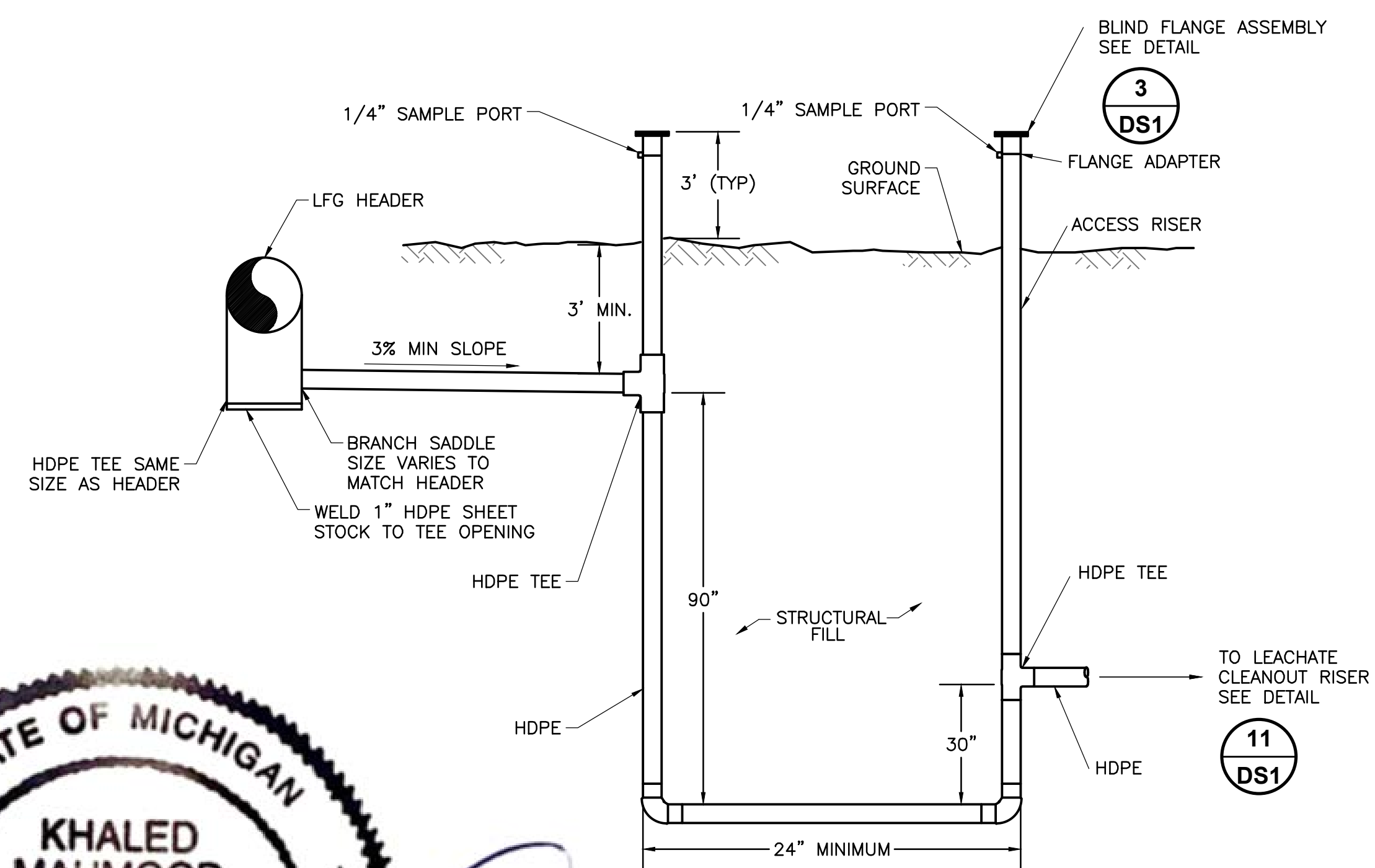
TYPICAL FINAL COVER

DETAIL 4
SCALE: NOT TO SCALE DS2



REMOTE WELLHEAD

DETAIL 6
SCALE: NOT TO SCALE DS2



CONDENSATE DRIPLEG

DETAIL 7
SCALE: NOT TO SCALE DS2

- NOTES:
1. DRIPLEG DESIGNED FOR 60" W.C. APPLIED VACUUM.
 2. FILL DRIPLEG WITH CLEAN WATER PRIOR TO OPERATION.

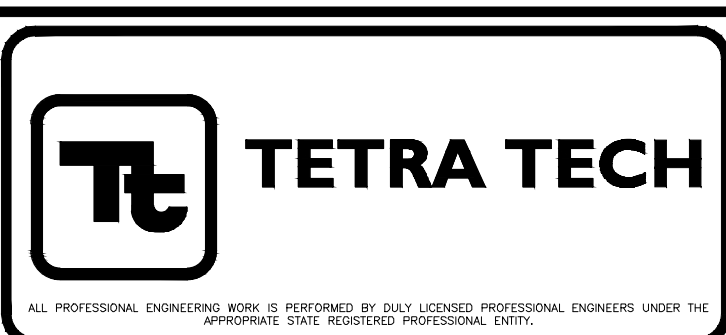


NOTE:

LAYOUT OF COMPONENTS AND DETAILS OF CONSTRUCTION ARE TYPICAL AND MAY VARY TO ACCOMMODATE FIELD CONDITIONS AND AVAILABLE TECHNOLOGY AT THE TIME OF CONSTRUCTION.

This drawing represents intellectual property of Tetra Tech, Inc. and is not to be reproduced, stored, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or by any information storage and retrieval system, without the prior written consent of Tetra Tech, Inc.

REV	DATE	DESCRIPTION	OWN BY	DES BY	CHK BY	APP BY
04/02/25			JRD	JPS	KM	



GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN
**GAS COLLECTION AND CONTROL SYSTEM
DETAILS**

SHEET NO.
DS2
PROJECT NO.
209-4251206

PROPOSED WEST UNIT WELL SCHEDULE – NOT FOR DRILLING

WELL ID	NORTHING (FT)	EASTING (FT)	APPROX. ELEVATION (FT MSL)		DEPTH OF WASTE (FT)	WELL DEPTH (FT) ⁽²⁾	BOTTOM OF WELL TO BOTTOM OF WASTE (FT) ⁽³⁾	PIPE LENGTH (FT)			PERFORATED AND SOLID PIPE DIAMETER AND MATERIAL
			PERMITTED TOP OF WASTE	PERMITTED BOTTOM OF WASTE				PERFORATED PIPE (FT)	SOLID WALL PIPE (FT)	SOLID WALL PIPE ABOVE GRADE (FT)	
GW-98	7,110	7,413	1,455	1,388	67	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-99	7,110	7,615	1,505	1,344	161	121	40	105	15	3	6" MIN. PVC, CPVC OR HDPE
GW-100	7,109	7,816	1,555	1,346	209	157	52	141	15	3	6" MIN. PVC, CPVC OR HDPE
GW-101	7,109	8,017	1,530	1,350	180	135	45	119	15	3	6" MIN. PVC, CPVC OR HDPE
GW-102	7,108	8,218	1,480	1,352	128	96	32	80	15	3	6" MIN. PVC, CPVC OR HDPE
GW-103	7,108	8,419	1,430	1,362	68	51	17	35	15	3	6" MIN. PVC, CPVC OR HDPE
GW-104	7,309	8,427	1,442	1,376	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-105	7,298	8,258	1,484	1,358	126	95	32	79	15	3	6" MIN. PVC, CPVC OR HDPE
GW-106	7,267	8,101	1,520	1,352	168	126	42	110	15	3	6" MIN. PVC, CPVC OR HDPE
GW-107	7,262	7,930	1,560	1,350	210	158	53	142	15	3	6" MIN. PVC, CPVC OR HDPE
GW-108	7,272	7,758	1,538	1,348	190	143	48	127	15	3	6" MIN. PVC, CPVC OR HDPE
GW-109	7,305	7,601	1,499	1,346	153	115	38	99	15	3	6" MIN. PVC, CPVC OR HDPE
GW-110	7,298	7,431	1,456	1,390	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-111	7,484	7,453	1,457	1,392	65	49	16	33	15	3	6" MIN. PVC, CPVC OR HDPE
GW-112	7,502	7,619	1,498	1,344	154	116	39	100	15	3	6" MIN. PVC, CPVC OR HDPE
GW-113	7,451	7,781	1,540	1,346	194	146	49	130	15	3	6" MIN. PVC, CPVC OR HDPE
GW-114	7,430	7,949	1,564	1,348	216	162	54	146	15	3	6" MIN. PVC, CPVC OR HDPE
GW-115	7,479	8,116	1,530	1,352	178	134	45	118	15	3	6" MIN. PVC, CPVC OR HDPE
GW-116	7,490	8,278	1,491	1,354	137	103	34	87	15	3	6" MIN. PVC, CPVC OR HDPE
GW-117	7,507	8,429	1,455	1,388	67	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-118	7,666	8,542	1,454	1,390	64	48	16	32	15	3	6" MIN. PVC, CPVC OR HDPE
GW-119	7,657	8,365	1,485	1,360	125	94	31	78	15	3	6" MIN. PVC, CPVC OR HDPE
GW-120	7,675	8,200	1,521	1,342	179	134	45	118	15	3	6" MIN. PVC, CPVC OR HDPE
GW-121	7,613	8,006	1,564	1,348	216	162	54	146	15	3	6" MIN. PVC, CPVC OR HDPE
GW-122	7,637	7,809	1,540	1,350	190	143	48	127	15	3	6" MIN. PVC, CPVC OR HDPE
GW-123	7,693	7,654	1,500	1,352	148	111	37	95	15	3	6" MIN. PVC, CPVC OR HDPE
GW-124	7,666	7,482	1,458	1,392	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-125	7,848	7,516	1,466	1,402	64	48	16	32	15	3	6" MIN. PVC, CPVC OR HDPE
GW-126	7,874	7,691	1,511	1,352	159	119	40	103	15	3	6" MIN. PVC, CPVC OR HDPE
GW-127	7,815	7,862	1,553	1,356	197	148	49	132	15	3	6" MIN. PVC, CPVC OR HDPE
GW-128	7,771	8,047	1,565	1,348	217	163	54	147	15	3	6" MIN. PVC, CPVC OR HDPE
GW-129	7,866	8,206	1,541	1,346	195	146	49	130	15	3	6" MIN. PVC, CPVC OR HDPE
GW-130	7,820	8,374	1,498	1,344	154	116	39	100	15	3	6" MIN. PVC, CPVC OR HDPE
GW-131	7,857	8,546	1,456	1,390	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-132	8,048	8,549	1,465	1,398	67	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-133	8,000	8,378	1,505	1,342	163	122	41	106	15	3	6" MIN. PVC, CPVC OR HDPE
GW-134	8,045	8,210	1,548	1,346	202	152	51	136	15	3	6" MIN. PVC, CPVC OR HDPE
GW-135	7,941	8,044	1,566	1,346	220	165	55	149	15	3	6" MIN. PVC, CPVC OR HDPE
GW-136	8,000	7,878	1,542	1,350	192	144	48	128	15	3	6" MIN. PVC, CPVC OR HDPE
GW-137	8,057	7,715	1,517	1,360	157	118	39	102	15	3	6" MIN. PVC, CPVC OR HDPE
GW-138	8,035	7,547	1,478	1,412	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-139	8,226	7,556	1,474	1,410	64	48	16	32	15	3	6" MIN. PVC, CPVC OR HDPE
GW-140	8,233	7,746	1,475	1,408	67	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-141	8,176	7,899	1,510	1,390	120	90	30	74	15	3	6" MIN. PVC, CPVC OR HDPE
GW-142	8,117	8,050	1,548	1,368	180	135	45	119	15	3	6" MIN. PVC, CPVC OR HDPE
GW-143	8,228	8,226	1,548	1,400	148	111	37	95	15	3	6" MIN. PVC, CPVC OR HDPE
GW-144	8,178	8,391	1,506	1,384	122	92	31	76	15	3	6" MIN. PVC, CPVC OR HDPE
GW-145	8,239	8,553	1,466	1,400	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-146	8,430	8,556	1,467	1,402	65	49	16	33	15	3	6" MIN. PVC, CPVC OR HDPE
GW-147	8,357	8,399	1,506	1,392	114	86	29	70	15	3	6" MIN. PVC, CPVC OR HDPE
GW-148	8,293	8,070	1,522	1,398	124	93	31	77	15	3	6" MIN. PVC, CPVC OR HDPE
GW-149	8,355	7,919	1,483	1,420	63	47	16	31	15	3	6" MIN. PVC, CPVC OR HDPE
GW-150	8,477	8,092	1,494	1,432	62	47	16	31	15	3	6" MIN. PVC, CPVC OR HDPE
GW-151	8,410	8,244	1,534	1,394	140	105	35	89	15	3	6" MIN. PVC, CPVC OR HDPE
GW-152	8,543	8,416	1,503	1,394	109	82	27	66	15	3	6" MIN. PVC, CPVC OR HDPE
GW-153	8,621	8,560	1,468	1,402	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-154	8,812	8,563	1,467	1,402	65	49	16	33	15	3	6" MIN. PVC, CPVC OR HDPE
GW-155	8,727	8,440	1,498	1,396	102	77	26	61	15	3	6" MIN. PVC, CPVC OR HDPE
GW-156	8,598	8,264	1,506	1,440	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-157	8,748	8,301	1,496	1,432	64	48	16	32	15	3	6" MIN. PVC, CPVC OR HDPE
GW-158	8,897	8,338	1,485	1,420	65	49	16	33	15	3	6" MIN. PVC, CPVC OR HDPE
GW-159	9,003	8,566	1,466	1,400	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-160	9,048	8,375	1,474	1,408	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-161	9,218	8,380	1,472	1,402	70	53	18	37	15	3	6" MIN. PVC, CPVC OR HDPE
GW-162	9,194	8,570	1,466	1,400	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-163	9,385	8,573	1,459	1,408	51	38	13	22	15	3	6" MIN. PVC, CPVC OR HDPE
GW-164	9,388	8,381	1,465	1,398	67	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE

NUMBER OF WELLS: 67 WELLS

NOTES:

1. THE ABOVE WELL SCHEDULES ARE CONCEPTUAL DESIGN FOR DISCUSSION PURPOSES ONLY AND SHALL NOT BE USED FOR DRILLING. WELLS MAY BE INSTALLED ON AN INTERIM BASIS AND WILL BE DESIGNED PER WASTE DEPTHS AT THE TIME OF INSTALLATION AND MAY VARY FROM THE WELL PARAMETERS IN THE ABOVE WELL SCHEDULE. ANY ALTERATION MADE TO THE WELL SCHEDULE DURING WELL INSTALLATION SHOULD NOT BE CONSIDERED AS A DEPARTURE FROM OR ADJUSTMENT TO ANY PREEXISTING OPERATING PERMIT ISSUED BY EGLE.

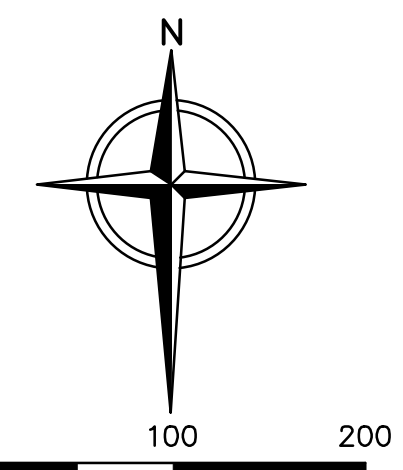
PROPOSED EAST UNIT WELL SCHEDULE – NOT FOR DRILLING

WELL ID	NORTHING (FT)	EASTING (FT)	APPROX. ELEVATION (FT MSL)		DEPTH OF WASTE (FT)	WELL DEPTH (FT) ⁽²⁾	BOTTOM OF WELL TO BOTTOM OF WASTE (FT) ⁽³⁾	PIPE LENGTH (FT)			PERFORATED AND SOLID PIPE DIAMETER AND MATERIAL
			PERMITTED TOP OF WASTE	PERMITTED BOTTOM OF WASTE				PERFORATED PIPE (FT)	SOLID WALL PIPE (FT)	SOLID WALL PIPE ABOVE GRADE (FT)	
GW-50	9,225	10,070	1,476	1,410	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-51	9,412	10,073	1,484	1,418	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-52	9,598	10,077	1,486	1,420	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-53	9,784	10,081	1,485	1,422	63	47	16	31	15	3	6" MIN. PVC, CPVC OR HDPE
GW-54	9,791	9,881	1,478	1,412	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-55	9,627	9,910	1,520	1,364	156	117	39	101	15	3	6" MIN. PVC, CPVC OR HDPE
GW-56	9,458	9,912	1,525	1,364	161	121	40	105	15	3	6" MIN. PVC, CPVC OR HDPE
GW-57	9,300	9,901	1,521	1,356	165	124	41	108	15	3	6" MIN. PVC, CPVC OR HDPE
GW-58	9,131	9,891	1,515	1,354	161	121	40	105	15	3	6" MIN. PVC, CPVC OR HDPE
GW-59	8,969	9,884	1,510	1,356	154	116	39	100	15	3	6" MIN. PVC, CPVC OR HDPE
GW-60	8,801	9,867	1,506	1,360	146	110	37	94	15	3	6" MIN. PVC, CPVC OR HDPE
GW-61	8,634	9,836	1,507	1,372	135	101	34	85	15	3	6" MIN. PVC, CPVC OR HDPE
GW-62	8,464	9,837	1,503	1,394	109	82	27	66	15	3	6" MIN. PVC, CPVC OR HDPE
GW-63	8,342	9,661	1,546	1,386	160	120	40	104	15	3	6" MIN. PVC, CPVC OR HDPE
GW-64	8,529	9,664	1,547	1,416	131	98	33	82	15	3	6" MIN. PVC, CPVC OR HDPE
GW-65	8,722	9,668	1,552	1,414	138	104	35	88	15	3	6" MIN. PVC, CPVC OR HDPE
GW-66	8,905	9,701	1,552	1,404	148	111	37	95	15	3	6" MIN. PVC, CPVC OR HDPE
GW-67	9,084	9,705	1,559	1,400	159	119	40	103	15	3	6" MIN. PVC, CPVC OR HDPE
GW-68	9,264	9,716	1,565	1,356	209	157	52	141	15	3	6" MIN. PVC, CPVC OR HDPE
GW-69	9,446	9,743	1,558	1,318	240	180	60	164	15	3	6" MIN. PVC, CPVC OR HDPE
GW-70	9,628	9,719	1,512	1,350	162	122	41	106	15	3	6" MIN. PVC, CPVC OR HDPE
GW-71	9,797	9,682	1,469	1,404	65	49	16	33	15	3	6" MIN. PVC, CPVC OR HDPE
GW-72	9,803	9,483	1,461	1,394	67	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-73	9,631	9,537	1,505	1,340	165	124	41	108	15	3	6" MIN. PVC, CPVC OR HDPE
GW-74	9,460	9,572	1,549	1,320	229	172	57	156	15	3	6" MIN. PVC, CPVC OR HDPE
GW-75	9,285	9,531	1,576	1,358	218	164	55	148	15	3	6" MIN. PVC, CPVC OR HDPE
GW-76	9,104	9,517	1,576	1,418	158	119	40	103	15	3	6" MIN. PVC, CPVC OR HDPE
GW-77	8,923	9,524	1,576	1,410	166	125	42	109	15	3	6" MIN. PVC, CPVC OR HDPE
GW-78	8,742	9,479	1,576	1,396	180	135	45	119	15	3	6" MIN. PVC, CPVC OR HDPE
GW-79	8,564	9,498	1,576	1,404	172	129	43	113	15	3	6" MIN. PVC, CPVC OR HDPE
GW-80	8,384	9,490	1,576	1,402	174	131	44	115	15	3	6" MIN. PVC, CPVC OR HDPE
GW-81	8,201	9,498	1,573	1,408	165	124	41	108	15	3	6" MIN. PVC, CPVC OR HDPE
GW-82	8,918	9,348	1,552	1,384	168	126	42	110	15	3	6" MIN. PVC, CPVC OR HDPE
GW-83	9,117	9,332	1,544	1,416	128	96	32	80	15	3	6" MIN. PVC, CPVC OR HDPE
GW-84	9,304	9,349	1,540	1,352	188	141	47	125	15	3	6" MIN. PVC, CPVC OR HDPE
GW-85	9,474	9,409	1,525	1,320	205	154	51	138	15	3	6" MIN. PVC, CPVC OR HDPE
GW-86	9,650	9,364	1,490	1,336	154	116	39	100	15	3	6" MIN. PVC, CPVC OR HDPE
GW-87	9,811	9,290	1,450	1,386	64	48	16	32	15	3	6" MIN. PVC, CPVC OR HDPE
GW-88	9,671	9,172	1,448	1,382	66	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE
GW-89	9,512	9,243	1,487	1,332	155	116	39	100	15	3	6" MIN. PVC, CPVC OR HDPE
GW-90	9,349	9,167	1,496	1,338	158	119	40	103	15	3	6" MIN. PVC, CPVC OR HDPE
GW-91	9,179	9,160	1,500	1,396	104	78	26	62	15	3	6" MIN. PVC, CPVC OR HDPE
GW-92	8,998	9,161	1,504	1,410	94	71	24	55	15	3	6" MIN. PVC, CPVC OR HDPE
GW-93	8,859	8,980	1,460	1,390	70	53	18	37	15	3	6" MIN. PVC, CPVC OR HDPE
GW-94	9,029	8,984	1,460	1,418	42	32	11	16	15	3	6" MIN. PVC, CPVC OR HDPE
GW-95	9,204	8,984	1,455	1,390	65	49	16	33	15	3	6" MIN. PVC, CPVC OR HDPE
GW-96	9,381	8,986	1,450	1,386	64	48	16	32	15	3	6" MIN. PVC, CPVC OR HDPE
GW-97	9,531	9,066	1,449	1,382	67	50	17	34	15	3	6" MIN. PVC, CPVC OR HDPE

April 2, 2025

APPENDIX D

SURFACE EMISSIONS MONITORING PLAN



- LEGEND**
- PROPERTY BOUNDARY
 - EXISTING SOLID WASTE BOUNDARY
 - EXISTING PHASE BOUNDARY
 - PROPOSED 10' CONTOUR
 - PROPOSED 2' CONTOUR
 - EXISTING LANDFILL GAS HEADER
 - EXISTING LANDFILL GAS HORIZONTAL COLLECTOR
 - EXISTING FORCEMAIN
 - EXISTING 3" FORCEMAIN
 - EXISTING 3"x6" DUAL CONTAINED FORCEMAIN LINE
 - EXISTING 4"x8" DUAL CONTAINED FORCEMAIN LINE
 - EXISTING AIRLINE
 - EXISTING 2" AIRLINE
 - EXISTING LFG EXTRACTION WELL
 - EXISTING LFG EXTRACTION WELL WITH CAISSON
 - EXISTING REMOTE WELLHEAD
 - EXISTING HEADER ACCESS RISER
 - EXISTING ISOLATION VALVE
 - EXISTING BLIND FLANGE
 - EXISTING FLANGE CONNECTION
 - EXISTING REDUCER FITTING
 - EXISTING CONDENSATE DRIPLEG
 - EXISTING MANHOLE
 - EXISTING FORCEMAIN ISOLATION VALVE
 - EXISTING AIRLINE ISOLATION VALVE
 - EXISTING FORCEMAIN CLEANOUT
 - EXISTING FLANGE CONNECTION
 - EXISTING CONDENSATE SUMP
 - EXISTING FINAL COVER AREA
 - PROPOSED SURFACE EMISSIONS MONITORING PATHWAY

- NOTES:**
- GRADES SHOWN REFLECT ANTICIPATED LANDFILL ELEVATIONS AT FINAL DEVELOPMENT.
 - FOR CLARITY, NOT ALL SITE FEATURES MAY BE SHOWN.
 - THIS SURFACE MONITORING PLAN REPRESENTS THE LANDFILL AT FINAL CLOSURE CONDITIONS. SURFACE MONITORING CONDUCTED WHILE THE SITE IS STILL ACTIVE MAY FOLLOW A DIFFERENT ROUTE.
 - PROPOSED SCAN ROUTE AROUND COVER PENETRATIONS (I.E. AROUND WELLHEADS AND HEADER ISOLATION VALVES) AT FINAL CLOSURE CONDITIONS, PER 40 CFR 63, SUBPART AAAA.
 - ACTUAL COVER PENETRATION LOCATIONS MAY VARY BASED UPON INSTALLED GCCS COMPONENTS AND LOCATIONS.
 - TYPICAL SPACING FOR SERPENTINE PATTERN IS APPROXIMATELY 30 METERS (100 FT). SOME LOCAL VARIATIONS MAY OCCUR TO ACCOMMODATE LANDFILL PHASING AND FIELD CONDITIONS AT THE TIME OF MONITORING.
 - THE FACILITY MAY EXCLUDE DANGEROUS AREAS, SUCH AS ROAD, THE ACTIVE FILL AREA, TRUCK TRAFFIC AREAS, AREAS WITH SNOW OR ICE COVER AND SLOPES STEEPER THAN OR EQUAL TO 3:1 FROM SURFACE TESTING.



REV	DATE	DESCRIPTION	DWN BY	DES BY	CHK BY	APP BY
04/02/25			JRD	JPS	KM	

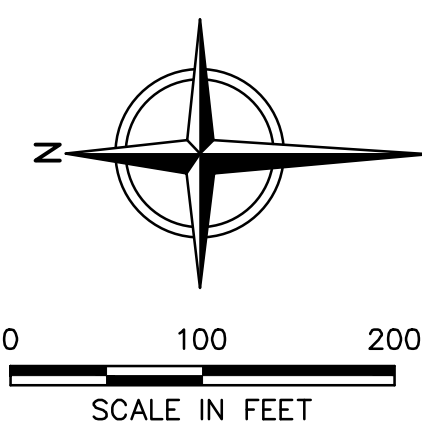


GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

**GAS COLLECTION AND CONTROL SYSTEM
SURFACE EMISSIONS MONITORING PLAN**

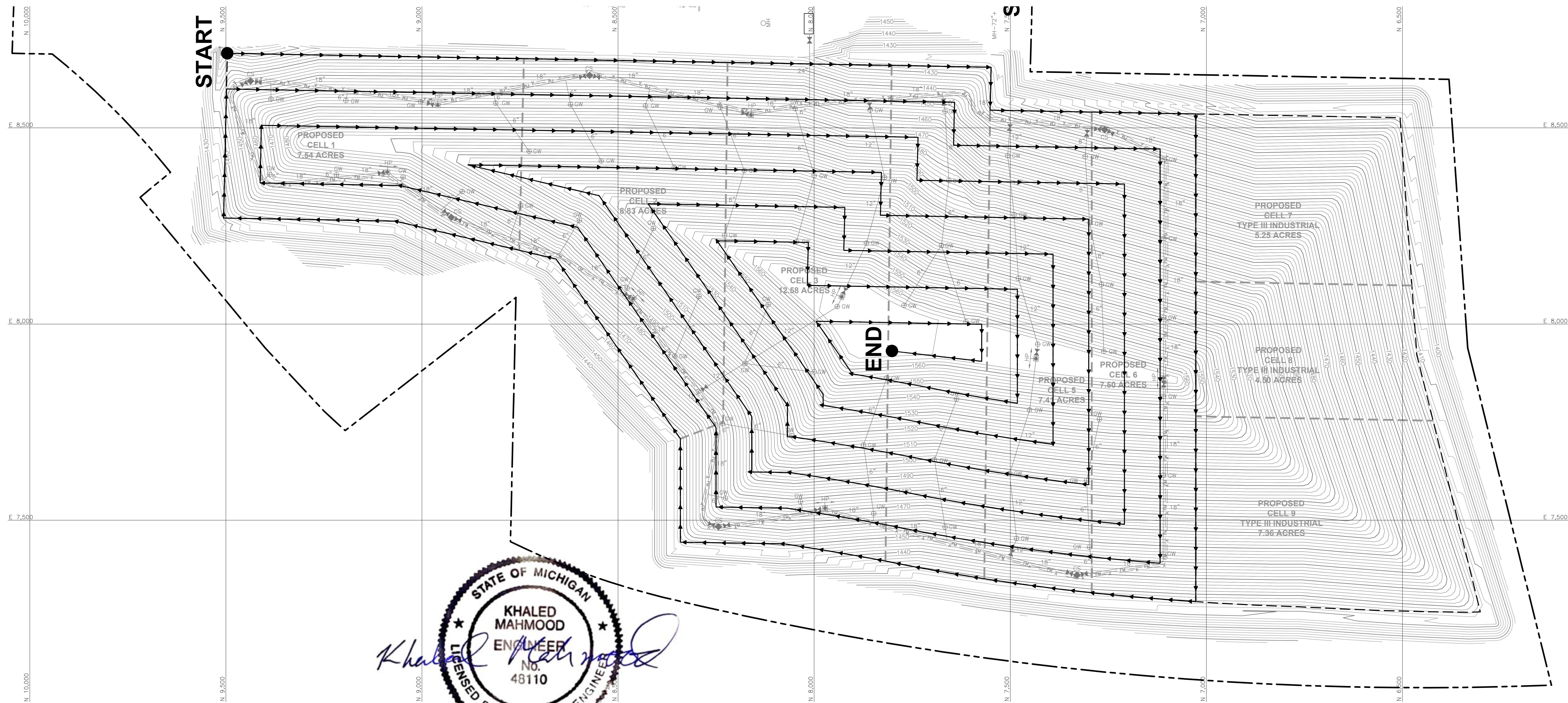
SHEET NO.
SEM A

PROJECT NO.
209-4251206



- LEGEND**
- PROPERTY BOUNDARY
 - EXISTING SOLID WASTE BOUNDARY
 - EXISTING PHASE BOUNDARY
 - PROPOSED 10' CONTOUR
 - PROPOSED 2' CONTOUR
 - 18"
 - 6"
 - EXISTING LANDFILL GAS HEADER
 - EXISTING LANDFILL GAS HEADER/LATERAL
 - EXISTING FORCEMAIN
 - EXISTING AIRLINE
 - EXISTING LFG EXTRACTION WELL
 - EXISTING REMOTE WELLHEAD
 - EXISTING HEADER ISOLATION VALVE
 - EXISTING FLANGE CONNECTION
 - EXISTING CONDENSATE SUMP
 - EXISTING HEADER ACCESS RISER
 - EXISTING HEADER HIGH POINT
 - PROPOSED SURFACE EMISSIONS MONITORING PATHWAY

- NOTES:**
- GRADES SHOWN REFLECT ANTICIPATED LANDFILL ELEVATIONS AT FINAL DEVELOPMENT.
 - FOR CLARITY, NOT ALL SITE FEATURES MAY BE SHOWN.
 - THIS SURFACE MONITORING PLAN REPRESENTS THE LANDFILL AT FINAL CLOSURE CONDITIONS. SURFACE MONITORING CONDUCTED WHILE THE SITE IS STILL ACTIVE MAY FOLLOW A DIFFERENT ROUTE.
 - PROPOSED SCAN ROUTE AROUND COVER PENETRATIONS (I.E. AROUND WELLHEADS AND HEADER ISOLATION VALVES) AT FINAL CLOSURE CONDITIONS, PER 40 CFR 63, SUBPART AAAA.
 - ACTUAL COVER PENETRATION LOCATIONS MAY VARY BASED UPON INSTALLED GCCS COMPONENTS AND LOCATIONS.
 - TYPICAL SPACING FOR SERPENTINE PATTERN IS APPROXIMATELY 30 METERS (100 FT). SOME LOCAL VARIATIONS MAY OCCUR TO ACCOMMODATE LANDFILL PHASING AND FIELD CONDITIONS AT THE TIME OF MONITORING.
 - THE FACILITY MAY EXCLUDE DANGEROUS AREAS, SUCH AS ROAD, THE ACTIVE FILL AREA, TRUCK TRAFFIC AREAS, AREAS WITH SNOW OR ICE COVER AND SLOPES STEEPER THAN OR EQUAL TO 3:1 FROM SURFACE TESTING.



REV	DATE	DESCRIPTION	DWN BY	DES BY	CHK BY	APP BY
04	02/25		JPS	JPS	JPS	JPS



GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

**GAS COLLECTION AND CONTROL SYSTEM
SURFACE EMISSIONS MONITORING PLAN**

SHEET NO.

SEM B

PROJECT NO.
209-4251206

APPENDIX E

UPLIFT RESISTANCE CALCULATIONS

WEXFORD COUNTY LANDFILL
Manton, Michigan
Geomembrane Uplift Resistance Calculations

Example Well Number	Example Cover Thickness (ft)	Average Soil Density (pcf)	Soil Load (psi)	Uplift Resistance (psi)	Uplift Resistance (inches of WC)	Factor of Safety	Allowable Positive Pressure - Flat Areas (inches of WC)	Allowable Positive Pressure - Sideslopes (inches of WC)
1	2.5	125	313	2.2	60	3.00	20	18
2	3.0	125	375	2.6	72	3.00	24	22
3	3.5	125	438	3.0	84	3.00	28	26
4	4.0	125	500	3.5	96	3.00	32	29

*Soil Density assumed based on standard industry values

**Cohesion between layers of the final cover is ignored to be conservative.

Sample Calculations:

Slope Angle (β) = 3:1 = 18.43 deg

Cover Thickness = 2.5 ft

Soil Density = 125 pcf

Load on Geomembrane = 2.5 ft * 125 pcf = 313 psf

Uplift Resistance in psi = Load / (144 psf / psi) = 313 psf / 144 (psf/psi)

Unit weight of water = 62.4 pcf

1 psi = 144 psf = 144 psf / 62.4 pcf = 2.31 feet of water column = 27.7 inches of water column

Uplift Resistance for Flat Areas (inches of water column) = psi * 27.7 inches of water column / psi

Uplift Resistance for Sideslopes (inches of water column) = Uplift Resistance for Flat Areas * $\cos(\beta)$

APPENDIX F

NSPS DETERMINATION LETTERS



UNITED STATES ENVIRONMENTAL PROTECTION AGENCY

REGION 5
77 WEST JACKSON BOULEVARD
CHICAGO, IL 60604-3590

RECEIVED
SCS ENGINEERS

MAY 17 2001

CINCINNATI, OH

REPLY TO THE ATTENTION OF:

MAY 14 2001

(AE-17J)

Jim Walsh
SCS Engineers
2060 Reading Road
Suite 200
Cincinnati, Ohio 45202

Dear Mr. Walsh:

The purpose of this letter is to respond to SCS Engineer's request for an applicability determination regarding the use of an alternative test method for oxygen or nitrogen monitoring for American Landfill in Waynesburg, Ohio. I apologize for the delay in responding. Under 40 C.F.R. § 60.753(c)(1) and (2), operational standards for collection and control systems, to demonstrate compliance with the regulations, municipal solid waste landfills are required to measure nitrogen by method 3C or oxygen by method 3A from gas wellheads on a monthly basis, unless an alternative test method has been approved by the Administrator.

In your March 6, 2001 facsimile, you stated that American Landfill requests the use of a hand-held gas extraction monitor to measure nitrogen or oxygen. The hand-held monitor is CES-LANDTEC's GEM-500 Gas Extraction Monitor. CES-LANDTEC has submitted information to the United States Environmental Protection Agency (U.S. EPA) stating that the GEM-500 meets the operational standards for collection and control equipment for the measurement of oxygen, 40 C.F.R. § 60.753(c)(2) and Method 3A, with the exception of the Intrinsically Safe rating. Information provided by CES-LANDTEC regarding the GEM-500 gas extraction monitor is enclosed.

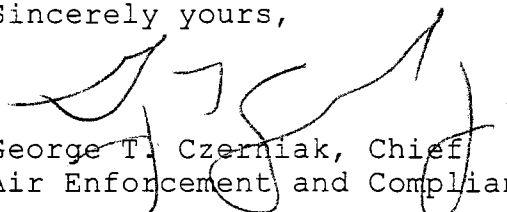
Method 3A is a performance-based sampling method which requires measurement system performance test procedures to be performed before the measurement of emissions. If American Landfill performs the required performance test procedures listed in Method 3A and 40 C.F.R. § 60.753(c)(2), then an alternative test method determination is not required to be submitted to U.S. EPA at this time. U.S. EPA or Ohio EPA may monitor emissions measurements performed at American Landfill as necessary to ensure that the standards of Method 3A and 40 C.F.R. §

60.753(c)(2) are being met.

The GEM-500 hand-held monitor does not meet the requirements of 40 C.F.R. § 60.753(c)(1), measurement of nitrogen levels in municipal solid waste landfill wellheads. U.S. EPA has confirmed with CES-LANDTEC that the GEM-500 does not meet the monitoring requirements of Method 3C. American landfill would be required to submit an alternative test method determination request to the nitrogen monitoring.

My staff has coordinated this response to your request with U.S. EPA's Office of Air Quality Planning and Standards, Office of Enforcement and Compliance Assurance, and its regional offices. If you have any additional questions, please contact Julie Monahan, of my staff, at (312) 886-0120.

Sincerely yours,



George T. Czerniak, Chief
Air Enforcement and Compliance Assurance Branch

cc: Patrick Shriver, Canton City Health Department
Robin Segall, OAQPS
Zofia Kosim, OECA

Enclosure



March 23, 2001

Re: U.S.E.P.A. Instrument Compliance

To Whom it May Concern:

"The CES-LANDTEC GEM™500 is in full compliance with final NSPS Operational Standards for Collection and Control Equipment 40 CFR 60.753(c)(2) and Method 3 A, as defined therein, with the exception of the Intrinsically Safe rating requirement."

"The CES-LANDTEC GEM™2000 is in full compliance with final NSPS Operational Standards for Collection and Control Equipment 40 CFR 60.753(c)(2) and Method 3 A, as defined therein, including the Intrinsically Safe rating requirement."

"The CES-LANDTEC SEM-500 is in compliance 40 CFR 755 (d) and Appendix A, Method 21 therein, with the exception of the Intrinsically Safe rating requirement. The basic unit, which is a Foxboro TVA 1000B, carries this rating. It has been determined that, due to the external changes of case color and modified software, the SEM 500 cannot be rated as such without being re-submitted for certification. This action is on hold, pending a review of this part of the methodology and its applicability to the application, i.e., the measurement of surface emissions in free air as opposed to confined space entry. The U.S.E.P.A. has assured CES-LANDTEC that this exception does not disqualify the SEM-500 as an acceptable instrument for use under NSPS Method 21."

If we can be of any further assistance, please contact me at (909) 783-3636, extension 6710.

Regards,
CES - LANDTEC

Michael Hardison
Regional Sales Manager



REGION 4
ATLANTA FEDERAL CENTER
61 FORSYTH STREET
ATLANTA, GEORGIA 30303-8960

IN. DIV. OF
AIR POLLUTION CONTROL

2007 JUN -5 AM 9:11

MAY 31 2007

4APT-ATMB

Barry R. Stephens, P.E., Director
Division of Air Pollution Control
TN Dept. of Environment & Conservation
9th Floor, L&C Annex
401 Church Street
Nashville, TN 37243-1531

RECEIVED

Dear Mr. Stephens:

This letter is in response to a letter in which Lacey J. Hardin of your staff requested a determination regarding an applicability issue that the Waste Management Company (WMC) raised regarding seven gas extraction wells at the Chestnut Ridge Landfill in Heiskell, Tennessee. This landfill is subject to 40 CFR Part 60, Subpart WWW (Standards of Performance for Municipal Solid Waste Landfills), and WMC requested that the seven wells in question be exempt from monthly pressure, temperature, and oxygen monitoring requirements until September 2010. Based upon our review of Subpart WWW and the information submitted by WMC, we have determined that exempting the wells in question from pressure, temperature, and oxygen content monitoring would be consistent with the intent of Subpart WWW if this area of the landfill remains active until September 2010 or later. Details regarding the basis for this determination are provided in the remainder of this letter.

Under provisions in 40 CFR §60.753(a), owners and operators of landfills subject to Subpart WWW are required to operate a gas collection and control system (GCCS) where waste has been in place for five years or more in active areas or where waste has been in place for two years or more in closed areas or in areas that have reached final grade. A number of operating limits for well heads in the GCCS are specified in 40 CFR §60.753(b) and 40 CFR §60.753(c). These provisions require that well heads be operated under negative pressure, that well head temperatures be kept below 55 °C, and that landfill gas nitrogen or oxygen levels be maintained within specified limits (less than 20 percent for nitrogen or less than 5 percent for oxygen). Under provisions in 40 CFR §60.755(a), owners and operators of GCCS subject to these operational limits are required to monitor well head pressure, temperature, and gas composition on a monthly basis. When deviations from any of the operating limits in the rule are noted, owners and operators must take corrective action to bring the collection system back into compliance.

According to the letter in which WMC requested a monitoring waiver, the seven wells in question are located in an active area of the Chestnut Ridge Landfill where waste was first placed in September 2005. Based upon the date when this area was first used for waste disposal, the company would not be required to operate gas extraction wells

Internet Address (URL) • <http://www.epa.gov>

Recycled/Recyclable • Printed with Vegetable Oil Based Inks on Recycled Paper (Minimum 30% Postconsumer)

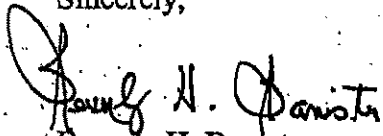
until September 2010. Since the wells in this area were installed prior to the applicable operating deadline in Subpart WWW, WMC asked that they be exempt from temperature, pressure, and oxygen monitoring until September 2010 if the area remains active or until two years after final grade is reached in the area where the wells are located.

Based upon our review of Subpart WWW, the operational requirements in 40 CFR §60.753 and the associated monitoring requirements in 40 CFR §60.755 are applicable in areas where a GCCS must be installed and operated under provisions in 40 CFR §60.753(a). Since the seven wells covered by WMC's request for a monitoring exemption are located in an active area that first received waste in September 2005, they are in an area where a GCCS would not be required under the provisions in 40 CFR §60.753(a). Because WMC is not currently required to operate a GCCS in this area of the landfill, the pressure, temperature, and oxygen limits in 40 CFR §60.753 would not apply to the seven well heads covered by the company's request for a monitoring waiver. Therefore WMC's proposal to delay pressure, temperature, and oxygen monitoring for these seven wells until September 2010 would be consistent with the intent of Subpart WWW if this area of the landfill remains active until September 2010 or later.

If any portion of the landfill covered by WMC's waiver request reaches final grade between September 2007 (i.e., two years after the area was first used for waste disposal) and September 2010, the requirement to operate a GCCS would apply at the time final grade is reached. This determination is based upon the provisions in 40 CFR §60.753(a)(2) that require operation of a GCCS where waste has been in place for two years or more in closed areas or areas at final grade. Therefore, WMC's request that the monitoring waiver for the seven wells in question be extended for two years if final grade is reached prior to September 2010 would not be consistent with the intent of Subpart WWW. If this area of the landfill reaches final grade prior to September 2007, the requirement to have a GCCS and implement the associated monitoring provisions would apply beginning in September 2007. If this area of the landfill reaches final grade after September 2007, but prior to September 2010, the requirement to have a GCCS and implement the associated monitoring would apply at the time final grade is reached.

If you have any questions about the issues addressed in this letter, please contact Mr. David McNeal of the U.S. Environmental Protection Agency Region 4 staff at (404) 562-9102.

Sincerely,

A handwritten signature in dark ink, appearing to read "Beverly H. Banister". The signature is fluid and cursive, with the first name being the most prominent.

Beverly H. Banister
Director
Air, Pesticides and Toxics
Management Division

cc: Lacey J. Hardin
Division of Air Pollution Control
TN Dept. of Environment & Conservation
9th Floor, L&C Annex
401 Church Street
Nashville, TN 37243-1531

fpub.epa.gov/adi/index.cfm?

1331068&CFTOKEN=40394580&jsessionid=4a30329d75bb9164ec30525a5f122849d269&requestid=2100



Compliance Monitoring

You are here: [EPA Home](#) [Compliance and Enforcement](#) [Compliance Monitoring](#) [Statutory Programs](#) [CAA](#) [Search ADI Database](#)

Search Applicability Determination Index

[Search ADI](#) | [Return to Search Results](#) | [What's New](#) | [Help](#) | [Technical Support](#) | [Recent ADI Updates](#) | [Related Links](#)

Determination Detail

Control Number: 0900059

Category: NSPS
EPA Office: Region 5
Date: 10/03/2008
Title: Alternative Compliance and Monitoring Timelines
Recipient: Maxheimer, Rachelle
Author: Czerniak, George T.
Comments:

Subparts: Part 60, WWW Municipal Solid Waste Landfills

References: 60.753
60.755

Abstract:

Q: Does EPA approve the request of the Roxana Landfill (Roxana) in Roxana, Illinois, for several alternative timelines to bring certain specified wells that were unable to perform the required monitoring for May 2008 into compliance under 40 CFR part 60, subpart WWW?

A: Yes. Roxana was unable to perform the required monitoring for May 2008 for landfill gas extraction wells 9, 36, 41, 44, and 47 due to unsafe conditions in the area of these wells. EPA approves the request to exempt these wells under NSPS subpart WWW for one month of monitoring only because of the safety issues and because the request covers a small percentage of the total wells at the site for a relatively short time period. EPA will grant Roxana alternative compliance timelines of various lengths to correct operating parameter exceedances at several other wells.

Letter:

(AE-17J)

Ms. Rachelle Maxheimer
Project Manager
Cornerstone Environmental Group, LLC
400 Quadrangle Drive, Suite E
Bolingbrook, Illinois 60440

Re: Wells 9, 25, 30, 31, 32, 33, 36, 38, 39, 40, 41, 44, 45, 46, 47, and 75

Dear Ms. Maxheimer:

Thank you for your June 30, August 23, and September 23, letters to the U.S. Environmental Protection Agency requesting approval of several alternative compliance timelines at the Roxana Landfill, Roxana, Illinois ("Roxana"). Roxana is subject to the New Source Performance Standards ("NSPS") for Municipal Solid Waste Landfills ("40 CFR Part 60, Subpart WWW"). The NSPS sets forth compliance provisions for gas collection and control systems under 40 CFR § 60.755.

Regulatory Background

Under the provisions of 40 CFR § 60.753(a), owners or operators of landfills subject to Subpart WWW are required to operate a gas collection and control system ("GCCS") where waste has been in place for five years or more in active areas or where waste has been in place for two years or more in closed areas that have reached final grade. A number of operating limits for wellheads in the GCCS are specified in 40 CFR § 60.753(b) and 60.753(c). These provisions require that wellheads be operated under negative pressure, with a landfill gas temperature less than 55 degrees Celcius and with either a nitrogen level less than 20 percent or an oxygen level less than 5 percent.

Per 40 CFR § 60.755(a), owners or operators of GCCS subject to these operational limits are required to monitor wellhead pressure, temperature and gas composition on a monthly basis. When exceedances of any of the operating limits in the rule are noted, owners or operators must take corrective action to bring the collection system back into compliance. If the exceedances cannot be remedied within 15 days, the GCCS system must be expanded to correct the deviations within 120 days of the initial exceedance. However, an alternative timeline for compliance can be requested from the Administrator for approval.

In addition, 40 CFR § 60.753(d) allows areas of steep slopes or other dangerous areas to be excluded from surface testing.

Roxana's Requests

The June 30, August 23 and September 23, 2008 letters together make five requests to EPA:

Request 1: Request for Approval of Alternative Monitoring Procedure or Alternative

Compliance Timeline for Wells 9, 36, 41, 44, and 47

Roxana was unable to perform the required monthly monitoring for landfill gas extraction wells 9, 36, 41, 44, and 47 in May 2008 due to landfill conditions in the areas of these wells. The above referenced wells are located in the active face of the landfill. According to Roxana, these conditions made it unsafe for operations and maintenance personnel to conduct the required monthly monitoring. Roxana explains that wells which are placed in active areas periodically need to be raised (i.e., the well casing extended 15 to 25 feet vertically) in order to not be buried under lifts of trash. When they are raised, the HDPE line which provides the applied vacuum is temporarily disconnected until the surrounding lift of trash is brought high enough to reconnect the well. The time frame between when a well is disconnected and raised, and when the waste height is high enough to reconnect the lateral, can often range from a few weeks to a few months. This can result in missed monthly readings at the well, since the well casing is too high for the technician to safely reach. Roxana originally requested until September 30, 2008 to allow for filling activities to cease in this area and allow safe access to the identified landfill gas monitoring wells. However, the September 23, 2008 letter states that the wells were able to be monitored in June 2008 and therefore only one month of monitoring was missed. Roxana has reported the failure to monitor to the Illinois EPA via a submittal dated May 30, 2008 of the required Excess Emissions, Monitoring, Equipment Downtime, and Miscellaneous Reporting Form under the Title 5 permitting program.

EPA's Determination on Request 1

Based upon our review of the information submitted by Roxana, we agree that the request to exempt wells 9, 36, 41, 44, and 47 from one month of monitoring is acceptable based upon the following factors:

1. Roxana's concerns about safety and the ability to make adjustments for wells that have been raised to between 15 to 25 feet in the air appears legitimate.
2. The number of wells that will be covered by this monitoring exemption is about 6.25 percent of the wells located at the site. Currently, there are more than 80 gas extraction wells at the landfill, and Roxana's request covers 5 wells.
3. Per 40 CFR § 60.752(b)(2)(v), the GCCS for a closed landfill cannot be removed until it has been in operation for at least 15 years. Since this provision also prohibits the GCCS from being removed until NMOC emissions are less than 50 MG per year, the actual operating life of the GCCS at the Roxana landfill could be considerably longer than 15 years. In the context of the operating life of the GCCS, the duration of the proposed monitoring exemption (one month) is relatively short.

Request 2: Request for Alternative Timeline to Fix Well 46

According to the June 30, 2008 letter, gas extraction well 46 was hit and became inoperable as a result of construction in the area. The well was capped and is being replaced as part of the installation of new landfill gas extraction wells. As of September 23, 2008, the well remained inoperable. Roxana requests until approximately December 23, 2008 to complete the installation of this well and bring the well online. Roxana also reported this event via the Excess Emissions, Monitoring Equipment Downtime, and

Miscellaneous Report Form under Title 5.

EPA's Determination on Request 2

EPA will grant Roxana an alternative compliance timeline of not more than 120 days from the date well 46 became inoperable to fix the well and bring it back into compliance. Since the well became inoperable in June, Roxana has until sometime in October to fix this well (120 day from the date in June when the well became inoperable). EPA will not grant Roxana until December to fix this well.

Request 3: Request for Alternative Compliance Timeline for Wells 25 and 36 for Oxygen Exceedances

The August 23 and September 23, 2008 letters request additional time to bring wells 25 and 36 back into compliance for oxygen exceedances which occurred in August 2008. Roxana suspects that well 25 is blocked and will require re-drilling. Well 36 experienced high liquid levels but was brought back into compliance in September 2008. Roxana first requested 90 days from August 23 to repair well 25 but now requests 90 days from September 23 to repair well 25.

EPA's Determination on Request 3

Since well 36 returned to compliance in September 2008, EPA will approve an alternative compliance timeline from the date of the monthly monitoring in August 2008 to the date of the monthly monitoring in September 2008 when the well returned to compliance. For well 25, EPA will only approve an alternative timeline of 90 days from August 23, 2008 to return this well to compliance.

Request 4: Request for Alternative Timeline for Wells 38, 39, 40, 41, 44, 45, 47, and 75 for Pressure Exceedances

The August 23 and September 23, 2008 letters request additional time to bring the following wells back into compliance: 38, 39, 40, 41, 44, 45, 47 and 75. The wells exhibited positive pressure during the August 2008 monthly monitoring event. The above reference wells are located in the area of the landfill undergoing construction activities for the installation of additional wells. Roxana originally requested three months to complete construction activities, including the installation, replacement and re-drilling of wells to bring all wells back into compliance. In the September 23, 2008 letter, however, Roxana explained that wells 38, 39, 40, 41, 44, 45 and 47 all returned to compliance in September 2008.¹ Well 75 still has a positive pressure and Roxana requests 90 days from September 23, 2008 to bring well 75 back into compliance.

EPA's Determination on Request 4

Since wells 36, 38, 39, 40, 41, 44, 45 and 47 returned to compliance in September 2008, EPA will approve an alternative compliance timeline from the monthly monitoring date in August 2008 to the monthly monitoring date in September 2008 (approximately 30 days) . For well 75, EPA will approve an alternative timeline of 120 days from the date of the initial exceedance in August 2008 to return this well to compliance.

Request 5: Request for Alternative Compliance Timeline for Wells 30, 31, 32, and 33

Wells 30, 31, 32 and 33 showed positive pressure during the monthly monitoring event on September 23, 2008. No vacuum is currently available for these wells because they are disconnected due to the construction activities. Roxana requests 90 days from September 23, 2008 to re-connect these wells and bring them back into compliance.

EPA's Determination on Request 5

EPA will approve an alternative compliance date of 90 days from September 23, 2008, which is the date of the initial pressure exceedance, to bring these wells back into compliance.

If you have any questions about the determination provided in this letter, please contact Ms. Linda H. Rosen, of my staff, at (312) 886-6810.

Sincerely,

George T. Czerniak
Chief
Air Compliance and Compliance Assurance Branch

cc: Ray Pilapil, Manager
Bureau of Air - Compliance and Enforcement Section
Illinois Environmental Protection Agency

Yasmine Keppner
Bureau of Air - Compliance and Enforcement Section Illinois Environmental Protection Agency

1 Although the September 23, 2008 letter did not list well 44 as returning to compliance, Ms. Rachelle Maxheimer informed Ms. Linda Rosen, EPA, during an October 1, 2008 phone call that well 44 did indeed return to compliance in September 2008.



Compliance Assistance

[Recent Additions](#) | [Contact Us](#) | [Print Version](#) EPA Search:**GO**[EPA Home](#) > [Compliance and Enforcement](#) > [Compliance Assistance](#) > [Applicability Determinations](#) > [Applicability Determination Index](#) > [Search ADI Database](#)

Search Applicability Determination Index

[Search ADI](#) | [Return to Search Results](#) | [Help](#) | [Technical Support](#) | [Recent ADI Updates](#) | [Related Links](#)

Determination Detail

Control Number: 0500087

Category: NSPS
EPA Office: Region 4
Date: 07/12/2004
Title: Alternative Monitoring Proposals for Landfill
Recipient: Capp, James A.
Author: Banister, Beverly H.
Comments:

Subparts: Part 60, WWW Municipal Solid Waste Landfills

References: 60.756(a)
60.753(c)
60.755(c)(4)
60.756(f)

Abstract:

Q1: Does EPA approve the proposed alternative oxygen concentration limit for 16 wells, under 40 CFR part 60, subpart WWW, at the Deans Bridge Road Landfill operated by the Augusta, Georgia Public Works and Engineering Department?

A1: Yes. EPA approves the proposed alternative concentration limit under NSPS subpart WWW because the temperature monitoring data for the wells in question indicate that oxygen levels greater than five percent have not poisoned methane producing bacteria.

Q2: Does EPA waive the requirement under 40 CFR part 60, subpart WWW to conduct methane surface concentration monitoring in a closed 52-acre section of the landfill?

A2: No. Because NSPS subpart WWW requires that methane surface concentration monitoring in closed areas be conducted at least annually, EPA concludes that the requirement to conduct this monitoring cannot be waived. However, the monitoring frequency can be reduced from a quarterly to an annual basis if none of the methane concentration readings in the closed section of the landfill were 500 parts per million or more during the June 2003 monitoring period.

Letter:

James A. Capp
Program Manager
Industrial Source Monitoring Program
Air Protection Branch

Environmental Protection Division
Georgia Department of Natural Resources
4244 International Parkway, Suite 120
Atlanta, Georgia 30354

Dear Mr. Capp:

The purpose of this letter is to provide you with a written determination regarding alternative monitoring proposals that the Augusta, Georgia Public Works and Engineering Department submitted for the Deans Bridge Road Municipal Waste Landfill located in August, Georgia. This landfill is subject to 40 CFR Part 60, Subpart WWW (Standards of Performance for Municipal Solid Waste Landfills), and Augusta submitted requests for two alternatives to requirements contained in this rule. The first of these requests is for approval of an alternative oxygen concentration limit for 16 of the wells included in the landfill's gas collection and control system (GCCS), and the second requests for a waiver of the requirement to conduct quarterly methane surface concentration monitoring in a closed section of the landfill.

Based upon our review, the alternative oxygen concentration limit proposed for 16 of the wells in the landfill's GCCS is acceptable, but the request for a waiver of the methane surface concentration monitoring in a closed section of the landfill cannot be approved. Although the requirement to conduct methane surface concentration monitoring in the closed section of the landfill cannot be completely waived, reducing the monitoring frequency from a quarterly basis to an annual basis may be acceptable. Details regarding the proposals submitted by Augusta and the basis for our conclusions are provided in the remainder of this letter.

The proposals submitted by Augusta cover three distinct areas in the Deans Bridge Road Landfill. One of these areas consists of two older, closed hills covering about six acres each (Area A), a large, closed hill covering approximately 52 acres (Area B), and a portion of the active area covering approximately 24 acres (Area C). Under provisions in 40 CFR 60.756(a), landfill owners and operators are required to monitor extraction wells on a monthly basis to verify compliance with the 55 C temperature limit and five percent oxygen concentration limit promulgated at 40 CFR 60.753(c). In its proposal, Augusta identified twelve wells in Area A (Numbers 1 through 10, 102, and 103), three wells in Area B (numbers 18, 101, and 104), and one well in Area 3 (Number 202) for which it requested an alternative monitoring approach that would allow oxygen concentration levels higher than five percent. According to Augusta, even the application of a relatively low vacuum in these wells leads to air intrusion within 24 hours because the wells are located in areas where the waste is old and shallow. Therefore, Augusta asked that the presence of oxygen levels greater than five percent be allowed provided that the temperature of the specified wells stays below the level (130 F) which would indicate that aerobic decomposition is occurring.

Under provisions in 40 CFR 60.753(c), landfill owners/operators can obtain approval for oxygen concentrations higher than five percent by supplying supporting data demonstrating that the elevated oxygen level does not cause fires or significantly inhibit anaerobic decomposition by killing methanogens. Included with Augusta's alternative monitoring request were temperature and oxygen concentration monitoring results for the first four months of 2003. Since monitoring was conducted on a monthly basis for each of the 16 wells covered by Augusta's request, 64 sets of temperature and oxygen concentration readings were summarized in the City's submittal. On nine occasions during this monitoring, oxygen concentration levels in the wells covered by Augusta's request exceeded five percent. The well temperature ranged from 49 F to 82 F during these times when the oxygen concentration exceeded the limit in Subpart WWW.

The 55 C (131 F) temperature limit promulgated at 40 CFR 60.753(c) is used to identify when the decomposition in a landfill has changed from an anaerobic mode to an aerobic mode due to the poisoning of methane production bacteria. The temperature and oxygen concentration monitoring results that Augusta provided for wells 1-10, 18, 101-104, and 202 at the Dean's Bridge Road Landfill, indicate that the temperature in these wells is significantly less than 55 C even when oxygen concentration levels exceed five percent. Based upon these results, the U.S. Environmental Protection Agency (EPA) Region 4 has concluded that air intrusion has not led to aerobic decomposition in the areas served by these wells. Therefore, the alternative monitoring approach that the City proposed for these wells is acceptable to Region 4.

In its second proposal, Augusta requested a waiver of the requirement to conduct quarterly methane surface concentration monitoring in the 52-acre closed section of the landfill designated as Area B. According to the City, the composition of the cover in this section of the landfill makes it highly unlikely that methane concentrations will ever exceed the 500 parts per million (ppm) limit in 40 CFR 60.755(C)

(4). Included with the City's request were results from methane surface concentration monitoring conducted in December 2002 and March 2003. During these two monitoring periods, a total of 520 readings were made in Area B, and the highest methane concentration detected by 44.8 ppm. Based upon these results, Augusta asked that the requirements to conduct quarterly methane surface concentration monitoring in Area B be waived if nothing is conducted around the perimeter of the area and in locations where visual observations indicate the presence of distressed vegetation.

Since provisions in 40 CFR 60.756(f) require at least annual methane surface concentration monitoring in closed landfills, Augusta's request for a complete waiver of the requirement to conduct monitoring in Area B of the Deans Bridge Road Landfill cannot be approved. If Augusta conducted methane surface concentration monitoring in June 2003 and none of the readings in Area B were 500 ppm or higher, an annual frequency for monitoring in Area B would be acceptable provided that there are no exceedances or the applicable 500 ppm limit. In the event that the methane surface concentration readings of 500 ppm or higher are detected during any of the annual checks in the future, Augusta would need to resume monitoring on a quarterly basis until there are three consecutive calendar quarters during which no exceedances of the 500 ppm limit occur.

If you have any questions about the determination provided in this letter, please contact Mr. David McNeal of the EPA Region 4 staff at (404) 562-9102.

Sincerely,

Beverly H. Banister
Director
Air, Pesticides and Toxics
Management Division

[Planning & Results](#) | [Compliance Assistance](#) | [Compliance Incentives & Auditing](#) | [Compliance Monitoring](#)
[Civil Enforcement](#) | [Cleanup Enforcement](#) | [Criminal Enforcement](#) | [Environmental Justice](#) | [NEPA](#)

[EPA Home](#) | [Privacy and Security Notice](#) | [Contact Us](#)

Last updated on Thursday July 13, 2006

URL: <http://cfpub.epa.gov/adi/index.cfm?CFID=1407598&CFTOKEN=18871620&requesttimeout=180>

From: Randy Frank <randy.frank@gflenv.com>
Sent: Wednesday, July 14, 2021 10:51 AM
To: Iniemann@eillc.com; 'Clark, Sherren'; 'Hammers, Mark'; Wes Younger; 'Paul Taylor'
Cc: Zwick, Ann M.; David Pepper; Antonio Testa; Bryan Wuester; Clarke Lundell; Eduardo Choquis Kollrich; Edward Hood; Gerald Allen Jr; Jennifer Glowacki; Kenneth Gann; Marcos Elizondo; Olivia Vogelsberg; Timothy Curry; Kari Rabideau; Selin Hoboy
Subject: FW: Subpart AAAA questions pertaining to well temperature
Attachments: 2021-06-21 LT Sheppard Request for Regulatory Interpretation.pdf

All – as most of you know, the attached letter was submitted to EPA on 6/21/21. EPA asked us to re-submit as an email rather than as a “regulatory interpretation request”. Below is the email submitted and EPA’s response. Will be incorporating this into the template for updated GCCS Design Plans accordingly. Questions, ask.

Thanks Ann and Laura for all your help with this! The EPA response turned out well.

Randy Frank | Regional Landfill Gas Program Manager

GFL Environmental

W144 S6350 College Ct., Muskego, WI, 53150

T (414) 422-9075 Ext. 102 | C (414) 801-0939 | randy.frank@gflenv.com | www.gflenv.com

Confidentiality Notice: This email message (including attachments, if any) is intended for the use of the individual or entity to which it is addressed and may contain information that is privileged, proprietary, confidential and exempt from disclosure. If you are not the intended recipient, you are notified that any dissemination, distribution or copying of this communication is strictly prohibited. If you have received this communication in error, please notify the sender and erase this email message immediately.

From: Sheppard, Andy <Sheppard.Andy@epa.gov>
Sent: Wednesday, July 14, 2021 9:21 AM
To: Zwick, Ann M. <azwick@freeborn.com>
Cc: Randy Frank <randy.frank@gflenv.com>
Subject: RE: Subpart AAAA questions pertaining to well temperature

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Hi Ann,

As mentioned in Subpart AAAA’s preamble (85 Fed. Reg. 17248 March 26, 2020), the increase in the wellhead temperature standard is expected to reduce the frequency of corrective actions for exceeding the temperature limit. An approved HOV for temperature replaces the 145 degrees Fahrenheit limit set by the rule. It is assumed that a landfill owner or operator had tried to properly address the corrective action for the initial exceedance of 145 degrees Fahrenheit, which ultimately led to the submittal of an HOV, therefore the landfill would not need to take any additional corrective action for exceedances of 145 degrees Fahrenheit, but would need to perform corrective action for exceedances above the approved HOV. Additionally, HOVs should be incorporated into the GCCS design plan.

An approved HOV higher than 165 degrees Fahrenheit would also replace the downwell temperature threshold, to where downwell temperature monitoring would be required if the temperature exceeded the approved HOV, rather than the 165-degree limit. Additionally, downwell temperature monitoring is only required for wells with temperature

above the 165-degree limit (see 85 Fed. Reg. 17249 March 26, 2020), so once a well falls back below the limit (or approved HOV), annual downwell temperature monitoring is no longer required.

At this time, we aren't planning to make any changes to the rule regulatory text in regards to your questions, but hopefully my explanation clears up any confusion.

Thanks!

Andy Sheppard
U.S. EPA, Sector Policies and Programs Division, OAQPS
Natural Resources Group
(919)541-4161

From: Zwick, Ann M. <azwick@freeborn.com>
Sent: Tuesday, July 13, 2021 1:05 PM
To: Sheppard, Andy <Sheppard.Andy@epa.gov>
Cc: randy.frank@gflenv.com
Subject: Subpart AAAA questions pertaining to well temperature

Andy,

Thank you for speaking with me and Randy Frank today regarding the recent revisions to 40 CFR Part 63 Subpart AAAA. As we discussed, we have the following questions specifically pertaining to well temperature at 40 CFR §63.1960(a)(4)(i) and 40 CFR §63.1961(a)(6):

40 CFR §63.1960(a)(4)(i)

The question is whether §63.1960(a)(4)(i) requires corrective action when a well exceeds 145 degrees Fahrenheit if a higher operating value has been approved for the well. As written, the corrective action provisions in §63.1960(a)(4)(i) do not appear to account for wells with approved higher operating values.

The following are the pertinent provisions in §63.1960(a)(4)(i) (see underlined text):

(i) Once an owner or operator subject to the provisions of this subpart seeks to demonstrate compliance with the operational standard for temperature in §63.1958(c)(1), the owner or operator must monitor each well monthly for temperature. If a well exceeds the operating parameter for temperature as provided in §63.1958(c)(1), action must be initiated to correct the exceedance within 5 days. Any attempted corrective action measure must not cause exceedances of other operational or performance standards.

(A) If a landfill gas temperature less than or equal to 62.8 degrees Celsius (145 degrees Fahrenheit) cannot be achieved within 15 days of the first measurement of landfill gas temperature greater than 62.8 degrees Celsius (145 degrees Fahrenheit), the owner or operator must conduct a root cause analysis and correct the exceedance as soon as practicable, but no later than 60 days after a landfill gas temperature greater than 62.8 degrees Celsius (145 degrees Fahrenheit) was first measured. The owner or operator must keep records according to 40 CFR 63.1983(e)(3).

(B) If corrective actions cannot be fully implemented within 60 days following the temperature measurement for which the root cause analysis was required, the owner or operator must also conduct a corrective action analysis and develop an implementation schedule to complete the corrective action(s) as soon as practicable, but no more than 120 days following the measurement of

landfill gas temperature greater than 62.8 degrees Celsius (145 degrees Fahrenheit). The owner or operator must submit the items listed in 40 CFR 63.1981(h)(7) as part of the next semi-annual report. The owner or operator must keep records according to 40 CFR 63.1983(e)(4).

(C) If corrective action is expected to take longer than 120 days to complete after the initial exceedance, the owner or operator must submit the root cause analysis, corrective action analysis, and corresponding implementation timeline to the Administrator, according to 40 CFR 63.1981(h)(7) and 40 CFR 63.1981(j). The owner or operator must keep records according to 40 CFR 63.1983(e)(5).

(D) If a landfill gas temperature measured at either the wellhead or at any point in the well is greater than or equal to 76.7 degrees Celsius (170 degrees Fahrenheit) and the carbon monoxide concentration measured, according to the procedures in 40 CFR 63.1961(a)(5)(vi) is greater than or equal to 1,000 ppmv the corrective action(s) for the wellhead temperature standard (62.8 degrees Celsius or 145 degrees Fahrenheit) must be completed within 15 days.

As written, §63.1960(a)(4)(i) requires corrective action if a well exceeds the operating parameter for temperature as provided in §63.1958(c)(1). Section 63.1958(c)(1) requires an interior wellhead gas temperature of less than 145 degrees Fahrenheit. Higher operating values can be established pursuant to §63.1958(c)(2), however, this section is not specifically referenced in §63.1960(a)(4)(i). As a result, owners and operators would presumably be required to initiate corrective actions for wells with temperatures greater than 145 degrees Fahrenheit, regardless of prior approval of higher operating values. This would effectively negate the approved higher operating values obtained pursuant to §63.1958(c)(2). Was this EPA's intent?

We note that the enhanced monitoring provisions in §63.1961(a)(5) specifically stop enhanced monitoring once a higher operating value is approved. There is no such exception specifically contained in §63.1960(a)(4)(i). Is there a reason that enhanced monitoring is not required for wells with higher operating values, but corrective action is required for these same wells?

Finally, §63.1960(a) states that, “[e]xcept as provided in §63.1981(d)(2),” the specified methods in paragraphs (a)(1) through (6) must be used. Section 63.1981(d)(2) provides that the GCCS design plan must include “alternatives to the operational standards” proposed by the owner or operator. Does the phrase, “[e]xcept as provided in §63.1981(d)(2),” provide a mechanism for an owner or operator to include the higher operating values in its GCCS design plan and, if approved, the corrective action requirements in §63.1960(a)(4)(i) would not apply to the wells with higher operating values? This would seem to be a reasonable interpretation, and would provide a resolution to the issue.

40 CFR §63.1961(a)(6)

These questions pertain to the annual measurement of temperature every 10 vertical feet of the well for wellheads that measure a landfill gas temperature greater than or equal to 73.9°C (165°F). The first question is what is the duration of the monitoring program, as there does not appear to be an end date to this annual measurement. Was it intended to be perpetual, i.e. for the life of the well even if the temperature greater than 165°F was only measured once? Or does it only apply to the calendar year in which the temperature greater than 165°F was measured at a particular well?

Secondly, similar to the first regulatory interpretation question, is this 10-foot annual temperature measurement required at wells that have an approved higher operating value? There is no language specifically excluding them, with the exception of the phrase at the beginning of 40 CFR §63.1961 that states “except as provided in §63.1981(d)(2).” As discussed above, this provision refers to GCCS design plan alternatives to the operational standards, test methods, procedures, compliance measures, monitoring, recordkeeping or reporting provisions of §§63.1957 through 63.1983 proposed by the owner or operator. Can both of these issues be resolved in a GCCS design plan alternative?

Suggested Rule Revisions

Based on our discussion, it is our understanding that each of the above temperature issues can be resolved in the GCCS design plan and the intent of the rules is to recognize approved higher operating values. However, to better clarify requirements, we would suggest the rules be revised as follows (see highlighted/italicized text for proposed additions):

40 CFR §63.1960(a)(4)

- (i) Once the owner or operator seeks to demonstrate compliance with the operational standard for temperature in 40 CFR 63.1958(c)(1), the owner or operator must monitor each well monthly for temperature for the purpose of identifying whether excess air infiltration exists. If a well exceeds the operating parameter for temperature as provided in 40 CFR 63.1958(c)(1) ***or a higher operating value established pursuant to 40 CFR 60.1958(c)(2), whichever is greater***, action must be initiated to correct the exceedance within 5 days. Any attempted corrective measure must not cause exceedances of other operational or performance standards.
- (A) If a landfill gas temperature less than or equal to 62.8 degrees Celsius (145 degrees Fahrenheit) ***or a higher operating value established pursuant to 40 CFR 60.1958(c)(2), whichever is greater***, cannot be achieved within 15 days of the first measurement of landfill gas temperature greater than 62.8 degrees Celsius (145 degrees Fahrenheit) ***or a higher operating value established pursuant to 40 CFR 60.1958(c)(2), whichever is greater***, the owner or operator must conduct a root cause analysis and correct the exceedance as soon as practicable, but no later than 60 days after a landfill gas temperature greater than 62.8 degrees Celsius (145 degrees Fahrenheit) ***or a higher operating value established pursuant to 40 CFR 60.1958(c)(2), whichever is greater***, was first measured. The owner or operator must keep records according to 40 CFR 63.1983(e)(3).

- (B) If corrective actions cannot be fully implemented within 60 days following the temperature measurement for which the root cause analysis was required, the owner or operator must also conduct a corrective action analysis and develop an implementation schedule to complete the corrective action(s) as soon as practicable, but no more than 120 days following the measurement of landfill gas temperature greater than 62.8 degrees Celsius (145 degrees Fahrenheit) *or a higher operating value established pursuant to 40 CFR 60.1958(c)(2), whichever is greater*. The owner or operator must submit the items listed in 40 CFR 63.1981(h)(7) as part of the next semi-annual report. The owner or operator must keep records according to 40 CFR 63.1983(e)(4).
- (C) If corrective action is expected to take longer than 120 days to complete after the initial exceedance, the owner or operator must submit the root cause analysis, corrective action analysis, and corresponding implementation timeline to the Administrator, according to 40 CFR 63.1981(h)(7) and 40 CFR 63.1981(j). The owner or operator must keep records according to 40 CFR 63.1983(e)(5).
- (D) If a landfill gas temperature measured at either the wellhead or at any point in the well is greater than or equal to 76.7 degrees Celsius (170 degrees Fahrenheit) *or a higher operating value established pursuant to 40 CFR 60.1958(c)(2), whichever is greater*, and the carbon monoxide concentration measured, according to the procedures in 40 CFR 63.1961(a)(5)(vi) is greater than or equal to 1,000 ppmv the corrective action(s) for the wellhead temperature standard (62.8 degrees Celsius or 145 degrees Fahrenheit) *or a higher operating value established pursuant to 40 CFR 60.1958(c)(2), whichever is greater*, must be completed within 15 days.

40 CFR §63.1961(a)

(6) For each wellhead with a measurement of landfill gas temperature greater than or equal to 73.9 degrees Celsius (165 degrees Fahrenheit) *or a higher operating value established pursuant to 40 CFR 60.1958(c)(2), whichever is greater*, annually monitor temperature of the landfill gas every 10 vertical feet of the well *until the well temperature decreases below 73.9 degrees Celsius (165 degrees Fahrenheit) or a higher operating value established pursuant to 40 CFR 60.1958(c)(2), whichever is greater*. This temperature can be monitored either with a removable thermometer, or using temporary or permanent thermocouples installed in the well.

We would appreciate a written response to the questions posed in this email, to confirm our understanding. Thank you again for your time. Please let me know if you have any questions or want to schedule a call to discuss this further.

Ann

ANN M. ZWICK
Attorney at Law



(312) 360-6254 office
(312) 952-1651 mobile
azwick@freeborn.com

Freeborn & Peters LLP
311 South Wacker Drive, Suite 3000
Chicago, IL 60606
www.freeborn.com



UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
REGION 5
77 WEST JACKSON BOULEVARD
CHICAGO, IL 60604-3590

REPLY TO THE ATTENTION OF:

AUG 04 2008

AE-171

Brian J. Ezyk, P.E.
Area Engineer
Republic Services, Inc.
P.O. Box 634
New Boston, Michigan 48164-0634

Re. Solar-powered flare determination
Brent Run Landfill

Dear Mr. Ezyk:

The U.S. Environmental Protection Agency is in receipt of your June 4, 2008, request for approval of the use of two solar-powered flares to control collected landfill gas emissions. EPA hereby conditionally approves your request as discussed below.

Republic Services, Inc. of Michigan, LLC, is the owner and operator of the Brent Run Landfill in Montrose, Michigan. It is subject to the New Source Performance Standards (NSPS) for municipal solid waste landfills found at 40 C.F.R Part 60, Subpart WWW, as amended.

In your letter you explain that the Brent Run Landfill has a landfill gas energy recovery plant that receives collected landfill gas; however, the gas collected in the header exceeds the capacity of the gas-to-energy plant, hence the need for flares to combust excess gas. The landfill has two Model CF-10 solar-powered gas flares constructed by Landfill Service Corporation.

In prior determinations cited in your letter, EPA had disapproved flares that relied on sparking devices alone to assure a continuous presence of a flame. EPA has determined that a sparking device on the tip of a flare is not a heat-sensing device that meets the requirement in the rules at 40 C.F.R. §§ 60.18(f)(2) and 60.756(c)(1) which require that the flares have pilot flames and heat sensors such as a thermocouple or ultraviolet beam sensor. It is our understanding that the two flares discussed in your request have thermocouples, and that there are automatic shutoffs should a flameout condition occur. We are less clear as to whether the flares have pilot flames, but pilots can be installed to burn landfill gas if necessary. EPA understands that Landfill Service Corporation has made available the option of installing a pilot on the CF-10 flares for the purpose of NSPS compliance.

Thank you for submitting the test report submitted to Michigan Department of Environmental Quality (DEQ) on November 29, 2007, that demonstrates compliance with the requirements that Flare 1 and 2 exit velocity and net heating values were 3.7 and 4.5 feet per second, and 19.57 and 19.72 megajoules per standard cubic meter, respectively. This demonstrates compliance with 40 C.F.R. § 60.18 for maximum tip velocity and minimum required gas net heating value.

EPA approves the use of two solar-powered flares to control collected landfill gas emissions as described above, on the condition that they are retrofitted with pilots and thermocouples. If you have any questions, feel free to contact Jeffrey Gahris, of my staff, at (312) 886-6794.

Sincerely,



George T. Czerniak
Branch Chief

Air Enforcement and Compliance Assurance Branch

cc: Michael Masterson, District Supervisor
Shiawassee District
Michigan DEQ

Rex Lane, Assistant District Supervisor
Kalamazoo District
Michigan DEQ

Khaled Mahmood
Cornerstone Environmental Group

Standard bcc's: official file copy w/attachment(s)
 originator's file copy w/attachment(s)
 originating organization reading file w/attachment(s)

Other bcc's: Sheila Desai
 Lynne Roberts

Creation Date:	August 1, 2008
Filename:	D:\EPAWork\landfills\Brent Run Passive Flare Determination.ltr.doc
Legend:	ARD:AECAB:AECAS(MI/WD):JGahris